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**COMPUTATIONAL EXPLORATIONS OF  
THE EVOLUTION OF ARTIFICIAL NEURAL NETWORKS  
IN PAVLOVIAN ENVIRONMENTS**

A Dissertation Presented

by

**JOSÉ E. BURGOS**

Submitted to the Graduate School  
of the University of Massachusetts Amherst in partial fulfillment  
of the requirements for the degree of

**DOCTOR OF PHILOSOPHY**

May 1996

Neuroscience and Behavior Program

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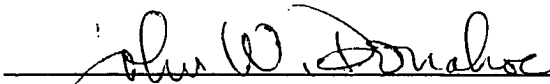
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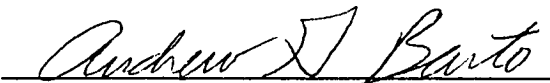
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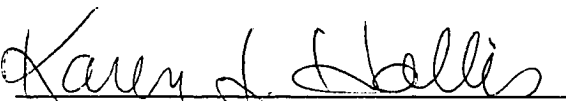
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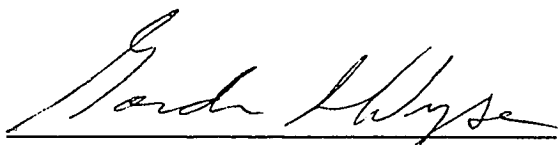
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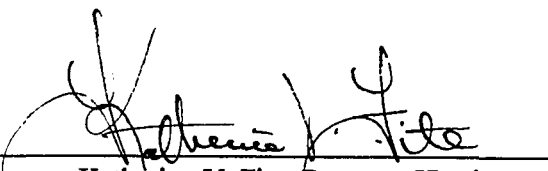
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**To my parents, Mercedes and José Gabriel**

**To my wife, Rocío**

## **ACKNOWLEDGEMENTS**

I am grateful to my graduate mentor, John W. Donahoe, for his guidance, encouragement, and insights in the present work and throughout my graduate career. I would also like to thank Andrew Barto, Karen Hollis, and Gordon Wyse for their willingness to serve as committee members for this intellectual voyage through uncharted waters.

# **ABSTRACT**

## **COMPUTATIONAL EXPLORATIONS OF THE EVOLUTION OF ARTIFICIAL NEURAL NETWORKS IN PAVLOVIAN ENVIRONMENTS**

MAY 1996

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The present work initiates a research line in the study of artificial life, through a preliminary characterization of a computational approach to evolutionary interpretations of Pavlovian-conditioning phenomena. The approach was implemented through a Neuro-Computational / Genetic-Algorithm (or NC-GA) hybrid model. The NC model described the functioning of neuron-like processing elements that were interconnected forming artificial neural networks (ANNs). The GA consisted of a set of rules for selecting ANNs for mating and reproduction. ANNs were developed from virtual chromosomes encoding variables determining the course of a neurodevelopmental program motivated by general concepts from developmental neuroscience. All chromosomes had a fixed length and encoded for the same set of variables. Also, that program involved a one-many relation, for most of the variables encoded by the chromosomes were probabilistic.

The NC-GA model was characterized through computer simulations. Four simulation experiments were performed, each consisting of two kinds of simulations,

evolution and test. In the evolution simulations, the GA was used to evolve ANNs that were trained in a Pavlovian procedure. ANNs with higher conditional-response proportions had a higher probability of being selected for mating and reproduction. In the test simulations, the behavioral competence of the evolved ANNs was determined by exposing them to conditions different from the ancestral ones.

In general, the results from the evolution simulations demonstrated that the mean chromosomal overlap and the mean population fitness increased as negatively accelerated functions of generations. Also, most ANNs at the beginning of evolution showed no learning, whereas ANNs by the end of evolution showed learning.

In Experiment 1, ANNs were selected for increased responding under forward-delay procedures in which the interstimulus interval (ISI), and the kind of CS were manipulated. After evolution, ANN sizes increased as a function of ISIs, and ANN performances in the test simulations were consistent with ISI functions, optimal-ISI noninvariance, and CS nonequipotentiality. In Experiment 2, two CSs were independently paired with the US. After evolution, ANN sizes increased as a nonmonotonic function of the ISI, and ANN performances in the test simulations showed generalization, discrimination, and blocking. In Experiments 3 and 4, ANNs were selected for orthogonal and nonorthogonal discrimination, respectively. Performances in the test simulations of both experiments also showed generalization, discrimination, and blocking. Collectively, these results are consistent with the pursuit of general-process approaches to learning. However, such approaches also allow for interpretations in which biological constraints on learning are seen as emerging from variations in general neurobiological processes, and as imposing limits on the range of variation of general biobehavioral processes.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Framework

The present work belongs in the area of computational neuroethology, a subdiscipline within the study of Artificial Life (AL). AL comprises all biological phenomena synthesized within alternative media (Langton, 1992), through hardware or software. At least three reasons justify the study of AL. First, the practical possibility of executing nontrivial experiments that take into account the multilevel complexity of actual biological entities is extremely small. The study of AL thus represents an *inquiry tool* that allows for the execution of virtual experiments whose counterpart in the real world would be practically impossible to perform, due to the inherent complexity of the entities and processes involved (Taylor & Jefferson, 1994). Second, any attempt to discover the lawful regularities underlying biological evolution must transcend the set of actual biological entities and explore the set of *possible* biological entities, given that the former are largely influenced by contingency and historical accident (Dennett, 1995; Gould, 1989). In this sense, the study of AL represents a methodological and theoretical approach to the study of possible biological entities (Emmeche, 1991; Langton, 1992, 1994). And, third, the study of AL through simulation experiments may be the only viable strategy for characterizing the dynamics of analytically intractable models that arise from an attempt to capture complex systems (Sober, 1992).



In the present effort, I shall adopt a conservative view in which AL represents primarily a *research tool* that may be useful to motivate new experimental research and to clarify our own theories, assumptions, and concepts about biological reality. In contrast, radical approaches to AL claim that the kinds of entities constituting AL are as alive as the those constituting biological reality (e.g., Langton, 1989). Such approaches are far more controversial than conservative ones (Sober, 1992), and will not be pursued here.

The kinds of AL studied in computational neuroethology involve a synthesis of the methods, concepts, and theories from two research areas, namely *computational neuroscience* and *computational evolutionism*. Both areas share two prescriptive assumptions: first, an understanding of adaptive behavior must arise from knowledge about biological systems, and, second, this knowledge must be expressible in mathematical models that allow for the design of *computer simulations* whose results increase our understanding of biological phenomena<sup>1</sup>.

### 1.1.1 Computational Neuroscience and Artificial Neural Networks

Computational neuroscience can be defined as the construction and application of mathematical models of the structure and functioning of nervous systems at the cellular, anatomical, and behavioral levels of organization (Churchland, Koch, &

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<sup>1</sup> An issue arises here regarding the scientific value of computer simulations for biological sciences. We can identify two extreme positions on this issue. On one extreme, all computer simulations are seen as a trivial oversimplifications of biological reality, and hence have no scientific value whatsoever. On the opposite extreme, computer simulations are seen as thinking tools that can be scientifically valuable, even if they say little about specific biological phenomena. The present dissertation is somewhere between these extremes. The study of AL, although inevitably simplifying, is a worthy *synthesizing* endeavor that must be constrained by *some kind* of knowledge about biological reality. This knowledge, however, need not be restricted to specific experimental data, but can also include principles, concepts, and inductive generalizations arising from such data.

Sejnowski, 1990; Churchland & Sejnowski, 1992; Perkel, 1990). In most cases, such models are expected to lead to computer simulations whose results contribute to a further understanding of nervous systems, to inspire new research, or to help us clarify current issues in the neurosciences.

Models inspired by the structure and functioning of nervous systems were already present during the early days of the serial/digital computers and symbolic artificial intelligence (e.g., Hebb, 1949; McCulloch & Pitts, 1943; Pitts & McCulloch, 1947; Rashevsky, 1948; Rosenblatt, 1958; Selfridge, 1959; Taylor, 1956; von Neumann, 1958; Widrow & Hoff, 1960). However, Minsky and Papert's (1969) work showed important limitations of some of the current models, diminishing interest in the neural metaphor, thus allowing symbolic models inspired by the Turing-machine metaphor to flourish unimpededly. Despite this situation, a few researchers continued working under the neural metaphor (e.g., Barto, 1976; Fukushima, 1975; Grossberg, 1982; Hopfield, 1982; Klopff, 1982; Sutton & Barto, 1981).

Later on, new developments revived interest in the neural metaphor (e.g., Barto, 1985; Carpenter & Grossberg, 1986; Changeux, 1985; Fukushima, 1988; Hinton & Sejnowski, 1986; Hopfield & Tank, 1985; Klopff, 1988; Kohonen, 1989; Rumelhart, Hinton, & Williams, 1986). Nowadays, computational neuroscience is a well-established and active research field, largely guided by advances in cellular neurobiology, neuroanatomy, animal learning, and behavioral ecology.

Some models formulated within computational neuroscience are implemented through systems known as *artificial neural networks* (ANNs)<sup>2</sup>. An ANN consists of a

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<sup>2</sup> The term *artificial-neural-network* typically refers to models that are motivated by *engineering* considerations, and hence, tend to be constrained only by very general properties of neurons and nervous systems. In the present work, I use that term in a broader sense, to include *any* implementation involving a number of interconnected *elements* whose structure and function are determined by a model that is *guided* or *inspired* by *neuron metaphors*, regardless of whether or

set of interconnected elements generally referred to as *neural processing elements*, or NPEs. NPEs are virtual neuron-like devices that receive input signals from their immediate environment, process them, and send output signals to other NPEs<sup>3</sup>. The functioning of an NPE over time is described by a *neurocomputational* (NC) model.

In NC models, the effect of the input signals to an NPE is determined not only by their values, but also by their associated *connection weights*. By taking connections into account, then, an NPE can be seen as being analogous to a *postsynaptic neuron*. An element sending an input signal to an NPE is the conceptual equivalent of a presynaptic neuron. A connection weight is analogous to the strength of a synapse.

Typical NC models describe two aspects of an NPE functioning, *activation* and *learning*. Activation is typically represented as the value of the NPE's output signal at a moment in time, and is a function of the activations of presynaptic input signals and their corresponding connection weights. The activation of an NPE can be interpreted in terms of membrane potential, probability of firing, or firing rate, depending on the assumptions of the model. Learning refers to changes in the values of the connection weights over time. The neurobiological analogues of such changes include the various cellular events involved in synaptic plasticity.

The number representing the output signal of an NPE at a moment in time is computed through a rule known as the *activation function*. The number representing the synaptic strength of a presynaptic element is computed through a rule known as the

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not such an implementation and model are motivated by engineering considerations or constrained by specific properties of real nervous systems.

<sup>3</sup> Strictly speaking, NPEs are entities too abstract and simple to be identified in a one-one fashion with the functioning of *real single* neurons. Thus, it is more precise to view NPEs as simulating *certain* aspects of single neurons or *groups* of neurons. In this sense, NPEs should be seen as being only *analogous* to neurons.

*learning function*. Different activation and learning functions reflect different modelling assumptions about neuronal activation and synaptic plasticity.

When a number of NPEs are interconnected to form an ANN, the realization of the NC model at each NPE allows the ANN's *overall behavior* to change as a result of external influences. Computational experiments have shown that some ANNs are capable of simulating aspects of changes observed in the behavior of real organisms under comparable conditions<sup>4</sup>. This feature of ANNs has been used as a justification for referring to them as *adaptive*, and is most relevant to the present work (see later).

The transition from an NPE to an ANN is analogous to that from a neuron to a neural circuit, and hence, from the neuronal to the neuroanatomical level of analysis. When the overall performance of ANNs is observed and qualified as adaptive or nonadaptive (see later for further discussion of this dichotomy), a third level of analysis is introduced. This level corresponds to that studied in the experimental analysis of behavior and behavioral ecology.

The NC model used in the present work is motivated by general principles about the structure and functioning of nervous systems at the cellular, neuroanatomical, and behavioral levels of analysis. At the cellular level, knowledge about the general functioning of neurons guides the definition of the activation and learning functions. At the neuroanatomical level, knowledge about the general organization of neural circuits and systems guide the construction of ANNs. At the behavioral level,

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<sup>4</sup> The similarity between data from NC simulations and data from experiments with real organisms is one criterion used to determine the plausibility of NC models. This is the *functional-isomorphism* criterion, according to which "good" models generate functional relations that resemble those generated through experimental observation of the real phenomenon. Some authors (e.g., Putnam, 1975; Smolensky, 1986, 1988) consider functional isomorphism as a necessary *and sufficient* criterion for model evaluation. However, it has been proved (Moore, 1956) that functional isomorphism may lead to a proliferation of competing models whose differences cannot be empirically resolved. Hence, functional isomorphism is necessary but not sufficient. Model evaluation must include an examination of the consistency of simulation data *and* the model's formal structure with knowledge arising from an experimental analysis of the phenomenon of interest (but see Footnote 1).

knowledge about animal-learning phenomena guide our interpretations of the functional significance of ANNs. At this level of analysis, the term *learning* refers to changes in an ANN's overall behavior as the result of training.

### **1.1.2 Computational Evolutionism and Genetic Algorithms**

Computational evolutionism can be defined as the construction and application of mathematical models of *evolutionary processes*, that allow for the design of *evolutionary algorithms* whose implementation through computer simulations may clarify current issues in evolutionary biology and pose new ones (De Jong, 1993; Fogel, 1995).

Evolutionary algorithms can be seen as formalizations of the principles of biological evolution, as formulated by Lewontin (1970):

1. Phenotypic variation: Phenotypic (i.e., morphological, physiological, and behavioral) differences exist among individuals within a population.
2. Differential fitness: Survival and reproductive differences that are correlated with those phenotypic differences.
3. Fitness heritability: Offspring tend to resemble their parents' phenotypes, and survival and reproductive abilities.

Different interpretations of these principles have produced three classes of evolutionary algorithms, namely evolution strategies (Rechenberg, 1965), evolutionary programming (Fogel, Owel, & Walsh, 1966), and genetic algorithms (Holland, 1975). The present work focuses on genetic algorithms (GAs).

A typical simulation based on an evolutionary algorithm consists of a *lineage* defined as a sequence of generations. A generation is defined as a development-

evaluation-reproduction sequence, applied to a population of individuals. In this context, an individual is defined as a virtual entity consisting of a genotype and phenotype. A distinctive feature of GAs, compared to other evolutionary algorithms, is how these two components are conceptualized and interrelated.

In a canonical GA, a genotype is a set of variables whose values are encoded in binary form and determine certain phenotypic traits of interest. A particular set of values takes the form of a string of binary digits, and is the conceptual analogue of a chromosome (in evolution strategies and evolutionary programming, chromosomes are sets of real values, rather than binary strings). In typical GA applications, an individual's genotype consists of only one chromosome. The position of a binary digit within the string corresponds to a locus. A binary digit within a string is analogous to a gene, and the value of a binary digit (either 0 or 1) is analogous to an allele. The decoding of a binary digit, as determined by its position and value, is the analogue of gene expression. A subset of contiguous, functionally related binary digits forms a chromosome *segment*. A chromosome segment consists of a set of genes that collectively determine a particular phenotypic trait. That is, a change in one gene's allele, within a chromosome segment, is sufficient to modify the phenotypic trait determined by the entire segment.

A definition of a virtual genotype inspired by concepts from molecular biology permits a further elaboration of Principle 3 and the introduction of two more principles:

- 3a. Fitness heritability is genetically determined.
4. Genotypic variation: Genetic differences exist among individuals within a population.

5. Central Dogma of molecular biology (Crick, 1970): Phenotypic traits acquired during an individual's life do not affect the genotype (i.e., information transfer occurs *only* from DNA to proteins).

Taken together, Principles 1 to 5 allow us to avoid the idea of Lamarckian inheritance, thus facilitating contact with neo-Darwinian formulations of evolutionary theory.

Development is conceptualized in typical GAs as the binary-to-decimal conversion of chromosome segments into quantities representing *phenotypic traits*. Several traits can be combined into a single quantity representing *a phenotype*. On this basis, phenotypic variation (Principle 1) can be expressed in terms of some statistical measure of variation of phenotypes across individuals.

After development, phenotypes are evaluated to determine their survival and reproduction possibilities, through a criterion known as a *fitness function*. A fitness function produces a value that determines an individual's probability of being selected for mating and reproduction, and hence, of passing its genes to the next generation. In general, the greater an individual's fitness, the greater the individual's chance of being selected for mating and reproduction. Like phenotypic differences, survival and reproductive differences (Principle 2) can be defined by some statistical measure of variation of fitness values across individuals.

Once the fitness of each individual in a population has been determined, a *selection scheme* is applied to generate new chromosomes that will constitute the genetic material of the next generation. Typically, a selection scheme is algorithmically defined as a mating-reproduction-mutation sequence. During mating, fitness values are used as quantitative criteria for selecting individuals from the population, and forming a *mating pool* or parent sample. During reproduction, contiguous fragments of chromosomes

from mating individuals or parents are randomly selected and combined to form new chromosomes representing the offspring's genotypes. In canonical GAs, such fragments do not necessarily correspond to chromosome segments (i.e., selected fragments typically consist of genes from several chromosome segments).

Reproduction is simulated in canonical GAs through a process that simulates genetic recombination or *crossover*. In this way, offspring tend to be genetically similar, although not identical to their parents. Also, offspring tend to resemble their parent's fitness, to the extent that fitness is genetically determined (Principles 3 and 3a). Finally, mutation involves changing alleles with a small probability value (typically less than .01). Thus, genotypic variation (Principle 4) in GAs is determined mainly by recombination. This feature represents a significant departure of GAs from neo-Darwinian approaches according to which mutation is the main source of genetic variation. In other evolutionary algorithms, genetic variation is determined mainly (evolution strategies) or only (evolutionary programming) by mutation. Once a new population of chromosomes is generated, a new generation starts, and the development-evaluation-reproduction cycle is repeated for as many generations as desired.

The general result observed in typical GA simulations is an increase in the average population fitness as a negatively accelerated function of evolutionary time (typically measured by the generation number), and a decrease in genotypic and phenotypic variation as negatively decelerated functions of evolutionary time (typically measured in generations). In this kind of simulation, such changes define evolution.



### 1.1.3 NC-GA Models

In computational neuroethology, the methods and concepts of computational neuroscience and computational evolutionism are combined to design computer simulations of the evolution of learning, behavior, and nervous systems (Cliff, 1991; Harvey, 1991; Hinton & Nowlan, 1987; Meyer & Guillot, 1991; Mühlenbein & Kindermann, 1989; Schaffer, Whitley, & Eshelman, 1992; Wilson, 1991). Such a combination has taken different forms. The form adopted in the present work is based on the idea of an *NC-GA model*.

In typical NC-GA models, the encoding structure consists of variables that determine certain architectural features of an ANN (i.e., number of elements, connectivity, initial connection weights, and activation and learning parameters). In simulations based upon typical models, an initial or founder population of chromosomes is randomly generated. Then, each chromosome is translated into an ANN through a model that simulates neural development. The resulting ANNs are placed in a virtual environment consisting of events whose occurrence is determined by some predefined learning task. After a number of learning epochs, each ANN's performance in the task is evaluated. This performance determines the ANN's fitness. A mating pool is then formed on the basis of the ANNs' fitness values. New chromosomes are generated through recombination and mutation of the parents' chromosomes. The new chromosomes constitute the genotypes of the next generation, and the development-learning-selection cycle begins anew.

The application of GAs to the evolution of ANNs assumes that architectural features impose substantial constraints on the type and number of tasks ANNs can learn. This assumption has been supported by computational investigations based on NC-GA models (e.g., Ackley and Littman, 1992; Belew, McInerney, & Schraudolph,

1992; French & Messinger, 1994; Gruau & Whitley, 1993; Harp & Samad, 1991; Hinton & Nowlan, 1987; Keesing & Stork, 1991; Miller, Todd, & Hegde, 1989; Nolfi, Elman, & Parisi, 1990; Paredis, 1991, 1994; Schaffer, Caruana, & Eshelman, 1990; Todd & Miller, 1991; Whitley, Starkweather, & Bogart, 1990). The NC models used in these studies include learning functions such as the Hebbian rule, perceptron rule, backpropagation, and reinforcement learning. Simulations using these models have shown that ANNs evolved through GAs perform significantly better than randomly-generated or many handcrafted ANNs, in a number of learning tasks.

## 1.2 The Present Approach

Appendices A and B present semiformal descriptions of the neurodevelopmental program and NC model, respectively, used in the present work. This section presents an informal description organized around the ontogeny-phylogeny dichotomy. My purpose in using this dichotomy is descriptive, rather than explanatory. Also, throughout the present work I use technical terms originally introduced to refer to real biological entities and processes (e.g., *locus*, *gene*, *allele*, *chròsome*, *neuron*, *synaptic-strength*, *synaptic-plasticity*, *postsynaptic-receptor*, *ontogeny*, *phylogeny*, *neural-development*, *synapse*, *fitness*, *organism*, *population*, etc.), to refer to virtual entities and processes. However, I acknowledge that the latter entities and processes are too abstract to merit literal applications of those terms. Consequently, such usage should be seen only as a *linguistic convention* to refer to virtual entities and processes that are *crude* representations of the real entities and processes named by such terms.

## 1.2.1 Ontogeny

The history of the term *ontogeny* reveals a diversity of meanings (Gould, 1977, 1992; Richards, 1992). I shall use the term to refer to two processes that can be fully described at the level of the *individual organism*, namely *development* and *learning*, although current uses in biology tend to restrict the term to the first process. In the present work, an individual organism is viewed as consisting of a *genotype*, a *structural phenotype*, and a *functional phenotype*. Development consisted of translating an individual's genotype into an "adult" ANN constituting the individual's structural phenotype, which remained unchanged throughout the individual's adult life. An individual's functional phenotype consisted of changes in an ANN's overall behavior over time, as a function of environmental conditions defined according to Pavlovian-conditioning procedures. I shall use the term *learning* to refer to such changes. Let me describe each component in more detail.

### 1.2.1.1 Genotypes and Neurodevelopmental Program

A chromosome was defined as a string consisting of 282 loci. Each locus contained a virtual gene taking a value of either '1' or '0'. Chromosomes were divided into 59 segments consisting of either 3, 4, or 5 contiguous loci. Each segment encoded a binary representation of a neurodevelopmental (ND) parameter. All chromosomes within and across populations had the same structure. However, different chromosomes could encode for different neurodevelopmental parameters (see Appendix A for relations between segments, parameters, and ND stages). All ND parameters were polygenic (i.e., determined by more than one gene), and there was no pleiotropy (i.e., any given gene determined one and only one ND parameter).

Development was simulated through an ND program whose course was determined by the parameters encoded in the individual's genotype. The ND program was defined

as a sequence of stages that simulated general aspects of the processes of cell proliferation, migration, differentiation, synaptogenesis, and element/connection elimination. The functional partition of neural development into these processes is a fundamental notion in developmental neurobiology (Brown, Hopkins, & Keynes, 1991). The outcome of all stages was determined by parameters encoded in the chromosome, and (with the exception of proliferation) by the outcome of the previous stage.

Proliferation simulated the production of hidden NPEs (see later for a neuroanatomical interpretation of the notion of a hidden NPE). The outcome of proliferation was consistent with that observed in the proliferation of neural cells in the developing mammalian brain (e.g., Allen, 1912; Altman, 1963; Kaplan, 1981; Lewis, 1968; Lewis & Lai, 1974; Luskin & Shatz, 1985; McDermott & Lantos, 1990; Sanderson & Weller, 1990).

Once a set of NPEs had been produced, they were assigned to layers according to probabilistic rules simulating cell migration (see Appendix A). These rules were guided by the kind of migration processes studied in the developing mammalian brain (e.g., Angevine & Sidman, 1961; Berry & Rogers, 1965; Butler & Calley, 1972; Gadisseux *et al.*, 1990; Lindner *et al.*, 1986; Rakic, 1972; Sidman & Rakic, 1973).

After migration, phenotypic differentiation determined certain features that were pertinent to the functioning of NPEs. The outcome of this process was analogous to that found in cell differentiation in vertebrate brain development (e.g., Austin & Cepko, 1989; Caviness & Sidman, 1973; Johnston, 1988; McConnell, 1988, 1989, 1991; Miller, 1988; Parnavelas, Papadopoulos, & Cavanagh, 1988; Rakic, 1988a).

Then, synaptogenesis determined the connectivity among the NPEs and the initial connection weights, using activity-dependent mechanisms analogous to those found in

mammalian nervous systems (e.g., Cooper & Rakic, 1983; Corner & Ramakers, 1992; Eckenhoff & Rakic, 1991; Rakic *et al.*, 1986). Finally, the excess elements and connections were eliminated through rules that simulated cell death in vertebrate nervous systems (e.g., Cowan, 1973; Holliday & Hamburger, 1976; Tanaka & Landmesser, 1986).

#### **1.2.1.1.1 Proliferation and Migration: The Radial-Unit Model**

The definition of the proliferation and migration routines was guided by the *radial-unit model* of neocortical histogenesis proposed by Rakic (1988b). This model attempts to account for neocortical histogenesis by integrating evidence obtained at the molecular, cellular, and neuroanatomical levels of analysis of neural development. The model begins with evidence indicating that neurons are produced through symmetrical cell divisions in a set of proliferative units located in the ventricular zone of the mammalian developing embryo, and whose number is strongly determined by genetic factors.

Proliferative units send radial glial fibers toward the cortical plate, whose terminals occupy the same relative positions as the proliferative units in the ventricular zone. Thus, those units form a kind of subcortical protomap of future cortical organization. The radial glial fibers serve as a supporting framework for future cell migration from the proliferative units to the cortical plate. After a new cell has been produced, it "climbs" the glial fiber, guided by local cell-glia biochemical interactions (e.g., Antonicek, Persohn, & Schachner, 1987; Gadisseux *et al.*, 1990; Hatten & Mason, 1986). The general effect of these interactions is a virtual migration of the newly produced cell, from its proliferative unit of origin to its corresponding glial-fiber terminal in the cortical plate.

After migration, cells follow an inside-out gradient, passing to occupy the most superficial position in the cortical plate, relative to cells that had already reached the terminal. This gradient produces sets of genealogically related cells stacked on top of one another, referred to as *ontogenetic columns*. This process, occurring at numerous ontogenetic columns, produces the well-known layered and columnar organization of the cerebral cortex. In neuroanatomical terminology, a column is a set of cells generated by the same proliferative unit at different moments in time, whereas a layer is a set of cells generated by different proliferative units at approximately the same time.

Figure 1.1 shows a simplified version of the radial-unit model, used in the present work to simulate the proliferation and migration of hidden NPEs. The number of proliferative units was directly encoded in the chromosomes, under the assumption that this feature is genetically conserved (Rakic, 1988b). All proliferative units produced the same number of NPEs (see Appendix A). After proliferation, NPEs migrated according to another rule, from their proliferative units to their corresponding ontogenetic column. NPEs produced by different kinds of proliferative units could migrate at different rates. Thus, differences in the number of NPEs across groups of ontogenetic columns depended upon migration. The result of the proliferation and migration routines was a set of hidden NPEs organized into ontogenetic columns. These columns eventually became the ANN's layers<sup>5</sup>.

#### 1.2.1.1.2 Differentiation

The radial-unit model is an account of neocortical histogenesis in terms of cell proliferation and migration. However, this model is also consistent with experimental evidence regarding neuronal differentiation. Basically, this evidence shows that

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<sup>5</sup> In ANN terminology, and in the context of the present ND program, I shall use the term *layer* in its ANN sense, to refer to what neuroanatomists would call *columns*.

phenotypic differentiation of neocortical cells is time-specific and occurs mainly during migration, before the cells reach their respective ontogenetic columns. Thus, neuroblasts (immature neurons) possess certain degree of differentiation when they reach their destinations after migration. The present model is consistent with this characteristic in that hidden NPEs had a certain degree of differentiation at the end of migration. Moreover, such a differentiation was determined at the moment of proliferation, and produced two classes of hidden NPEs, namely *sensory association* and *motor association* (more on this distinction later).

Further NPE differentiation was determined by probabilistic rules for deciding which NPEs were excitatory and which were inhibitory. Further differentiation also involved determining certain parameters used in the NC model (see later). Different kinds of NPEs could have different values for the same NC parameter.

#### **1.2.1.1.3 Synaptogenesis and Cell Death**

After a set of completely differentiated (i.e., mature) NPEs were available, they were connected through the synaptogenesis routine. The basic parameter here was the probability of a connection being formed between two coactive elements. Synaptogenesis probabilities for different kinds of connections were encoded by separate chromosome segments. Once a connection was formed, its strength was determined by the spontaneous levels of activation of the pre- and postsynaptic elements, according to a Hebbian rule (see Appendix A). This strength represented the initial weight for that connection. The present ND program thus implemented activity-dependent synaptogenesis, which has been shown to occur extensively in mammalian neocortical development.

In contrast with other NC-GA models, chromosomes in the present model encoded no initial connection weights or particular connections. In this way, the model allowed

for phenotypic variation across ANNs, using a *fixed-length* chromosome. As a restriction, however, only feedforward connections between adjacent layers were allowed. That is, no recurrent connections from NPEs in later layers back to NPEs in earlier layers were allowed. This restriction represented only a simplifying device, and not a theoretical proposition about the architecture of real nervous systems. It is clear that recurrent connections are ubiquitous in real nervous systems. However, I show later that such connections were not necessary for simulating the kinds of behavioral phenomena studied in the present work, although allowing for those connections may be required for simulating more complex learning phenomena (e.g., operant chaining). Also, only lateral connections between excitatory and inhibitory NPEs within a layer were allowed, thus simulating a form of lateral inhibition. No lateral connections between inhibitory NPEs or feedforward connections between excitatory and inhibitory NPEs were allowed.

After synaptogenesis, all input elements and hidden NPEs that did not send any connections to any other NPEs were eliminated. The connections being received by the eliminated NPEs were eliminated as well. Also, all hidden and output NPEs that did not receive connections from any other elements, as well as the connections sent by those NPEs, were eliminated. Hence, NPE and connection elimination depended only upon the *available* NPEs and connections (i.e., no variables were genetically encoded for this stage).

In the present model, the differentiation and synaptogenesis routines were defined as *probabilistic* rules whose specific realizations could be different from one developing ANN to the other. A consequence of this strategy was that the very same chromosome could generate different ANNs (i.e., neural development was a *one-many relation*, from chromosomes to ANNs). This strategy had the potential problem of the



recoverability of "good" ANNs. Indeed, if the same genotype was capable of generating different ANNs, then "good" ANNs could be lost during the simulations. However, as demonstrated later, ANN recoverability did not represent an insurmountable problem, at least for the kind of problems investigated here.

### 1.2.1.2 Structural phenotypes

The end-product of an individual's development was an ANN whose features determined a *structural phenotype*. In the present work, an ANN was a set of interconnected NPEs, whose functioning was determined by an NC model that I shall describe later. For the moment, I want to focus upon the *anatomical* level of description of individuals.

Figure 1.2 shows a generic ANN depicting some of the architectural traits that determined a structural phenotype. Information in all ANNs flowed either in a feed-forward manner (i.e., from left to right in Figure 1.2), or laterally. All ANNs consisted of two subnetworks, namely *sensory* and *motor*. The sensory subnetwork consisted of a layer of input elements and a one or more sensory association (*sa*) layers, each serving a set of *cal* NPEs. The motor subnetwork consisted of one or more motor association (*ma*) layers and one output layer. Each *ma* layer served a set of *vta* NPEs. Functionally, the difference between *sa* and *ma* layers relates to how synaptic plasticity of their constituting NPEs was modulated by activity in the *cal* and *vta* NPEs (see later). Let me describe each component in more detail.

#### 1.2.1.2.1 The Sensory Subnetwork: Input Elements

The input layer consisted of three kinds of elements, namely  $s_1$  (represented as small empty squares),  $s_2$  (represented as small empty circles), and  $us$  (represented as a small empty triangle). The first two are analogous to afferent presynaptic inputs from

primary to sensory association cortex. The *us* (or *unconditional stimulus*) element represented a primary sensory element sensitive to the kinds of environmental events used in Pavlovian procedures as primary reinforcers (e.g., food, water, sex, etc.). Activations of input elements were directly assigned according to a training protocol representing the ANN's environment. In the present work, all training protocols were defined as Pavlovian procedures (see later). Hence, the temporal distribution of stimuli was unaffected by an ANN's behavior.

Input elements were connected to NPEs in the second layer through feedforward plastic connections (thin lines extending from the small squares and circles). The *us* element sent nonplastic connections (thicker lines) directly to a reinforcement-sensitive system consisting of *vta* NPEs (represented as small filled circles) simulating the *ventral tegmental area*, a dopaminergic subcortical nucleus known to be involved in reinforcement (see later). In turn, *vta* NPEs sent nonplastic connections directly to the output NPEs, whose activity determined the individual's overall behavior or *functional phenotype* (see later). The *vta* and output NPEs were thus *unconditionally activated* at the activation of the *us* element. Hence, the paths from the *us* element to the *vta* NPEs to the output NPEs are analogous to those paths that mediate *unconditional responses* of the kinds used in Pavlovian-conditioning procedures.

#### **1.2.1.2.2 The Sensory Subnetwork: *sa* and *cal* NPEs**

The *sa* layers consisted of NPEs analogous to polysensory neurons of the kind found in sensory-association cortex. Synaptic plasticity in these NPEs was modulated by a signal arising from the *cal* NPEs (represented as small filled squares). The magnitude of this modulatory signal depended partially upon changes in the activation of *cal* NPEs over time. The existence of such a modulatory role of *cal* NPEs is consistent with neuroanatomical evidence suggesting the existence of multisynaptic

pathways that originate in the CA1 area of the hippocampus and project diffusely to the sensory-association cortex (Amaral, 1987). A modulatory role of the hippocampus is also suggested by evidence suggesting that this structure may be involved in a number of Pavlovian conditioning phenomena (e.g., Berger, Alger, & Thompson, 1976; Berger & Orr, 1983; Berger *et al.*, 1983; Berger & Swain, 1989; Berger & Thompson, 1978; Moore & Solomon, 1984; Schmajuk, Spear, & Isaacson, 1983; Schmaltz & Theios, 1972; Solomon, 1977; Weikart & Berger, 1986; Weisz *et al.*, 1982; Weisz, Solomon, & Thompson, 1980). On the basis of this evidence, models have been proposed in which hippocampal activity is viewed as having a modulatory role in Pavlovian-conditioning phenomena such as blocking and reversal discrimination (e.g., Schmajuk & DiCarlo, 1991; Schmajuk & Moore, 1985, 1988, 1989). Also, the hippocampus has been shown to be involved in learning tasks that require polysensory integration (e.g., Morris *et al.*, 1982; O'Keefe & Dostrovsky, 1971; O'Keefe & Nadel, 1978; Olton, 1983; Olton & Samuelson, 1976; Sutherland & Dyck, 1983).

As the result of the synaptogenesis routine, *sa* NPEs could be connected to *ca1* NPEs through plastic synapses whose strengths changed as a result of experience. Such changes eventually allowed some of the *sa* NPEs to activate the *ca1* NPEs, and hence, contribute to the production of the *ca1* modulatory signal. This process implemented a mechanism through which an individual's learning could be enhanced not only by external reinforcement (i.e., the activation of the *us* input element) but also through a form of *sensory* reinforcement in which the activation of the *sa* NPEs by the sensory input elements could be sufficient to activate the *ca1* NPEs, and hence, to produce the *ca1* modulatory signal (see Appendix B). This sensory-reinforcement mechanism is consistent with neuroanatomical evidence showing that the primate hippocampus receives the great majority of its inputs from sensory-association areas

(e.g., Amaral, 1987; Squire, Shimamura, & Amaral, 1989; Squire & Zola-Morgan, 1988).

The magnitude of the *ca1* modulatory signal depended not only upon the magnitude of the activation of the *ca1* NPEs by the *sa* NPEs, but also upon a modulatory signal arising from the *vta* NPEs. The latter dependence is consistent with neuroanatomical evidence showing that pathways originating in subcortical dopaminergic nuclei project to areas of the hippocampus (e.g., Swanson, 1982; Verney *et al.*, 1985). Evidence at the cellular level has also shown that synaptic strength, as well as activity and cell membrane properties in hippocampal CA1 cells *in vitro* are modulated by dopamine (e.g., Stein & Belluzzi, 1985, 1988, 1989; Stein, Xue, & Belluzzi, 1993). Finally, LTP in the CA1 region has been shown to be prevented by dopamine antagonists (e.g., Frey, Schroeder, & Matthies, 1990).

#### **1.2.1.2.3 The Motor Subnetwork: *ma*, *vta*, and Output NPEs**

The motor subnetwork consisted of *ma* NPEs, analogous to neurons found in motor association cortex, the *vta* NPEs, and the output NPEs. Synaptic plasticity in both the three kinds of NPEs was modulated by a signal arising from the *vta* NPEs. The magnitude of this signal depended upon changes in the activation of the *vta* NPEs over time. This modulatory role of the *vta* NPEs is consistent with neuroanatomical evidence showing that VTA dopaminergic neurons diffusely project to portions of the motor association cortex through a set of connections known as the *medial forebrain bundle* (e.g., Berger, Gaspar, & Verney, 1991; Fallon, 1988; Fuster, 1989; Lindvall, 1979; *cf.* Parnavelas & Papadopoulos, 1989). Also, behavioral evidence shows that activity in dopaminergic subcortical systems (e.g., VTA and nucleus accumbens) and the medial forebrain bundle is involved in the kinds of behavioral-strengthening effects of external stimuli used as primary reinforcers in Pavlovian and instrumental

procedures (e.g., food, water, sex, and electrical shocks.) (e.g., Beninger, 1983; Hernandez & Hoebel, 1990; Hoebel, 1988; Kelley & Delfs, 1991a, 1991b; Mora & Myers, 1977; Olds & Milner, 1954; Pierce *et al.*, 1990; Salamone, 1991; White, Packard, & Hiroi, 1991; Wise, 1989; Wise & Rompré, 1989).

In the present approach, the magnitude of the *vta* modulatory signal was simulated as depending upon factors that determine the dynamics of the activation of cells in the VTA, such as neural refractory periods and dopamine depletion, reuptake, and inhibitory autoreceptors. This dependence is supported by evidence showing that dopamine neurons in areas A8, A9, and A10 of monkeys are activated by biologically significant stimuli (e.g., food, water, sex, electrical shocks) and by unexpected visual, auditory, and somatosensory stimuli. Evidence has also shown that originally neutral stimuli become capable of activating the *vta* after reinforced training in which those stimuli reliably precede a reinforcer (e.g., Schultz, 1992).

After synaptogenesis, *ma* NPEs could be connected to the *vta* NPEs through plastic synapses whose strengths could then increase as a result of reinforced training. Such an increase (modulated by the *vta* signals) eventually allowed some of the *ma* NPEs to activate the *vta* NPEs, and hence, contribute to the production of the *vta* modulatory signal. This process implemented a kind of *internal*-reinforcement mechanism in the motor subnetwork, similar to that described above for *sa* and *cal* NPEs (see Appendix B). This mechanism is consistent with neuroanatomical evidence suggesting the existence of corticostriate projections from motor association cortex to dopaminergic subcortical nuclei (e.g., Carman, Cowan, & Powell, 1963; Faull, 1982; Glees, 1944; Johnson, Rosvold, & Mishkin, 1968; Kemp & Powell, 1970; McGeorge & Faull, 1989). Also, behavioral evidence shows that after a sensory stimulus has acquired the ability to evoke an unconditional response, that stimulus can be used as a

reinforcer (e.g., Pavlov, 1927; Rescorla, 1980). This is the so-called *second-order conditioning*.

Finally, NPEs in the last layer of *ma* NPEs were connected to the output NPEs. These NPEs could be unconditionally activated by the *us* input element through the *vta* NPEs or conditionally by the  $s_1$  and/or  $s_2$  input elements through the *sa* and *ma* NPEs. The first kind of activation determined an *unconditional response* (UR), whereas the second kind determined a *conditional response* (CR). The collective activations of the output NPEs determined the ANN's overall behavior or functional phenotype (see later).

The *maximum* numbers of  $s_1$  and  $s_2$  input elements, *cal* NPEs per *sa* layer, and VTA NPEs per motor-association layer, were not probabilistically determined. Rather, these numbers were genetically constrained, under the assumption that the corresponding parts in the mammalian brain (i.e., thalamus, hippocampus, VTA, and primary motor cortex) show more intraspecific similarities, relative to sensory and motor association cortex, and hence tend to be genetically conserved (Allman, 1990; Arbas, Meinertzhagen, & Shaw, 1991; Eccles, 1989). Of course, after cell death, an adult ANN could have fewer elements than the genetically determined maximum.

#### 1.2.1.2.4 The Functioning of NPEs: The NC Model

The functioning of an NPE was determined by an NC model defined by an activation function and a synaptic-plasticity function<sup>6</sup>. Appendix B presents a formal description of the NC model used in the present work (see also Donahoe, Burgos, & Palmer, 1993; Donahoe & Palmer, 1994). This section presents an informal

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<sup>6</sup> This latter function is more often referred to as a *learning* function in ANN research. However, I shall use the less traditional term, for I want to reserve the term *learning* for referring to changes in an ANN's *overall* output-NPE activation over time, under certain environmental conditions.

description of some of the general aspects of the activation and synaptic-plasticity functions.

The NPE-activation function had the following features:

- An activation was a number between 0.0 and 1.0 representing the probability of occurrence of an action potential. This feature is consistent with the idea that stochastic models seem more appropriate and feasible than deterministic models for describing the functioning of neurons and synapses (e.g., Abeles, 1991; del Castillo & Katz, 1954; Faber *et al.*, 1992).
- NPEs could be either excitatory or inhibitory, depending upon how their signals affected postsynaptic or target NPEs. This distinction is consistent with that drawn between pyramidal (glutamergic) and smooth-stellate (GABAergic) cortical cells (e.g., Abeles, 1991; Crick & Asanuma, 1986).
- An NPE could receive excitatory and/or inhibitory input signals. If an NPE received both types of signals, then its activation at a given moment in time was a joint function of such signals. Inhibitory signals had a *subtractive* effect on the NPE's activation. When the level of postsynaptic inhibition was equal to or larger than the level of excitation, the NPE's activation was set to zero. These features are consistent with the idea of inhibition as "sculpting" postsynaptic activation (Eccles, 1957).
- Postsynaptic excitation could be summated in time. There was no temporal summation of inhibition.
- Activations could decay in time.
- Whether there was summation of excitation or decay of activation was determined by an excitation threshold.

The synaptic-plasticity function had the following features:

- A connection weight was a number between 0.0 and 1.0 representing the proportion of postsynaptic membrane receptors controlled by or committed to a given presynaptic input at a given moment on time.
- Presynaptic terminals of the same type (either excitatory or inhibitory) competed for their respective types of transmitter receptors on the postsynaptic membrane<sup>7</sup>.
- Synaptic competition did not occur among presynaptic terminals of different types. This feature is consistent with the fact that excitatory and inhibitory inputs affect different types of neurotransmitter receptors on the membrane of a postsynaptic process. Indeed, it is known that glutamate affects a type of receptors (NMDA, quisqualate, and kainate) different from those affected by GABA (GABA-A and GABA-B receptors). By separating competition between excitatory and inhibitory inputs, it was assumed that NMDA and non-NMDA receptors are unaffected by GABA, and that GABA-A and GABA-B receptors are unaffected by glutamate.
- Synaptic competition and plasticity were modulated by a signal that simultaneously affected the synaptic efficacies of all *active* inputs connected to the NPE. The effect of this modulatory signal was to increment connection weights, for which it can be referred to as a *reinforcing* signal. This signal can be seen as the outcome of mechanisms such as axo-axonic synapses that regulate presynaptic neurotransmitter release, axo-somatic synapses that regulate the kinetics of postsynaptic membrane receptors, etc. The model did not include any explicit assumptions in this regard.

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<sup>7</sup> This form of competition is different from that in which output NPEs compete for being the one to be active (e.g., Fukushima, 1975; Grossberg, 1987; Kohonen, 1989; Rumelhart & Zipser, 1986). The kind of competition simulated here is more akin those thought to occur in synaptic rearrangement and stabilization (e.g., Changeux, Courge, & Danchin, 1973; Changeux & Danchin, 1976).



- Changes in synaptic efficacies of *sa* and *ma* NPEs were modulated by different signals, namely  $d_s$  (sent by the *cal*) and  $d_m$  (sent by *vta* NPEs), respectively. A reinforcing signal was the difference in the activation of these NPEs between two successive time-steps.
- Two distinct learning mechanisms were implemented, one for increasing (acquisition) and one for decreasing connection weights (extinction). The acquisition mechanism was initiated every time the reinforcing signal was larger than zero. Otherwise, the extinction mechanism was initiated. Only the acquisition mechanism was influenced by the magnitude of the reinforcing signal. Also, the rates at which synaptic strength increased or decreased were independent from each other, and hence, could be different. These were called, respectively, *acquisition* and *extinction rates*. Different kinds of NPEs could have different acquisition rates. Also, different NPEs could have different extinction rates.

Figure 1.3 shows the generic NPE that implemented the NC model. The NPE was the structural and functional unit of analysis of an ANN. In the present work, the NPE (large circle labeled as  $j$ ) was conceptualized as an abstract postsynaptic element receiving information from a number of presynaptic elements, which could be either primary-sensory input elements (i.e.,  $s_1$  and/or  $s_2$  elements) or other NPEs.

Activations of presynaptic excitatory NPEs are labeled as  $a_1^+$ ,  $a_2^+$ , and so on, whereas activations of presynaptic inhibitory NPEs are labeled as  $a_1^-$ ,  $a_2^-$ , and so on, where the subindexes identify the presynaptic elements. Connection weights are represented as medium-sized circles. Weights for excitatory connections are labeled as  $w_{1j}^+$ ,  $w_{2j}^+$ , and so on, whereas weights for inhibitory connections are labeled as  $w_{1j}^-$ ,  $w_{2j}^-$ , and so on, where the first subindex identifies the presynaptic element and subindex  $j$  represents the postsynaptic NPE. Changes in connection weights were

modulated by the reinforcing signal (either  $d_s$  or  $d_m$ ). The thick lines pointing down toward the NPE are intended to represent the diffuse nature of this signal, and to leave open its synaptic interpretation (see Frey, Huang, & Kandel, 1993, and Stein & Belluzi, 1988, 1989 for some possible interpretations).

The NPE works as a summing junction that computes the quantities  $exc_j$  and  $inh_j$  separately, from the dot products  $\mathbf{a}^+ \cdot \mathbf{w}^+$  and  $\mathbf{a}^- \cdot \mathbf{w}^-$ , respectively, where  $\mathbf{a}^+$  denotes the set of activations of excitatory presynaptic excitatory inputs,  $\mathbf{w}^+$  denotes the set of synaptic strengths of these inputs,  $\mathbf{a}^-$  denotes the set of activations of inhibitory presynaptic inputs, and  $\mathbf{w}^-$  denotes the set of synaptic strengths of these inputs. The quantities  $exc_j$  and  $inh_j$  can be neurobiologically interpreted as the amount of excitation and inhibition produced on the NPE, respectively. These quantities are then used (again, separately) as the arguments of the *logistic* function (denoted as  $L$  in the figure), to compute quantities that can be interpreted as the probability of occurrence of an excitatory postsynaptic potential and the probability of occurrence of an inhibitory postsynaptic potential, respectively. The effect of the former also depended upon two thresholds (viz.,  $\theta$  and  $inh_j$ ). Finally, an NPE's output resulted from multiplying those probabilities by certain parameters, and combining them through the activation function (see Appendix B).

### 1.2.1.3 Functional Phenotypes

An ANN's functional phenotype was determined by its overall behavior after training under certain environmental conditions. In an effort to have ecological validity, current NC-GA models have conceptualized the environment as a set of objects randomly scattered across a two-dimensional space (e.g., Ackley and Littman, 1992; 1994; French & Messinger, 1994; Miller, Todd, & Hegde, 1989; Nolfi, Elman, &

Parisi, 1990; Paredis, 1991; Todd & Miller, 1991). Such objects represent the kinds found in natural environments (e.g., food, water, mates, mountains, fire, predators, competitors, etc.).

In contrast, the present approach conceptualizes the environment more abstractly, in terms of the kinds of *temporal* and *probabilistic* relations among environmental events, as defined by *Pavlovian procedures*. Such a conceptualization was motivated by the fact that much of the theoretical and experimental work on the relations between learning and nervous systems have been inspired by Pavlovian conditioning phenomena (e.g., Buenomano & Byrne, 1990; Donegan, Gluck, & Thompson, 1989; Grossberg, 1991; Hawkins, 1989; Kandel, 1985; Klopff, 1988; Klopff & Morgan, 1990; Moore, 1991; Moore, Berthier, & Blazis, 1990; Schmajuk & DiCarlo, 1991; Sutton, 1988, 1991; Sutton & Barto, 1981; Thompson, 1986; Thompson, Berger, & Madden, 1983; Wagner & Donegan, 1989).

#### **1.2.1.3.1 The Meaning of the Term *Adaptive***

Before describing the kinds of environmental conditions simulated here, I need to clarify the sense in which I use the term *adaptive* throughout the present work. Typically, an ANN is said to be adaptive if it simulates the kinds of behavioral changes that are observed in *real* organisms under *comparable* environmental changes. For example, if an ANN behaves in a way that simulates an interstimulus or ISI function (see later), then that ANN is said to be "adaptive", because this function is a phenomenon observed in real organisms under conditions similar to the simulated ones. In this sense, then, the term *adaptive*, as applied to ANNs, signifies *consistent-with-real-plastic-behavior*.

In evolutionary biology, however, a trait is adaptive if and only if it confers a *fitness advantage* to an individual *in the present*, without necessarily having conferred

a fitness advantage to the individual's ancestors (Sober, 1993). Thus, strictly speaking, to refer to an ANN as "adaptive" we must determine if the kinds of plastic behaviors it simulates confer a fitness advantage to the real organisms in which they are observed. In most cases, such a determination is logically established, mostly by the way of counterfactual argument (i.e., by referring to events or situations that *might have occurred*, although they did *not* actually occur), rather than empirically demonstrated. For example, if an ANN behaves in a way consistent with that observed in NMR conditioning, then we consider that ANN as adaptive under the tacit argument that had the organisms been in their natural environments, their learning to *anticipate* the occurrence of certain stimuli would have increased their reproductive success.

Since Pavlov's (1927) seminal work, evolutionary interpretations of conditioning have referred to the fitness advantage of anticipatory behavior. The logic underlying these considerations is that strategies involving such behavior increase individual fitness. Very little experimental work has been inspired by this idea. Studies of the relationship between Pavlovian conditioning and territorial behavior in *Betta splendens* and *Trichogaster trichopterus* represent a notable exception (e.g., Domjan & Hollis, 1988; Hollis, 1984, 1990; Hollis, Cadieux, & Colbert, 1989; Hollis, Martin, Cadieux, & Colbert, 1984).

In those studies, a nonspecific stimulus, such as a light (the CS), is paired with a specific stimulus, such as the presence of a conspecific male or the subject's mirror image (the US), which elicits a behavioral pattern that is part of the organism's territorial behavior related to reproductive success. Results have shown that individuals that learn to anticipate a potential territorial situation based upon nonspecific environmental cues, are *better prepared* to deal with real situations (i.e., when the CS is presented, trained individuals are significantly better fighters than

untrained ones). This preparedness, in turn, is thought to increase the learner's fitness<sup>8</sup>. The main implication is that any genetically determined phenotypic trait that favors Pavlovian learning of the territorial response (e.g., the structure of the individual's nervous system), will be selected and passed to future generations.

These experiments suggest that Pavlovian conditioning may affect the course of evolution by increasing the probability of occurrence of anticipatory behaviors that confer a potential fitness advantage to the individual. However, contrary to standard ANN interpretations, I shall not assume that an ANN's behavior is adaptive because it is consistent with some behavior that hypothetically confers some fitness advantage. Rather, I shall consider an ANN as adaptive if and only if its overall behavior truly confers a fitness advantage within a given simulation, *regardless of whether or not such a behavior is consistent with that observed in real organisms*. The external validity of this interpretation, of course, requires not only consistency with real behavior, but also the assumption that the kinds of temporal and probabilistic relationships studied in Pavlovian conditioning also occur in natural environments. This assumption is fundamental to the present work (cf. Gray, 1985; Reed, 1985; Schleidt, 1985).

I shall thus use the term *adaptive* in close relation to the term *fitness*. In evolutionary algorithms, the latter term does not refer to actual reproductive success. Rather, fitness is defined by some aspect of an individual's phenotype that determines (and hence precedes) actual reproductive success. This is the *propensity interpretation* of fitness, introduced by Mills and Beatty (1979), and further elaborated by Brandon (1990).

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<sup>8</sup> This sense of the term *preparedness* should not be confused with that arising from the notion of biological constraints on learning (see later). In the present context, the term refers to *anticipation*, whereas in that notion, the term refers to the extent to which an organism's biological endowment *permits* certain kinds of learning or performance under certain environmental conditions.

In an effort to be consistent with this interpretation, individual fitness was defined in the present work exclusively in terms of an ANN's *overall behavior* (i.e., collective output-NPE activation) after a number of training trials. Under this definition, the higher an ANN's responding to a CS, the higher its chance was of being selected for mating and reproduction. Thus, the present scheme simulated a form of *directional selection*, for it favored performance toward the upper limit of the fitness range.

In contrast to other NC-GA models, the present model allows for the possibility of more than one fitness function. That is, different kinds of arrangements may involve different fitness functions. For example, a situation in which ANNs were selected for increased responding to only one CS (Chapter 3), versus a situation in which ANNs were selected for increased responding to two CSs (Chapter 4), involved different fitness functions. Thus, different functional phenotypes were associated with different fitness functions in different experiments.

### 1.2.2 Phylogeny

In the present work, phylogeny was simulated through a GA. In this section, I present an informal description of the main components of the GA.

After a fitness value was computed for each ANN in the population, a selection scheme was applied in which the ANNs with high fitness values were selected for mating and reproduction to form a new chromosome population constituting the next generation. This scheme consisted of four steps, namely formation of a mating pool through fitness-dependent tournament selection, random parent coupling among members of the mating pool, genetic crossover, and mutation.

Mating pools were formed for each generation by applying a rule that simulates *tournament selection*, which is computationally more efficient and biologically

plausible (Goldberg & Deb, 1991). Tournament selection involved sampling a fixed number of contestant ANNs randomly from the population. The contestant ANN showing the largest fitness value was declared the winner, and selected to become a member of the mating pool. Then, all contestants were returned to the population (i.e., ANNs were given more than one chance to enter the mating pool). Tournaments occurred until the total number of winners was equal to a predefined mating-pool size.

Once a mating pool was complete, matings were simulated by selecting pairs of parent ANNs randomly from the mating pool. When a mating occurred, new chromosomes were formed by crossing-over randomly chosen fragments of the parent chromosomes. After crossover, the new chromosomes were mutated and parents were returned to the mating pool. This routine was repeated until the number of new chromosomes was equal to the population size, which marked the beginning of a new generation.

In the above scheme, individual learning (an ontogenetic process) represents a guiding force in evolution (a phylogenetic process). This consideration is potentially problematic, for it could be interpreted as meaning that nondevelopmental changes at the ontogenetic level (i.e., learned behavior) affect evolutionary changes at the phylogenetic level. Such an interpretation has a Lamarckian flavor, for it allows for the possibility of the inheritance of acquired characters. However, the relationship between learning and evolution can be interpreted in a non-Lamarckian way. The first author to anticipate this idea was Baldwin (1896), in what has come to be known as the "Baldwin Effect" (Simpson, 1953).

The key idea underlying the Baldwin Effect is that certain nondevelopmental ontogenetic changes such as learning may have an adaptive value (i.e., increase individual fitness). Genetic hereditary factors impose global constraints on the

environmental conditions under which such changes are likely to occur. The presence of these factors in an individual's genotype does not guarantee the occurrence of the changes in question. Rather, these factors make such an occurrence more *probable*, *given* the presence of the appropriate environmental conditions. Therefore, if an environment remains relatively constant over generational time, then genetic factors that make more probable the acquisition of adaptive changes *in that environment*, as well as the changes themselves, will become more frequent in subsequent generations.

In computational-neuroethology investigations, the Baldwin Effect has been observed as an increase in the rate of learning across generations, to the point that, at the end of the evolutionary process, individuals need virtually no learning (e.g., Ackley & Littman, 1992; French & Messinger, 1994; Gruau & Whitley, 1993; Hinton & Nowlan, 1987; Keesing & Stork, 1991). This effect is largely due to the fact that in these models the initial connection weights (i.e., the synaptic strengths *before* training) are directly encoded in the genotype. Evolution in these simulations thus consists of a search for "good" initial connection weights. Under this interpretation, the Baldwin Effect refers to the extent to which increased responding to a stimulus is *innate*.

In the present model, no initial connection weights were directly encoded in the genotype. Rather, initial connection weights were determined during the neurodevelopmental stage of synaptogenesis by the spontaneous activation of pre- and postsynaptic elements, which, in turn, was determined by a genetically encoded parameter used in the activation rule (see Appendix B). This strategy raises the issue of whether or not the Baldwin Effect, defined in the present work as the extent to which ANNs show high responding to nonreinforced *novel* CSs, would be observed even in the case in which initial connection weights were only indirectly determined by the genotype. This issue is also explored in the present work.



## 1.3 General Research Strategy

The general research strategy adopted in the present work has been philosophically inspired by the view elaborated by Salthe (1993) that biological phenomena are too complex to be understood in terms of models driven by the traditional experimentalist approach to science associated with names such as Roger Bacon (1214-1292), John Duns Scotus (1265-1308), William of Ockham (1280-1349), Galileo (), Descartes (1596-1650), and Newton (1642-1727). In Salthe's words:

"[C]omplex systems do not yield to prediction in satisfactory ways. [Traditional experimentalist] science has been, in part, based on the art of simplifying problems by isolating measurable aspects of the world, ascertaining their regularity, and then, on the basis of interpretative models, trying experimental inputs to see if desirable alterations could be produced. This depends crucially on being able to isolate aspects of the world from their contexts without altering their nature, on the notion that these aspects of the world do not change their behavior as a result of observation, and on the idea that these aspects of the world do not have so much internal coherence that interference with one part would unpredictably alter the behaviors of other parts (as is the case with machines). None of these things turns out to be true with interesting subject matter —say the brain, or ecosystems, or social systems, or indeed any biological system. (pp. 2-3).

These considerations can be extended to modelling strategies that are closely constrained by specific experimental data. Indeed, the external and ecological validity of a model will depend heavily upon the external and ecological validity of the experimental data upon which the model is based. An implication of Salthe's criticism is that external and ecological validity are severely limited in models that are guided too closely by specific experimental data have been generated under the kinds of simplifying assumptions adopted by traditional experimentalist science.

An alternative strategy (which was adopted in the present work) is to *start* with complex models that are constrained by general principles that cut across different kinds of experimental data, rather than by specific experimental data. This strategy is

justified under the assumption that such models will produce simulation results that are sufficiently general to be applicable to a variety of situations, but sufficiently rich to be appealing to researchers across different areas. Only after interesting simulation results have been produced, we can focus upon more specific aspects of the model that are thought to underlie some of these results, and then construct models that are more formally tractable and applicable to specific experimental data. Of course, such data may be used to revise the smaller model so that its assumptions and concepts are consistent with some of the details of the phenomenon of interest.

On the basis of this strategy, then, the present work must be seen as strictly *exploratory, heuristic, and empirical*. By *exploratory* I signify *nonexhaustive, nonfinal, mostly-interpretative* (as opposed to *heavily-constrained-by-or-evaluated-through-specific-experimental-data*) and *scientifically-oriented* (as opposed to *pragmatically-or-engineering-oriented*). By *heuristic* I signify *question-seeking* (as opposed to *answer-seeking*). By *empirical* I signify *emphasis-on-simulation-results* (as opposed to *emphasis-on-model-formal-properties*).

The present work involved computer simulations that were extremely complex by most standards found in the AL literature. Such complexity arose not only from the modeling strategy adopted here (i.e., the construction of a *supermodel* consisting of a *hierarchy* of submodels inspired by *general* neurobiological, neuroanatomical, behavioral, and evolutionary *principles*), but also from the considerable number of conditions explored here. Also, due to the complexity of the model, the number of dependent variables was unusually large, again by AL-simulation standards.

This strategy inevitably led to more questions than answers, to more interpretation than proof, to more description than explanation. Consequently, the present work focused upon the *competence and range of variation* of a computational approach to

the evolution of Pavlovian conditioning, rather than upon exhaustive characterizations of specific aspects of that approach. However, I believe that many of the results I am about to describe are sufficiently rich and intriguing to compensate for the interpretative uncertainties caused by the complexities of the models and simulations.

## 1.4 Research Objectives and Reader's Guide

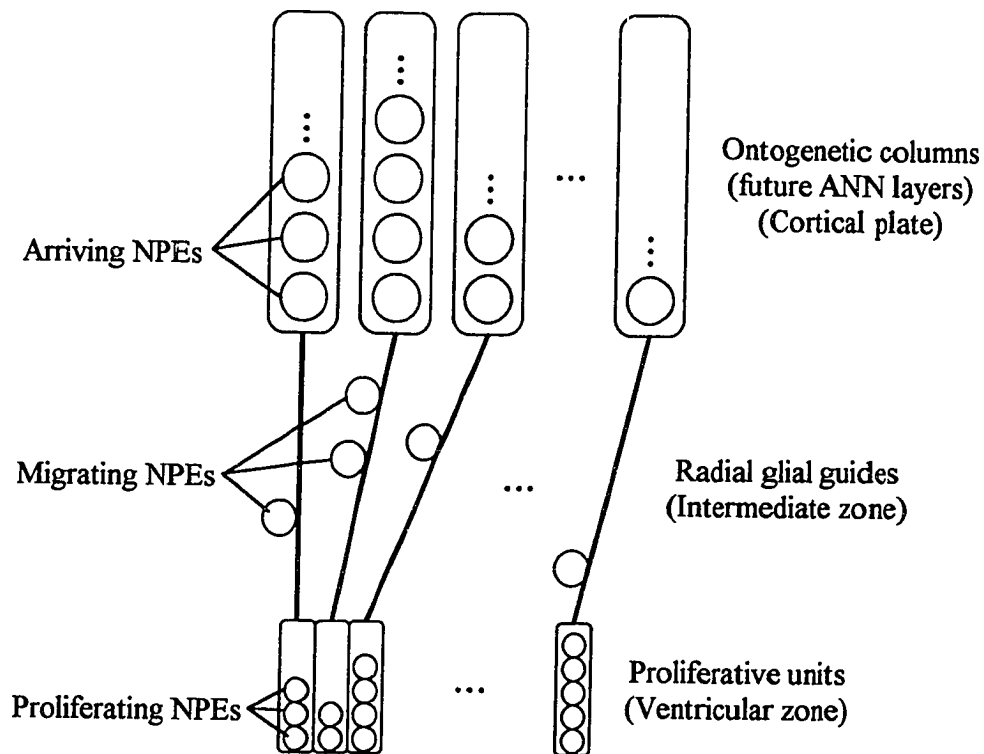
The main objective of the present work was to determine if phenotypically (i.e., structurally and functionally) different types of ANNs would emerge from selection for increased responding under different kinds of environmental conditions. Four simulation experiments were performed. Each experiment consisted of two kinds of simulations, namely evolution and test. Evolution simulations involved the application of the NC-GA model described in the present chapter, to the evolution of ANNs trained according to some Pavlovian procedure. During the test simulations, only the NC model was operative. Test simulations involved exposing ANNs from the last generation of an evolution simulation to environmental conditions different from the ancestral ones. The purpose of these simulations thus was to assess the behavioral competence of ANNs under environmental conditions that differed from those under which ancestral selection took place, and hence to test for the *emergence* of functional phenotypes. In this context, I considered a functional phenotype as emergent if and only if it was observed in a population of ANNs *without having been explicitly selected for throughout phylogeny* (i.e., during the evolution simulations).

The distinction between evolution and test simulations can be conceptually aligned with the distinction between traits that are adaptations and traits that are adaptive, emphasized by Sober (1993). According to this distinction, a trait is an adaptation if its

presence is due to selective processes accumulated over phylogenetic time, regardless of whether or not such a trait increases individual fitness in the present. In contrast, a trait is adaptive if it increases individual fitness in the present, but its current presence is not the result of a selective process occurred over phylogenetic time. Under this distinction, then, a trait can be an adaptation without being adaptive, and *vice versa*. Of course, a given trait can be adaptive as well as an adaptation. The adaptation-adaptive dichotomy can be applied to the distinction between evolution and test simulations by saying that the former simulate the emergence of traits that are adaptations, whereas the latter allow for the study of traits that are adaptive. Because individual fitness in the present work was directly determined by functional phenotypes, the adaptation-adaptive dichotomy applies within this context only to behavioral traits.

The next chapter describes the general methodology used in all of the simulations. Chapter 3 describes Experiment 1, in which individuals were selected for increased responding to a particular ISI and CS. ANNs from the last generation were then tested with CSs and ISI values different from the ancestral ones. Chapter 4 describes Experiment 2, in which ANNs were selected for responding with high CR proportions to *two* independently reinforced CSs. ANNs from the last generation were then tested for discrimination (orthogonal and nonorthogonal), generalization, and blocking. Chapter 5 describes Experiment 3, in which ANNs were selected for increased responding to the reinforced CS in an orthogonal differential reinforcement procedure, in which the reinforced and nonreinforced CSs did not share any input elements. Then, ANNs from the last generation were tested for nonorthogonal discrimination, generalization, and blocking. Chapter 6 describes Experiment 4, in which ANNs were selected for responding with high CR proportions in the presence of a reinforced CS and in the presence of a nonreinforced CS, in a nonorthogonal differential

reinforcement procedure, in which the CSs shared some input elements. ANNs from the last generation were then tested in orthogonal discrimination, generalization, and blocking. Finally, Chapter 7 discusses general conclusions about the implications of the above experiments for the theoretical and empirical study of the evolution of learning and behavior, and suggests directions for future research.



**Figure 1.1: Two-dimensional version of the radial-unit model, after Rakic (1988b).** This model was used to simulate the proliferation and migration stages of the ND program. During proliferation, relatively undifferentiated elements were produced in proliferative units located in the ventricular zone of the developing individual. Migration involved the movement of elements from their respective proliferative units to the cortical plate on glial fibers that served as supporting framework and were located in the intermediate zone. Once in the cortical plate, elements were organized into ontogenetic columns that would eventually become the ANN's layers. A central feature of this model is the one-one relation between proliferative units and ontogenetic columns, as determined by the glial fibers. That is, elements constituting a particular ontogenetic column have proliferated in and migrated from one and only one proliferative unit. In the present work, that relation arose from genetic coding of separate migration probabilities for elements produced in two kinds of proliferative units, namely *sa* and *ma*. This strategy allowed *sa* and *ma* layers to have different numbers of elements.

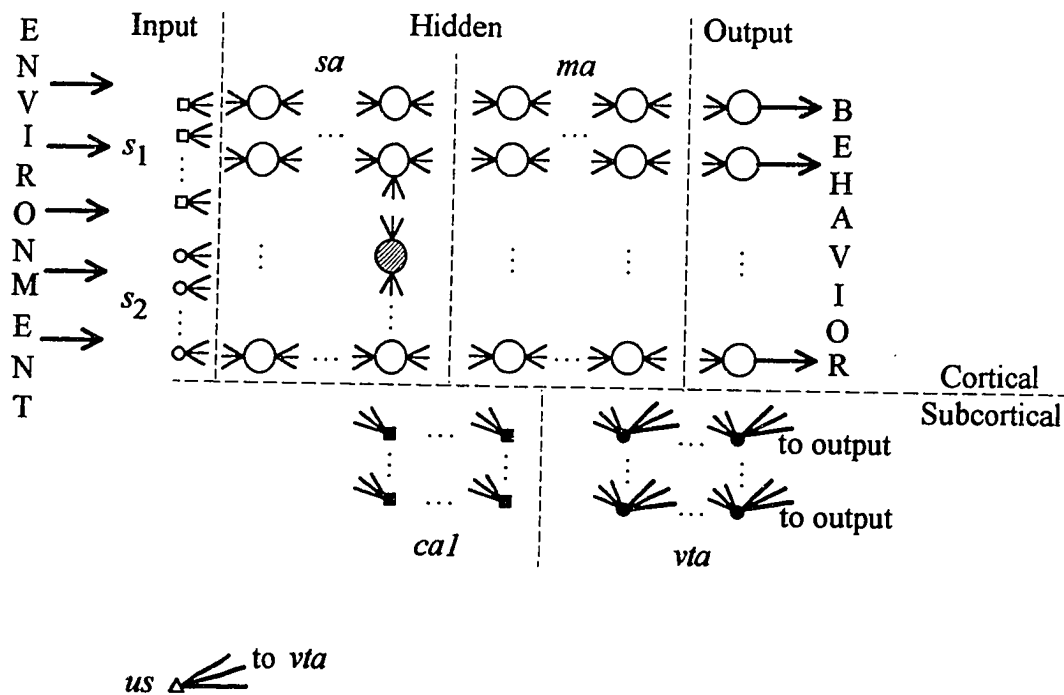
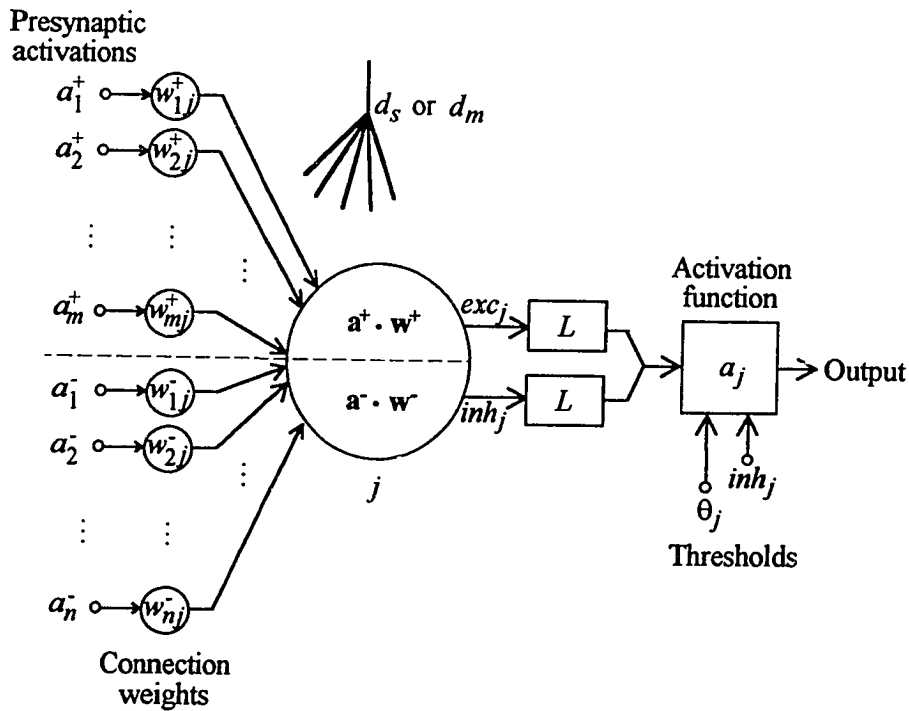


Figure 1.2: **Generic ANN.** All ANNs consisted of one input layer, two or more hidden layers, and one output layer. The three kinds of layers are delimited in the figure by vertical dashed lines. The input layer consisted of three kinds of input elements labeled as  $s_1$  (represented as small empty circles),  $s_2$  (represented as small empty squares), and  $us$  (for *unconditional stimulus*, represented as a small empty triangle). Input elements were activated by events constituting the ANN's environment (i.e., stimuli), according to a training protocol defined by a Pavlovian procedure (see later). See text for details.



**Figure 1.3: Generic NPE.** The generic NPE was the structural and functional unit of analysis of an ANN. The NPE (large circle labeled as  $j$ ) was conceptualized as a postsynaptic element receiving signals from a number of presynaptic elements (represented as small empty circles). Presynaptic elements, represented as small empty circles, could be either excitatory or inhibitory. Changes in connection weights were modulated by either  $d_s$  or  $d_m$ , depending upon whether the NPE was in the  $sa$  or in the  $ma$  subnetwork, respectively. See text for details.



# **CHAPTER 2**

## **GENERAL METHODOLOGY**

The present chapter describes methodological aspects common to all of the simulations described in subsequent chapters. These aspects include hardware, software, general procedures, and dependent variables. Methodological aspects specific to each experiments are discussed in the corresponding chapters.

### **2.1 Hardware and Software**

Simulations were run on a Gateway 2000 IBM-compatible PC, with a 486DX CPU running at 66 MHz, a math coprocessor running at 16 MHz, 8 MB of RAM, and 550 MB of hard-disk space. Simulation programs were designed and written by the author in Turbo Pascal, v. 7.0 for DOS (Borland International, 1992). Data analysis and graphing were facilitated by SigmaPlot for Windows, v. 2.0 (Jandel Scientific, 1994).

### **2.2 General Procedure**

A total of four experiments were run. Each experiment consisted of two kinds of simulations, namely evolution and test. Evolution simulations involved using the NC-GA model (see Chapter 1) to generate lineages. Test simulations involved the operation of the NC model only (i.e., no evolution was simulated). The present section, I shall concentrate on describing the general procedural aspects of the evolution simulations.

## 2.2.1 Evolution Simulations

An evolution simulation consisted of a lineage of nonoverlapping generations. A generation was defined as a development-training-selection sequence applied to each member of a population of 100 individuals. In the present chapter, I focus on the general procedural aspects of training and selection.

For any given evolution simulation, the maximum possible number of generations produced by the GA was 100. However, this number could be smaller if a fitness-stability criterion was satisfied (see later). An individual was represented as a chromosome-ANN pair. A chromosome determined an individual's genotype. Structural traits of an ANN determined an individual's structural phenotype. An ANN's overall performance in a Pavlovian learning task determined the individual's functional phenotype.

Different lineages evolved independently of one another, and corresponded to different environmental conditions. Environmental conditions could be different in the temporal or probabilistic relations between CS and US, or the kind of CS. These differences can be viewed as analogous to those determining reproductive isolation and speciation (e.g., continental drift, migration, niche adaptation, etc.).

### 2.2.1.1 Training and evaluation

Training was defined as a sequence of trials of certain duration. Time was conceptualized as a discrete variable. Therefore, the passage of time was represented as a sequence of discrete *time-steps*, or *ts*. No assumptions were made about the specific real-time duration of a *ts*. A trial was thus defined as the activation of certain input elements for a number of *ts*, according to a predefined training protocol. At each *ts*, the activations and connection weights of all NPEs of an ANN were updated.

The updating procedure consisted of generating a random sequence of *all* NPEs at a ts, and then changing their activations and connection weights *in that order*. This procedure implemented a form of *asynchronous updating* (i.e., the activation and connection weights of the NPE currently being updated could be affected by a mixture of updated and unupdated NPEs). Asynchronous updating has been considered to be biologically more plausible than synchronous updating (Amit, 1989)<sup>1</sup>. Also, asynchronous updating has been found to induce stability in certain AL models (e.g., Bersini & Detours, 1994).

Trials were reinforced according to a Pavlovian forward-delay procedure. That is, in reinforced trials, the *us* input element was activated with a value of 1.0 only at the *last* CS ts. In all reinforced trials, the duration of the US was 1 ts and the interstimulus interval (ISI) was equal to the CS duration minus 1 (i.e., the ISI was the number of ts from the first to the penultimate ts of a reinforced CS trial). In nonreinforced trials, the *us* input element was not activated at any ts. The intertrial interval (ITI) was not directly simulated. Rather, the ITI was assumed to be sufficiently long to allow the activations of all NPEs to decay to near-zero spontaneous levels of activation.

Immediately after training, ANNs were given a number of evaluation trials during which individual fitness was determined. Fitness was defined as the CR proportion

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<sup>1</sup> Amit does not provide any explicit argument or empirical evidence supporting this claim. However, it is intuitively reasonable to assume that differences must exist across individual neurons (i.e., no two neurons are identical in *all* of their properties, although, of course, they may share many properties), and that such differences may give rise to differences in the rates at which neurons change their states. One could argue that this assumption holds only at the cellular level (where time scales are relatively small), breaking down at the neuroanatomical, microcircuit level of organization (which is perhaps the level most closely represented by NPEs in the present NC model, and where time scales are substantially larger). This argument arises from the idea that individual differences at the cellular level cancel out statistically at the neuroanatomical level. However, this latter argument is no better supported by empirical evidence than the idea that individual differences across neurons cause asynchronic changes at the cellular *and* at the neuroanatomical levels. Thus, parsimony seems to be the only criterion we have to decide among the two arguments. And, it is more parsimonious to assume that asynchrony occurs at both levels than assuming that asynchrony occurs at one level while synchrony occurs at a higher level.

over a number of evaluation trials. For any individual, the occurrence of a CR was defined as a mean activation across output NPEs of .5 or more at the penultimate ts of a trial. That is, if at the penultimate ts of a trial the mean output-NPE activation of an ANN was .5 or more, then the occurrence of a CR was counted for that trial. At the end of evaluation, the individual's fitness was determined by the proportion of evaluation trials in which a CR had occurred.

### 2.2.1.2 Selection

After all individuals within a generation were evaluated, a mating pool was formed through a series of *one-winner-only tournaments*. A one-winner-only tournament involved sampling five *different* participants randomly from the population (i.e., individuals could not compete against themselves at any given tournament), and comparing their fitness values. The participant with the *highest* fitness value was declared the winner of the tournament and was selected as a member of the mating pool. Then, all participants were returned to the population (i.e., a given individual could, in principle, participate in more than one tournament). For simplicity, no tied winners were selected from any given tournament (i.e., the highest fitness value found in the tournament was required to correspond to one and only one individual). If the highest fitness value was shared by two or more individuals, then the tournament was dismissed and another tournament occurred. Tournament selection ended after 100 winners were selected.

The above tournament selection procedure provided a convenient fitness-stability criterion for the evolution simulations. Indeed, for any given generation, the larger the number of individuals having the same fitness value, the larger the number of tournaments required until the no-tied-winners condition was satisfied, and hence the

longer the duration of tournament selection. On this basis, an evolution simulation was considered to have reached fitness stability if tournament selection was not complete within one hour or more. Also, a minimum of two different individuals with a fitness greater than 0.0 were required for tournament selection to end, reproduction to occur, and a new generation to begin.

After a mating pool was formed, reproduction consisted of 50 mating trials. A mating trial involved sampling two individuals randomly from the mating pool (i.e., individuals were not allowed to mate with themselves), and recombining their respective genetic materials with a probability of .8. Recombination consisted of selecting two loci randomly (the same in both parent chromosomes) and translocating the middle segment of both parent chromosomes. This is the so-called *two-point crossover*, anticipated by Cavicchio (1970), and further characterized by DeJong (1975), Holland (1975), and Schaffer, Eshelman, and Offutt (1991).

After recombination, the parent chromosomes were returned to the population (i.e., members of the mating pool were allowed to mate more than one time) and alleles of the new chromosomes were mutated. Mutation consisted of running a Bernoulli trial for each gene in both chromosomes, and then changing its value with a probability of .001. The result of a mating trial, then, was two chromosomes consisting of genetic material (possibly mutated) from both parents. The end result of reproduction was a set of 100 chromosomes constituting a new generation. Then, the development-training-selection sequence was repeated for the individuals of the new generation.

# CHAPTER 3

## EXPERIMENT 1

The simulations described in the present chapter were motivated by two aims: first, to determine if phenotypically (i.e., structurally and functionally) different types of ANNs would emerge from selection for increased responding under environments that differed in the ISI and the kind of CS, and second, to determine how the resulting ANNs would perform under ISIs and CSs different from those present when ancestral selection took place. Performance under different ISIs was expected to simulate ISI functions and optimal-ISI noninvariance, whereas performance under different CSs was expected to simulate CS nonequipotentiality.

The ISI function is a relation between the ISI and some behavioral measure, typically CR percentage. An ISI function resembles an inverted U-shaped curve whose peak indicates high responding and whose corresponding ISI value is considered as *optimal*. ISI functions arise from a decrease in CR percentage as the ISI departs from the optimal ISI. The ISI function is the phenomenon that supports the view that temporal contiguity is necessary for excitatory Pavlovian conditioning.

Optimal-ISI noninvariance refers to the fact that experimental preparations, such as nictitating membrane response (e.g., Frey & Ross, 1968; Schneiderman, 1966; Smith, Coleman, & Gormezano, 1969), autoshaping (e.g., Gibbon, Baldock, Locurto, Gold, & Terrace, 1977), conditioned suppression (Libby, 1951; Yeo, 1974), and taste aversion (e.g., Etscorn & Stephens, 1973; Garcia, Ervin, & Koelling, 1966) require very different optimal ISIs.

This kind of evidence represents a refutation of the idea of *optimal short ISIs*, a fundamental premise of *general-process* approaches to learning, according to which *all* Pavlovian conditioning occurs only when the ISI is very short (e.g., Kimble, 1961). An implication of this premise is that the range of optimal ISIs is very small (i.e., that optimal ISIs tend to be invariant; e.g., Hull, 1943, 1952). In this sense, the noninvariance of optimal ISIs, as observed across different preparations, represents a refutation of the premise of the optimal-ISI invariance.

Another premise that has been refuted is the notion of stimulus equipotentiality, the idea that *any* kind of CS is *equally effective* in evoking a response after being paired with a US. Among these phenomena can be mentioned preparedness in taste aversion (e.g., Garcia & Koelling, 1966), avoidance learning (Bolles, 1970), auto-shaping (e.g., Jenkins & Moore, 1973; Peterson, Ackil, Frommer, & Hearst, 1972; Squier, 1969), and discrimination (e.g., Lawicka, 1964). These and other related phenomena have shown that, in a number of species and under a variety of circumstances, different CSs are *nonequipotential* (i.e., different stimuli are *differentially effective* in evoking a response after being paired with a given US).

The above evidence has suggested that the optimal-ISI invariance and stimulus-equipotentiality premises, as adopted by general-process approaches to learning, must be rejected. Acknowledgement of this situation has yielded to the notion of *biological constraints on learning* (e.g., Boakes, Poli, Lockwood, & Goodal, 1978; Hinde, 1973; Lockard, 1971; Rozin & Kalat, 1971; Seligman, 1970), according to which organisms possess *inherent differences* on what they can and cannot learn. Such differences are viewed as being *biologically determined*. This view has facilitated contact between animal-learning laboratory research, behavioral- ecology research, and the neurosciences, leading to evolutionary interpretations of learning, behavior, and the

structure and function of nervous systems. Although some evidence has been reported on some of the aspects of those relationships (see Chapter 1), performing experiments that take into account ontogenetic (neurodevelopmental and behavioral) *and* phylogenetic (genetic, populational, and evolutionary) aspects of the evolution of real behavior and nervous systems in real organisms becomes exceedingly difficult as we increase the complexity of organisms, behaviors, and environments, which is the case in most situations that are interesting for many neuroscientists, learning psychologists, and behavioral ecologists. Such a practical limitation makes AL an ideal strategy for the study of the evolution of learning, behavior, and nervous systems<sup>1</sup>.

The premise underlying the following simulations, then, is that applications of the notion of biological constraints to Pavlovian conditioning must account for ISI functions, optimal-ISI noninvariance, and CS nonequipotentiality in terms of interrelated *emergent* functional properties of individuals that have *evolved* under *different selection pressures* throughout phylogeny. And these pressures must arise from environments whose critical events differ in their temporal relationships and kind. Thus, in a computational approach to that notion, ISI functions should emerge from selection for increased responding under a *particular* ISI, optimal-ISI noninvariance should emerge from selection for increased responding under *different* ISIs, and CS nonequipotentiality should emerge from selection for increased responding to a particular kind of CS.

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<sup>1</sup> Comparative approaches represent another strategy, in which differences and similarities in learning and behavior across species are studied. However, even these approaches are inherently limited to the study of *present organisms*, so that no direct evidence can be gathered on the *evolutionary processes as such*.



## 3.1 Evolution Simulations

### 3.1.1 Procedure

Before starting the simulations, a founder chromosome population was generated. This population was designated as G0<sup>2</sup>. G0 was chosen from several randomly-generated chromosome populations, on the basis of the average population fitness in a forward delay Pavlovian conditioning procedure in which two hundred trials were randomly distributed between CS1 and CS2. A CS1 trial involved the activation of all  $s_1$  input elements with a value of 1.0 and no activation of any  $s_2$  elements, for 5 ts. A CS2 trial involved no activation of any  $s_1$  input elements and the activation of all  $s_2$  elements with a value of 1.0, also for 5 ts. Both kinds of trials were reinforced by activating the  $us$  input element at the last ts (i.e., the ISI was 4 ts). After this training, 50 more training trials, randomly distributed into CS1 and CS2, were given to compute fitness. If the average population fitness was greater than 0.0, then the chromosome population was chosen as G0. This strategy was adopted after preliminary simulations showed that completely random founder populations resulted in very little or no evolution.

After generating G0, eight environmental conditions were defined by a combination of four ISIs (2, 4, 8, or 16 ts) and two CSs (CS1 or CS2). A lineage of ANNs evolved for each condition. As a first step, eight ANNs were developed from each chromosome in G0, to obtain 800 ANNs, which were divided into groups of 100. A group constituted the ANN founder population of a condition. ANNs in each group were then trained in a forward-delay procedure using a particular ISI and CS. For ANNs in the CS1 lineages, the CS involved the activation of all  $s_1$  input elements with

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<sup>2</sup> As a notational convention, I shall refer to the generations constituting a given lineage as G0 (for Generation 0), G1 (for Generation 1), G2 (for Generation 2), and so on.

a value of 1.0 and no activation of any  $s_2$  input element. For ANNs in the CS2 lineages, the CS involved no activation of any  $s_1$  element and the activation of all  $s_2$  elements with a value of 1.0. All ANNs were given 125 CS-US pairings, the last 25 of which served as evaluation trials to determine individual fitness. In the present simulations, individual fitness was identical to the proportion of evaluation trials that elicited a CR (see Chapter 2 for a definition of a CR).

After evaluation, a new chromosome population was created through the selection scheme of the GA. This new population constituted G1 (the next generation). Once the new chromosomes were created, new ANNs were developed and trained in the *same* procedure as their ancestors. The same operation was repeated generation after generation, until the termination criterion (either 100 generations after G0 or fitness stability) was satisfied. This strategy resulted in eight lineages, which I shall label as 2-CS1, 2-CS2, 4-CS1, 4-CS2, 8-CS1, 8-CS2, 16-CS1, 16-CS2, where the beginning number represents the ISI, and 'CS1' and 'CS2' represent the kind of CS involved in training and evaluation. All lineages evolved independently of one another.

### 3.1.2 Results and Discussion

The results of the evolution simulations are grouped into three parts. First, genotypic evolution is described in terms of changes in mean chromosomic overlap over generations. Second, effects on functional phenotypes are described in terms of changes in population fitness over generations, mean output-NPE activations across trials for the first and last generation, and changes in connection weights before and after training for the last generation. Finally, the effects on structural phenotypes are described in terms of differences in structural traits among ANNs in the last generation of each lineage. For convenience, I have omitted error bars in the graphs describing the

effects of structural phenotypes. However, standard errors for all structural phenotypes were negligible, ranging from 0.002 for the mean number of input elements of a given kind to 3.45 for the mean number of connections per element. Standard errors for NC parameters ranged from 0.001 to 0.05.

### 3.1.2.1 Genotypic Evolution

Genotypic evolution was measured by changes in mean chromosomal overlap across generations. Chromosomal overlap is a measure of similarity between two chromosomes, defined as the ratio  $(A - D) / N$ , where  $A$  denotes the number of agreeing loci (i.e., loci containing genes that have the same allele from one chromosome to the other),  $D$  denotes the number of differing loci (i.e., loci containing genes that have different alleles), and  $N$  denotes the total number of loci in a chromosome. Naturally, this measure required that the two chromosomes being compared have the same number of loci, which was the case in the present GA. The chromosomal overlap between two chromosomes could range from -1.0 (the two chromosomes had no agreeing loci) to 1.0 (the two chromosomes had no differing loci). A chromosomal overlap of 0.0 indicated that the two chromosomes agreed (or differed) in 50% of their loci, which is the expected value for randomly generated chromosome populations. In the present work, the mean chromosomal overlap in a generation was computed from all the possible combinations of 2 chromosomes in a population of 100.

Figure 3.1 shows that the mean chromosomal overlap increased as a negatively accelerated function of generations, for all lineages. Indeed, the overlap for G0 was close to 0.0, which is consistent with the expected value for a randomly generated population. Overlaps increased rapidly within a few generations, and then became stable at relatively high values. Individuals *within lineages* thus became genetically

more similar as generations progressed. For all lineages, the mean overlap of the last generation was less than 1.0, which signifies that some genetic variation remained at the end of evolution. Also, in general the greater the number of generations in a lineage, generally the greater the mean overlap of the last generation, for both CS1 and CS2. This last result indicates that the amount of genetic variation in a population depended, at least in part, on how many generations had passed until the last generation.

### **3.1.2.2 Effects on Functional Phenotypes**

#### **3.1.2.2.1 Effects on Fitness**

One kind of effect on functional phenotypes refers to *behavioral evolution*, as expressed by changes in mean population fitness over generations. Figure 3.2 shows that mean population fitness increased as a negatively accelerated function of generations, for all lineages. Indeed, all lineages started with low-fitness (close to 0.0) and ended with high-fitness (close to 1.0) ANNs. Also, ANNs in the last generation of all lineages tended to have similar fitness. Hence, comparably fit ANNs emerged from selection for increased responding under different environmental conditions. Finally, lineages satisfied the fitness-stability termination criterion well before 100 generations, except 2-CS1, 2-CS2, and 16-CS2. Fitness values in these lineages rarely reached 1.0. This situation produced sufficient functional-phenotypic variation to ensure a winner at all tournaments throughout generations, and hence a continuation of the selection process until the maximum number of generations.

Figures 3.3 through 3.6 show 3D mesh plots depicting changes in the frequency distributions of fitness values across generations, for all lineages. These figures give an idea of how fitness values varied within and between generations. Each figure shows

the CS1 lineage (top panel) and CS2 lineage (bottom panel) for a given ISI. All graphs show the same overall tendency, that is, a high number of low-fitness individuals at G0, followed by an increase in the number of high-fitness individuals after a few generations. Indeed, the high frequency peak at the origin of the generation-fitness plane indicates that most individuals in G0 had a low fitness (i.e., between 0.0 and 0.05). However, as generations passed, the number of individuals with a high fitness (i.e., between 0.95 and 1.00) increased, rapidly for the ISI-2, ISI-4, and ISI-8 lineages, and more slowly for the ISI-16 lineages. Thus, for all lineages, the number of low-fitness individuals decreased while the number of high-fitness individuals increased, as a function of generations. Finally, fitness variability at the end of evolution was relatively low (i.e., the majority of individuals tended to have the same fitness), except for 2-CS1 (see top panel, Figure 3.3) and 16-CS2 (see bottom panel, Figure 3.6).

#### **3.1.2.2.2 Effects on Learning**

The above results show that mean population fitness increased across generations. However, a high fitness only indicated a high proportion of evaluation trials in which the mean output-NPE activation satisfied the CR criterion (i.e., 0.5 or more). Thus, fitness gives a very rough idea of output-NPE activations. A second kind of effect upon functional phenotypes, then, relates to how these activations changed across trials, for the first and last generation of each lineage. Figures 3.7 through 3.14 show 3D mesh plots of changes in frequency distributions of mean output-NPE activations across training trials, for the first (top panel) and last (bottom panel) generation of each lineage.

All figures show the same overall trend. In G0, the number of ANNs with low output-NPE activations (i.e., between 0.0 and .05) remained relatively constant across trials. Only a few ANNs responded with activations greater than the CR criterion of

0.5 toward the end of training, a condition that was necessary and sufficient to ensure the minimum number of tournament winners required for the GA to generate G1. However, in the last generation, the number of individuals with mean output-NPE activations greater than 0.5 tended to increase as a function of trials. Moreover, most ANNs far exceeded the CR criterion towards the end of training, despite the fact that their ancestors did not need to respond with such high activations to win selection tournaments.

Note that in the last generation of ISI-2, ANNs tended to respond with output-NPE activations above 0.5 *on the first trial* (see bottom panels of Figures 3.7 and 3.8). In fact, the frequency distributions did not change substantially across trials. This result indicates that ANNs in the last generation of 2-CS1 and 2-CS2 tended to respond *innately* to CS1 and CS2, respectively. A tendency to respond with high output-NPE activations on the first trial can also be observed (although to a smaller degree) in some of the ANNs for ISI-4 (see bottom panels of Figures 3.9 and 3.10) and ISI-16 (see bottom panel of Figure 3.14).

Finally, a third measure of functional phenotypes relates to changes in connection weights before and after training. Figure 3.15 shows the mean weight per connection before (top panel) and after training (bottom panel) for the last generation, as a function of ISIs and CSs. Overall, the initial connection weights were relatively constant across ISIs and CSs. This result suggests that ANNs across lineages tended to have similar initial connection weights, and hence similar spontaneous-activation parameters.

In contrast, after training, the mean weight per connection tended to increase as a function of ISIs, indicating that differences between initial and final weights were relatively small in ANNs that evolved under shorter ISIs. This result is consistent with

the high responding shown by these ANNs at the beginning of training (e.g., see bottom panels of Figures 3.7 and 3.8). In contrast, the difference between initial and final weights in ANNs that evolved under longer ISIs was substantially larger. These results suggest that whether or not initial connection weights were sufficient to allow for high responding without reinforcement depended on the ISI.

Overall, the above results show that selection for increased responding under certain environmental conditions generated ANNs whose behavior changed from low to high conditioned responding under those conditions. And this result was accomplished within a strictly neo-Darwinian model of evolution. Indeed, the definition of the present NC-GA model was consistent with the Central Dogma of molecular biology, in that an individual's genotype was not affected by the individual's *own* functional phenotype. Rather, an individual's genotype depended upon the functional phenotypes of its *ancestors*. This is the sense in which we can say, from a neo-Darwinian perspective, that learning affects evolution.

#### 3.1.2.2.3 Control Simulations

The above results show that learning and behavior-dependent fitness were sufficient for evolution to occur. However, it is unclear whether or not learning and behavior-dependent fitness were *necessary*. A set of control simulations addressed this issue. Two kinds of control simulations were run, namely *No-Learning* (NL) and *Random-Fitness* (RF). Both kinds of simulations were run with the same ISIs and CSs as before. However, in the NL simulation, the synaptic-plasticity algorithm was disabled, so that ANNs could not learn but fitness was still behavior-dependent, whereas in the RF simulation, ANNs could learn but fitness was behavior-independent (i.e., individual fitness was a random number, rather than CR proportion). In both simulations, fitness remained close to 0.0 throughout 100 generations, for all conditions. This result

suggests that learning *and* selection for increased responding were *both* necessary for evolution to occur.

### 3.1.2.3 Effects on Structural Phenotypes

The above results show that different selective pressures produced *functionally equivalent* ANNs (i.e., comparably fit under environmental conditions equivalent to those under which ancestral selection occurred). This result raises the issue of whether or not such ANNs were also *structurally* equivalent. In the following sections, I address this issue by showing differences in structural phenotypes across ANNs in the *last* generation of different lineages.

#### 3.1.2.3.1 General Effects: Examples of ANNs

Before describing specific effects, let me give a general idea of the kinds of differences I shall describe. Figure 3.16 shows one typical ANN from the last generation of each lineage. All the ANNs had a fitness above .95. Different kinds of elements are represented using the same code shown in Figure 1.2. The empty spaces within layers indicate elements that were eliminated at the end of development, because of their lacking input and/or output connections. For simplicity, inhibitory NPEs are shown without connections. However, the presence of an inhibitory NPE signifies that it received *and* sent at least one connection within the layer. The *us* input element is not shown for any ANN.

Some general architectural differences can be observed. For example, selection under CS1 tended to produce ANNs with more  $s_1$  input elements than ANNs evolved under CS2, while the latter tended to have more  $s_2$  input elements. The number of excitatory NPEs was substantially larger than the number of inhibitory NPEs, for all ANNs. ANNs evolved under longer ISIs tended to have more layers, particularly *sa*



layers, although the number of *ma* layers increased for ISI 16. The number of *sa* and *ma* layers remained relatively constant across CSs, although some differences can be observed in the number of NPEs per layer, mostly in the ISI-8 and ISI-16 ANNs. The number of *via* NPEs also remained relatively constant across ISIs and CSs. In contrast, CS1 ANNs tended to have far more *cal* NPEs than CS2 ANNs. Finally, the number of output NPEs remained relatively constant across ISI, although substantial variation was observed across CSs.

While selecting these ANNs, some structural variability *within the same population* were observed, especially in the ISI-8 and ISI-16 conditions. Indeed, although most ANNs within the last generation of the ISI-2 and ISI-4 lineages seemed to be instances of the same basic architecture, the last generation of ISI-8 and ISI-16 included a few high-fitness ANNs with architectures very different from those shown in Figure 3.16. Figure 3.17 illustrates this result by showing two atypical examples of 8-CS1 and 16-CS1 ANNs, both of which had a fitness higher than .95. Clearly, these ANNs possess architectures very different from those represented by the corresponding ANNs shown in Figure 3.16. Thus, selection for increased responding under a particular environmental condition did not produce a unique kind of architecture, but a *class* of architectures.

Also, not all ANNs at the end of evolution had a high fitness. Figure 3.18 shows a sample of low-fitness ANNs. All of these ANNs had a fitness between 0.0 and 0.16. Some general differences with respect to the high-fitness ANNs can be observed. For example, the 2-CS2 low-fitness ANN had two *ma* layers, whereas its high-fitness counterpart had only one. Also, the 4-CS1 low-fitness ANNs had far fewer *sa* NPEs per layer than its high-fitness counterpart. Finally, the 8-CS1 and 16-CS1 low-fitness ANNs had only one *sa* layer, whereas their high-fitness counterparts had four.

### 3.1.2.3.2 Effects on Hidden Layers

Perhaps the most salient structural feature of an ANN is the number of hidden layers. Figure 3.19 shows the mean number of *sa* (top panel) and *ma* (bottom panel) layers as a function of ISIs and CSs. The mean number of *sa* layers increased over ISIs 2 to 8, but then decreased for ISI 16, remaining relatively constant across CSs. In contrast, the mean number of *ma* layers remained constant at 1.0 over ISIs 2 to 8, but increased to 2.0 for ISI 16. Hence, the overall number of hidden layers (*sa* and *ma* layers combined) increased as the ISI increased from 2 to 8 ts, and then remained relatively unchanged as the ISI increased from 8 to 16. In sum, ANNs that evolved under longer ISIs tended to have more hidden layers than ANNs that evolved under shorter ISIs, whereas ANNs that evolved under different CSs tended to have approximately the same number of hidden layers.

### 3.1.2.3.3 Effects on Excitatory and Inhibitory NPEs

One outcome of the differentiation stage of the ND program was the division of hidden NPEs into excitatory and inhibitory. This division applied to *sa* as well as *ma*. Figure 3.20 shows the mean number of excitatory (top panel) and inhibitory (bottom panel) NPEs, as a function of ISIs and CSs. The number of excitatory NPEs tended to increase over ISIs for both CSs, and then decreased abruptly for 16-CS1. In contrast, the number of inhibitory NPEs remained low across ISIs and CSs, with 16-CS2 ANNs having slightly more inhibitory NPEs. Comparisons with Figure 3.19 show that the number of excitatory NPEs is positively related to the number of *sa* layers across ISIs, except for 16-CS2 ANNs, which tended to have a higher number of excitatory NPEs but a lower number of *sa* layers. These differences resulted from the fact that these ANNs tended to have far more excitatory *ma* NPEs than other ANNs.

Overall, then, ANNs that evolved under different ISIs and CSs had substantially more excitatory than inhibitory NPEs. Also, ANNs that evolved under longer ISIs tended to have more excitatory NPEs. Finally, ISI-16 ANNs that evolved under CS2 had more excitatory NPEs than ISI-16 ANNs that evolved under CS1. In general, then, inhibitory NPEs did not seem to play a major role in the evolution of high-fitness ANNs, at least under the conditions simulated here.

#### **3.1.2.3.4 Structural Phenotypes: Architectural versus Parametric Traits**

The above descriptions give only a general idea of some of the effects observed on structural phenotypes. However, details on different kinds of elements cannot be adequately appreciated in those descriptions. I shall thus spend the rest of this section describing more specific differences and similarities in various structural traits for different kinds of elements, across ANNs that evolved under different selective pressures. I have divided structural traits into two kinds, namely *architectural* and *parametric*.

Architectural traits refer to the numbers of different types of elements and their connectivity. For input elements, *divergent* connectivity was measured in terms of the number of *postsynaptic* connections per element. For *sa*, *cal*, *ma*, *vta*, and output NPEs, *convergent* connectivity was measured in terms of the number of *presynaptic* connections per NPE.

Parametric traits refer to the NC parameters that affected NPE activations and weights. I shall concentrate on temporal summation, activation decay, and weight-acquisition and weight-extinction rates. In addition to providing information about structural phenotypes, NC parameters also give an idea of genetic variation across different populations at the end of evolution, for more than 30% of an individual's genotype encoded for NC parameters. I shall not describe spontaneous-activation

parameters, for they were set to 0.1 for all ANNs after development and before training, to eliminate, as a possible outcome of evolution, ANNs consisting of NPEs with high spontaneous activations. Differences in spontaneous-activation parameters played a role only during the synaptogenesis stage of the ND program by affecting the initial connection weights, which I have already discussed (see Section 3.1.2.2.2).

I have adopted the following organization. First, I describe effects on structural phenotypes (architectural traits followed by parametric traits) in the sensory subnetwork (i.e., input elements, and *sa* and *caI* NPEs; no parametric traits are described for input elements, for they were not NPEs). Then, I describe effects on structural phenotypes in the motor subnetwork (i.e., *ma*, *vta*, and output NPEs). I shall focus upon excitatory NPEs, for they represented the vast majority of NPEs in most ANNs.

Although my emphasis will be upon structural comparisons among ANNs that evolved under different environmental conditions, I shall also point at general interrelations among traits. However, because such interrelations are very likely to be complex, and hence, to require statistical analyses that are beyond the scope of the present work, I shall describe only a few interrelations and interpret them only by offering very general conjectures.

### **3.1.2.3.5 Effects on the Sensory Subnetwork: Input Elements**

One architectural feature of ANNs was the number of input elements of a given kind. Figure 3.21 shows the mean number of  $s_1$  (top panel) and  $s_2$  (bottom panel) input elements, as a function of  $\log_2$ ISIs and CSs (the effects of CS1 and CS2 are represented as solid and dashed lines, respectively). There was no systematic trend in the number of input elements of either kind as the ISI increased. Indeed, the number of  $s_1$  elements decreased over ISIs for CS1 ANNs and increased for CS2 ANNs. These

differences may have been due to genetic biases in the chromosome founder population across segments that determined traits related to input elements.

More systematic effects were observed across CSs. Indeed, CS1 ANNs tended to have a far more  $s_1$  input elements than CS2 ANNs, with the exception ISI-16 ANNs. In contrast, CS2 ANNs tended to have far more  $s_2$  elements than CS1 ANNs. This latter difference was most noticeable in ISI-2 and ISI-16 ANNs.

It is puzzling that CS1 ANNs tended to have more  $s_1$  than  $s_2$  elements while CS2 ANNs tended to have more  $s_2$  than  $s_1$  elements, in view of the fact that the ancestors of CS1 ANNs were not exposed to instances of CS2, nor the ancestors of CS2 ANNs were exposed to instances of CS1, *during training*. However, the spontaneous activations of input elements during synaptogenesis were randomly determined, elements of *both* kinds may have been activated (see Appendix A). This strategy may have introduced some amount of competition between different kinds of input elements for connection weights in NPEs in the first *sa* layer. And such a competition may have been sufficient for ANNs having few (or none) of the unnecessary input elements to have a higher fitness.

ANNs' architectures were also determined by their connectivity. Figure 3.22 shows changes in divergent connectivity from input elements, measured as the mean number of postsynaptic connections per input element, over ISIs and CSs. Again, there was no consistent overall trend in divergent connectivity in  $s_1$  elements across ISIs. However, substantial differences were observed between CS1 and CS2 ANNs. Indeed, CS1 ANNs had more connections per  $s_1$  element than CS2 ANNs, with the exception of ISI-4 ANNs. In contrast CS2 ANNs had more connections per  $s_2$  input element than CS1 ANNs, with the exception of ISI-8 ANNs, and with the largest difference being between 2-CS1 and 2-CS2 ANNs.

These differences are generally consistent with those shown in Figure 3.21 (i.e., CS1 ANNs tended to have more  $s_1$  elements *and* postsynaptic connections per  $s_1$  element than CS2 ANNs, whereas CS2 ANNs tended to have more  $s_2$  elements and postsynaptic connections per  $s_2$  element than CS1 ANNs). The exceptions were 4-CS2 ANNs (which had more connections per  $s_1$  element but fewer  $s_1$  elements than 4-CS1 ANNs), 8-CS1 ANNs (which had more connections per  $s_2$  element but fewer  $s_2$  elements than 8-CS2 ANNs), and 16-CS1 ANNs (which had more connections per  $s_1$  element but fewer  $s_1$  elements than 16-CS2 ANNs). Finally, note that 16-CS2 ANNs tended to have a relatively high number of  $s_1$  elements, but relatively few connections per  $s_1$  element (i.e., there was less divergent connectivity for  $s_1$  elements in these ANNs). This last result can be appreciated in the example of a 16-CS2 ANN depicted in Figure 3.16.

In summary, ANNs that evolved under different ISIs had approximately the same number of input elements and connections of a given kind. However, ANNs that evolved under CS1 had more  $s_1$  elements and connections per  $s_1$  element, but fewer  $s_2$  elements and connections per  $s_2$  element than ANNs that evolved under CS2. A functional implication of these effects is that ANNs having different numbers of  $s_1$  and  $s_2$  elements may be differentially sensitive to different kinds of CSs, which could be a basis for CS nonequipotentiality. Later, in the test simulations, I describe results that address this possibility.

### **3.1.2.3.6 Effects on the Sensory Subnetwork: *sa* and *cal* NPEs**

#### **3.1.2.3.6.1 Effects on Numbers of NPEs**

Another architectural measure referred to the number of hidden NPEs of different types, regardless of how they were distributed among layers. Figure 3.23 shows the mean number of *sa* (top panel) and *cal* (bottom panel) NPEs as a function of ISIs and

CSs. Here, the changes were generally consistent across ISIs for both CSs. Indeed, the number of both kinds of NPEs increased over ISIs 2 to 8, but then decreased for ISI 16. Also, the number of *cal* NPEs was far greater for ANNs in the CS1 than the CS2 conditions. Comparisons with Figures 3.19 show that the numbers of *sa* and *cal* NPEs tended to change in the same direction as the number of *sa* layers, across ISIs.

In sum, then, ANNs that evolved under longer ISIs tended to have more *sa* and *cal* NPEs, whereas ANNs that evolved under different CSs tended to have the same number of *sa* NPEs. However, ANNs that evolved under CS1 had substantially more *cal* NPEs than ANNs that evolved under CS2. This and other differences between the effects of CS1 and CS2 (see later) may have been due to biases in the chromosome founder population.

#### **3.1.2.3.6.2 Effects on Connectivity**

The architecture of the sensory subnetwork can also be characterized in terms of its connectivity. Figure 3.24 shows the amount of *convergent* connectivity, measured as the mean number of presynaptic inputs per *sa* (top panel) and *cal* (bottom panel) NPE, across ISIs and CSs. Both measures remained relatively constant at about 6 inputs across ISIs and CSs. Therefore, and relating to Figure 3.23, the number of presynaptic connections per *sa* and *cal* NPE changed approximately as fast as the number of *sa* and *cal* NPEs, across ISIs and CSs. In general, then, ANNs that evolved under different ISIs and CSs had relatively the same proportion of presynaptic inputs to *sa* and *cal* NPEs.

#### **3.1.2.3.6.3 Effects on Activation Parameters**

Structural phenotypes were also determined by the NC parameters that affected changes in NPE activations. Figure 3.25 shows the mean temporal summation in *sa* (top panel) and *cal* NPEs (bottom panel), across ISIs and CSs. Temporal summation

determined what fraction of an NPE's excitation at  $t-1$  was summed to the NPE's excitation at  $t$ . Overall, temporal summation in *sa* NPEs remained relatively constant at about 0.3 across ISIs and CSs. Temporal summation in *cal* NPEs remained relatively constant at about 0.5 over ISIs 2 to 8, but then decreased to approximately 0.25 for ISI 16. However, *cal* NPEs in CS1 ANNs tended to have a substantially larger temporal summation than *cal* NPEs in CS2 ANNs. This effect is consistent with that observed for the number of *cal* NPEs, which was also substantially greater in CS1 than CS2 ANNs (see bottom panel of Figure 3.23), although the peaks of the CS1 functions in the bottom panels of Figures 3.23 and 3.25 corresponded to different ISIs (4 and 8, respectively). These similarities suggest a relation between *cal* temporal summation and the number of *cal* NPEs.

Also, temporal summation in *cal* NPEs decreased for CS1 ANNs across ISIs. This effect is consistent with that observed on the number of  $s_1$  input elements, which also tended to decrease across ISIs (see top panel of Figure 3.21). Thus, in CS1 ANNs, higher *cal* temporal summations tended to correspond to more  $s_1$  input elements, which may have conferred a fitness advantage by increasing the range of variation of *cal* NPEs, and hence the overall magnitude of *cal* discrepancies. However, lower *cal* temporal summations tended to correspond to fewer  $s_1$  input elements. The functional effect of this last correspondence is unclear. Perhaps, having few  $s_1$  input elements and low temporal summations conferred a fitness advantage by decreasing the range of variation of *cal*-NPE activations, which may have been too large due to the effect of other interrelations among architectural and parametric traits. These interpretations, of course, arise from the assumption that there may have existed an optimal range of variation of discrepancy signals, which could be achieved in many different ways, all of them involving different kinds of interrelations among different kinds of traits.



Summarizing, then, *sa* NPEs in ANNs that evolved under different ISIs and CSs tended to have similar temporal summations. Also, *cal* NPEs in ANNs that evolved under different ISIs tended to have similar temporal summations. However, *cal* NPEs in ANNs that evolved under CS1 had substantially larger temporal summations than *cal* NPEs in ANNs that evolved under CS2. The former also tended to have larger temporal summations as the ISI increased from 2 to 8. Therefore, *cal* temporal summation seemed to have played a more substantial role than *sa* temporal summation, at least in the kinds of learning tasks studied here. Everything else being equal, an increase in the temporal summation of *cal* NPEs produces an increase in the magnitude of the *cal* discrepancy. Thus, adaptation in 2-CS1 and 4-CS1 ANNs (the ANNs whose NPEs had the largest *cal* temporal summation parameters) may have required substantially greater *cal* discrepancies than adaptation in any other ANNs.

Another activation parameter was activation decay, which determined what fraction of an NPE's activation at  $t-1$  was subtracted from the NPE's activation at  $t$ . Figure 3.26 shows the mean activation decays in *sa* (top panel) and *cal* (bottom panel) over ISIs and CSs. For ANNs in CS1 lineages, *sa* as well as *cal* activation decays changed as a U-shaped function of ISIs. Thus, for these ANNs, the largest activation-decay parameters (about 0.85) in the sensory subnetwork corresponded to the shortest and the longest ISIs. For CS2 ANNs, *sa* activation decays tended to decrease across ISIs (here, the lowest decay parameter was approximately 0.3, and corresponded to 8-CS2 ANNs; the largest parameter was about 0.75, and corresponded to 4-CS2 ANNs), whereas *cal* activation decays also tended to change as a U-shaped function of ISIs. Also, *sa* NPEs in CS1 ANNs tended to have higher activation decays than *sa* NPEs in CS2 ANNs (except for ISI-4 ANNs), whereas *cal* NPEs in CS1 and CS2 ANNs

tended to have similar activation decays (except for ISI-8 ANNs). Finally, overall, *sa* and *cal* NPEs tended to have similar decay parameters.

Comparisons with previous effects show that, in general, *cal* activation decays in CS2 ANNs tended to change in the same direction as *cal* temporal summations, across ISIs (see CS2 function in bottom panel of Figure 3.25). However, activation decays tended to change in directions opposite to those observed in the number of *sa* and *cal* NPEs across ISIs (see Figure 3.23). That is, generally, *sa* and *cal* NPEs in larger ANNs tended lower activation decays than *sa* and *cal* NPEs in small ANNs.

#### **3.1.2.3.6.4 Effects on Synaptic-Plasticity Parameters**

Besides activation parameters, NPEs could also differ in synaptic-plasticity parameters, which affected changes in connection weights over time. Figure 3.27 shows the mean acquisition rates in *sa* (top panel) and *cal* NPEs (bottom panel) over ISIs and CSs<sup>3</sup>. Acquisition rate determined how fast connection weights increased over time. Acquisition rates tended to increase in *sa* and decrease in *cal* NPEs across ISIs. The former tendency is approximately consistent with the increase in the number of *sa* NPEs across ISIs (see top panel of Figure 3.23), whereas the second tendency is consistent with the decrease in temporal summations in *cal* NPEs across ISIs (see bottom panel of Figure 3.25). Thus, in general, the larger the number of *sa* NPEs, the higher the *sa* acquisition rates. And, the lower the *cal* temporal summations, the lower the *cal* acquisition rates.

Also, *sa* NPEs in CS1 and CS2 ANNs tended to have similar acquisition rates (except in 2-CS1 ANNs, whose *sa* acquisition-rate parameters were substantially

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<sup>3</sup> As Figure 1.3 suggests, a weight was considered as a property of a connection from a presynaptic element to a postsynaptic NPE. Thus, when I talk about acquisition rates in *sa* NPEs, for example, I am referring to the weight-increase rates for connections from any kind of presynaptic input (i.e., input elements or other *sa* NPEs) to *sa* NPEs. The same convention applies to other kinds of NPEs. This convention arises from defining NPE connectivity in terms of convergence.

higher), whereas *cal* NPEs in CS1 ANNs tended to have substantially larger acquisition-rate parameters than *cal* NPEs in CS2 ANNs. The latter effect corresponds approximately to differences between *cal* temporal summations in CS1 and CS2 ANNs (see bottom panel of Figure 3.25) and in the number of *cal* NPEs (see bottom panel of Figure 3.23). However, the decrease observed in *cal* acquisition rates is in contrast to the increase observed in the number of *cal* NPEs across ISIs, except for ISI-16 ANNs. Thus, in general, the larger the number of *cal* NPEs, the lower the *cal* acquisition rates, at least for ISI-2, ISI-4, and ISI-8 ANNs.

Finally, Figure 3.28 shows the mean extinction rate in *sa* (top panel) and *cal* NPEs (bottom panel) over ISIs and CSs. This parameter determined how fast connection weights declined over time. Extinction rates tended to decrease in *sa* and increase in *cal* NPEs across ISIs, although the latter effect showed considerable variation. Changes in *sa* extinction rates were negatively related to those observed in *sa* acquisition rates (see top panel of Figure 3.27) and in the number of *sa* NPEs across ISIs (see top panel of Figure 3.23). Similarly, changes in *cal* extinction rates were negatively related those observed in *cal* temporal summations across ISIs (see bottom panel of Figure 3.25). Thus, the larger the number of *sa* NPEs and the higher the *sa* acquisition rates, the lower the *sa* extinction rates. And the lower the temporal summation in *cal* NPEs, the higher the *cal* extinction rates.

In general, these relations suggest a trade-off between acquisition and extinction rates in *sa* NPEs, and between temporal summation and extinction rates in *cal* NPEs. The first trade-off relation may have had the effect of compensating the decremental effects of extinction rates in larger ANNs, whose ancestors were selected under longer ISIs. Such a compensating relation may have represented a mechanism of reinforcement-delay tolerance in these ANNs. The functional effect of the second

trade-off relation is less clear. Perhaps, low temporal summations and high extinction rates in *cal* NPEs conferred a fitness advantage by reducing a larger-than-optimal range of variation of *cal*-NPE activations, produced by other parameters in other kinds of NPEs (e.g., high *sa* acquisition rates; see top panel of Figure 3.27). Again, these interpretations arise from the assumption that a larger-than-optimal range of variation of *cal* activations may have decreased overall *cal* discrepancy signals.

Also, *sa* NPEs in CS1 and CS2 ANNs tended to have similar extinction rates (except in 8-CS2 ANNs, whose *sa* extinction rates were substantially higher than those in 8-CS1 ANNs). In contrast, *cal* NPEs in CS1 ANNs tended to have larger extinction rates than *cal* NPEs in CS2 ANNs. This last effect corresponds to differences between CS1 and CS2 ANNs in the number of *cal* NPEs (see bottom panel of Figure 3.23), as well as in *cal* acquisition rates (see bottom panel of Figure 3.27) and *cal* temporal summations (see bottom panel of Figure 3.25). In contrast, changes in *cal* extinction rates related negatively to those observed in the number of *cal* NPEs across ISIs, except for the ISI-16 ANNs. Thus, in general, the larger the number of *cal* NPEs, the lower the *cal* acquisition rates. This relation may have conferred a fitness advantage by decreasing a larger-than-optimal range of variation of *cal* activations, again, under the assumption that such activations had an optimal range of variation that determined an optimal overall magnitude of *cal* discrepancy signals.

### **3.1.2.3.7 Effects on the Motor Subnetwork: *ma* and *vta* NPEs**

#### **3.1.2.3.7.1 Effects on the Number of NPEs**

Figure 3.29 shows the mean number of *ma* (top panel) and *vta* (bottom panel) NPEs across ISIs and CSs. The number of *ma* NPEs remained relatively constant at about 10, across ISIs from 2 to 8 and for both CSs, but with a substantial increase for 16-CS2. This last effect relates to the fact that these ANNs also had more *ma* layers

than any other ANN (see bottom panel of Figure 3.19). However, note that 16-CS1 ANNs tended to have more *ma* layers than 16-CS2 ANNs, although the latter had substantially more *ma* NPEs. In fact, 16-CS1 ANNs had practically the same number of *ma* NPEs as ANNs in other ISIs. These apparent inconsistencies relate to the fact that 16-CS1 ANNs tended to have substantially fewer NPEs per *ma* layer than any other ANNs. This effect can be appreciated in the 16-CS1 ANN shown in Figure 3.16. Indeed, although this ANN has two more *ma* layers, it has only three NPEs per *ma* layer, for a total of nine *ma* NPEs, which is approximately equal to (and, in some cases, less than) the overall number of *ma* NPEs in other ANNs. Finally, the bottom panel of Figure 3.29 also shows that the number of *vta* NPEs tended to remain relatively constant over ISIs and CSs.

#### 3.1.2.3.7.2 Effects on Connectivity

Regarding convergent connectivity within the motor subnetwork, Figure 3.30 shows the mean number of presynaptic inputs per *ma* (top panel) and *vta* (bottom panel) NPE, for ISIs and CSs. The number of connections per *ma* NPE remained relatively constant at about 6 for ISIs 2 to 8 and both CSs, with a substantial increase for 16-CS2. This increase suggests that 16-CS2 ANNs had considerably more *ma* NPEs, as well as more connections per *ma* than any other ANNs by the end of evolution. This effect is in contrast to that observed in the sensory subnetwork, in which the amount of *sa* convergent connectivity remained relatively constant across ISIs and CSs (see top panel of Figure 3.24), despite the fact that the number of *sa* NPEs changed substantially across ISIs (see top panel of Figure 3.23).

Also, the number of connections per *vta* NPE remained relatively constant across ISIs and CSs at about 3 (see bottom panel of Figure 3.30), which is substantially smaller than the number of presynaptic connections per *cal* NPE (which was about 9;

see bottom panel of Figure 3.24). This difference indicates that synaptic competition among *ma* NPEs for *via* NPEs was less substantial than synaptic competition among *sa* NPEs for *cal* NPEs, which may have compensated for the fewer number of *ma* NPEs, and hence increased the overall magnitude of *via* discrepancies.

An examination of the example of a 16-CS2 ANN shown in Figure 3.16, and the number of inhibitory elements in 16-CS2 ANNs (see bottom panel of Figure 3.20), suggests that the increase in *ma* convergent connectivity in these ANNs may relate to an increase in the number of inhibitory connections in the motor subnetwork. Indeed, the 16-CS2 ANN shown in Figure 3.16 does not seem to have a richer feedforward connectivity within the *ma* subnetwork than other ANNs. However, the increase in the number of *ma* inhibitory NPEs in these ANNs is apparent. Such an increase might have been sufficient to increase substantially the amount of *ma* convergent connectivity in these ANNs.

### 3.1.2.3.7.3 Effects on Activation Parameters

Figure 3.31 shows the mean temporal summation in *ma* (top panel) and *via* NPEs (bottom panel), across ISIs and CSs. Temporal summation in *ma* NPEs decreased steadily over ISIs and remained relatively constant from CS to the other. Note that *ma* temporal summation decreased faster for CS2 ANNs, ISI-16 ANNs showing the lowest *ma* temporal summations. In the latter ANNs, low *ma* temporal summations corresponded to a high number of *ma* NPEs and a high convergent connectivity (see top panels of Figures 3.29 and 3.30, respectively).

A reexamination of previous effects shows that *sa* temporal summations (see top panel of Figure 3.25) as well as *sa* connectivity (see top panel of Figure 3.24) remained relatively constant, despite the fact that the number of *sa* NPEs changed substantially across ISIs (see top panel of Figure 3.23). These interrelations, when compared to the

corresponding ones observed in *ma* NPEs, suggest that temporal summation may have been more related to connectivity than to the number of NPEs. Perhaps, variations in the number of NPEs tended to magnify certain detrimental effects of temporal summation (e.g., increase in the range of variation of NPE activations above the optimal range), which in turn may have decreased fitness.

In the case of *vta* NPEs, temporal summation remained relatively constant across ISIs in CS1 ANNs. However, substantial differences were observed over ISIs for CS2. First, *vta* temporal summation increased rapidly as the ISI increased from 2 to 4, and then, decreased rapidly as the ISI increased from 4 to 16. This pattern seems to relate positively to that observed in the *vta* connectivity (see bottom panel of Figure 3.30). Indeed, the number of presynaptic inputs per *vta* NPE increased as the ISI increased from 2 to 4. Then, this number decreased as the ISI increased from 4 to 16. 4-CS2 ANNs had the highest *vta* temporal summation, as well as the largest number of presynaptic connections per *vta* NPE, suggesting, again, that temporal summation may have been more related to connectivity than to the number of NPEs.

Figure 3.32 shows the mean activation decays in *ma* (top panel) and *vta* NPEs (bottom panel) over ISIs and CSs. Overall, *ma*-activation decays showed no consistent trend across ISIs, although *ma*-activation decays in CS2 ANNs tended to be larger than that those observed in CS1 ANNs, except for 8-CS1 ANNs. In contrast, *vta*-activation decay tended to remain relatively constant across CSs, although varying somewhat across ISIs. In general, for CS2 ANNs, *ma*- and *vta*-activation decays tended to change in the same direction, suggesting that adaptedness in these ANNs may have required similar *ma*- and *vta*-activation decays. In contrast, *ma*- and *vta*-activation decays for CS1 ANNs changed in opposite directions, suggesting that adaptedness in these ANNs may have required very different *ma*- and *vta*-activation decays. The

increase in the *vta*-activation decay for 16-CS2 ANNs may also relate to that observed in the number of *ma* NPEs and presynaptic inputs per *ma* NPE for the same ANNs (see to panels of Figures 3.29 and 3.30, respectively).

#### 3.1.2.3.7.4 Effects on Synaptic-Plasticity Parameters

Regarding synaptic-plasticity parameters within the motor subnetwork, Figure 3.33 shows the mean acquisition rates in *ma* (top panel) and *vta* NPEs (bottom panel) over ISIs and CSs. Acquisition rates tended to decrease in *ma* and increase in *vta* NPEs across ISIs, specially for CS2 ANNs. The former effect was positively related to that observed in *ma* temporal summation (see top panel of Figure 3.31), suggesting that less *ma* temporal summation may have required less *ma* acquisition rate.

A comparative reexamination of previous effects shows that, overall, changes in acquisition rates within the motor subnetwork tended to be negatively related to those observed within the sensory subnetwork. Indeed, while *ma* acquisition rates tended to decrease, *sa* acquisition rates tended to increase across ISIs (compare top panels of Figures 3.33 and 3.27, respectively). However, while *vta* acquisition rates tended to increase, *cal* acquisition rates tended to decrease across ISIs (compare bottom panels of Figures 3.27 and 3.33). These relations suggest a trade-off between acquisition rates in the sensory subnetwork and acquisition rates in the motor subnetwork. Such a trade-off may have conferred a fitness advantage by allowing variations in *cal* and *vta* activations to approach an optimal range.

Figure 3.34 shows the mean extinction rates in *ma* (top panel) and *vta* NPEs (bottom panel) over ISIs and CSs. Extinction rates remained quite constant in *ma* NPEs, at values close to 0.0. This effect is in sharp contrast to those observed in all other NC parameters, and may have been related to the number of *ma* layers. Indeed, most ANNs had only one *ma* layer, which connected directly to the output layer, an



architectural feature that may have magnified the detrimental effects of *ma* extinction rates on output-NPE activations, and hence on fitness. ANNs with high *ma* extinction rates may have had low fitness, and hence may have been selected against. In this sense, then, ANNs tended to have low extinction rates maybe because they tended to have only one *ma* layer. The issue, then, is why these ANNs tended to have only one *ma* layer. I shall address this issue and offer a possible interpretation in the last chapter.

Finally, *vta* extinction rates in CS1 ANNs remained quite constant at about 0.5 across ISIs, whereas substantial (although unsystematic) differences were observed in CS2 ANNs. Indeed, *vta* extinction rates in these ANNs decreased substantially as the ISI increased from 2 to 4, then increased more substantially as the ISI increased from 4 to 8, and decreased again as the ISI increased from 8 to 16. A similar pattern was observed in the same ANNs for *vta* acquisition rates (see bottom panel of Figure 3.33), whereas the opposite pattern was observed for *vta* and *ma* decay rates (see bottom panel of Figure 3.32).

### 3.1.2.3.8 Effects on the Motor Subnetwork: Output NPEs

To conclude the presentation of the effects on structural phenotypes, I will discuss the effects upon output NPEs, whose activation determined most directly the behavior (and hence fitness) of ANNs. Figure 3.35 shows that, overall, the mean number of output NPEs tended to remain relative constant across ISI, but changed in opposite directions for CS1 and CS2 ANNs. Indeed, the number of output NPEs tended to change as an inverted U-shaped function of ISI for CS1 and as a U-shaped function for CS2 ANNs. 2-CS2 ANNs had the highest number of output NPEs, followed by 8-CS1 and 4-CS1. 4-CS2 ANNs had the smallest number of output NPEs. For CS2 ANNs, these changes observed were positively related to those observed in the number of  $s_2$

input elements (see bottom panel of Figure 3.21), but negatively related to those observed in *via* temporal summations (see bottom panel of Figure 3.31).

Figure 3.36 shows that the number of presynaptic inputs per output NPE decreased across ISIs, for both CSs. This effect tended to be negatively related to that observed in the number of *ma* NPEs and number of presynaptic inputs per *ma* NPE, which suggests that adaptedness may have required a trade-off between the number of *ma* NPEs, *ma* convergent connectivity, and synaptic competition at the level of output NPEs. A functional effect of these relations may have been to reduce the amount of synaptic competition among *ma* NPEs for the output NPEs, in ANNs that evolved under longer ISIs. This reduction may have provided a mechanism for reinforcement-delay tolerance in these ANNs.

Figure 3.37 shows that the mean temporal summation in output NPEs changed as a U-shaped function of the ISI, for both CSs, remaining at values equal to or above 0.5. For CS1 ANNs, this function was negatively related to the number of output NPEs (see Figure 3.35). Thus, a trade-off seems to have taken place between the number of output NPEs and output-NPE temporal summations in CS1 ANNs. Perhaps, the detrimental effects of high temporal summations under CS1 (e.g., an increase in the range of variation of NPE activations, due to the presence of high activation decays) were magnified either by an increase or a decrease in both the number of output NPEs *and* temporal summations. In this way, ANNs with few output NPEs and high temporal summations, as well as ANNs with many output NPEs and low temporal summations may have been favored.

Figure 3.38 shows that the mean activation decays in output NPEs also changed as a U-shaped function of ISIs for CS1 ANNs, but tended to decrease unsystematically across ISIs for CS2 ANNs. The first function is positively related to *sa*- and *cal*-

activation decays (see Figure 3.26), and to output-NPE temporal summations (see Figure 3.37). In general, these relations suggest that, for CS1 ANNs, having similar *sa*, *cal*, and output decays either had no substantially detrimental effects on fitness (and hence, were not selected against), or conferred a fitness advantage by compensating for the detrimental effects of variations in other parameters. A similar interpretation may apply to the positive relations between changes in output-NPE activation decays, *cal* temporal summations (see bottom panel of Figure 3.26), *vta* acquisition rates (see bottom panel of Figure 3.33), and *vta* extinction rates (see bottom panel of Figure 3.34) in CS2 ANNs.

Figure 3.39 shows that output-NPE acquisition rates in output NPEs changed as a U-shaped function of ISIs, for both CSs. This pattern is similar to that observed in output-NPE temporal summations (see Figure 3.37), and hence, to those observed in *sa*-, *cal*- (see Figure 3.26), and output-NPE activation-decay rates (see Figure 3.38). In the case of CS2 ANNs, changes in output-NPE acquisition rates across ISIs related positively to those observed in the number of *ma* NPEs (see top panel of Figure 3.29) and presynaptic inputs per *ma* NPE (see top panel of Figure 3.30).

Finally, Figure 3.40 shows that output-NPE extinction rates decreased across ISIs for CS1 ANNs, but changed unsystematically for CS2 ANNs. The first effect was positively related to the number of presynaptic inputs per output NPE (see Figure 3.26). The second effect was positively related to those observed in *ma*- and *vta*-activation decays (see Figure 3.32), but negatively to those observed in output-NPE activation decays (see Figure 3.38). In fact, output NPEs in 16-CS2 ANNs had the highest extinction and the lowest decay rates. Also, these ANNs tended to have very low *vta* temporal summations, but very high output-NPE extinction rates. Overall,

these extreme differences suggest that the 16-CS2 condition may have imposed selective pressures toward major parametric trade-offs at the level of output NPEs.

## 3.2 Test Simulations

In the previous sections I showed that ANNs that evolved under different environmental conditions were functionally equivalent (although structurally different) under the *same* environmental conditions under which ancestral selection took place. This result raises the issue of how the same ANNs would behave under environmental conditions different from those under which ancestral selection occurred. The test simulations addressed this issue. These simulations determined if structurally different ANNs were also functionally different under environmental conditions that were changed from the ancestral ones.

### 3.2.1 Procedure

A series of simulations were run in which the *same* ANNs described in Section 3.1.2.3 were exposed to environmental conditions different from those under which ancestral selection occurred. The data-storage routines of the NC-GA simulator were designed so that two versions of all ANNs were always stored, namely *naïve* and *trained*. These versions represented two different times in an ANNs life, namely *before* and *after* training, respectively. Both versions had exactly the same structural traits (architectural as well as parametric). However, connection weights in the naïve versions were identical to the initial connection weights after development and before training, whereas connection weights in the trained versions were identical to the final connection weights, after learning. Each test simulation started with the naïve versions.

The first test simulation involved a CS-alone procedure. The purpose of this simulation was to determine the extent of the Baldwin Effect (i.e., the extent to which high responding to the CS in those ANNs did not require external reinforcement). In this simulation, ANNs from the last generation of each lineage were given 125 presentations of CS-alone trials (i.e., without activating the *us* input element at any ts), the last 25 of which were used as evaluation trials to determine CR proportion. ANNs were given the same CS type and duration experienced by their ancestors.

The rest of the test simulations involved exposing each ANN from the last generation of each lineage to *each one* of the environmental conditions used in the evolution simulations. All ANNs were given 125 trials of each condition, the last 25 of each were used as evaluation trials to compute CR proportion. The purpose of these simulations was to determine if ANNs from different populations were functionally different when exposed to environmental conditions different from those under which ancestral selection occurred. All the results are expressed in terms of changes in the mean CR proportion across  $\log_2$ ISI and CSs.

## 3.2.2 Results and Discussion

### 3.2.2.1 CS-Alone Tests and the Baldwin Effect

Figure 3.41 shows the results of the CS-alone test. ISI-2 ANNs showed very high responding to nonreinforced CSs. This result is consistent with those shown Figures 3.7 and 3.8, which indicated that most of these ANNs responded with high output-NPE activations from the beginning of training. The result is also consistent with that shown in Figure 3.15, which indicated that the connection weights in these ANNs were very similar before and after training. These consistencies suggest that ISI-2 ANNs had initial connection weights that were sufficiently high to allow for high responding to

CSs without external reinforcement. This finding can be interpreted in Pavlovian-conditioning terminology by saying that evolution under ISI 2 resulted in ANNs that mediated *unconditional reflexes*.

However, responding without reinforcement decreased rapidly as the ISI increased, (down to 0.0 for ISI-8 and ISI-16 ANNs). This last effect is consistent with those observed in Figures 3.11 through 3.14, which indicated that most of ISI-8 and ISI-16 ANNs tended to respond with high output-NPE activations only after a number of CS-US pairings. The results are also consistent with those shown in Figure 3.15, which indicated that the connection weights in those ANNs increased substantially after learning. Together, these findings suggest that those ANNs had initial connection weights that were not sufficiently high to allow for high responding to CSs without external reinforcement.

Overall, these results show that the Baldwin Effect, as instantiated in the present simulations, depended upon how CSs were temporally related to the US during ancestral selection. ANNs whose ancestors were selected under shorter ISIs (i.e., 2 and 4) required relatively little or no external reinforcement for increased responding to CS-alone trials. In contrast, ANNs whose ancestors were selected under longer ISIs (i.e., 8 and 16) showed CR proportions close to 0.0 in the absence of external reinforcement.

The fact that initial connection weights were not directly encoded in the genotype may have contributed to these effects. Indeed, because initial connection weights were determined only indirectly by the genotype, evolution in the present simulations did not involve a search in weight space. Rather, evolution involved a search in architecture and parameter space. One of the parameters determined NPE spontaneous activations, which in turn determined initial connection weights. One could argue that ANNs

evolved under ISIs 2 and 4 showed high responding to CS-alone trials because they had larger initial connection weights than ANNs that evolved under ISIs 8 and 16. However, as Figure 3.15 (top panel) shows, ANNs that evolved under different ISIs had similar initial connection weights. Therefore, similar initial weights must have played different roles in different structural phenotypes, thus giving rise to the functional differences observed in the CS-alone test. For example, similar initial connection weights may have been sufficiently high to allow for high responding to CS-alone trials in small ANNs, but too low to allow for such responding in large ANNs.

### **3.2.2.2 ISI and CS Tests: ISI Functions, optimal-ISI Noninvariance, and CS Nonequipotentiality**

Figure 3.42 shows the ISI and CS tests of 2-CS1 (top panel) and 2-CS2 ANNs (bottom panel). The arrows indicate the condition under which ancestral selection occurred for each lineage. In general, the highest proportion of CRs occurred precisely under that condition. Indeed, overall, the highest proportion CRs were observed under ISI 2 (i.e., 2 ts was an optimal ISI), except for 2-CS2 ANNs, which responded with slightly higher CR proportions under an ISI of 4 ts. Also, for 2-CS1 ANNs, responding to CS1 was substantially higher than responding to CS2, which is consistent with the fact that most of these ANNs had no  $s_2$  input elements (see Figure 3.21, bottom panel). In contrast, for 2-CS2 ANNs, responding to CS2 was substantially higher than responding to CS1, which is consistent with the fact that most of these ANNs had no  $s_1$  input elements (see Figure 3.21, top panel). Hence, CS1 and CS2 were nonequipotential for ISI-2 ANNs.

Figure 3.43 shows the ISI and CS tests of 4-CS1 (top panel) and 4-CS2 ANNs (bottom panel). The arrows indicate the conditions under which ancestral selection

took place. Responding in these ANNs changed as an inverted U-shaped function of ISIs, resembling an ISI function whose peak coincided with 4 ts. Responding was substantially higher under ISI 8 than under ISI 2, indicating that the closest longer ISI was more effective than the closest shorter one. Also, responding to CS2 in 4-CS1 ANNs was close to zero, which is consistent with the fact that most of these ANNs had no  $s_2$  input elements (see Figure 3.21, bottom panel). In contrast, 4-CS2 ANNs responded very similarly to CS1 and CS2, although slightly higher to the latter. Therefore, CS1 and CS2 were nonequipotential for 4-CS1 ANNs, but equipotential for 4-CS2 ANNs.

Figure 3.44 shows the ISI and CS tests of 8-CS1 (top panel) and 8-CS2 ANNs (bottom panel). The arrows indicate the conditions under which ancestral selection took place. Again, responding in these ANNs changed as an inverted U-shaped function of ISIs, resembling an ISI function whose peak coincided with 8 ts (i.e., 8 ts was the optimal ISI for these ANNs). Also, these ANNs responded very similarly to CS1 and CS2, although slightly higher to the ancestral CS, indicating that CS1 and CS2 were approximately equipotential for both 8-CS1 and 8-CS2 ANNs.

Finally, Figure 3.45 shows the ISI and CS tests of 16-CS1 (top panel) and 16-CS2 ANNs (bottom panel). The arrow indicates the condition under which ancestral selection occurred for each lineage. Responding in 16-CS1 ANNs changed as an inverted U-shaped function of ISIs, resembling an ISI function whose peaks coincided with ISIs 8 and 16 (i.e., 8 and 16 ts were optimal ISIs in these ANNs). Also, for these ANNs, responding to CS1 was substantially higher than responding to CS2, indicating that CS1 and CS2 were nonequipotential. In contrast, 16-CS2 ANNs responded with very high CR proportions under 4, 8, and 16, under both CSs, indicating CS1 and CS2 stimuli were equipotential for these ANNs.



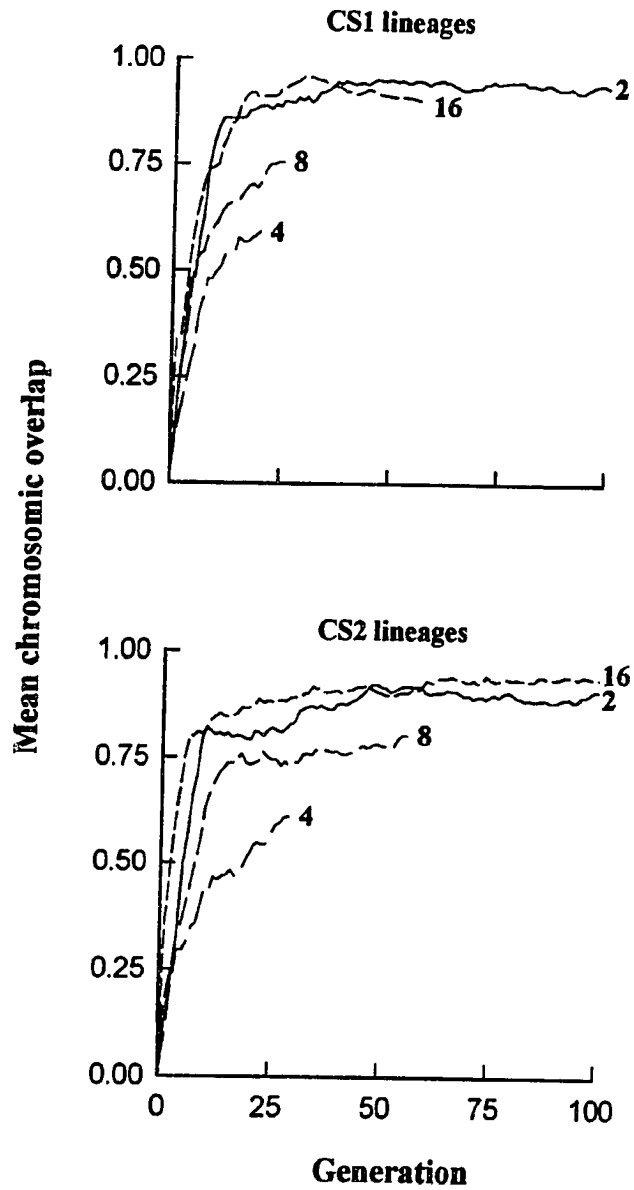


Figure 3.1: Mean chromosomal overlap as a function of generations.

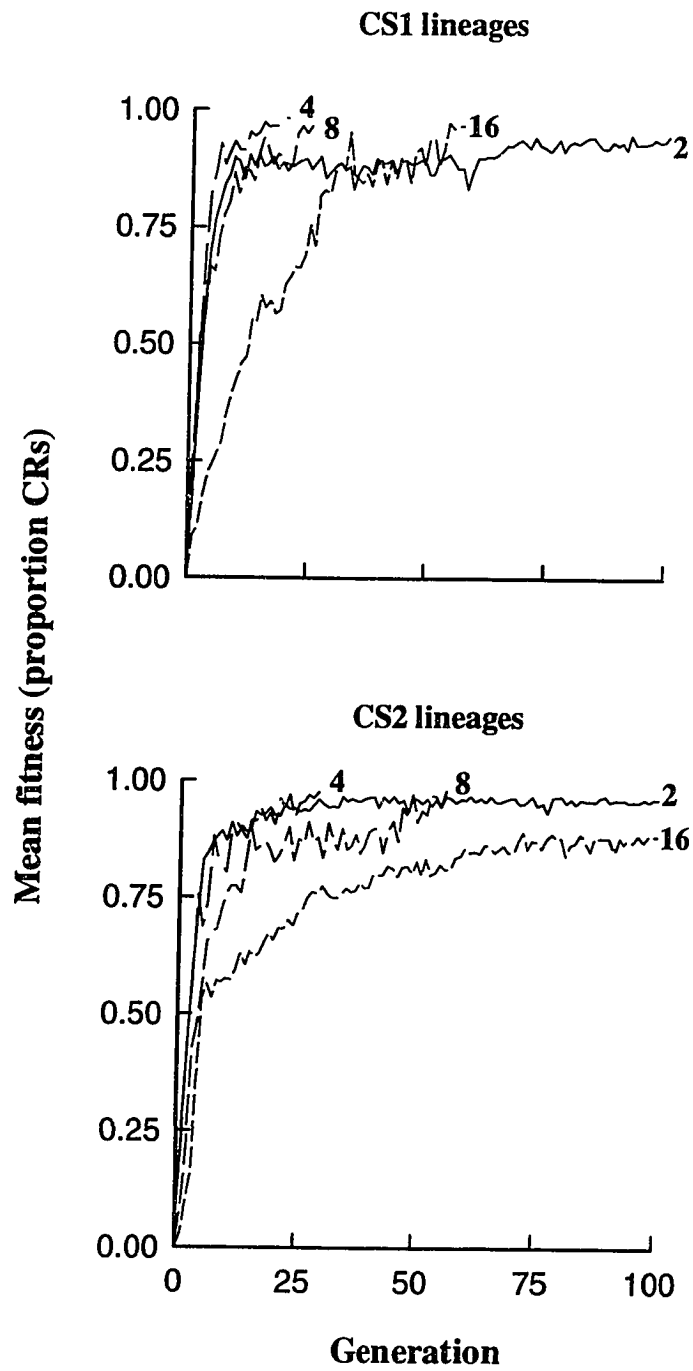
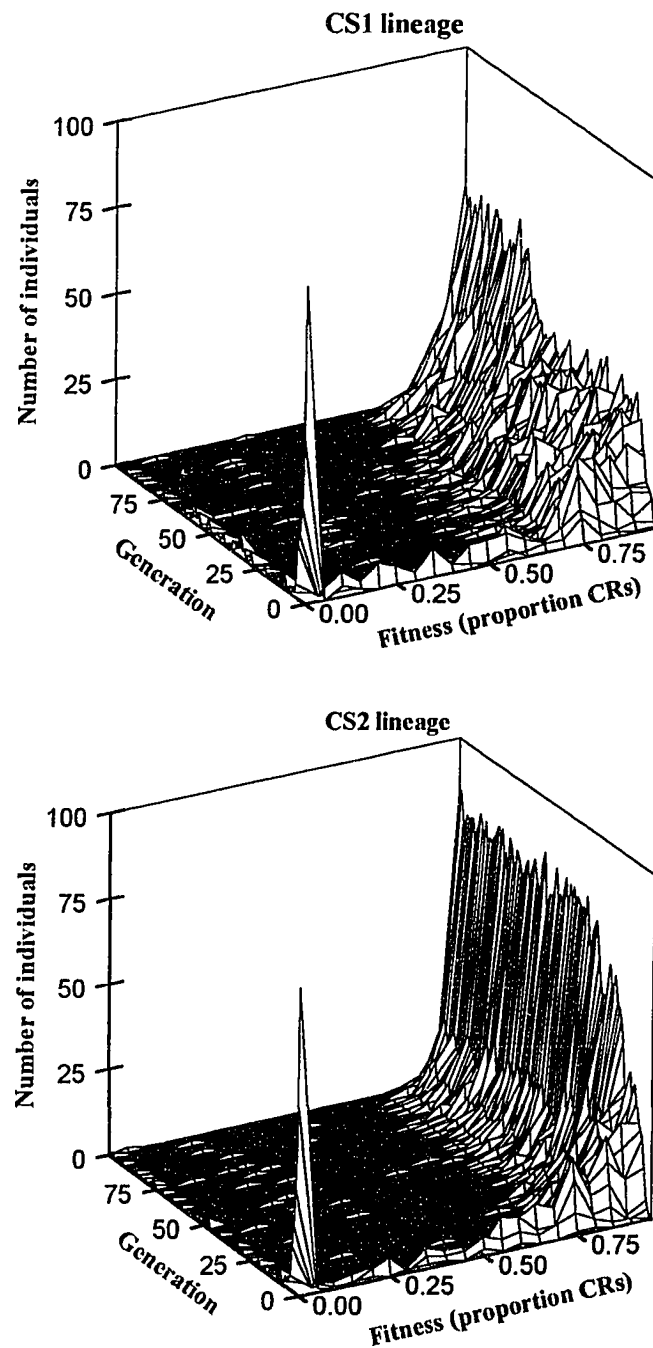


Figure 3.2: Mean population fitness as a function of generations.



**Figure 3.3: Frequency distributions of fitness values over generations, for ISI-2 lineages.**

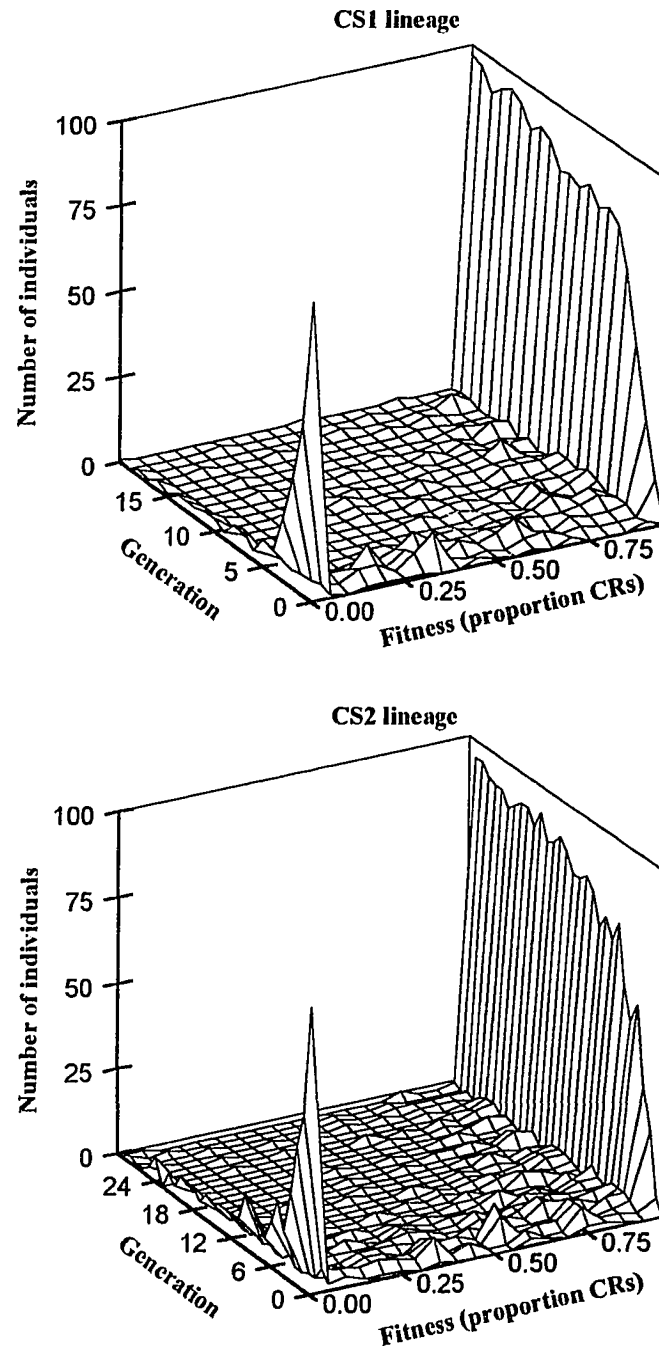
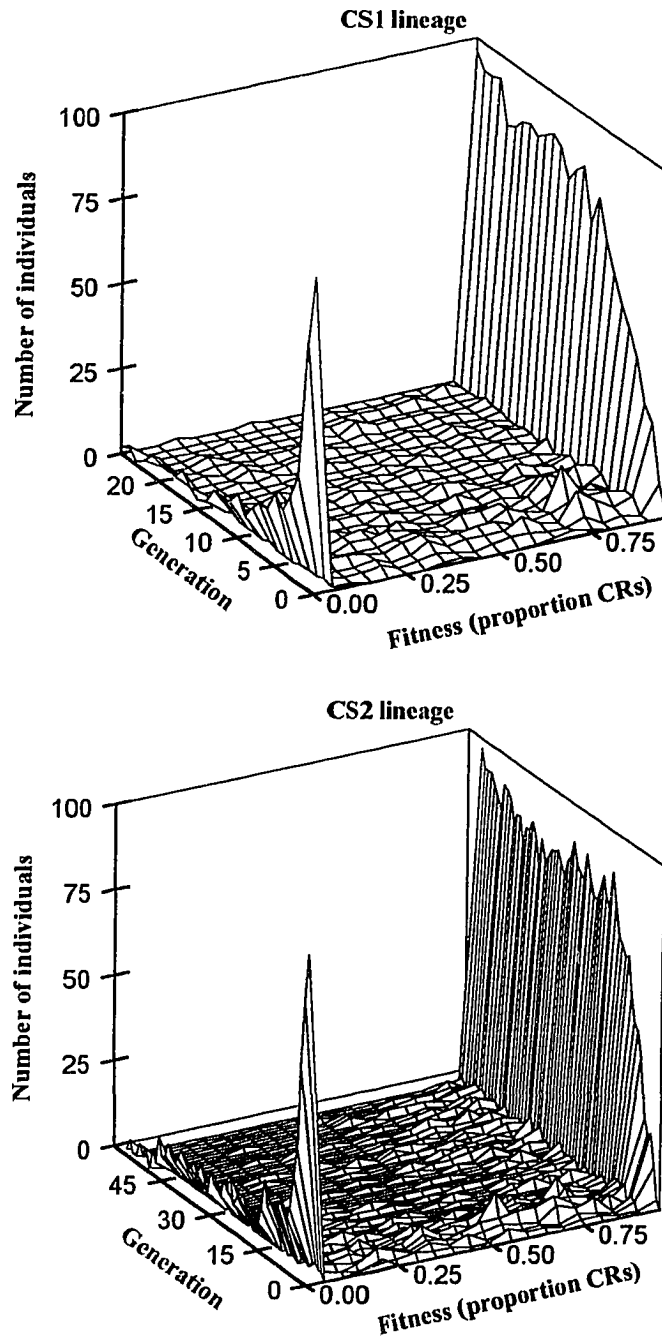


Figure 3.4: Frequency distributions of fitness values over generations, for ISI-4 lineages.



**Figure 3.5: Frequency distributions of fitness values over generations, for ISI-8 lineages.**

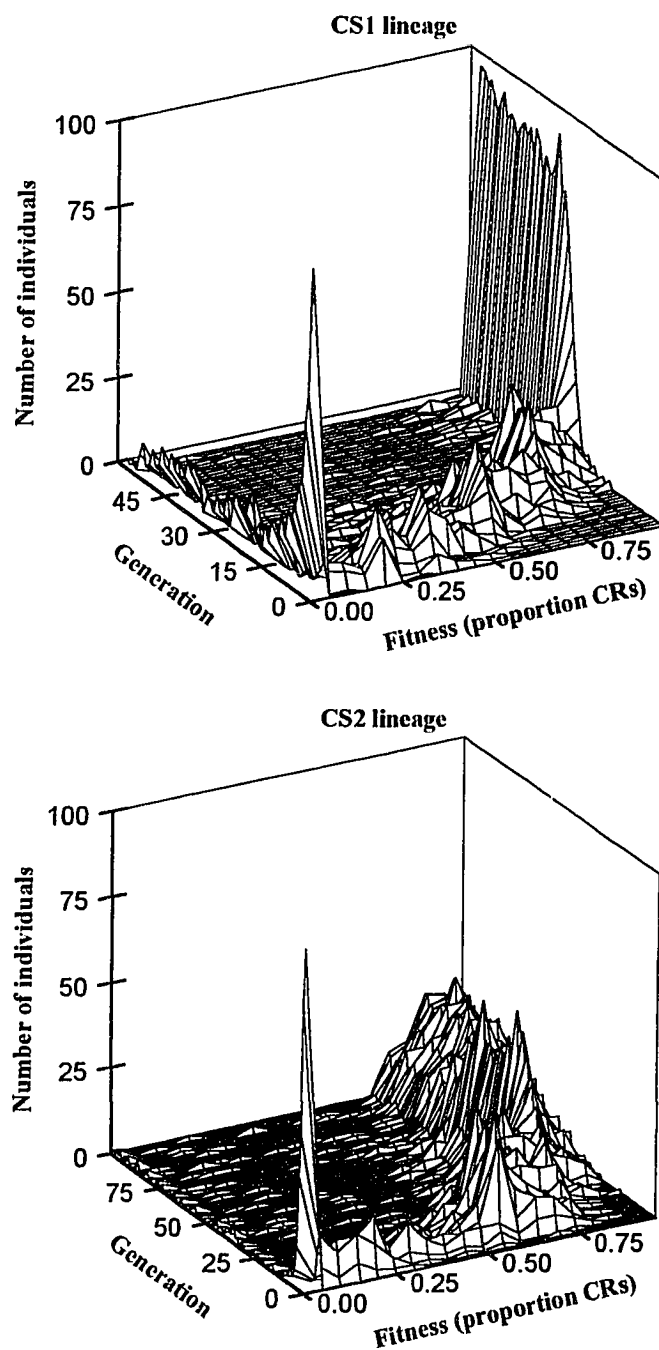
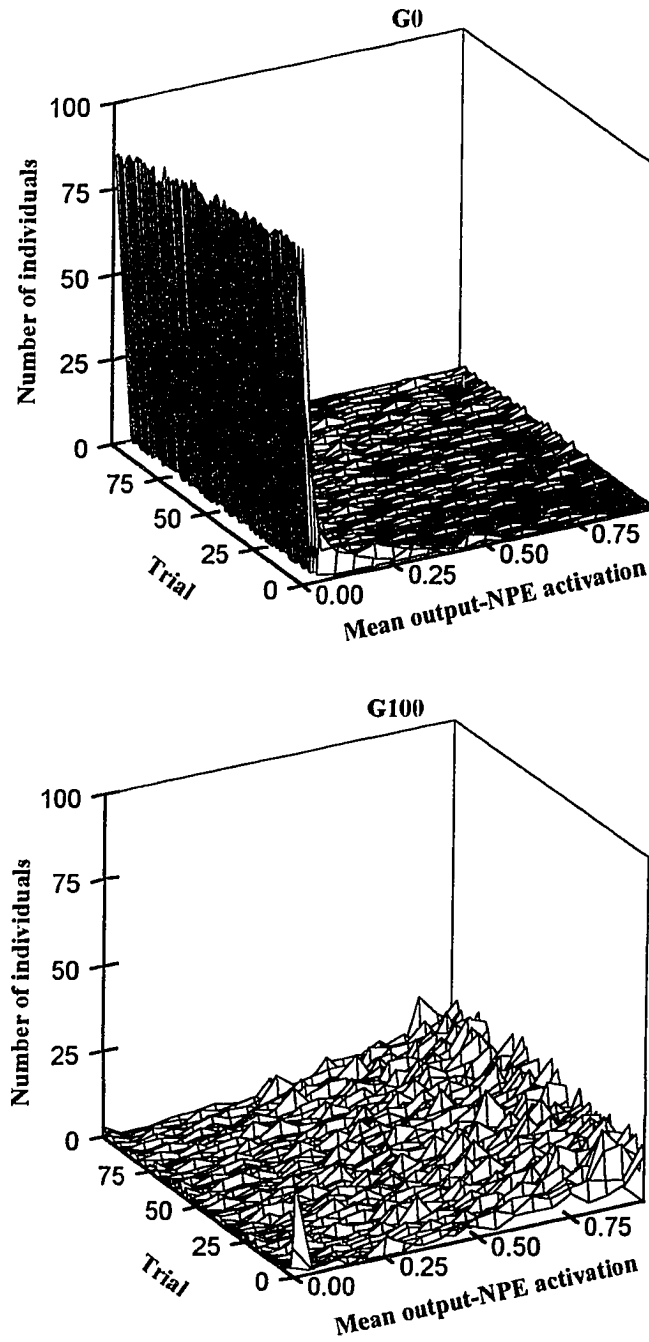
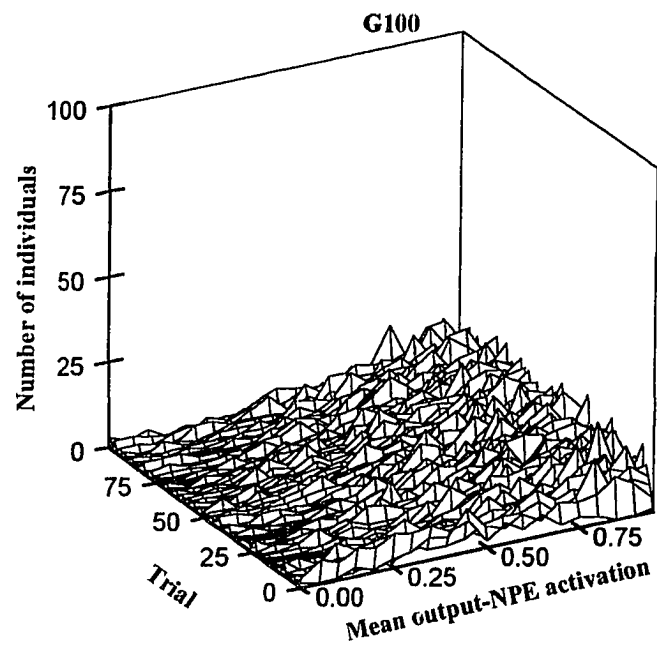
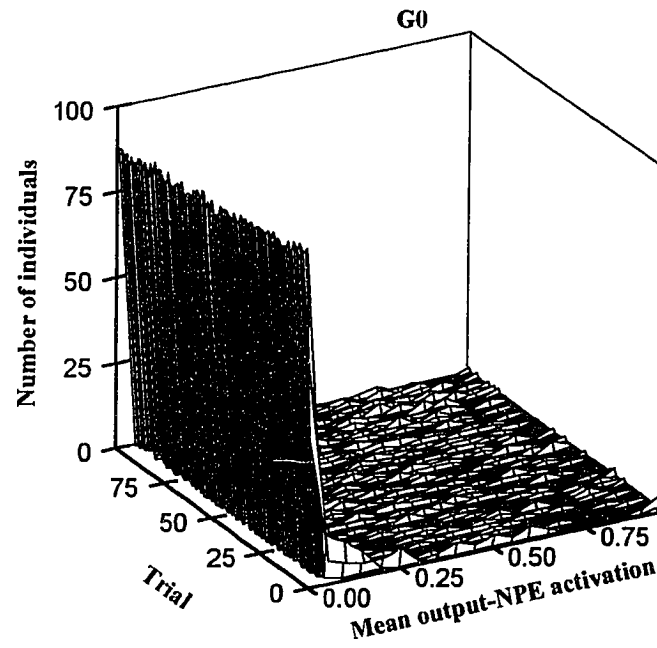


Figure 3.6: Frequency distributions of fitness values over generations, for ISI-16 lineages.



**Figure 3.7: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 2-CS1.**



**Figure 3.8: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 2-CS2.**



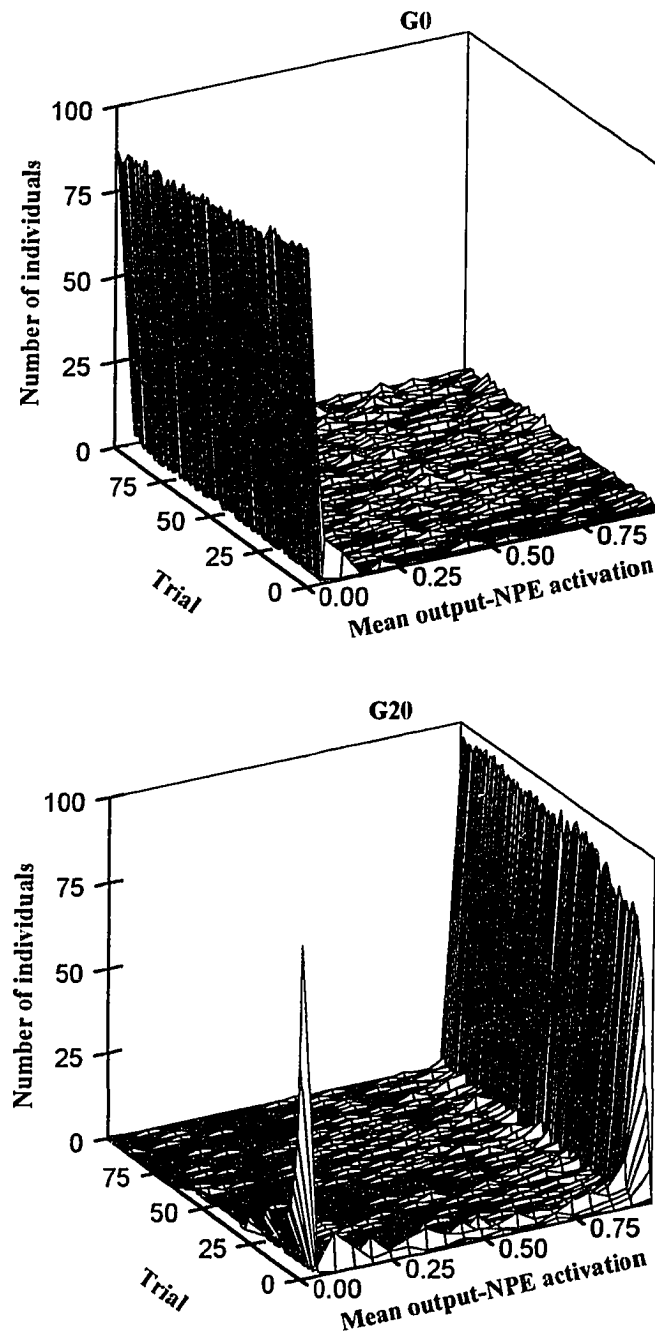
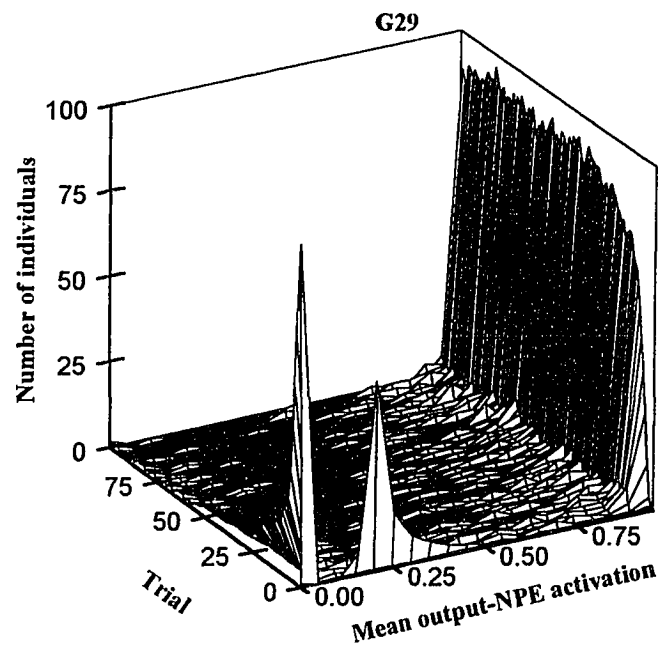
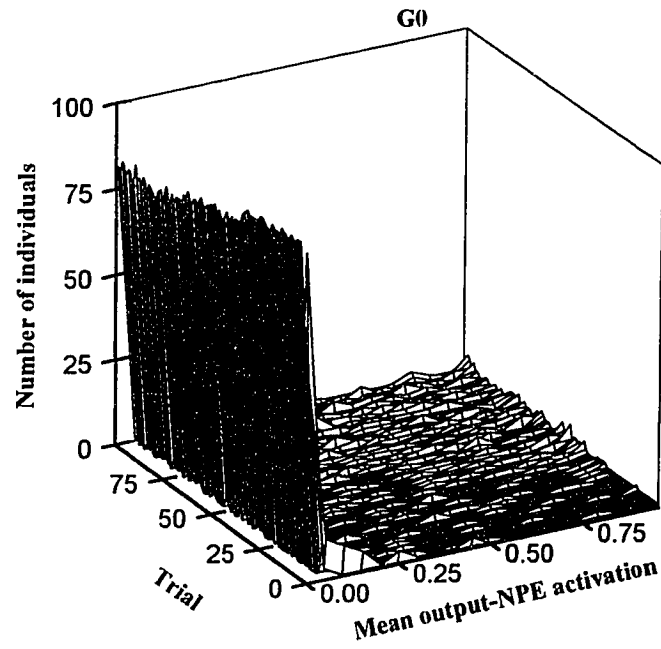
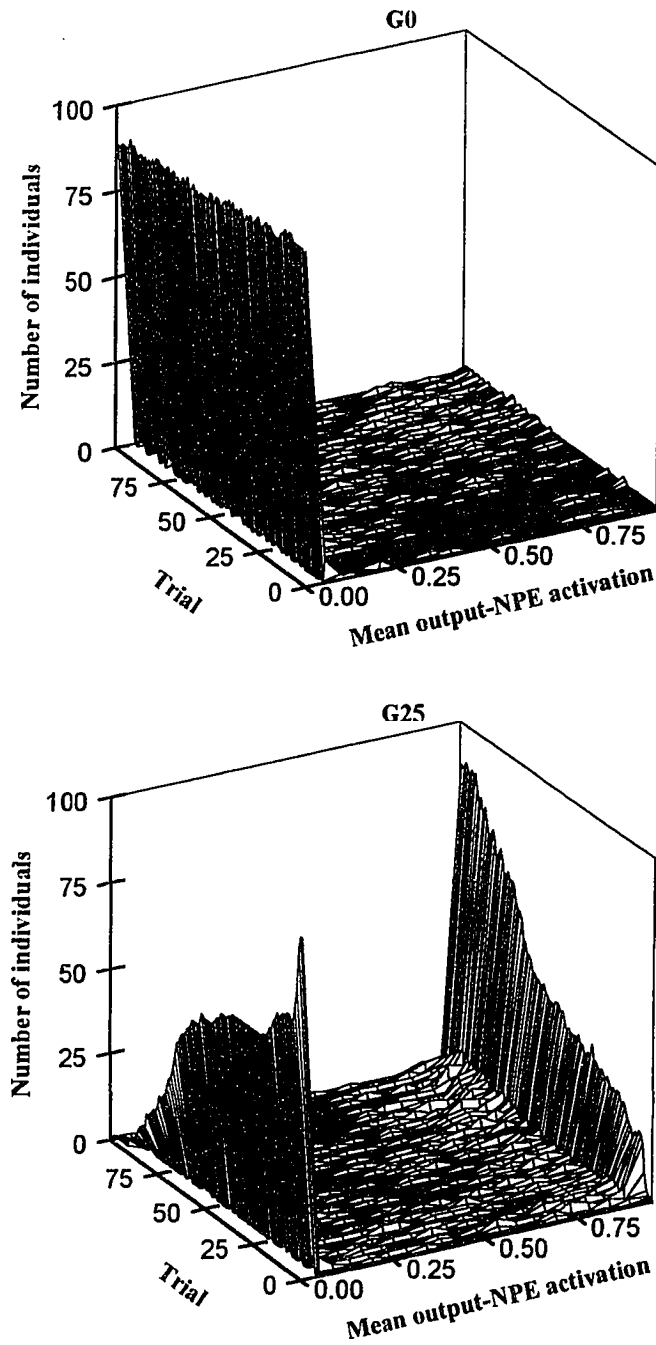


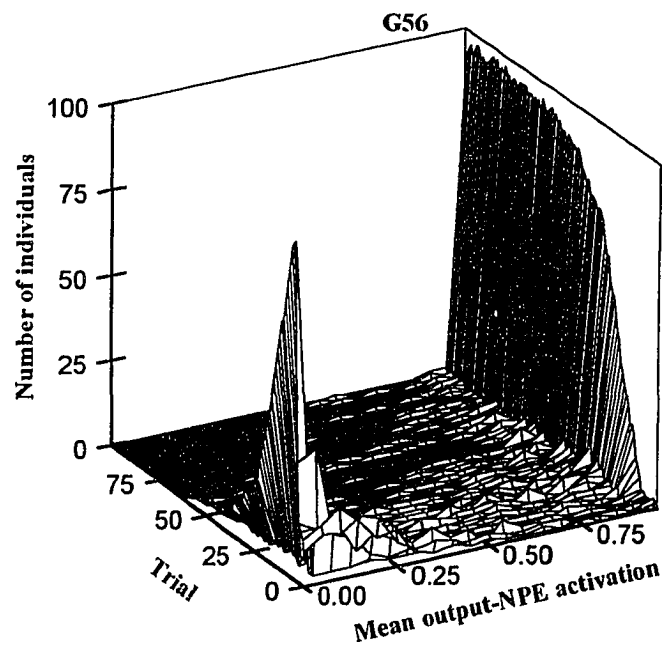
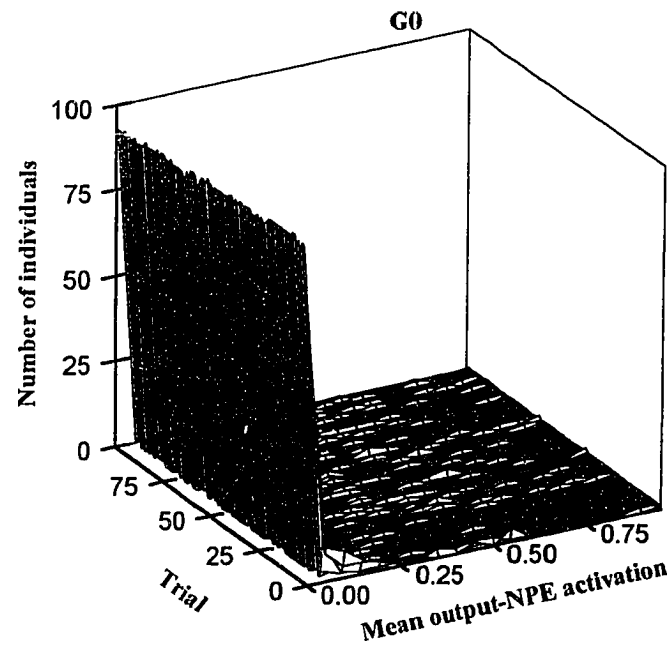
Figure 3.9: **Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 4-CS1.**



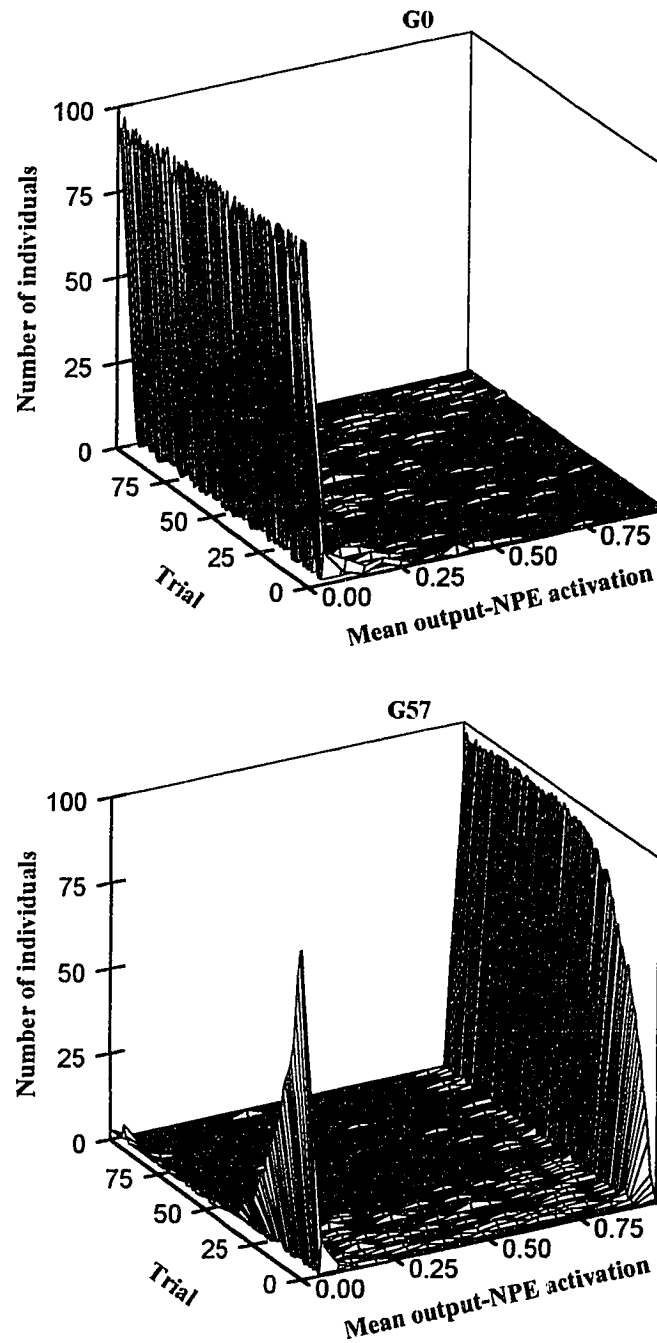
**Figure 3.10: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 4-CS2.**



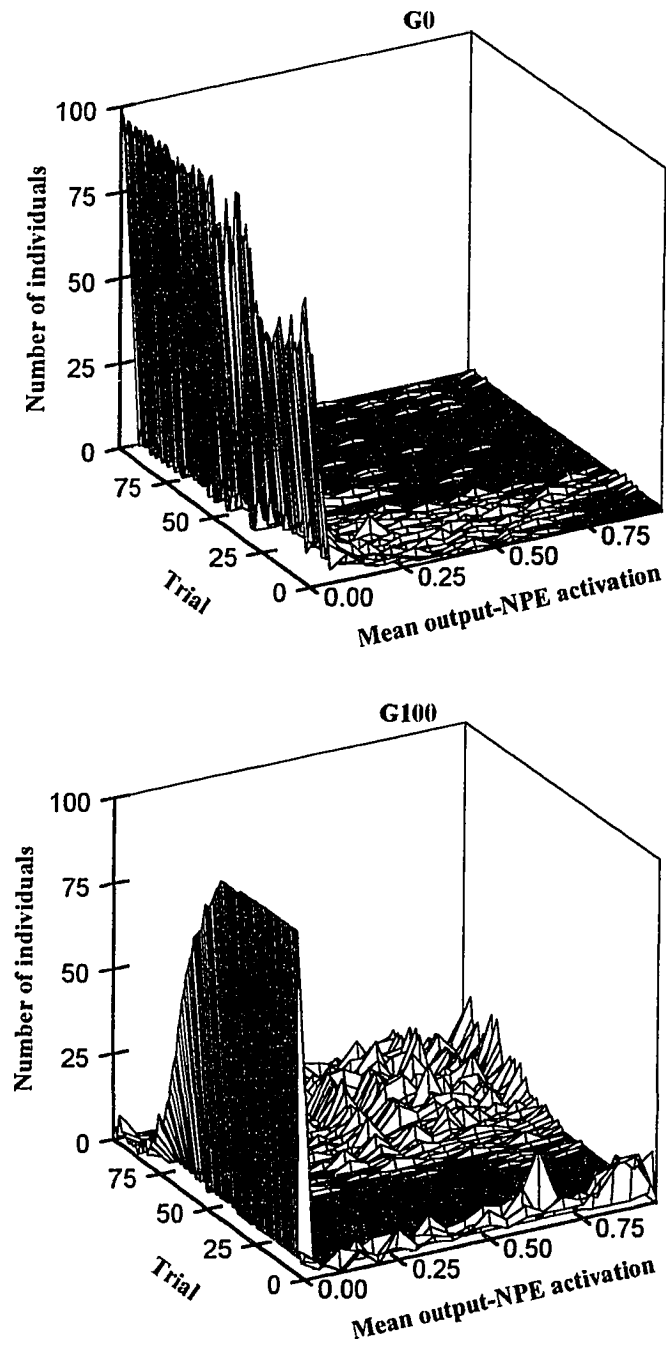
**Figure 3.11: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 8-CS1.**



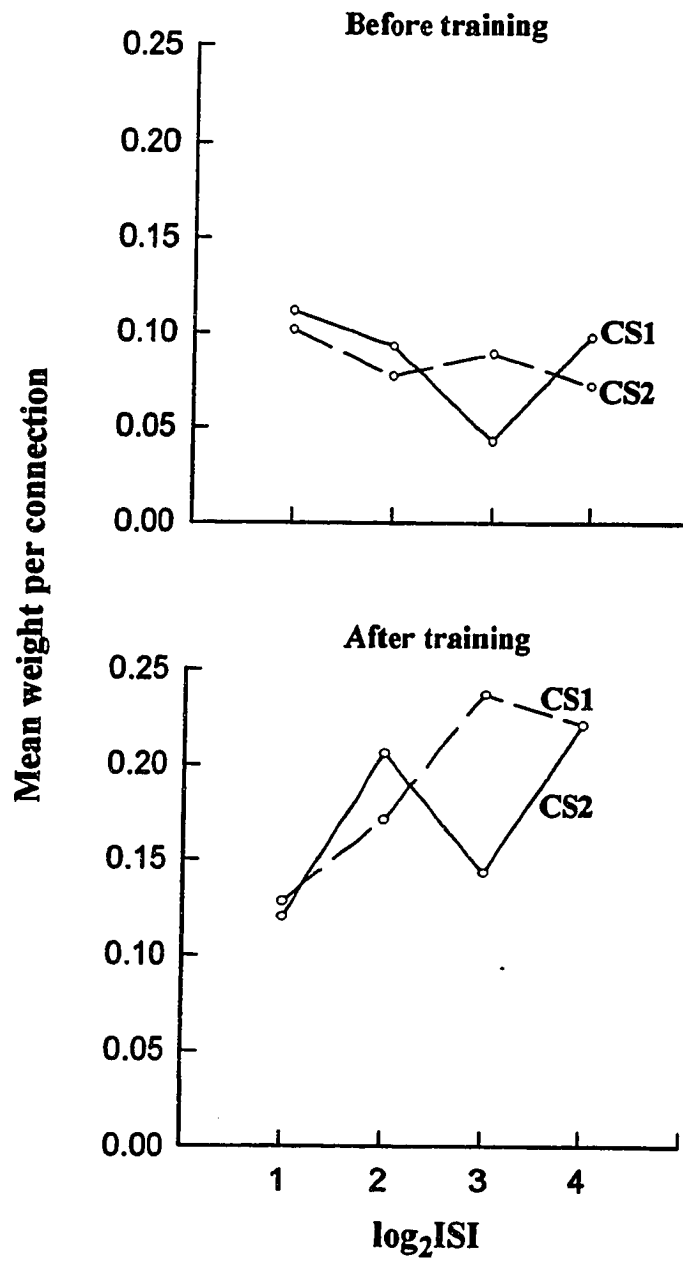
**Figure 3.12: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 8-CS2.**



**Figure 3.13: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 16-CS1.**



**Figure 3.14: Frequency distributions of mean output-NPE activations over trials, for the first and last generations of 16-CS2.**



**Figure 3.15: Mean weight per connection in last generation of each lineage, before and after training.**

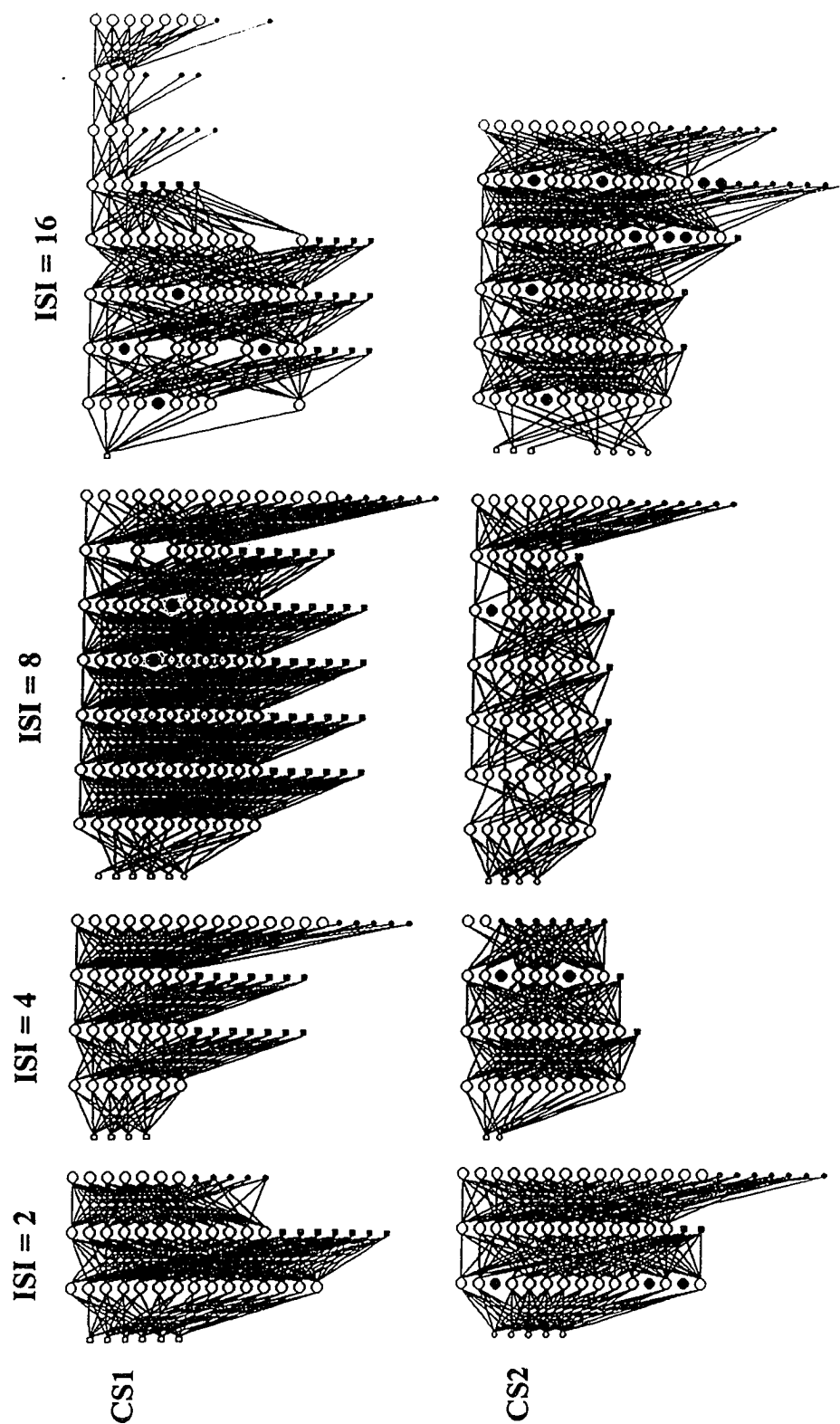
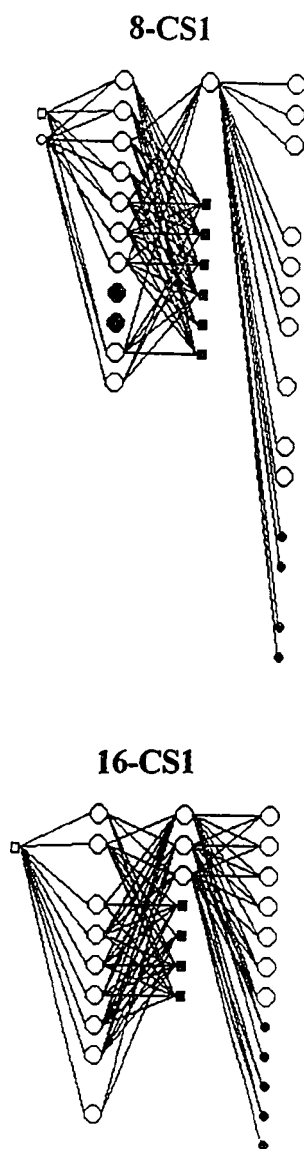


Figure 3.16: Sample of high-fitness ANNs from the last generation of each lineage. The fitness of all ANNs was above 0.95.





**Figure 3.17: Sample of high-fitness ANNs from the last generation of 8-CS1 and 16-CS1.** The fitness of both ANNs was 1.00.

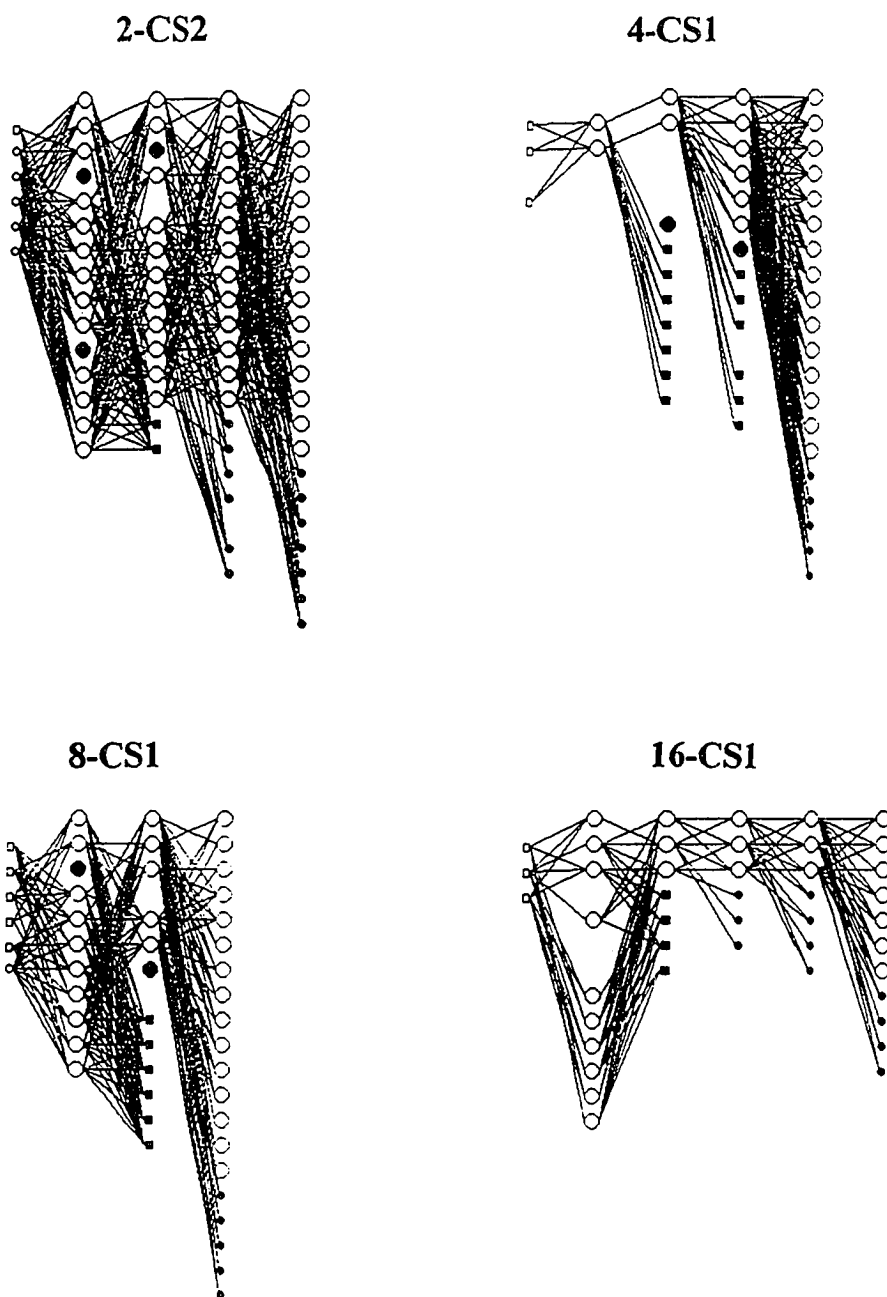


Figure 3.18: **Sample of low-fitness ANNs.** The fitness of all ANNs was below 0.2

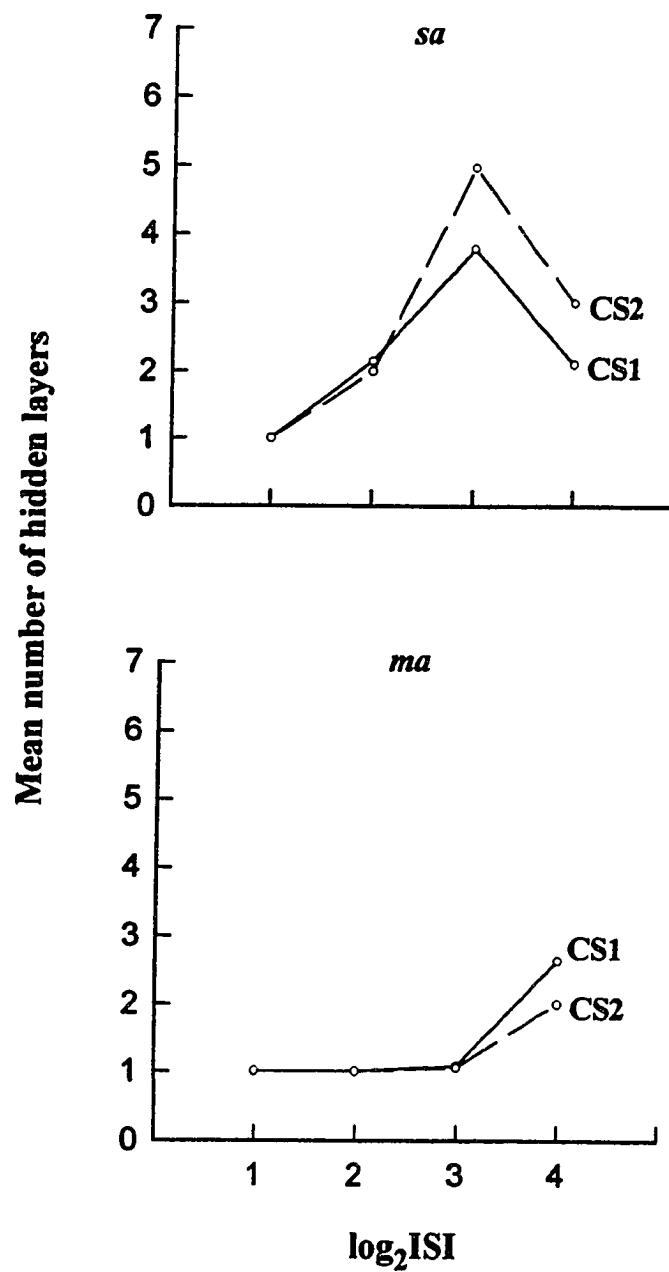
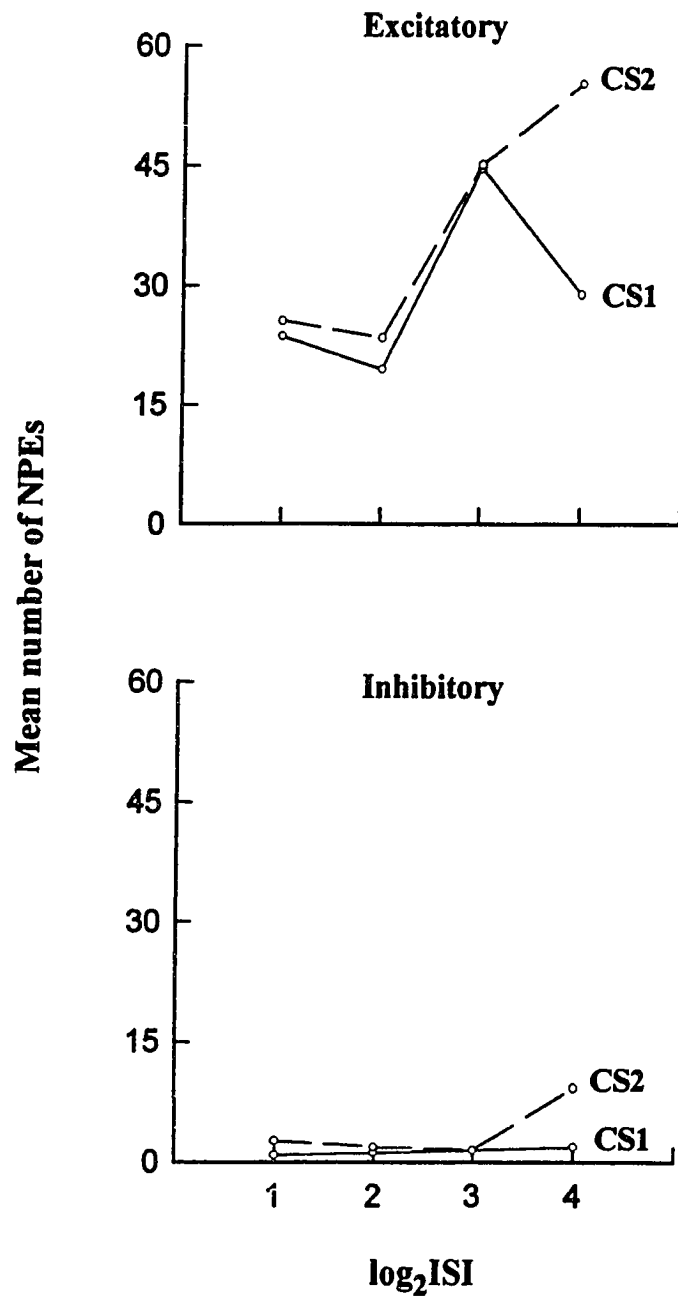


Figure 3.19: Mean number of hidden layers in last generation of each lineage.



**Figure 3.20: Mean number of excitatory and inhibitory NPEs in last generation of each lineage.**

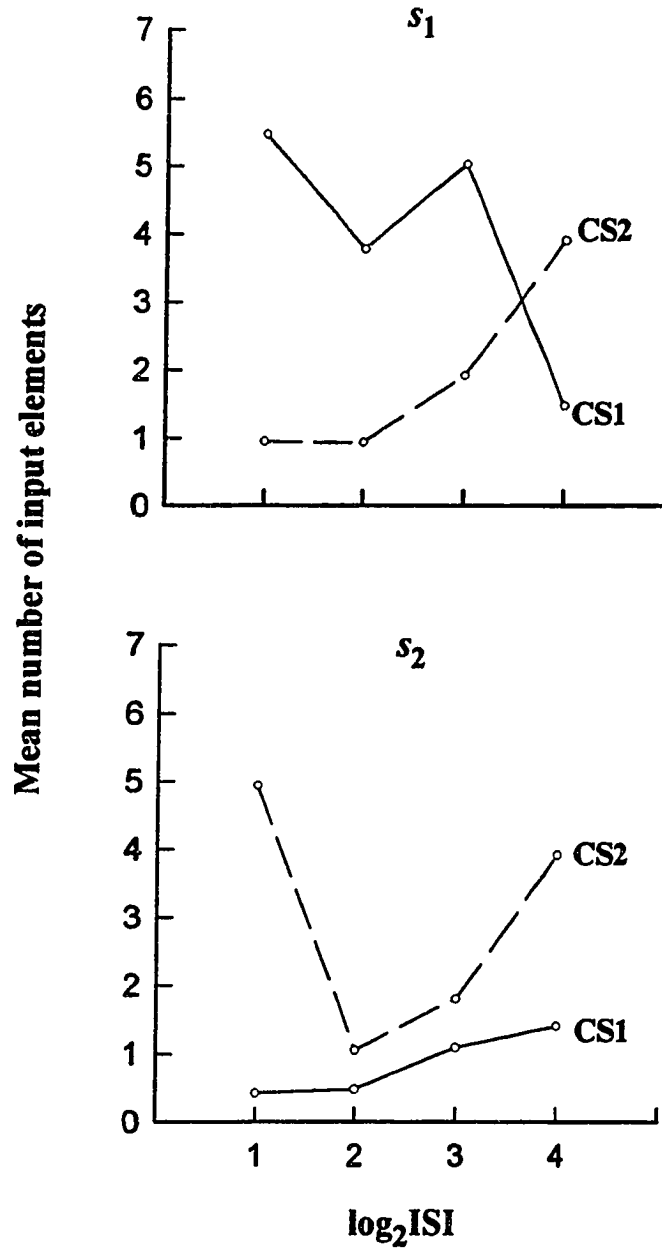
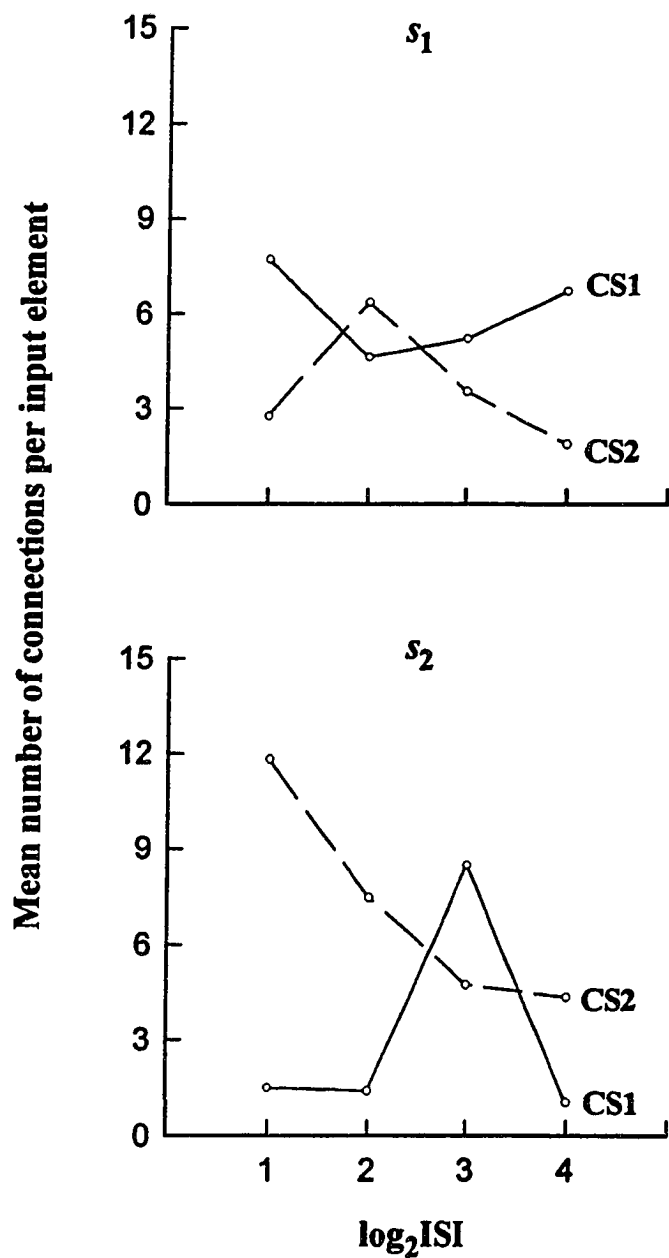


Figure 3.21: Mean number of input elements in last generation of each lineage.



**Figure 3.22: Mean number of connections per input element in last generation of each lineage.**

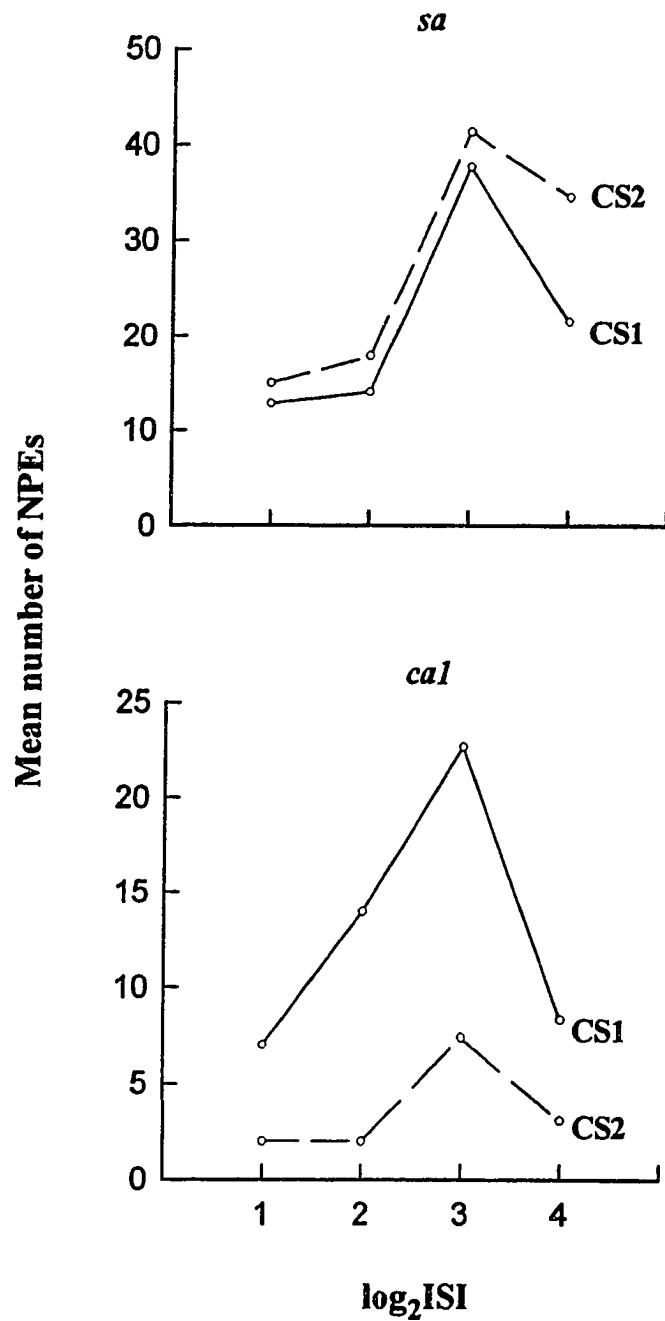
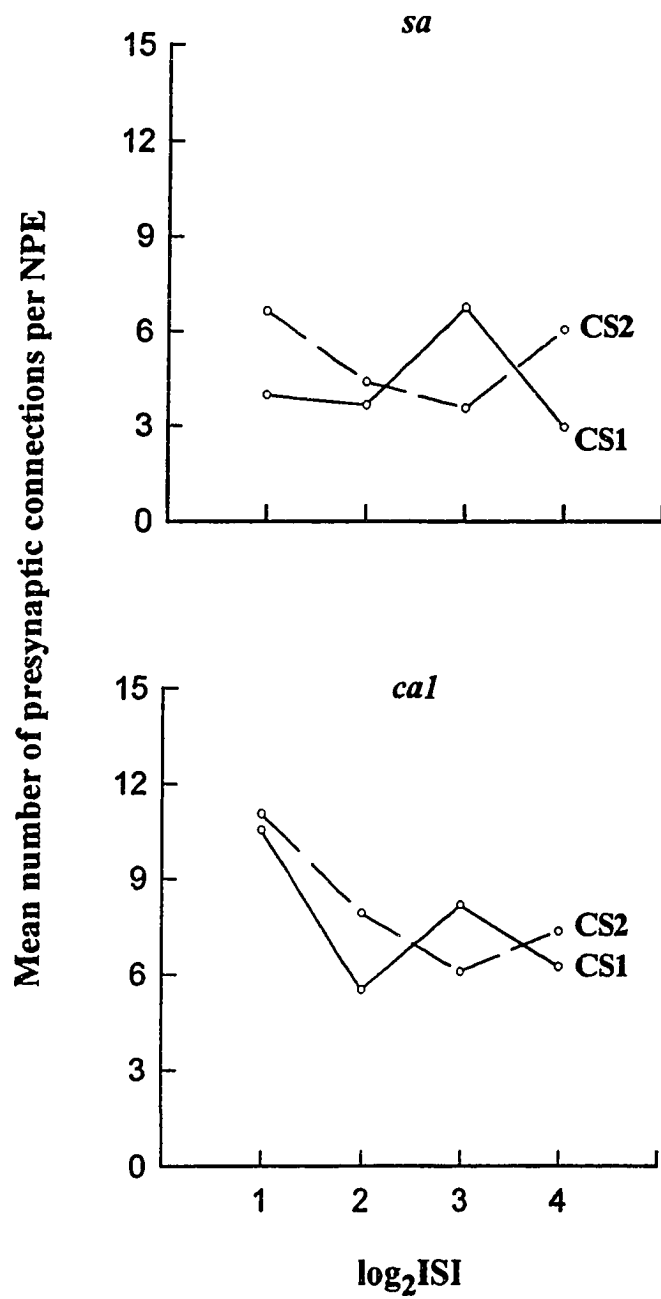


Figure 3.23: Mean number of *sa* and *cal* NPEs in last generation of each lineage.



**Figure 3.24: Mean number of presynaptic connections per *sa* and *cal* NPE in last generation of each lineage.**



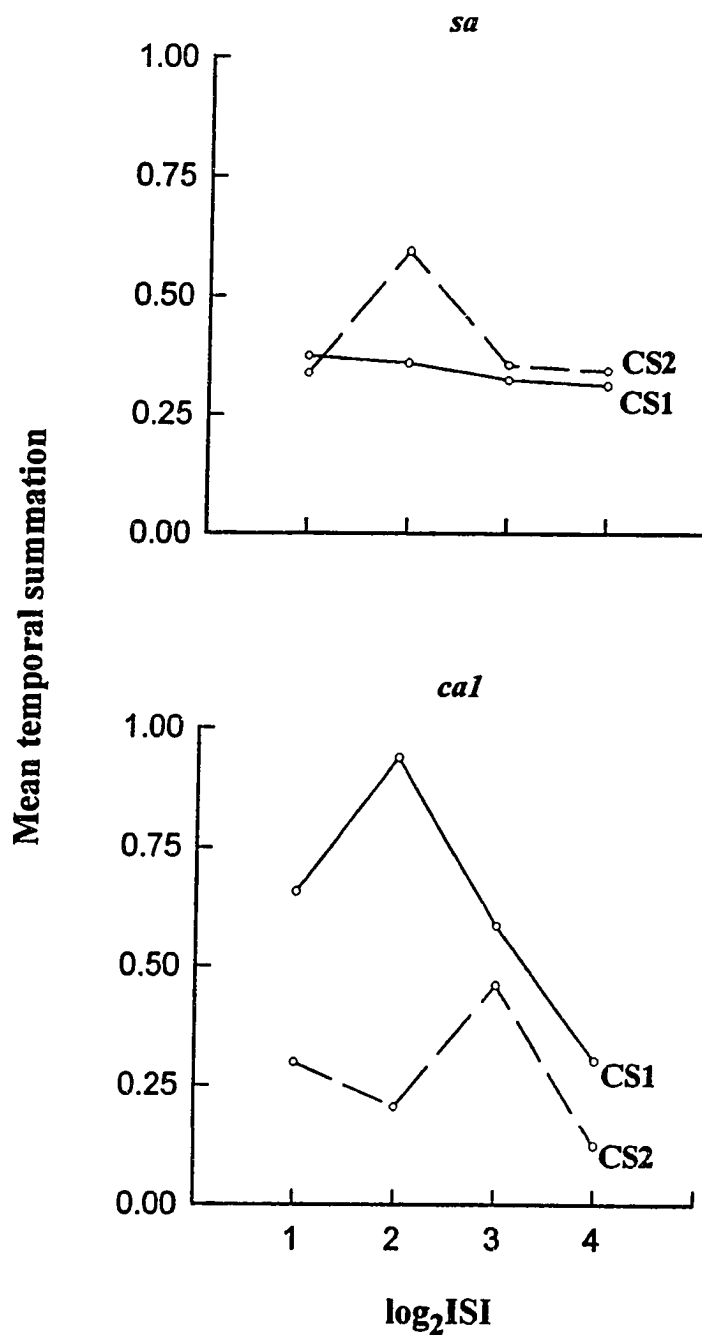


Figure 3.25: Mean temporal summation in *sa* and *cal* NPEs in last generation of each lineage.

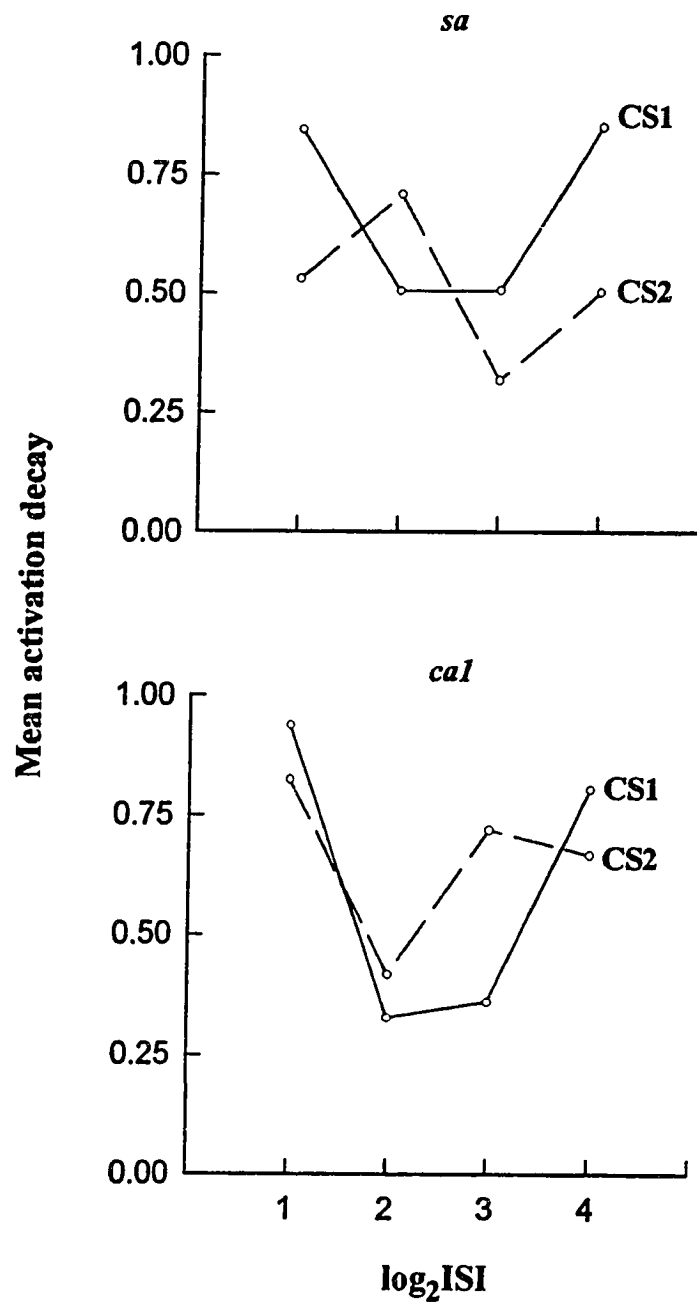


Figure 3.26: Mean activation decay in *sa* and *cal* NPEs in last generation of each lineage.

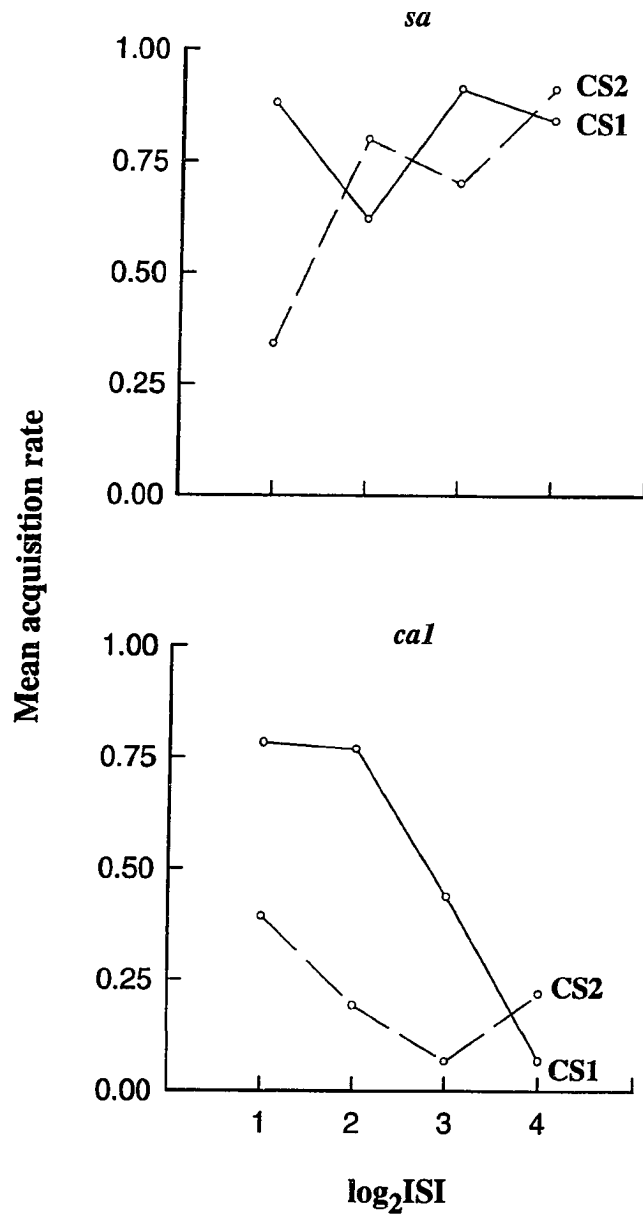


Figure 3.27: Mean acquisition rate in *sa* and *cal* NPEs in last generation of each lineage.

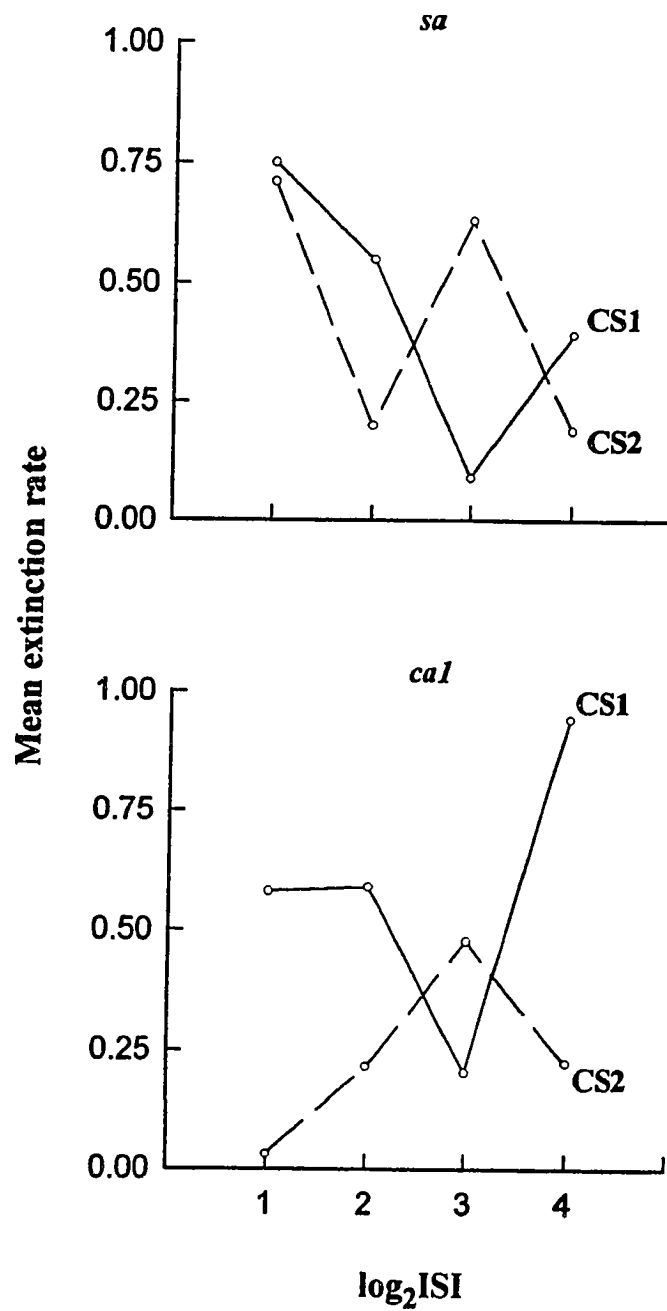


Figure 3.28: Mean extinction rate in *sa* and *cal* NPEs in last generation of each lineage.

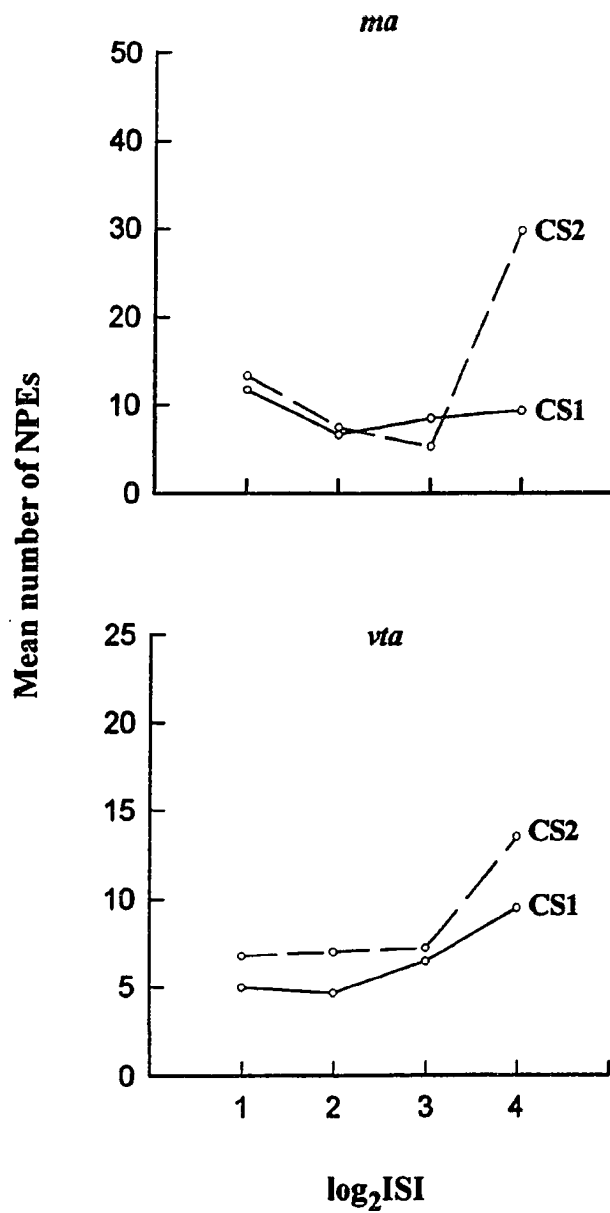


Figure 3.29: Mean number of *ma* and *vta* NPEs in last generation of each lineage.

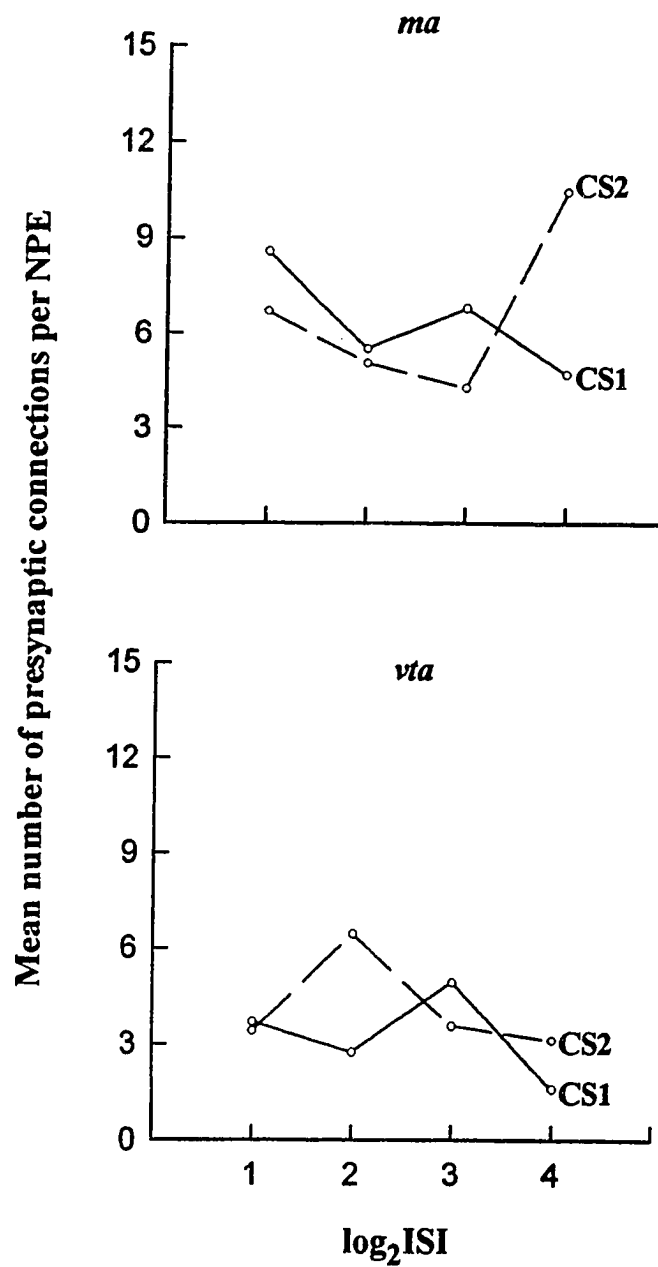


Figure 3.30: Mean number of presynaptic connections per *ma* and *vta* NPE in last generation of each lineage.

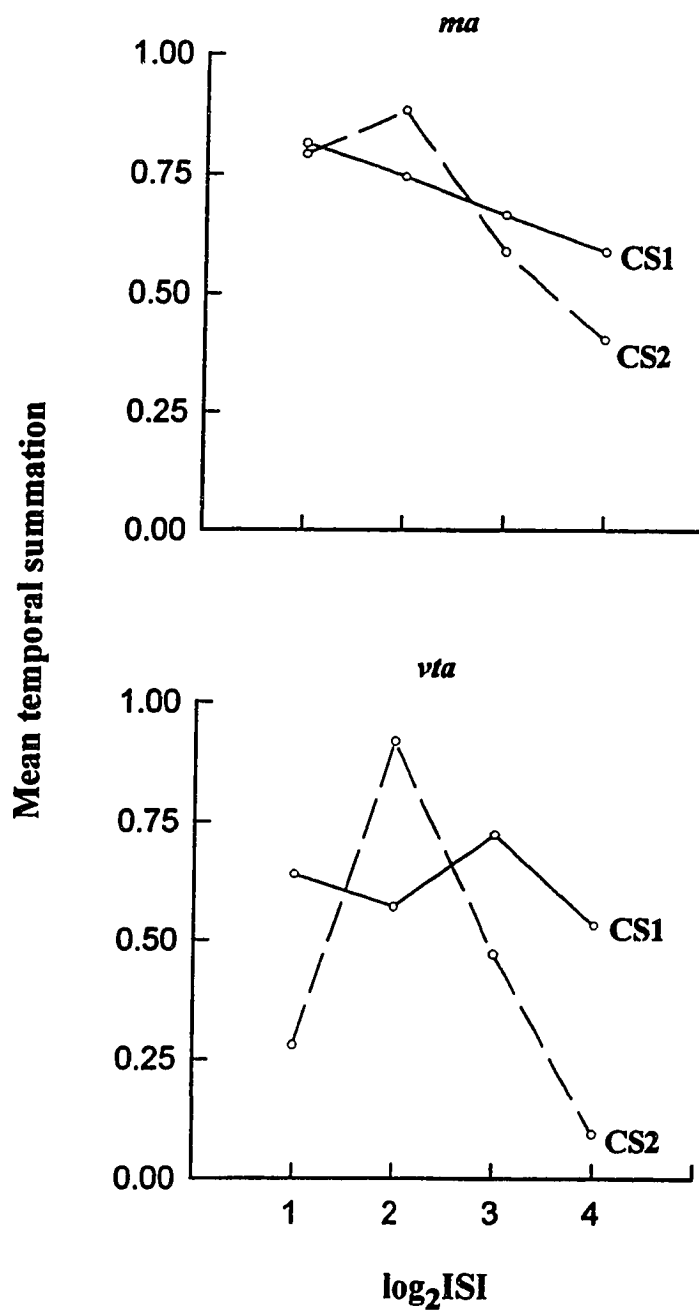
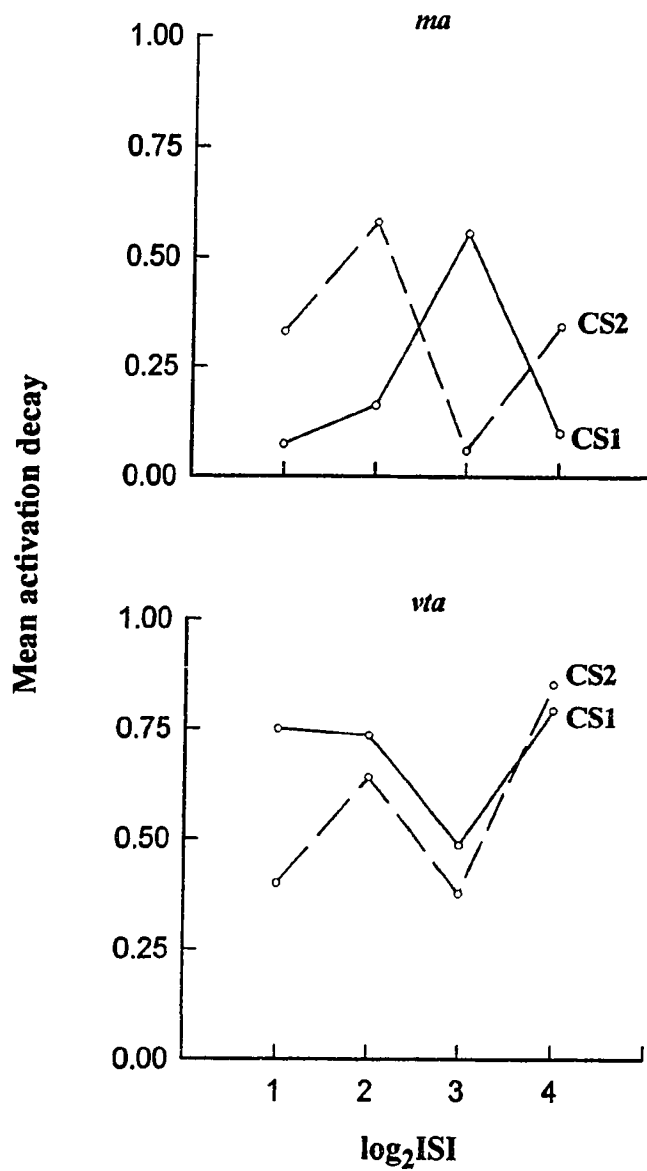


Figure 3.31: Mean temporal summation in *ma* and *vta* NPEs in last generation of each lineage.



**Figure 3.32: Mean activation decay in *ma* and *vta* NPEs in last generation of each lineage.**



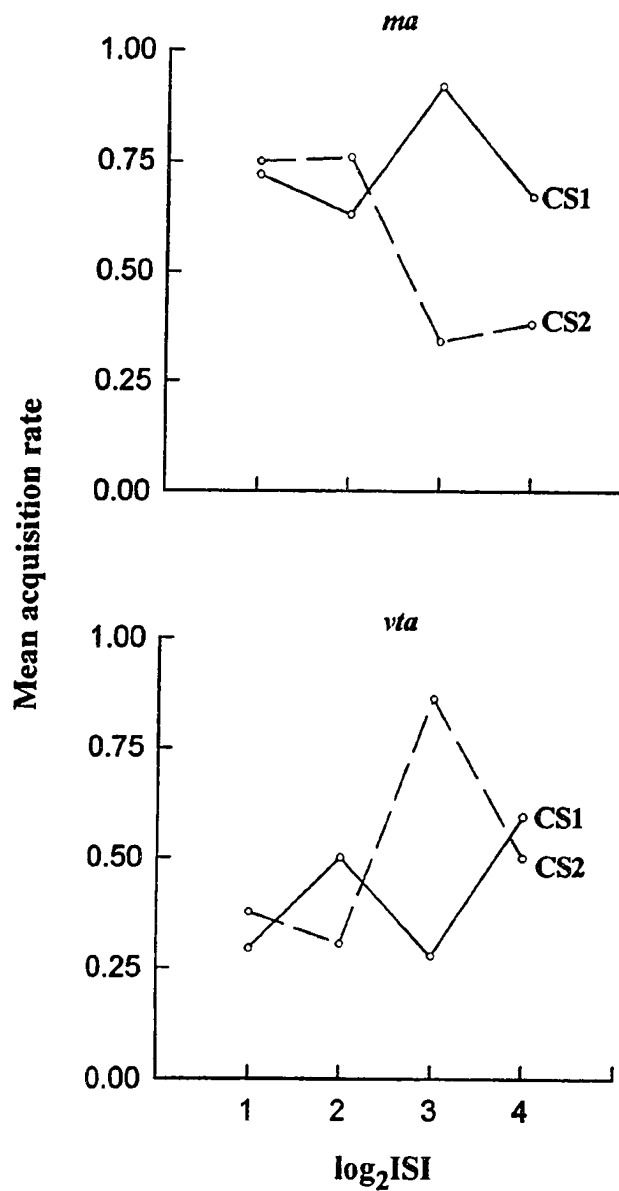


Figure 3.33: Mean acquisition rate in *ma* and *vta* NPEs in last generation of each lineage.

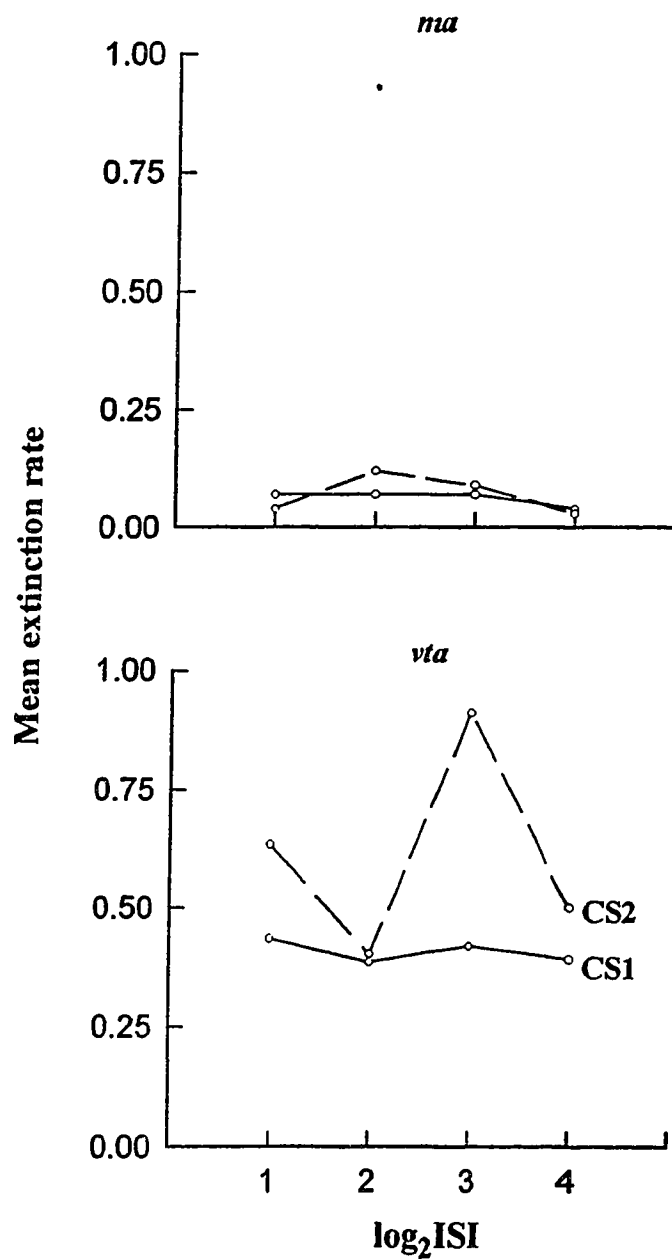


Figure 3.34: Mean extinction rate in *ma* and *vta* NPEs in last generation of each lineage.

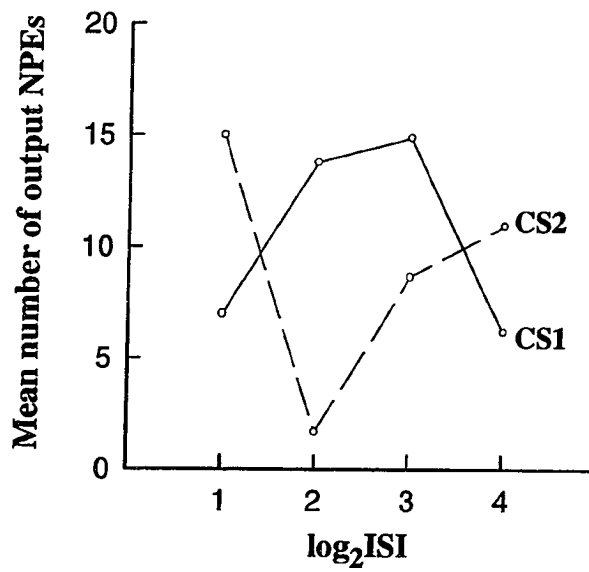


Figure 3.35: Mean number of output NPEs in last generation of each lineage.

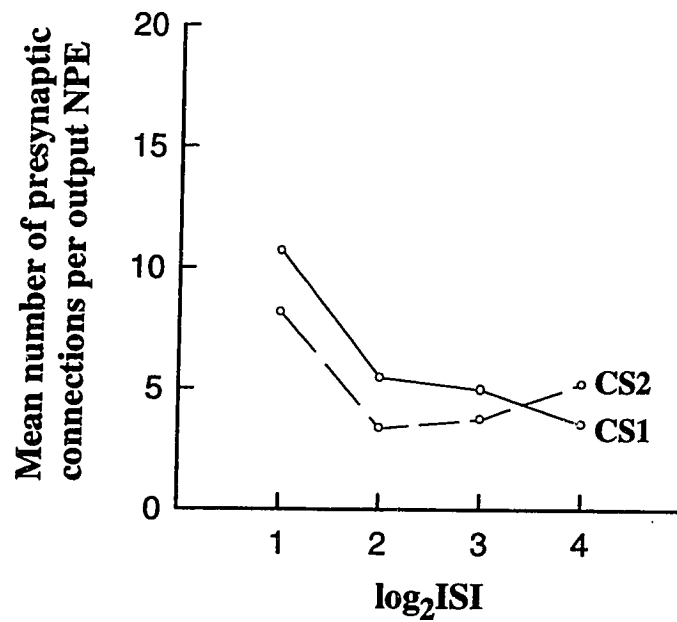
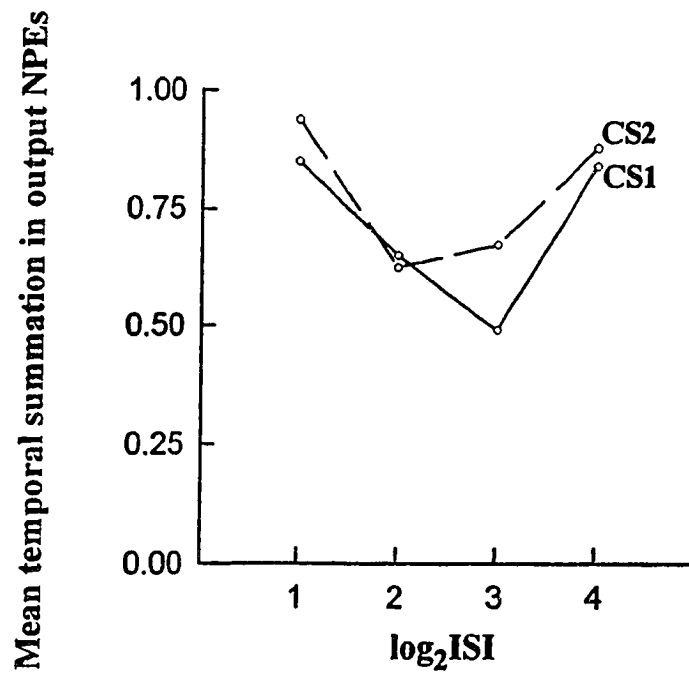
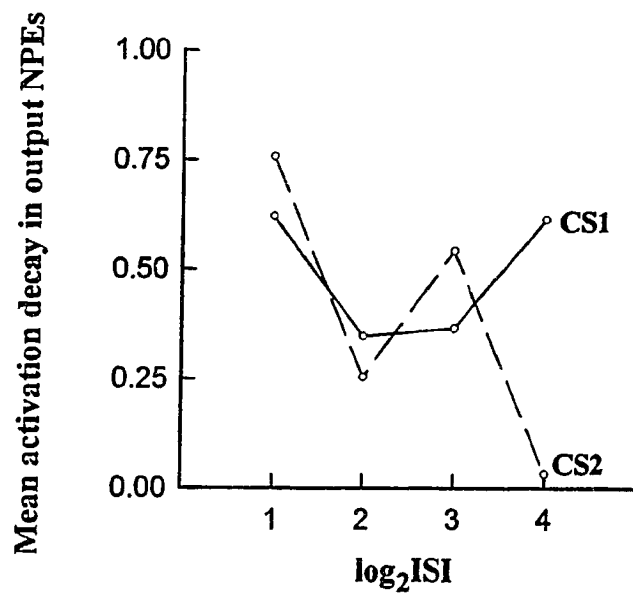


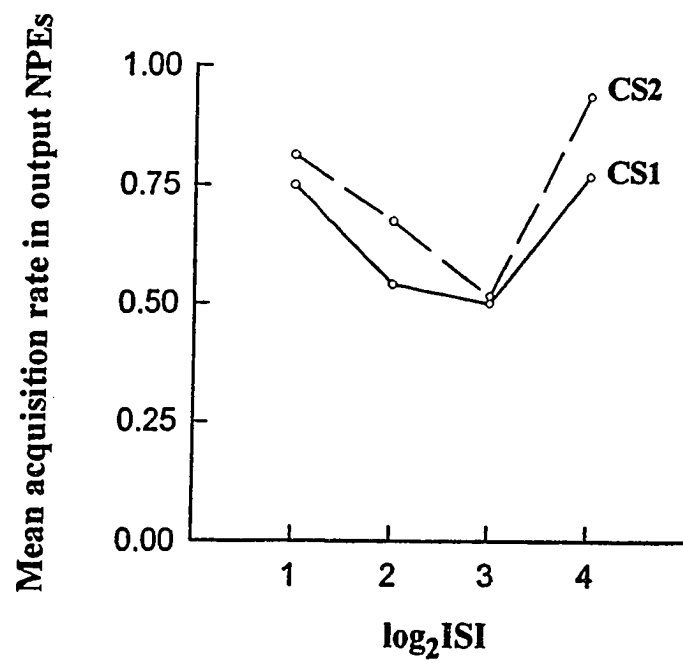
Figure 3.36: Mean number of presynaptic connections per output NPE in last generation of each lineage.



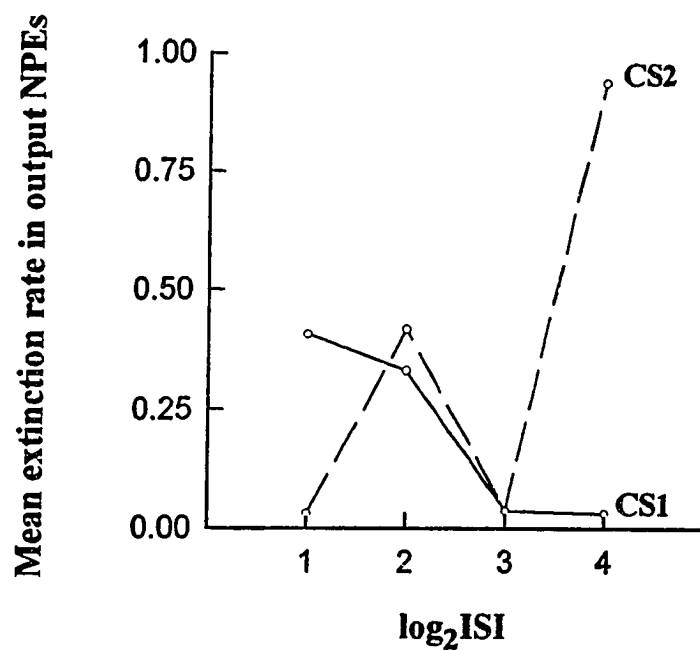
**Figure 3.37: Mean temporal summation in output NPEs in last generation of each lineage.**



**Figure 3.38: Mean activation decay in output NPEs in last generation of each lineage.**

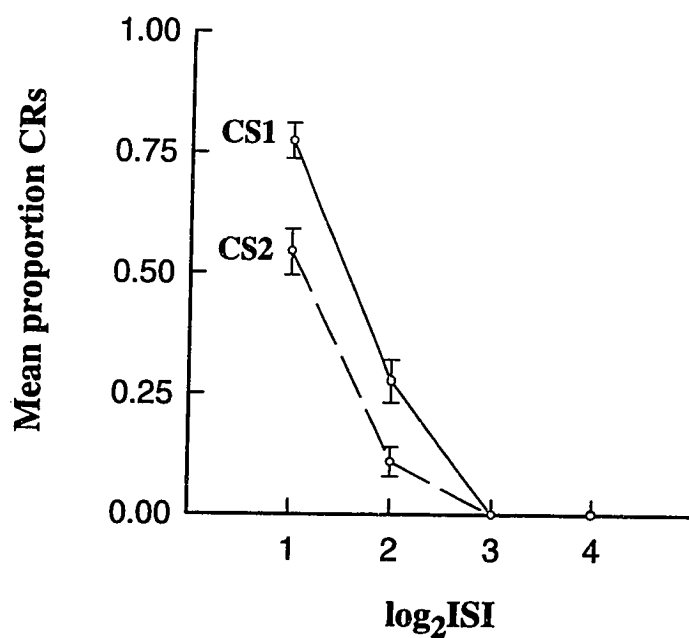


**Figure 3.39: Mean acquisition rate in output NPEs in last generation of each lineage.**

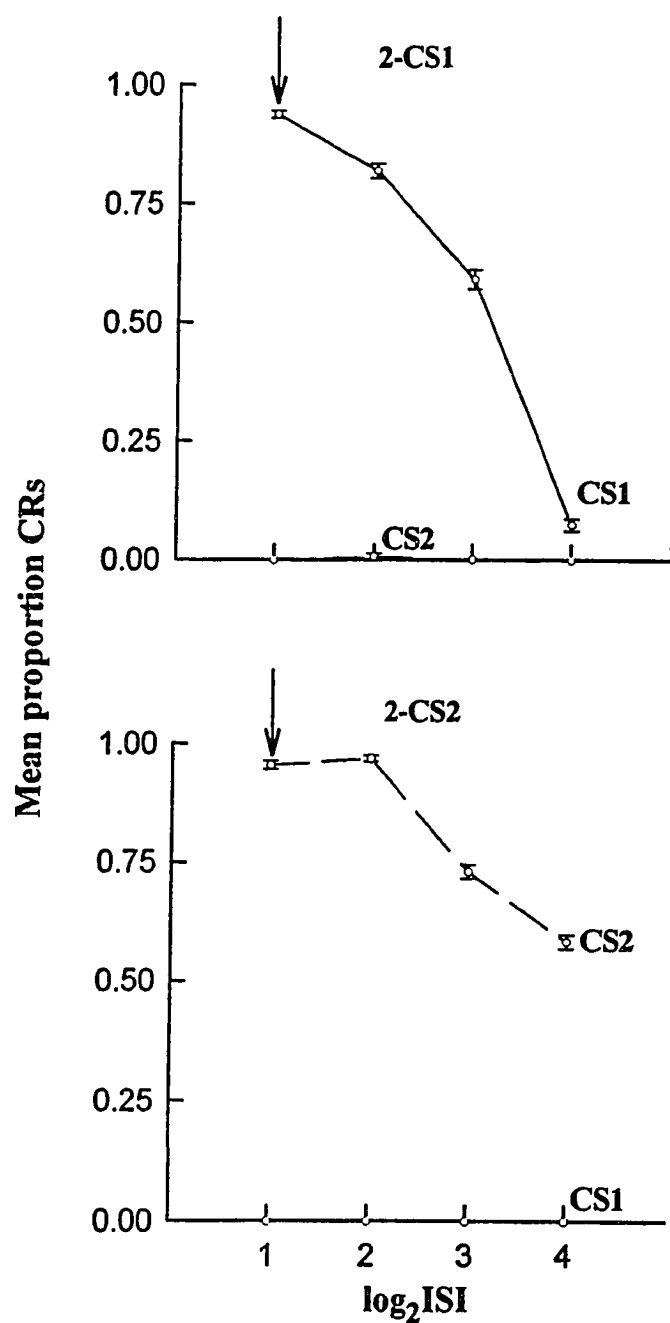


**Figure 3.40: Mean extinction rate in output NPEs in last generation of each lineage.**

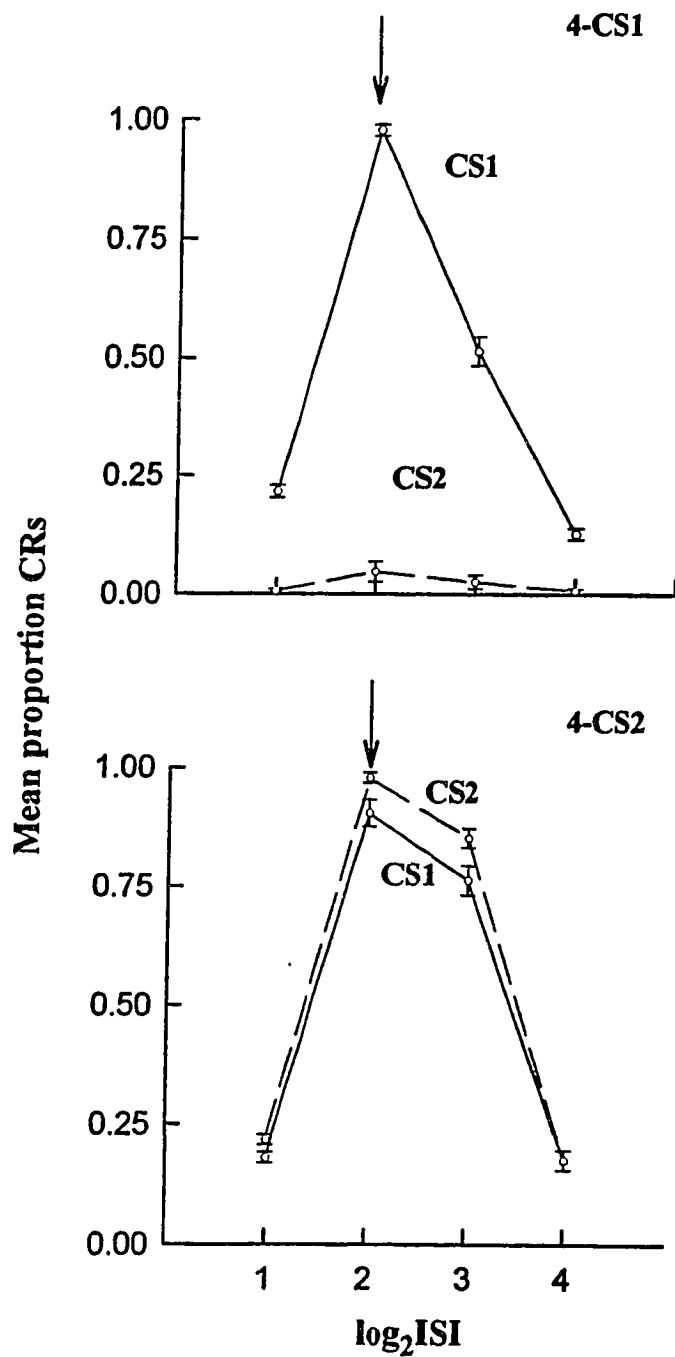




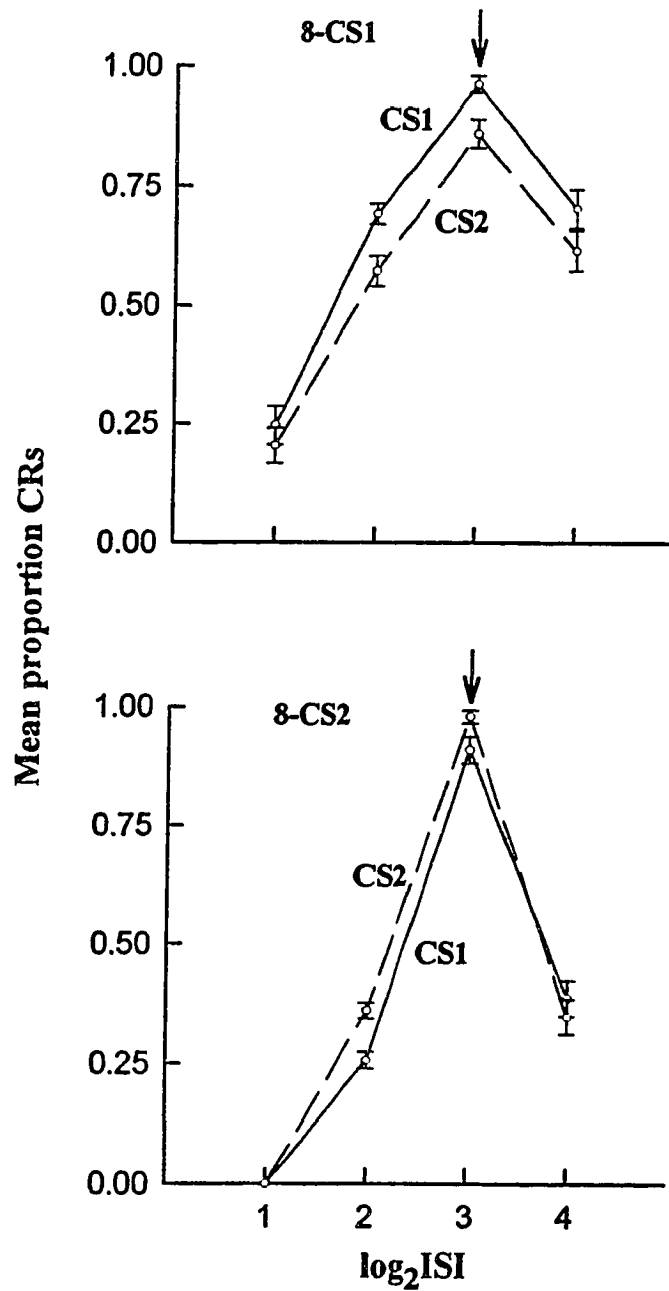
**Figure 3.41: Mean proportion CRs in CS-alone test of ANNs in last generation of each lineage.**



**Figure 3.42: Mean proportion CRs in ISI and CS tests of ANNs in last generation of 2-CS1 and 2-CS2. The arrows indicate the conditions under which ancestral selection took place.**



**Figure 3.43: Mean proportion CRs in ISI and CS tests of ANNs in last generation of 4-CS1 and 4-CS2. The arrows indicate the conditions under which ancestral selection took place.**



**Figure 3.44: Mean proportion CRs in ISI and CS tests of ANNs in last generation of 8-CS1 and 8-CS2. The arrows indicate the conditions under which selection took place.**

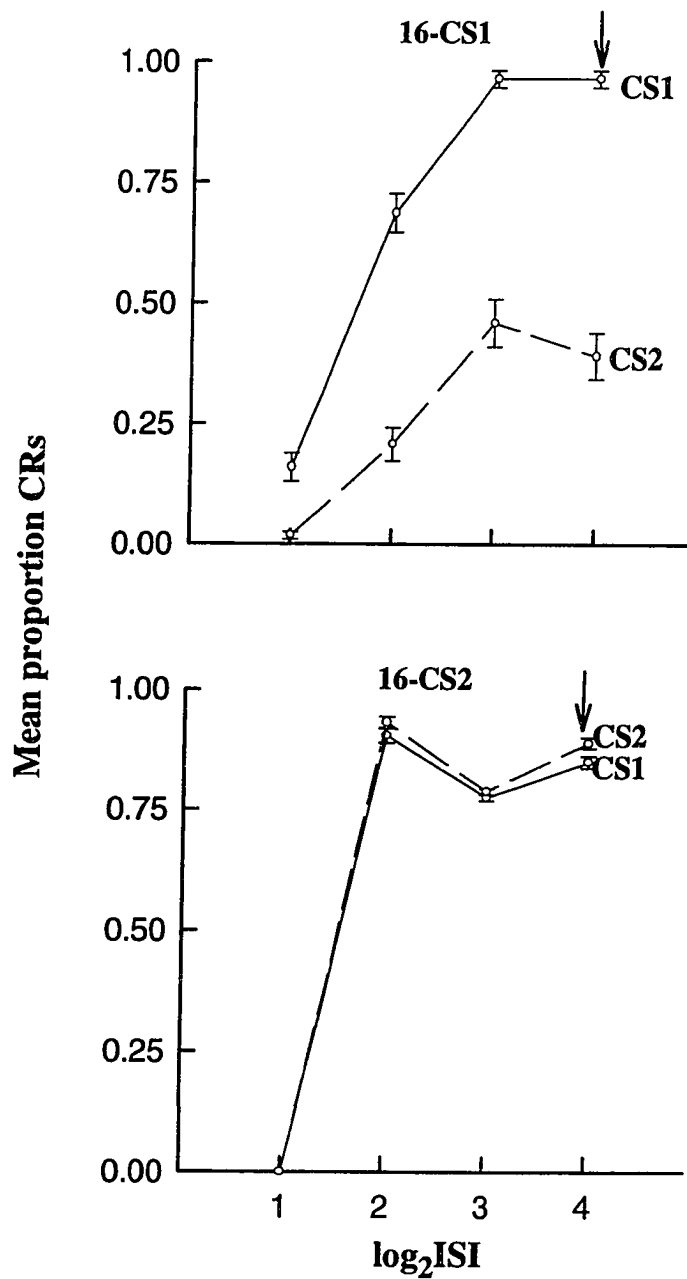


Figure 3.45: Mean proportion CRs in ISI and CS tests of ANNs in last generation of 16-CS1 and 16-CS2. The arrows indicate the conditions under which selection took place.

# CHAPTER 4

## EXPERIMENT 2

In Experiment 1, ANNs evolved under environmental conditions in which instances of only *one* CS class (either CS1 or CS2) occurred. That is, individual ANNs experienced instances of one and only one kind of CS during training. An outcome of this arrangement was that in half the conditions of the experiment CS1 and CS2 were nonequipotential in that, at the end of evolution, ANNs learned to respond with high CR proportions to instances of only one kind of CS (the kind present when ancestral selection took place). In general, ANNs that evolved under long ISIs tended to be larger than ANNs that evolved under short ISIs. Also, ANNs that evolved under CS1 tended to have more  $s_1$  but fewer  $s_2$  input elements than ANNs that evolved under CS2. ISI functions, optimal-ISI noninvariance, and CS nonequipotentiality emerged as functional properties of these structural differences.

However, real organisms are exposed to instances of more than one kind of stimulus. Evolutionarily speaking, such a complexification implies that individual fitness, and hence evolution may depend upon learning to respond to different kinds of CSs. The purpose of the present experiment, then, was twofold. First, to determine if structurally different types of ANNs would emerge from selection for increased responding under environmental conditions that differed only in the ISI, but comprised the presence of instances of *two* CS classes. Second, to determine performance of the resulting ANNs under three kinds of Pavlovian-conditioning

phenomena involving more than one kind of CS, namely generalization, discrimination, and blocking.

Generalization involves high responding to novel CSs that are physically distinct in some aspects but similar in others to previously reinforced instances of a given CS class (e.g., Hoffman & Flesher, 1961; Hovland, 1937; Moore, 1972; Pavlov, 1927; Siegel, Hearst, George, & O'Neal, 1968). In contrast, discrimination (Pavlov, 1927) entails *differential responding* to present instances of at least two CS classes after *differential reinforcement* of past instances of those classes (i.e., reinforcement of instances of one class and nonreinforcement of instances of the other class). Some authors (e.g., Honig, 1965) have distinguished between *interdimensional* and *intradimensional* or, more formally, *orthogonal* and *nonorthogonal* discrimination<sup>1</sup>. Orthogonal (or interdimensional) discrimination involves CSs that affect different sensory modalities (e.g., lights vs. tones), and hence, are noticeably different from each other. Hence, in this case, generalization from one stimulus to the other tends to be minimal or nonexistent. Nonorthogonal (or intradimensional) discrimination involves CSs that are somehow similar (i.e., affect the same modality), for which they allow for some amount of generalization. In both kinds of discrimination, responding after training is substantially stronger in the presence of the reinforced than in the presence of the nonreinforced CS. Finally, blocking (Kamin, 1968) is the phenomenon of low responding to instances of a CS class that have been reinforced in the concurrent presence of instances of another, previously reinforced CS class.

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<sup>1</sup> These two latter terms arise from the possibility of representing the reinforced-CS and the nonreinforced-CS input-pattern activations as *vectors*. If the two CSs have no elements in common, then the inner (or dot or scalar) product of the vector representations of the corresponding input patterns is equal to zero. In this case, the input-pattern vectors are said to be *orthogonal*. In contrast, if the two CSs have no elements in common, then the inner product of the vector representations of the corresponding input patterns is different from zero. In this case, the input-pattern vectors are said to be *nonorthogonal*.

A common feature of these phenomena is that they presuppose ANNs that can sense and react to instances of more than one CS class. This feature made testing for these procedures an unfeasible task for half the conditions of Experiment 1. In the present experiment, however, selection for increased responding to instances of two separate, independent CS classes was expected to generate ANNs with a sufficient number of input elements of both kinds, so that testing for generalization, discrimination, and blocking would be possible in all conditions.

As in Experiment 1, two kinds of simulations were run, namely evolution and test. In the evolution simulations, ANNs were selected for high responding to separate, randomly intermixed instances of both CS1 and CS2. In the test simulations, ANNs at the end of evolution were tested for generalization, discrimination, and blocking.

## **4.1 Evolution Simulations**

### **4.1.1 Procedure**

Preliminary simulations showed that G0 resulted in the evolution of ANNs that could not learn to respond to more than one kind of CS. By the end of evolution, some ANNs learned to respond to CS1 only, and other ANNs learned to respond to CS2 only. This difficulty was thought to be due to genetically-determined structural differences between  $s_1$  and  $s_2$  input elements that favored the evolution of ANNs that learned to respond to only one kind of CSs. Before starting the present evolution simulations, thus, a new chromosome founder population designated as G0' was generated. G0' was built from G0 by making, for each member of the population, all ND parameters that determined the number and connectivity of  $s_2$  input elements equal



to the ND parameters that determined the number and connectivity of  $s_1$  input elements.

After generating  $G_0'$ , four different environmental conditions were defined by four ISIs (either 2, 4, 8, or 16 ts). A lineage of generations of ANNs evolved for each condition. As a first step, four ANNs were developed from each one of the 100 chromosomes in  $G_0'$ , to obtain 400 ANNs, which were divided into groups of 100, one group corresponding to the ANN founder population of a condition. ANNs in each group were trained in a forward-delay procedure in which the ISI was either 2, 4, 8, or 16 ts, and the CS was either CS1 or CS2. However, in contrast to Experiment 1, each ANN in the present experiment was given randomly intermixed trials of *both* kinds of CS. As in Experiment 1, CS1 trials involved the activation of all  $s_1$  input elements with a value of 1.0 and no activation of any  $s_2$  input element, whereas CS2 trials involved the activation of all  $s_2$  elements with a value of 1.0 and no activation of any  $s_1$  element. All ANNs were given a total of 250 trials of randomly intermixed CS1-US and CS2-US pairings, the last 50 of which were used as evaluation trials to determine individual fitness for each kind of CS. Fitness for a given CS class was the proportion of evaluation trials of that class in which a CR was elicited. An individual won a selection tournament if and only if the individual had *both* the highest CS1-fitness *and* CS2-fitness.

After evaluation, a new population of chromosomes was created through the selection scheme of the GA. This new population constituted  $G_1$  (the next generation). Once the new chromosomes were created, new ANNs were developed and trained in the same procedure as their ancestors. The same operation was repeated generation after generation, until the termination criterion (either 100 generations after  $G_0'$  or fitness stability) was satisfied. This strategy resulted in a total of four lineages, which I

shall label as ISI-2, ISI-4, ISI-8, and ISI-16. All lineages evolved independently of one another.

Simulations showed that ANNs evolved under the ISI-2 condition did not learn to respond to both stimuli by the end of evolution. These ANNs learned to respond with high CR proportions to either CS1 or CS2. The effects on functional and structural phenotypes for these ANNs were comparable to those observed in Experiment 1. Therefore, I shall present and discuss results only for ISI-4, ISI-8, and ISI-16.

## **4.1.2 Results and Discussion**

### **4.1.2.1 Genotypic Evolution**

Figure 4.1 shows that the mean chromosomal overlap increased as a negatively accelerated function of generations, for all lineages. The overlap for G0' were close to 0.0, which is consistent with the expected value for a randomly generated population of chromosomes. Overlaps increased rapidly within a few generations, and then became stable at relatively high values by the end of evolution. Hence, individuals within lineages became genetically more similar as generations progressed. For all lineages, overlaps in the last generation were close to 1.0. Thus, a small amount of genetic variation was still present at the end of evolution. Finally, no substantial differences were observed among the last-generation overlaps across lineages.

### **4.1.2.2 Effects on Functional Phenotypes**

#### **4.1.2.2.1 Effects on Fitness**

Figure 4.2 shows that mean population fitness increased as a negatively accelerated function of generations, for all lineages. Indeed, lineages started with low-fitness (close

to 0.0) and ended with a high-fitness ANNs (between 0.8 and 1.0). Also, at the end of evolution, ANNs across lineages were comparably fit. ISI-4 and ISI-8 showed the highest fitness values (close to 1.0) after relatively few generations (34 and 38, respectively), whereas ISI-16 ANNs showed the lowest fitness values (around .75) after 100 generations. Fitness values for ISI-16 ANNs rarely reached 1.0. This situation produced sufficient functional-phenotypic variation among these ANNs to ensure a winner at all tournaments across generations, and hence a continuation of the selection process until the maximum number of generations.

Figures 4.3 through 4.5 show 3D mesh plots depicting changes in the frequency distributions of fitness values across generations. Each figure shows the CS1 (top panel) and CS2 condition (bottom panel) for a given ISI lineage. All graphs show a similar overall tendency, that is, a high number of low-fitness individuals at G0', followed by an increase in the number of high-fitness individuals after a number of generations. Indeed, the high frequency peak at the origin of the generation-fitness plane indicates that most individuals in G0' had a fitness between 0.0 and 0.05. As generations passed, the number of individuals with fitness values between 0.95 and 1.00 increased rapidly for all lineages. Also, functional-phenotypic variability at the end of evolution was relatively low (i.e., the majority of individuals had the same fitness), except for ISI-16 (see Figure 4.5), in which the number of individuals with low-fitness values remained substantial across generations. This latter result is consistent with the fact that ISI-16 took 100 generations.

#### **4.1.2.2.2 Effects on Learning**

Figures 4.6 through 4.11 show 3D mesh plots depicting changes in frequency distributions of mean output-NPE activations across the first 75 CS1 and 75 CS2 trials,

for the first (top panel) and last (bottom panel) generation of each lineage<sup>2</sup>. In G0', the number of ANNs with output-NPE activations between 0.0 and .05 remained relatively constant across CS1 and CS2 trials. Only a few ANNs responded with activations greater than the CR criterion of 0.5 at the end of training, a condition that was necessary and sufficient to ensure the minimum number of tournament winners required for the GA to generate G1. In contrast, in the last generation, the number of individuals with high output-NPE activations (between 0.95 and 1.0) increased as a function of trials. Moreover, as in Experiment 1, most ANNs far exceeded the CR criterion towards the end of training, despite the fact that their ancestors did not need to respond with such high activations to win selection tournaments.

Note that, by the end of evolution, many ISI-4 and ISI-16 ANNs tended to respond with output-NPE activations between 0.25 and 0.50 to CS1 and CS2 very early in training (see bottom panel of Figure 4.6 and 4.7). However, the mean output-NPE activations in these ANNs decreased after approximately 10 training trials, after which ANNs with low output-NPE activations represented the vast majority. Then, after 10 more training trials, the number of low-activation ANNs decreased and the number of high-activation ANNs increased. Also, the number of ISI-16 ANNs with low mean output-NPE activations at the end of evolution remained substantially higher than those observed for ISI-4 or ISI-8 ANNs.

Finally, Figure 4.12 shows the mean connection weights before and after training for the last generation of each lineage. For all lineages, connection weights were

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<sup>2</sup> Only the first 75 training trials of each CS kind are shown because CS1 and CS2 trials were randomly intermixed. Consequently, different ANNs experienced not only different sequences of trials, but also different numbers of trials of each kind, although the average number of CS1 and CS2 trials was approximately 100. Seventy-five was the smallest number of trials of given kind (either CS1 or CS2) presented to an ANN in the last generation.

substantially higher after than before training, most specially for ISI-4 and ISI-16 ANNs. Note also that the latter had initial connection weights very close to 0.0.

The results described above indicate that selection for increased responding to two independent CSs generated ANNs whose behavior changed from low to high conditioned responding to each CS. Moreover, ANNs seemed not to be sensitive to the *order* in which CS1-US and CS2-US trials were presented. Indeed, ANNs by the end of evolution showed high output-NPE activations after training, despite the fact that those trials were randomly intermixed throughout training. This result is significant in that previous simulations using the present NC model (e.g., Donahoe, Burgos, & Palmer, 1993) showed that handcrafted ANNs were very sensitive to random sequences of reinforced trials. That is, in different simulations, the same ANN would learn to respond either with high output-NPE activations to only one CS (either CS1 or CS2), or to both CSs but with medium output-NPE activations (between 0.3 and 0.5). In contrast, ANNs generated through the present GA learned to respond to each CS with high output-NPE activations, regardless the order of presentation of CS1-US and CS2-US trials. Also, handcrafted ANNs would not learn to respond to one CS after receiving a number of reinforced trials of the other CS. However, in a separate test (results not shown here), the present ANNs learned to respond with high output-NPE activations to one CS after having received 100 reinforced trials of the other CS.

### **4.1.2.3 Effects on structural phenotypes**

#### **4.1.2.3.1 General Effects**

Figure 4.13 shows a sample of typical ANNs from the last generation of each lineage. All the ANNs had a fitness above .95. The differences in the ANNs' sizes are

clear. Selection under ISI 8 generated the largest ANNs obtained throughout the present work. Also, note that each ANN had the same number of  $s_1$  and  $s_2$  input elements, although the ISI-4 and ISI-8 ANNs had substantially more input elements of each kind than the ISI-16 ANN. All ANNs had only one *ma* layer, substantially more excitatory than inhibitory NPEs, and approximately the same number of output NPEs. Although the ISI-4 and ISI-16 ANNs had only one *sa* layer, the former had substantially more *sa* NPEs than the latter. Finally, the number of connections per output NPE in the ISI-8 ANN was substantially smaller than that observed in the ISI-4 and ISI-16 ANNs.

As in Experiment 1, some within-population variability among ANN architectures was observed, especially in the ISI-8 and ISI-16 conditions. For the ISI-4 condition, most ANNs within the last generation seemed to be instances of the same basic architecture. In contrast, the last generation of ISI-8 and ISI-16 included some high-fitness ANNs whose architectures are notably different from those shown in Figure 4.13. Figure 4.14 illustrates this result by showing another sample of high-fitness ANNs from ISI-8 and ISI-16. The ISI-8 ANN was different in that it had substantially fewer *sa* layers. The ISI-16 ANN was different in that it had only one *ma* NPE. Thus, selection for increased responding to two CSs under a particular ISI generated a *class* of architectures, rather than a unique architecture. Also, as in Experiment 1, not all ANNs in the last generation responded with high CR proportions to both CSs. Figure 4.15 shows a sample of low-fitness ANNs from each lineage. All of these ANNs had a fitness below 0.3 for one of the CSs (either CS1 or CS2).

It is unclear why these ANNs had such low fitness, but we can offer some conjectures. An examination of the NC parameters for the ISI-4 ANN did not reveal any unusual values, for which it is reasonable to believe that its performance was more

determined by architectural traits. Perhaps, the presence of two *sa* inhibitory NPEs in this ANN was critical in determining its performance. A similar interpretation can be given for the ISI-16 ANN, for an examination of the NC parameters of its constituting NPEs also revealed no unusual values. In the case of the ISI-8 ANN, an examination of the NC parameters revealed that the extinction rate of its output NPEs was 0.53, which is substantially larger than the average output-NPE extinction rate in the last generation of this lineage (which was approximately 0.25; see later). On this basis, it is reasonable to believe that this structural trait was critical in determining this ANN's performance.

General quantitative effects showed that ISI-4 and ISI-16 ANNs tended to have only one *sa* layer, whereas ISI-8 ANNs tended to have six, although all ANNs had one *ma* layer (see Figure 4.16). Also, as in Experiment 1, ANNs in all conditions tended to have substantially more excitatory than inhibitory NPEs, with ISI-8 having substantially more excitatory NPEs than ISI-4 or ISI-16 ANNs (see Figure 4.17). In general, these effects are consistent with the ISI effects described in Experiment 1, although differences among lineages were far more prominent in the present experiment.

Indeed, ISI-8 ANNs in the present experiment were substantially larger than those described in Experiment 1 (e.g., compare the ISI-8 ANNs shown in Figure 3.16 to the ISI-8 ANN shown in Figure 4.15). In contrast, ISI-4 and ISI-16 ANNs, tended to be smaller than their counterparts in Experiment 1, especially the latter. Thus, differences in the effects of the ISI throughout selection for increased responding to one CS were somehow magnified by selection for increased responding to two CSs.

In the case of ISI-8 ANNs, this result seems to be intuitively evident if we argue that these ANNs were larger in the present experiment because their ancestors were required to learn double the number of CS-US associations, relative to ANNs in

Experiment 1. However, the decrease in ANN size for ISI 4 and ISI 16 in the present experiment defies our intuition that larger ANNs should be necessary to learn more CS-US associations. Perhaps, genotypic constraints were such that adequate combinations of NC parameters did not exist (or were extremely improbable) that allowed small ANNs to learn two CS-US associations under ISI 8, but adequate combinations of NC parameters existed (or were far more probable) that allowed small ANNs to learn two CS-US associations under ISIs 4 and 16. Obviously, these conjectures must also take into consideration the fact that a different chromosome founder population was used in the present experiment, although the specific effects of this difference remain unclear.

The rest of the section presents the specific effects on those aspects of structural phenotypes related to the sensory and motor subnetworks. The organizing scheme is similar to that used in the previous chapter, with some minor omissions of subheadings. Also, as in the previous chapter, the graphs presented in this section do not include error bars. However, as in Experiment 1, standard errors were negligible for all conditions and structural traits, ranging from 0.001 for the mean number of input elements to 1.563 for the mean number of connections per element. Standard errors for NC parameters ranged from 0.0003 to 0.02.

#### **4.1.2.3.1 Effects on the Sensory Subnetwork**

In general, ISI-8 ANNs had the largest number of both kinds of input elements, whereas ISI-16 ANNs had the smallest (see Figure 4.18), although ANNs *within lineages* tended to have the same number of  $s_1$  and  $s_2$  input elements. Also, ANNs within lineages tended to have the same number of connections per kind of input element (see Figure 4.19). ISI-8 ANNs tended to have the largest number of connections per input element, whereas ISI-16 ANNs had the smallest.



Figure 4.20 shows the mean number of *sa* (top panel) and *cal* (bottom panel) NPEs as a function of ISIs. ISI-8 ANNs had substantially more *sa* and *cal* NPEs than ISI-4 and ISI-16 NPEs. However, ISI-4 and ISI-8 ANNs had approximately the same number of connections per *sa* NPE, and considerably more than ISI-16 ANNs (see top panel of Figure 4.21). Also, ISI-8 ANNs tended to have more connections per *cal* NPE than ISI-4 and ISI-16 ANNs, although all ANNs had substantially fewer connections per *cal* than per *sa* NPE (see bottom panel of Figure 4.21).

Figure 4.22 through 4.25 show changes in parametric traits across ISIs, in the sensory subnetwork. Overall, temporal summation in *sa* NPEs remained constant across ISIs at values close to 1.0 (see top panel of Figure 4.22). In contrast, *cal* temporal summation was close to 0.0 in ISI-4 and ISI-8 ANNs, but close to 1.0 in ISI-16 ANNs (see bottom panel of Figure 4.22). Thus, although the sensory subnetworks in ISI-4 and ISI-16 ANNs tended to have very similar architectures, they were very different in their respective *cal* temporal summation parameters. These results indicate that there was a substantial trade-off between *sa* and *cal* temporal summations for ISI-4 and ISI-8 ANNs, but not for ISI-16 ANNs. In the latter, having both high *sa* and *cal* temporal summations may have conferred a fitness advantage by increasing the overall *cal* discrepancy. However, it is unclear how fitness could have been increased in ISI-4 and ISI-8 ANNs by having near-zero *cal* temporal summations. Perhaps other NC parameters must be taken into consideration, before any attempt at an interpretation.

Indeed, activation decays in *sa* NPEs decreased as the ISI increased, reaching values close to 0.0 in ISI-16 ANNs (see top panel of Figure 4.23). In contrast, activation decays in *cal* NPEs tended to increase across ISIs, with ISI-16 ANNs having the highest *cal* activation decays (see bottom panel of Figure 4.23). Therefore, *sa* and *cal* NPEs in ISI-4 and ISI-8 ANNs tended to have both high temporal

summations *and* activation decays, which may have increased the range of variation of *sa* and *cal* NPEs, and hence, the likelihood of low *cal* activations and discrepancies. By decreasing the *cal* temporal summations, this effect may have been reduced. An analogous effect may have been accomplished by ISI-16 ANNs, but by having near-zero *sa* activation decays.

Acquisition rates in *sa* NPEs remained relatively constant at about 0.3 across ISIs, although they were somehow lower in ISI-8 ANNs (see top panel of Figure 4.24). However, *cal* NPEs in these ANNs had substantially higher (close to 1.0) acquisition rates (see bottom panel of Figure 4.24). This trait may have allowed for a relatively fast increase in *cal* connection weights, and hence, in *cal* discrepancies, allowing ISI-8 ANNs to learn faster. The rather large size of these ANNs may have been critical for these ANNs to need higher *cal* acquisition rates to learn as fast as the smaller ANNs.

Finally, *sa* NPEs in ISI-4 ANNs had near-zero extinction rates, relative to ISI-8 or ISI-16 ANNs, whose *sa* NPEs tended to have extinction rates between 0.25 and 0.35 (see top panel of Figure 4.25). In contrast, *cal* NPEs in ISI-4 ANNs had substantially higher extinction rates than ISI-8 or ISI-16 ANNs (see bottom panel of Figure 4.25). In fact, ISI-4 ANNs tended to have higher *cal* extinction than acquisition rates, whereas the opposite tendency was observed in ISI-8 ANNs. In the latter case, the trade-off may have conferred a fitness advantage by allowing *cal* connection weights to increase even faster. The high *cal* extinction rates for ISI 4 are more difficult to interpret, for one might think they would have caused ANNs to learn very slowly. However, having near-zero *sa* extinction rates may have compensated for the potentially detrimental effects of high *cal* extinction rates. In this way, high *cal* extinction rates may have not been selected against in these ANNs.

#### 4.1.2.3.2 Effects on the Motor Subnetwork: *ma* and *vta* NPEs

Figure 4.26 shows that the mean number of *ma* NPEs remained relatively constant at about 10 (see top panel), whereas *vta* NPEs remained at about 5 for all lineages (see bottom panel). In general, these effects are roughly consistent with those observed in Experiment 1 for CS1 lineages, indicating that ANNs whose ancestors were selected for increased responding under CS1 and CS2 had no more *ma* and *vta* NPEs than ANNs whose ancestors were selected for increased responding under CS1. This conclusion also applies to ISIs (i.e., ANNs whose ancestors were selected for increased responding under a particular ISI tended to have no more *ma* and *vta* NPEs than ANNs whose ancestors were selected for increased responding under a different ISI).

Apparently, then, relatively small numbers of *ma* and *vta* NPEs were sufficient to generate *vta* discrepancies high enough to allow for increased responding, and hence, for evolving high-fitness ANNs across a variety of environmental conditions, although, of course, differences may have evolved among NC parameters for these NPEs (see below). In fact, comparisons with Figure 4.20 reveal that ISI-8 ANNs had substantially fewer *ma* and *vta* than *sa* and *cal* NPEs, an effect similar to that observed in Experiment 1. This consistency indicates that learning under two CSs, like learning under one CS, may have depended more heavily on the sensory than on the motor subnetwork, at least for ISI 8. I shall offer one possible explanation of this finding in the final chapter.

Regarding connectivity, ISI-8 ANNs had substantially more connections per *ma* NPE than ISI-4 or ISI-16 ANNs (see top panel of Figure 4.27), which may have been necessary for the former to show comparably high *ma* activations, and hence, *vta* discrepancies. In contrast, the number of connections per *vta* NPE remained roughly constant at about 6 across ISIs, which is consistent with the corresponding effect

observed in Experiment 1. These effects suggest that a certain number of connections per *vta* were sufficient to achieve comparable levels of *vta* discrepancy across a variety of environmental conditions, an effect analogous to that described above for the number of *vta* NPEs.

However, the number of connections per *vta* in the present experiment tended to be substantially higher than the number of connections per *cal* NPE (compare bottom panels of Figures 4.27 and 4.21, respectively). This effect is in contrast to that observed in Experiment 1, where the number of connections per *vta* NPE was considerably lower than the number of connections per *cal* NPE. In general, the functional effect of a higher *vta* connectivity in the present ANNs may have been to increase the overall magnitude of *vta* discrepancies, which may have been necessary for these ANNs to learn two CS-US associations.

Regarding activation parameters, *ma* NPEs in ISI-8 ANNs had considerably lower temporal summations (at about .6) than *ma* NPEs in ISI-4 or ISI-16 ANNs (see top panel of Figure 4.28). In fact, *ma* NPEs in ISI-8 ANNs tended to have lower temporal summations than *sa* NPEs (see top panel of Figure 4.22). However, the former may still have been sufficiently high to have no detrimental effects on *ma*-NPE activations. In contrast, *vta* NPEs in ISI-8 ANNs had substantially higher temporal summations (close to 1.0) than *vta* NPEs in ISI-4 or ISI-16 ANNs (see bottom panel of Figure 4.28). This effect deviates sharply from that observed in *cal* NPEs (see bottom panel of Figure 4.22). Such high *vta* temporal summations in ISI-8 ANNs may have compensated for the detrimental effects of low *cal* temporal summations on *cal* discrepancies.

Comparisons with Figure 4.27 (to panel) show that higher temporal summations in *ma* NPEs corresponded to smaller numbers of connections per *ma* NPE (ISI-4 and ISI-

16 ANNs), whereas lower *ma* temporal summations corresponded to larger numbers of connections per *ma* NPE (ISI-8 ANNs). These relations suggest the presence of a trade-off between *ma* temporal summations and the number of connections per *ma* NPE. The functional significance of this trade-off may have been related to synaptic competition. That is, in general, as the number of input connections to an NPE decrease, the less likely it is that the NPE would be activated at a given *ts*, everything else being equal. Thus, higher *ma* temporal summations may have compensated for a smaller number of connections by increasing the likelihood of *ma* NPE being activated across *ts*. In contrast, the larger the number of input connections to an NPE, the more likely it is that the NPE would be activated at a given *ts*. However, high temporal summations here would make it more likely that a few presynaptic inputs would have very high connection weights, producing a situation similar to that in which the NPE receives connections from only a few presynaptic elements. By decreasing temporal summations, the likelihood of this situation occurring can be reduced, thus giving more presynaptic inputs more chance to have larger weights. The net effect of this counterbalance, again, would be to increase the likelihood that the NPE will be activated at a given *ts*.

Activation decays in *ma* NPEs decreased as the ISI increased, reaching values close to 0.0 for ISI-8 and ISI-16 ANNs (see top panel of Figure 4.29). The functional effect of this result may have been to increase the tolerance of *ma* NPEs to reinforcement delay. This effect may have been especially relevant in the case of *ma* NPEs, given the small number of NPEs of this type found across ANNs. In the case of *vta* NPEs, activation decays remained below 0.25 across ISIs (see bottom panel of Figure 4.29). This result suggests that, like the number of *vta* NPEs and connections per *vta* NPE, similar *vta*-activation decays may have been adequate to achieve comparable levels of

*vta* discrepancies across a variety of environmental conditions. Comparisons with Figure 4.23 show that *ma* and *vta* NPEs had substantially lower activation decays than *sa* and *caI* NPEs, for all ISIs, which is consistent with the idea of low decays as facilitating tolerance of *ma* NPEs to reinforcement delay.

Regarding synaptic-plasticity parameters, acquisition rates in *ma* NPEs remained relatively constant at about 0.75 across ISIs (see top panel of Figure 4.30), indicating that similar (although high) acquisition rates may have been sufficient to achieve comparable levels of *ma* activations, and hence, *vta* discrepancies across different ISIs. In contrast, *vta* NPEs in ISI-8 ANNs had substantially lower acquisition rates than *vta* NPEs in ISI-4 or ISI-8 ANNs (see bottom panel of Figure 4.30).

Comparisons with Figure 4.28 (bottom panel) suggest a trade-off between temporal summations and acquisition rates in *vta* NPEs for ISI-8 ANNs. This trade-off may have reduced the range of variation of connection weights to *vta* NPEs, and hence, *vta* activations and discrepancies. Such a reduction may have conferred a fitness advantage by decreasing the behavioral flexibility of these ANNs, which may have been necessary to adapt to an environment whose reinforcement contingencies were fixed (i.e., in which all *all* CSs were reinforced). This effect may have been especially relevant to ISI-8 ANNs, due to their large sizes. Perhaps, the detrimental effects of a high behavioral flexibility under fixed reinforcement conditions were "magnified" in large ANNs. In this way, high *vta* acquisition rates may have pushed large ancestral ANNs toward *local minima* (i.e., low-performance points), and hence, toward low fitness.

Finally, extinction rates in *ma* NPEs remained relatively low (below 0.25) for all ISIs (see top panel of Figure 4.31). Low *ma* extinction rates were also observed in Experiment 1 (see bottom panel of Figure 3.34), indicating that connection weights in *ma* NPEs tended to decrease quite slowly across a variety of environmental conditions,

relative to connection weights in other kinds of NPEs. Low *ma* extinction rates may have compensated for a small number of *ma* NPEs in a way analogous to that in which high *ma* temporal summations may have compensated for a small number of connections per *ma* NPE. However, *vta* NPEs in ISI-8 ANNs had substantially higher extinction rates than ISI-4 or ISI-16 ANNs (see bottom panel of Figure 4.31), which may have compensated for the detrimental effects of low *cal* acquisition rates observed in these ANNs (see bottom panel of Figure 4.25). In this way, *cal* discrepancies may have been greater than they would have been otherwise.

#### **4.1.2.3.3 Effects on the Motor Subnetwork: Output NPEs**

Figures 4.32 through 4.37 show changes across ISIs in structural traits that relate to output NPEs, as a function of ISIs. The mean number of output NPEs remained relatively constant at approximately 10 (see Figure 4.32). However, ISI-8 ANNs tended to have substantially fewer connections per output NPE than ISI-4 or ISI-16 ANNs (see Figure 4.33). Therefore, synaptic competition among *ma* NPEs in the last hidden layer for output-NPE weights was greatly reduced, which could have facilitated the ability of these ANNs to respond with high output-NPE activations to each CS separately. The mean temporal summation of output NPEs remained close to 1.0 for all ISIs (see Figure 4.34). Also, mean activation decays remained constant at values below .5 (see Figure 4.35). High temporal summations and low activation decays may have conferred a fitness advantage by increasing the tolerance of output NPEs to reinforcement delay. Output NPEs in ISI-8 ANNs had substantially lower acquisition rates than ISI-4 or ISI-16 ANNs (see Figure 4.36). Lower acquisition rates in output NPEs corresponded to fewer connections per output NPE (see Figure 4.33), for ISI-8 ANNs. Such a correspondence may have had the effect of decreasing the range of variation of connection weights to output NPEs, and hence decreasing the likelihood of

inappropriate connections being strengthened. Finally, extinction rates remained roughly constant across ISIs, at values below 0.25 (see Figure 4.37), which may have increased the tolerance of output NPEs for reinforcement delay.

## 4.2 Test Simulations

A series of simulations were run in which the *same* ANNs described in Section 4.1.2.3 were tested in four Pavlovian procedures, namely generalization, orthogonal discrimination, nonorthogonal discrimination, and blocking. These procedures involved environmental conditions other than those present when selection took place. Hence, they allowed for testing whether or not those ANNs showed emergent functional phenotypes consistent with the effects of such procedures on learning and behavior in real organisms.

### 4.2.1 Generalization

In the evolution simulations, ANNs showed high CR proportions to each CS *separately*. However, it is not clear whether or not these ANNs are competent to learn to respond to concurrent reinforced presentations of *both* CSs (*compound* conditioning), and then to respond more (i.e., to generalize) or less to untrained CSs that are more or less similar to the compound CS, respectively. To address this issue, a generalization simulation was run. During a training phase, naïve ANNs were given 100 reinforced trials in which the ISI was identical to the ancestral one, and the CS involved the activation of *all* input elements with a value 1.0 (i.e., trials consisted of *compound* CSs). During a testing phase, the trained ANNs were given 175 probe trials of seven randomly intermixed input-activation patterns. To avoid extinction during this phase, the synaptic-plasticity function of the NC model was disabled.



Figure 4.38 depicts typical examples of the kind of input-activation patterns used in this simulation. Filled squares and circles represent  $s_1$  and  $s_2$  input elements, respectively, that were activated with a value of 1.0. Pattern 3, labeled as CS+, indicates the pattern that was reinforced during the training phase (i.e., the compound CS). In general, for any given pattern, the number of activated input elements of a given kind depended upon the number of elements of that kind found in the ANN. For example, Pattern 1 involved the activation of two  $s_1$  elements for ANNs having five  $s_1$  elements (which was the typical case in the last generation of ISI-4). However, Pattern 1 involved the activation of three  $s_1$  elements for ANNs having six  $s_1$  elements (which was the typical case in the last generation of ISI-8), and the activation of one  $s_1$  element for ANNs having two  $s_1$  elements (which was the typical case in the last generation of ISI-16). As Figure 4.38 shows, test input patterns differed from the trained pattern as the total number of activated input elements decreased relative to the reinforced pattern. Also, note that Patterns 4 and 5 involved the activation of all input elements of one kind *and* some input elements of the other kind (i.e., these patterns had some input elements in common).

Figures 4.39, 4.40, and 4.41 show generalization gradients obtained during the test phase of ISI-4, ISI-8, and ISI-16 ANNs, respectively. Overall, it can be observed that ANNs showed generalization in that they responded with higher CR proportions to input patterns that were more similar to the trained pattern. ISI-8 ANNs showed a substantial amount of generalization to all input patterns except 1 and 7. In contrast, ISI-16 ANNs showed the steepest generalization gradient, with CR proportions falling below 0.5 for Patterns 1, 2, 6, and 7. These results indicate that selection for increased responding to two independent CSs generated ANNs that were capable of showing generalization as an *emergent* functional phenotype.

## 4.2.2 Orthogonal Discrimination

In the evolution simulations, ANNs were selected for increased responding to two *reinforced* CSs that involved the activations of either all  $s_1$  or all  $s_2$  input elements. Although the resulting ANNs showed high CR proportions in the presence of each CS, it is not clear whether or not these ANNs are competent to learn to respond to only one reinforced CS, when nonreinforced instances of a different CS are given.

To address this issue, an orthogonal discrimination test simulation was run. Naïve ANNs were trained in two different conditions, each one consisting of a training and a testing phase. In the training phase of the first condition, ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials (an average of 100 trials per CS kind). CS1 involved the activation of all  $s_1$  input elements with a value of 1.0 (see Pattern 2 in Figure 4.38), whereas CS2 involved the activation of all  $s_2$  input elements, also with a value of 1.0 (see Pattern 6 in Figure 4.38). In the training phase of the second condition, ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 probe trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled.

Figures 4.42, 4.43, and 4.44 show the CR proportions obtained during the testing phase of each condition (conditions are separated by a dotted line), for ISI-4, ISI-8, and ISI-16 ANNs, respectively. In general, ANNs showed orthogonal discrimination in that they responded with substantially higher CR proportions to the reinforced CS (CS1 in the first condition and CS2 in the second condition) than to the nonreinforced CS. ISI-4 and ISI-16 ANNs showed higher discrimination ratios than ISI-8 ANNs. Also, ISI-16 ANNs showed lower CR proportions to the reinforced CS than ISI-4 or

ISI-8 ANNs. These results indicate that selection for increased responding to two independent CSs generated ANNs that were capable of simulating orthogonal discrimination as an emergent functional phenotype.

### 4.2.3 Nonorthogonal Discrimination

The orthogonal-discrimination test simulations showed that ANNs whose ancestors were selected for increased responding to two reinforced CSs were capable of learning an orthogonal discrimination, regardless of whether the reinforced CS was CS1 or CS2. However, in that simulation, CS1 and CS2 had no input elements in common (i.e., affected different sensory modalities, and hence, were relatively easy to discriminate). This feature was simulated by defining CS1 and CS2 as patterns involving the activation of *either* all  $s_1$  *or* all  $s_2$  input elements. This result raises the issue of whether or not the same ANNs would learn a nonorthogonal discrimination, in which the reinforced and nonreinforced CSs involve the activation of some input elements in common.

To address this issue, a nonorthogonal discrimination test simulation was run. Naïve ANNs were trained in two different conditions, each one consisting of a training and a testing phase. In the training phase of the first condition, ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials (an average of 100 trials per CS kind). CS1 and CS2 were identical to Patterns 3 and 5 respectively, as described in the generalization test (see Figure 4.38a). In the training phase of the second condition, the ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled.

Figures 4.45, 4.46, and 4.47 show the CR proportions obtained during the testing phase of each condition (conditions are separated by a dotted line), for ISI-4, ISI-8, and ISI-16 ANNs, respectively. ANNs did not show nonorthogonal discrimination in that they responded with equally high CR proportions to the reinforced CS (CS1 in the first condition and CS2 in the second condition) and the nonreinforced CS. ISI-4 and ISI-8 ANNs showed higher CR proportions than ISI-16 ANNs. These results indicate that selection for increased responding to two independent CSs generated ANNs that did not show nonorthogonal discrimination as an emergent functional phenotype.

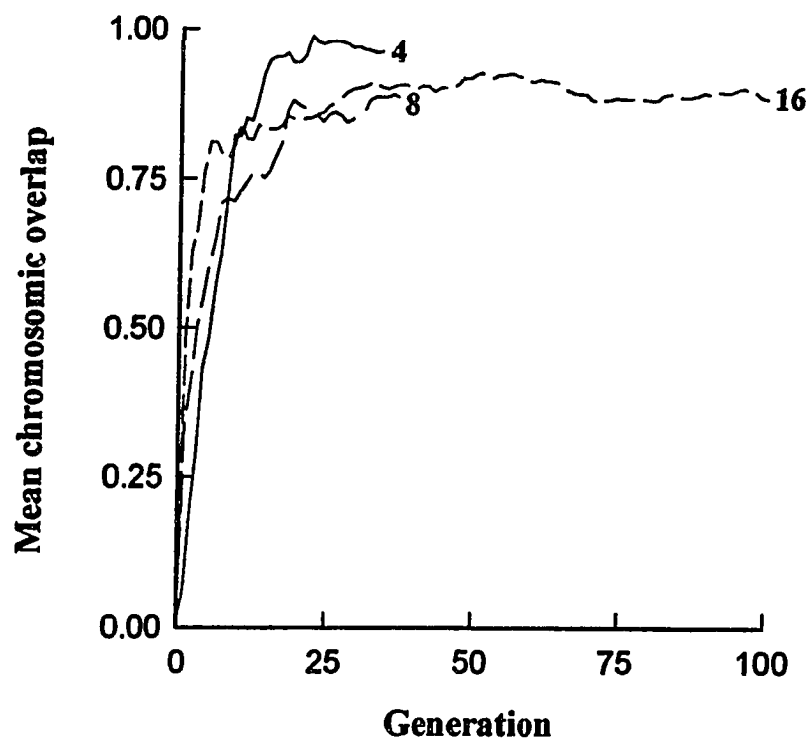
#### 4.2.4 Blocking

The above test simulations show that ANNs whose ancestors were selected for increased responding to two reinforced orthogonal CSs were capable of showing generalization and orthogonal discrimination. However, these ANNs did not show nonorthogonal discrimination. Generalization and discrimination represent basic Pavlovian conditioning phenomena. Another basic phenomenon is blocking. Are ANNs that evolved under two reinforced CSs, capable of showing blocking?

To address this issue, naïve ANNs were trained in two conditions consisting of three phases. During the *Single-CS* phase, ANNs were given 100 reinforced presentations of a CS. Then, during the *Compound-CS* phase, those trained ANNs were given 100 reinforced concurrent presentations of the previously trained CS and another, novel, previously nonexperienced CS. In one condition, the CS reinforced during the *Single-CS* phase was CS1, whereas the novel CS presented during the *Compound-CS* phase was CS2. Conversely, in the other condition, the CS reinforced during the *Single-CS* phase was CS2, whereas the novel CS presented during the *Compound-CS* phase was CS1. Finally, during the *Test* phase, ANNs were given 75 randomly intermixed trials of

CS1, CS2, and the CS1/CS2 compound (an average of 25 trials per CS kind). For all ANNs, CS1 involved the activation of all  $s_1$  input elements with a value of 1.0, whereas CS2 involved the activation of all  $s_2$  input elements, also with a value of 1.0. The CS1/CS2 compound involved the activation of all input elements. The synaptic-plasticity function of the NC model was disabled during the Test phase.

Figures 4.48, 4.49, and 4.50 show the mean CR proportions per CS kind during Phase 3 of the blocking test of ISI-4, ISI-8, and ISI-16 ANNs, respectively. The top panels show the results for the first condition, in which CS1 was reinforced during the Single-CS phase (marked as CS+ in the graphs) and CS2 was the novel CS introduced in the Compound-CS phase. The bottom panels show the results for the second condition, in which CS2 was reinforced during the Single-CS phase and CS1 was the novel stimulus introduced in the Compound-CS phase. ANNs in both conditions showed blocking in that they tended to respond substantially more to the previously reinforced CS, as well as to the CS1/CS2 compound, than to the novel CS introduced in the Compound-CS phase, the strongest effect being observed in ISI-16 ANNs (see Figure 4.49). These results indicate that selection for increased responding to two independent CSs generated ANNs that were capable of simulating blocking as an emergent functional phenotype.



**Figure 4.1: Mean chromosomal overlap as a function of generations.**

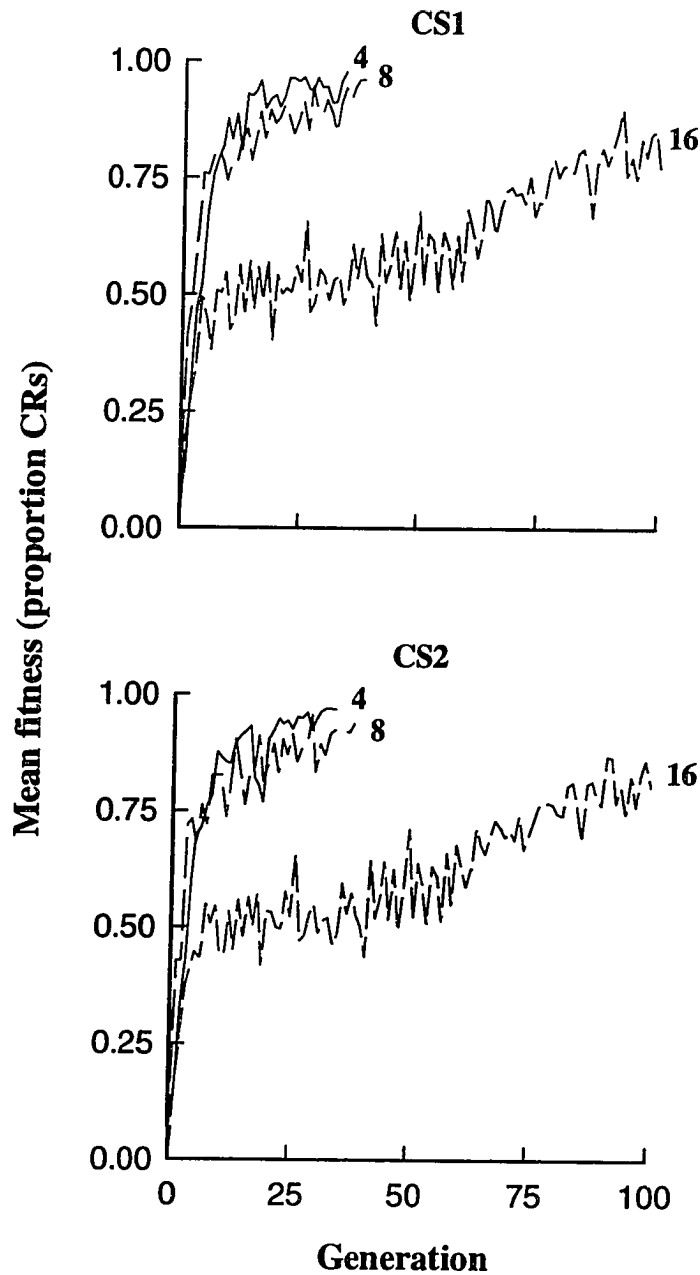
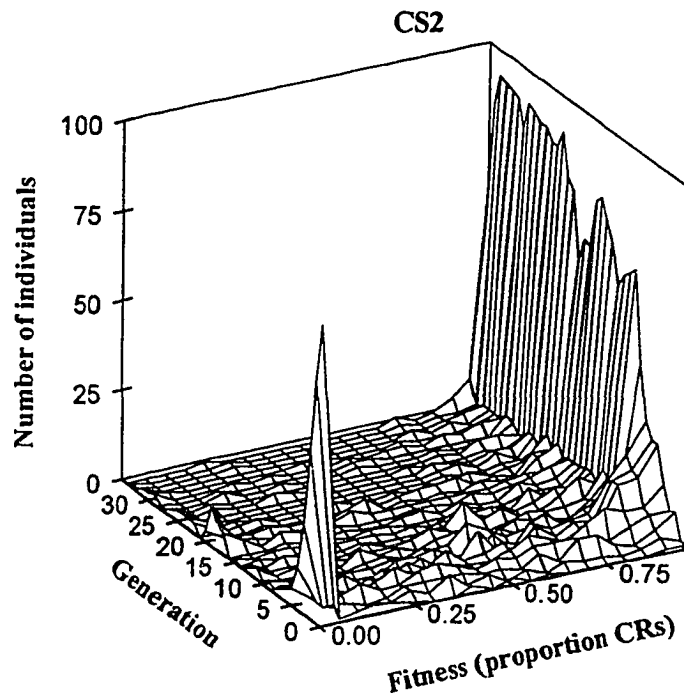
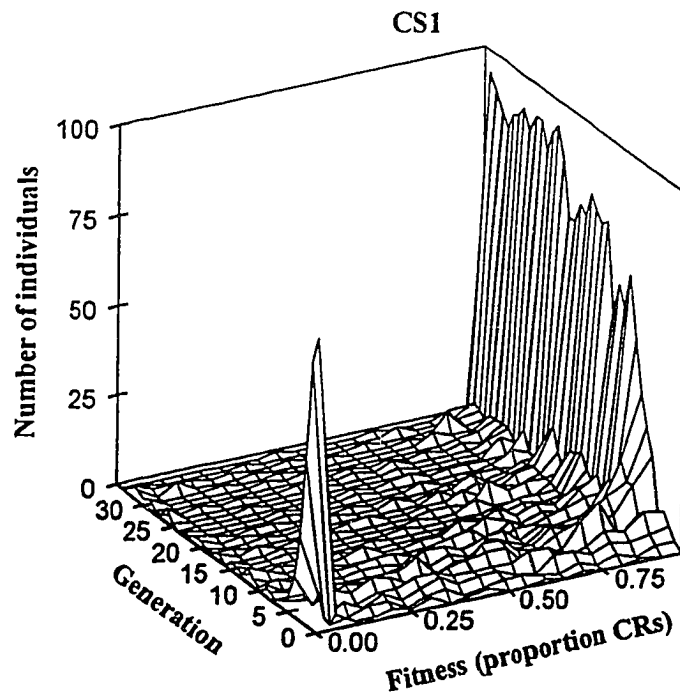
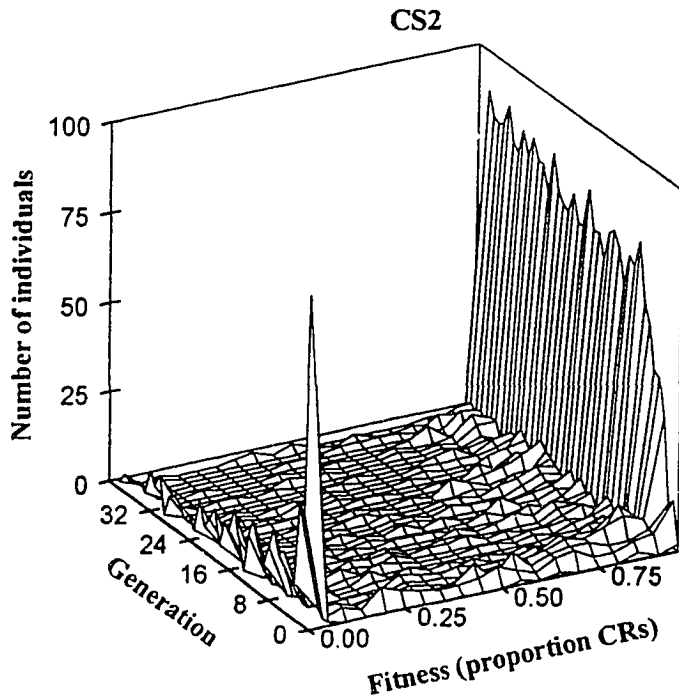
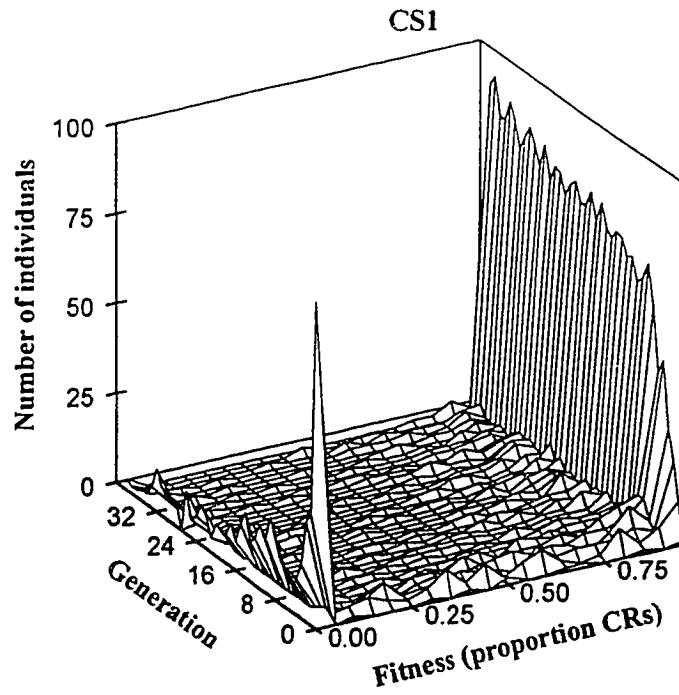


Figure 4.2: **Mean population fitness as a function of generations.** CS1 and CS2 correspond to different kinds of CSs given to all ANNs, rather than different lineages.

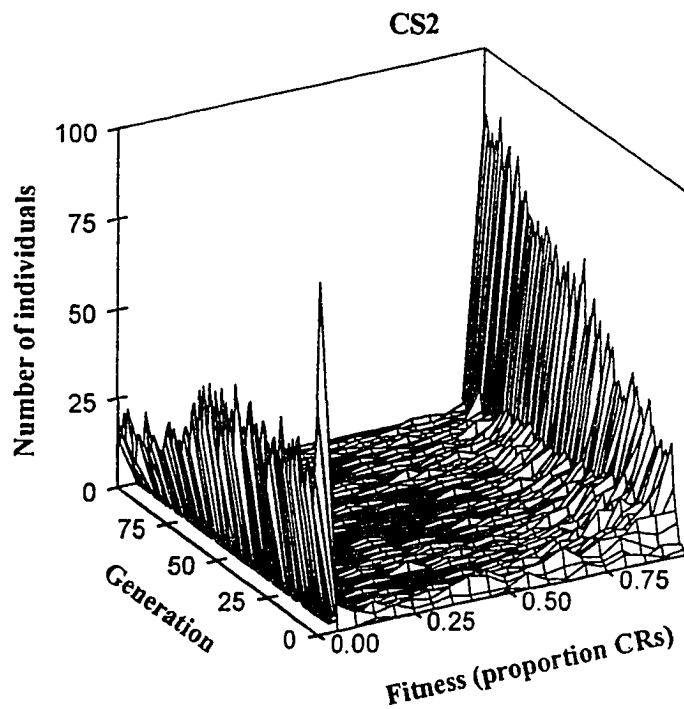
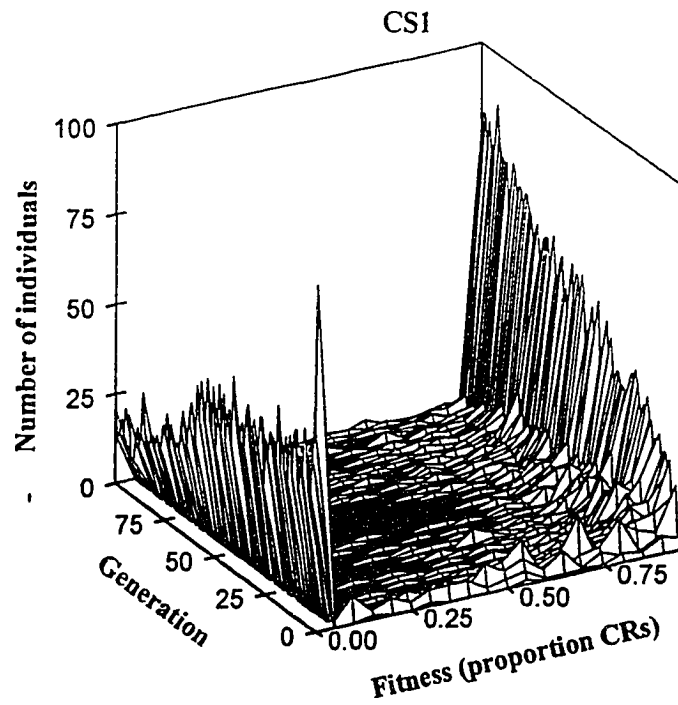


**Figure 4.3: Frequency distributions of fitness values over generations, for ISI-4.**

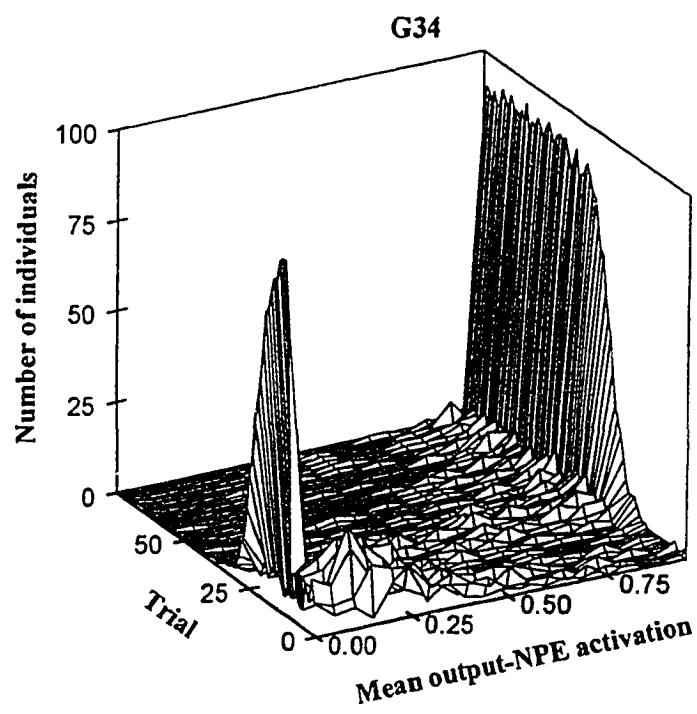
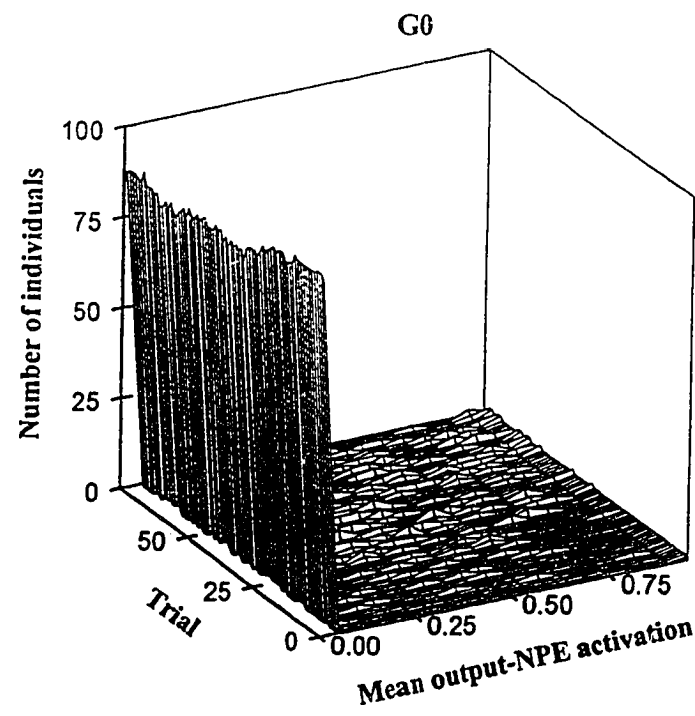




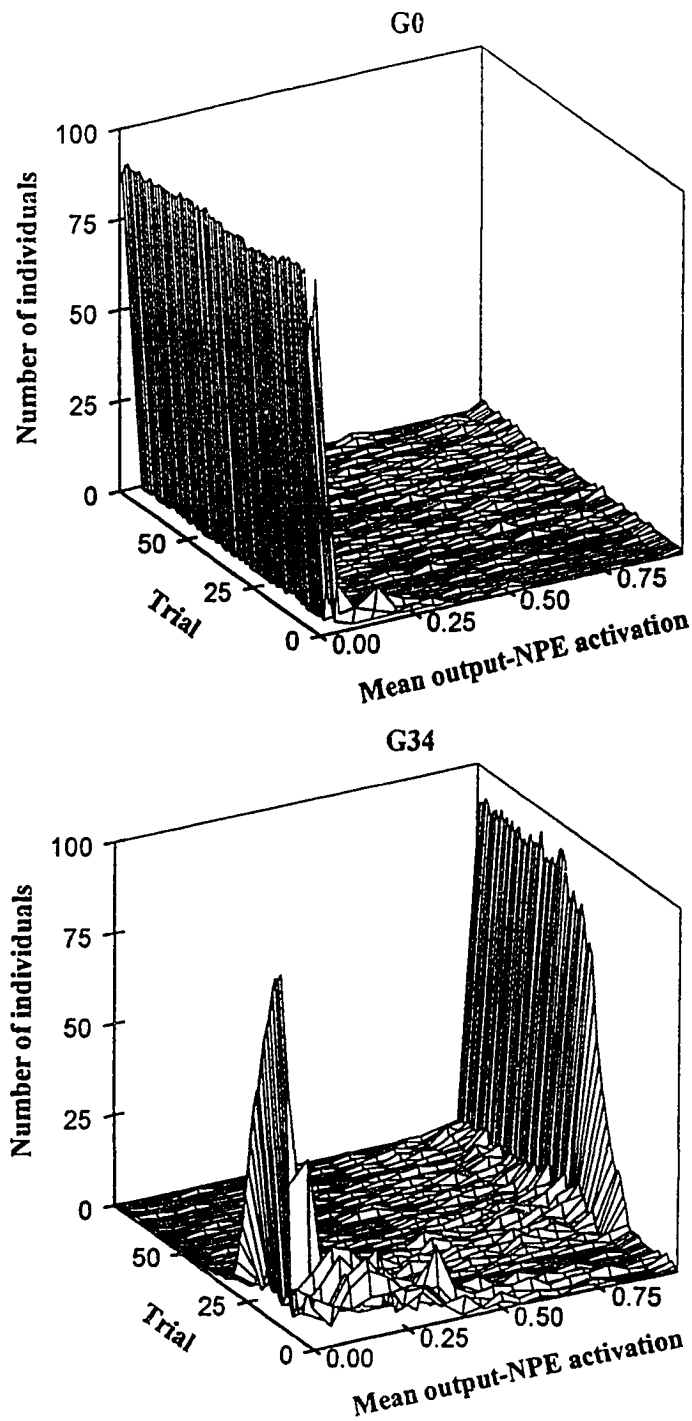
**Figure 4.4: Frequency distributions of fitness values over generations, for ISI-8.**



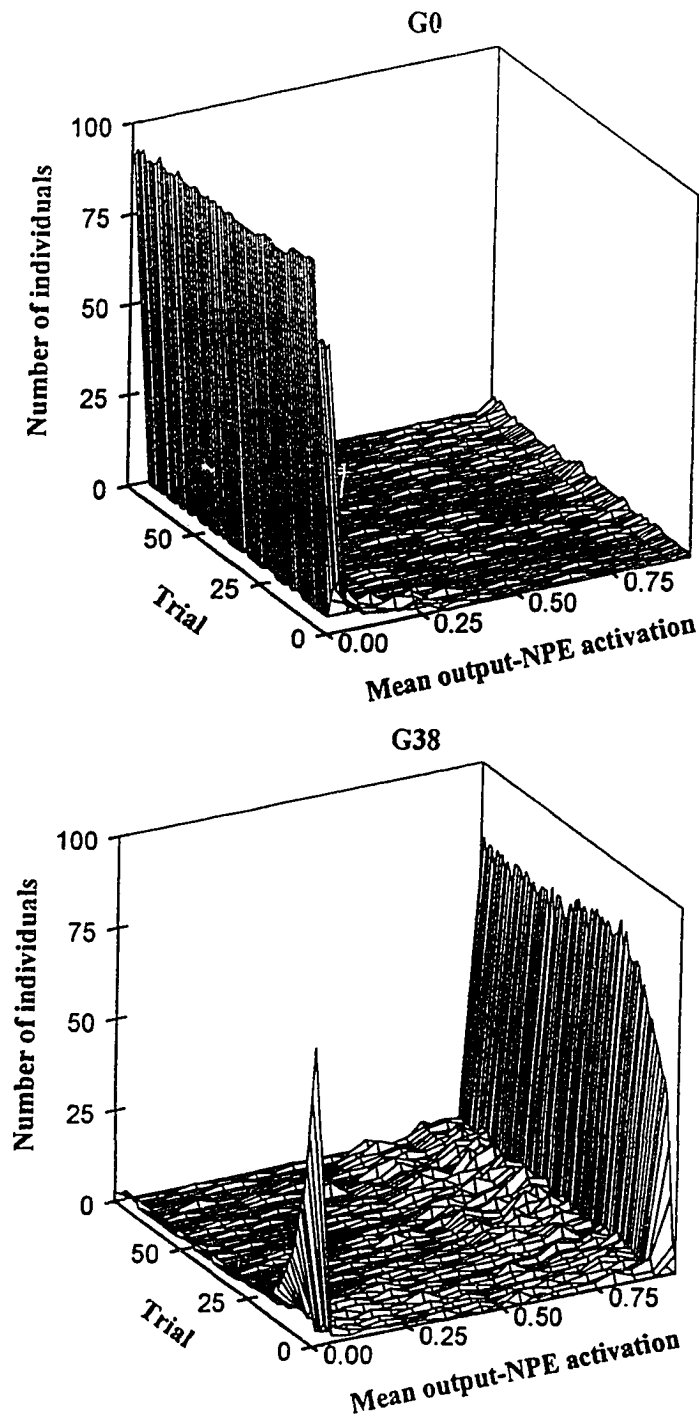
**Figure 4.5: Frequency distributions of fitness values over generations, for ISI-16.**



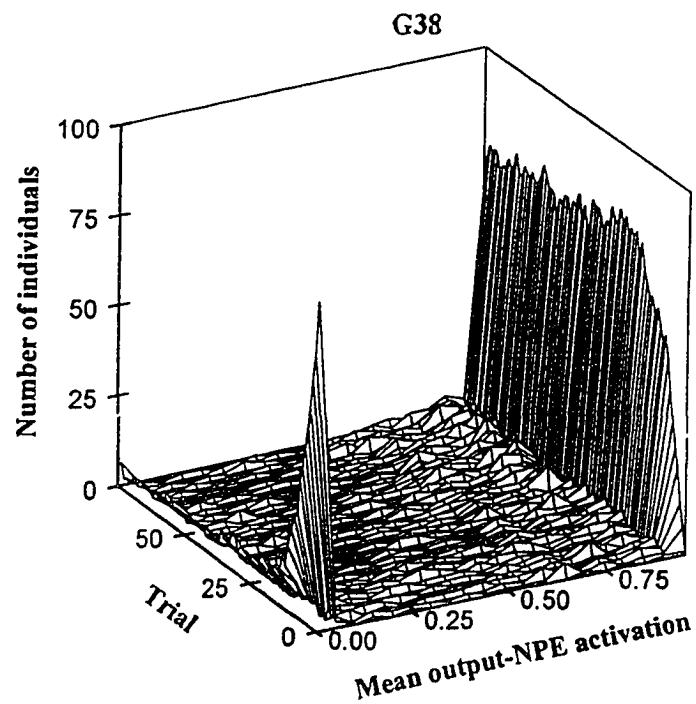
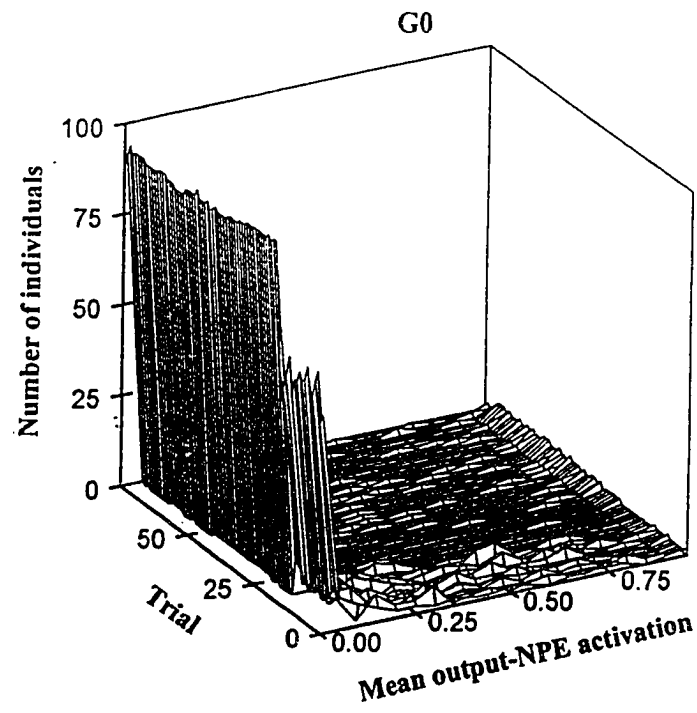
**Figure 4.6: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of ISI-4.**



**Figure 4.7: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of ISI-4.**



**Figure 4.8: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of ISI-8.**



**Figure 4.9: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of ISI-8.**

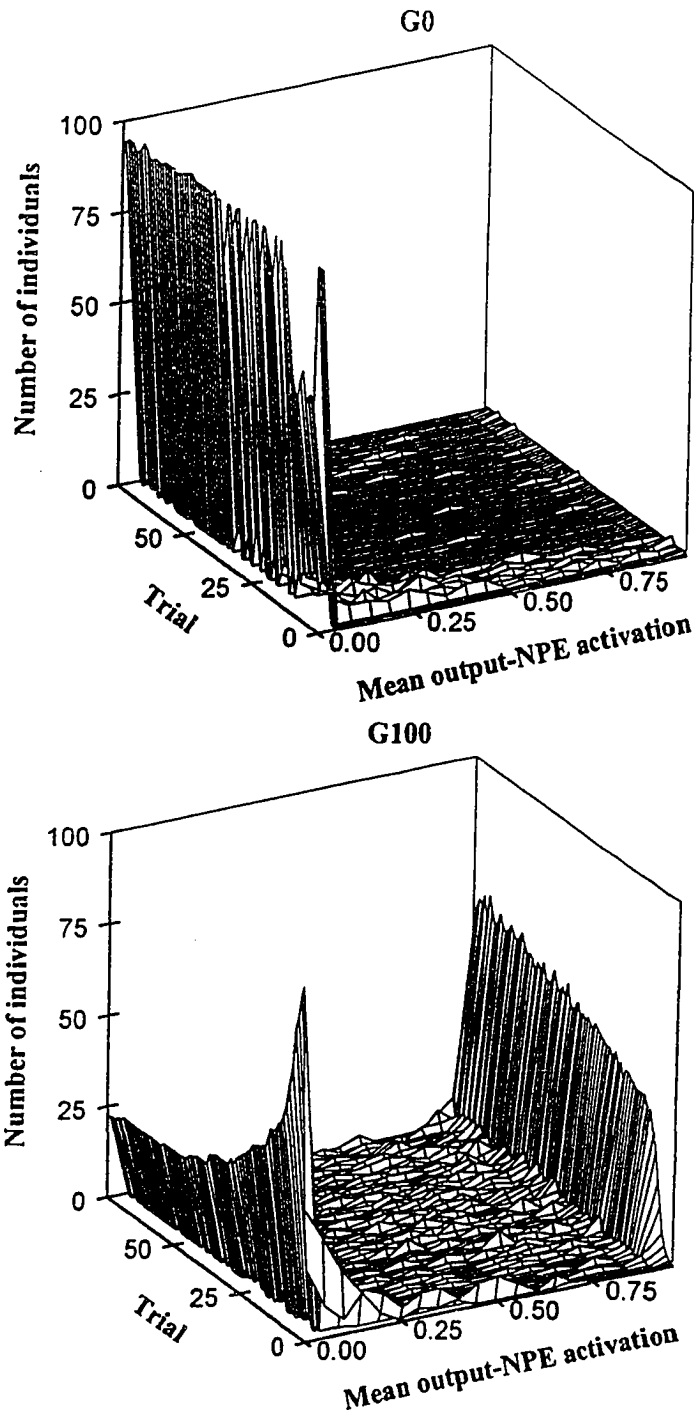
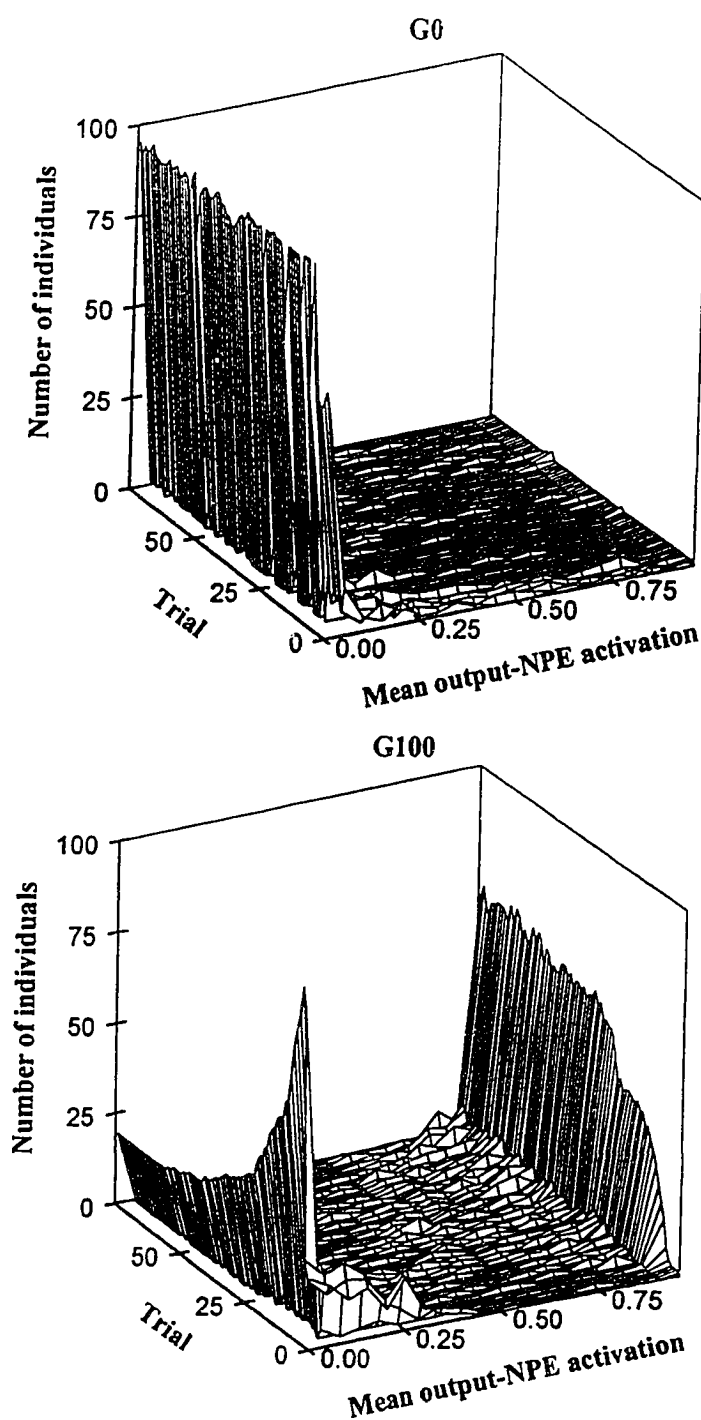
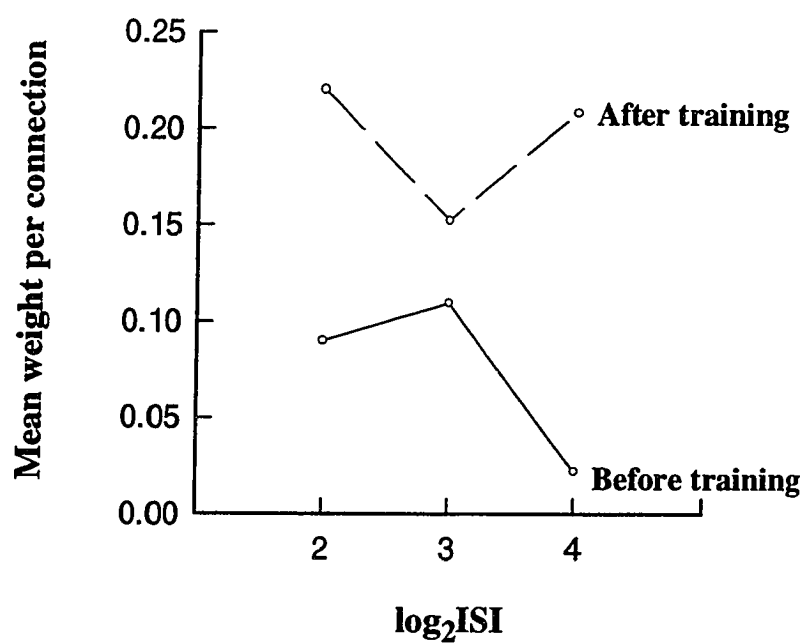


Figure 4.10: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of ISI-16.



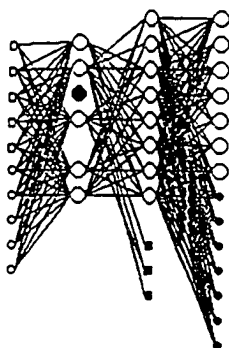
**Figure 4.11: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of ISI-16.**



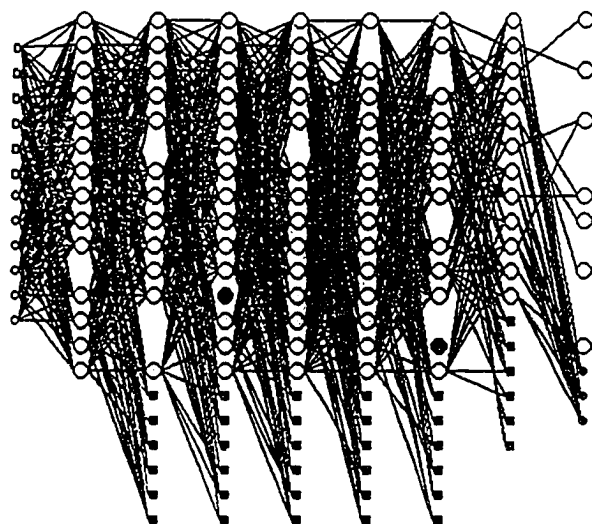


**Figure 4.12: Mean weight per connection in last generation of each lineage, before and after training.**

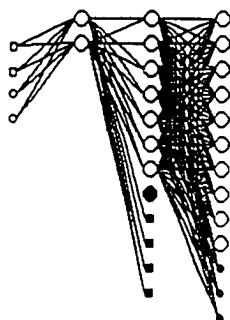
**ISI-4**



**ISI-8**

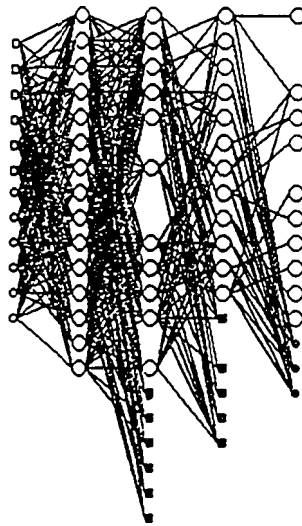


**ISI-16**

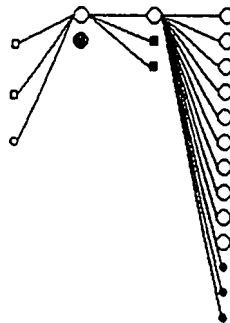


**Figure 4.13: Sample of high-fitness ANNs from the last generation of each lineage. The fitness of all ANNs was above 0.95.**

**ISI-8**

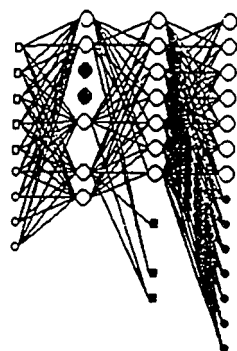


**ISI-16**

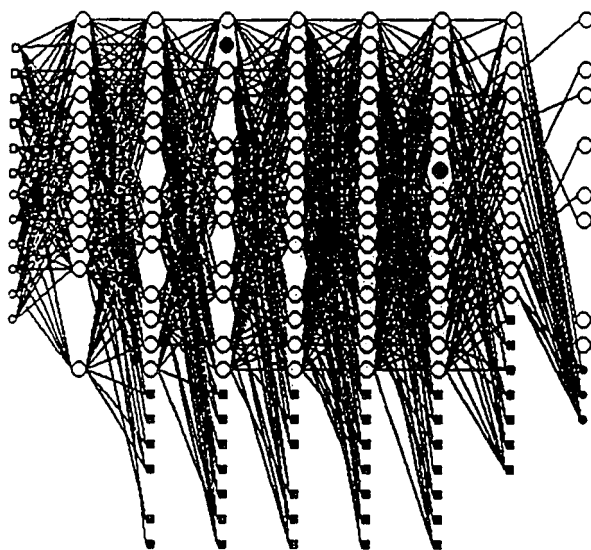


**Figure 4.14: Other high-fitness ANNs from the last generation of ISI-8 and ISI-16.**  
The fitness of both ANNs was above 0.95.

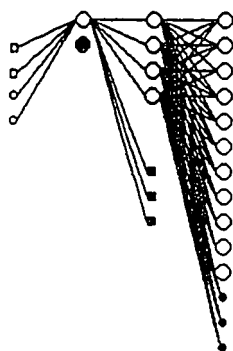
**ISI-4**



**ISI-8**



**ISI-16**



**Figure 4.15: Sample of low-fitness ANNs from the last generation of each lineage. The fitness of all ANNs for one CS (either CS1 or CS2) was below 0.3.**

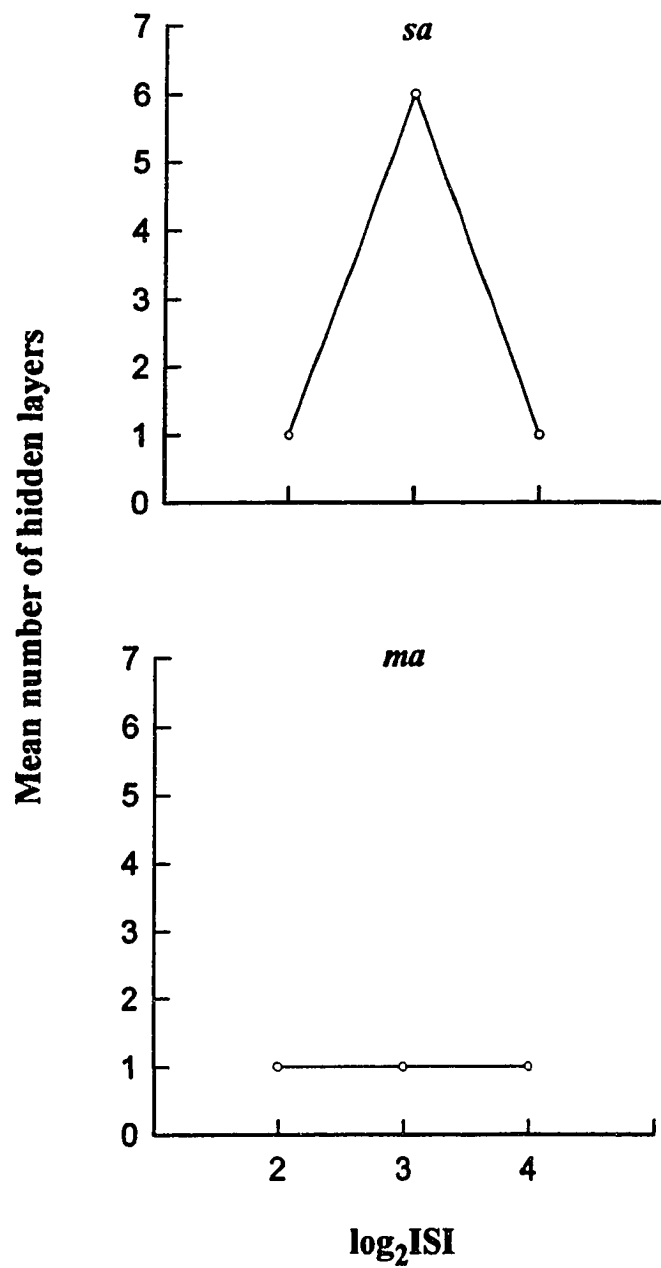
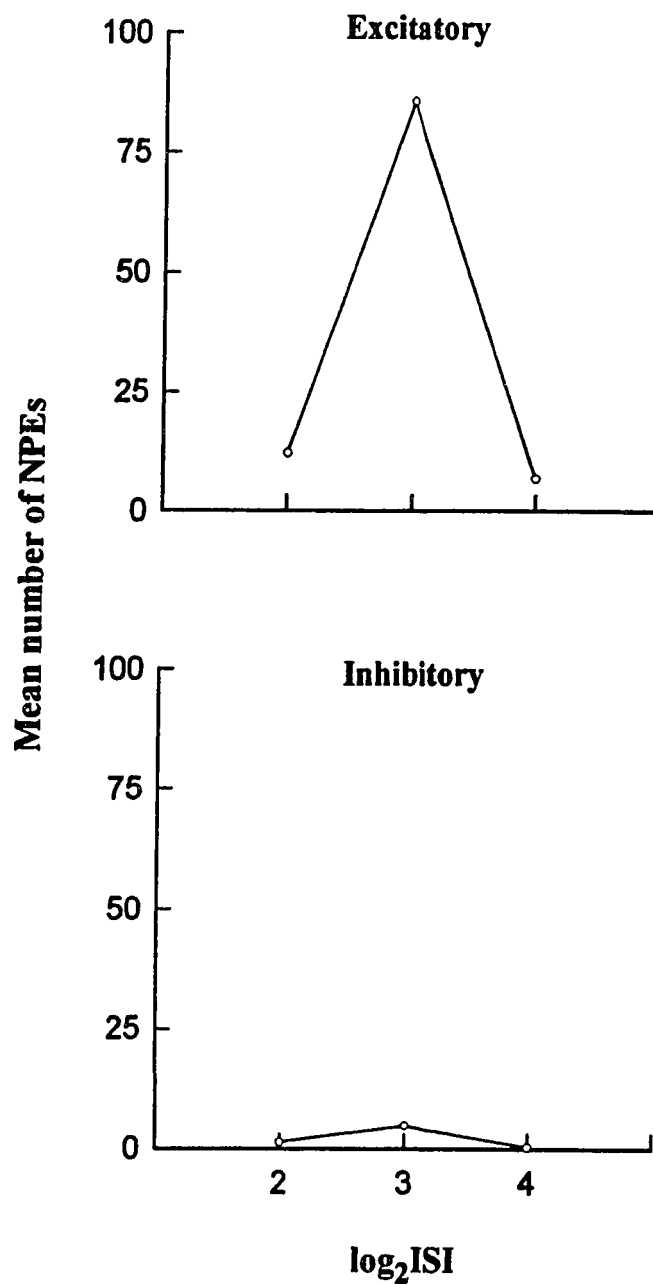
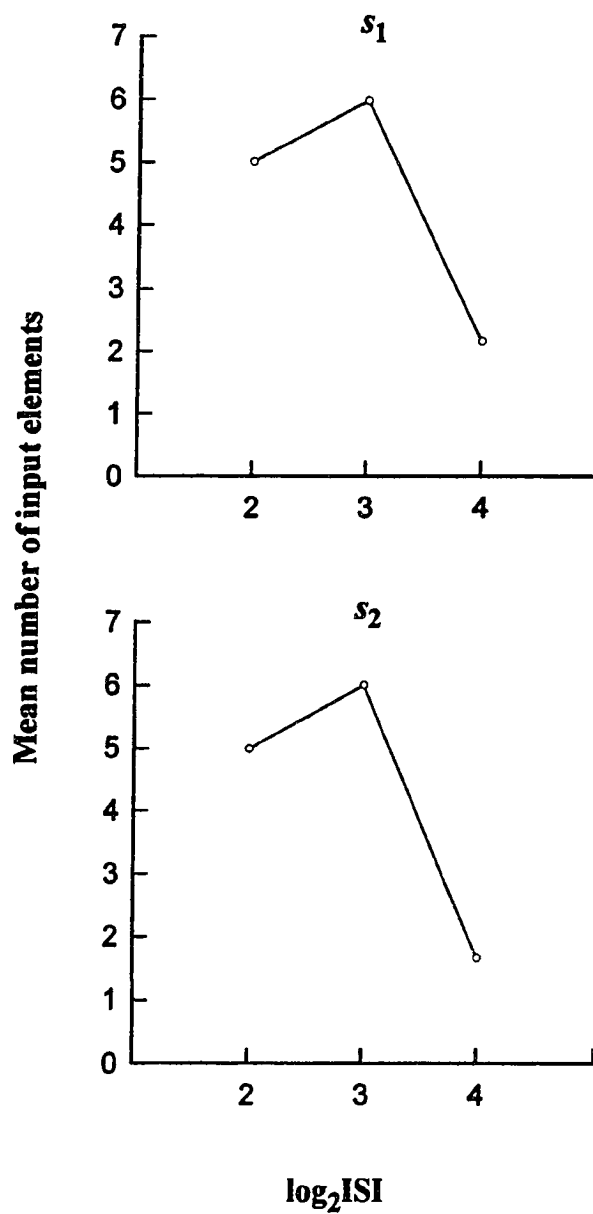


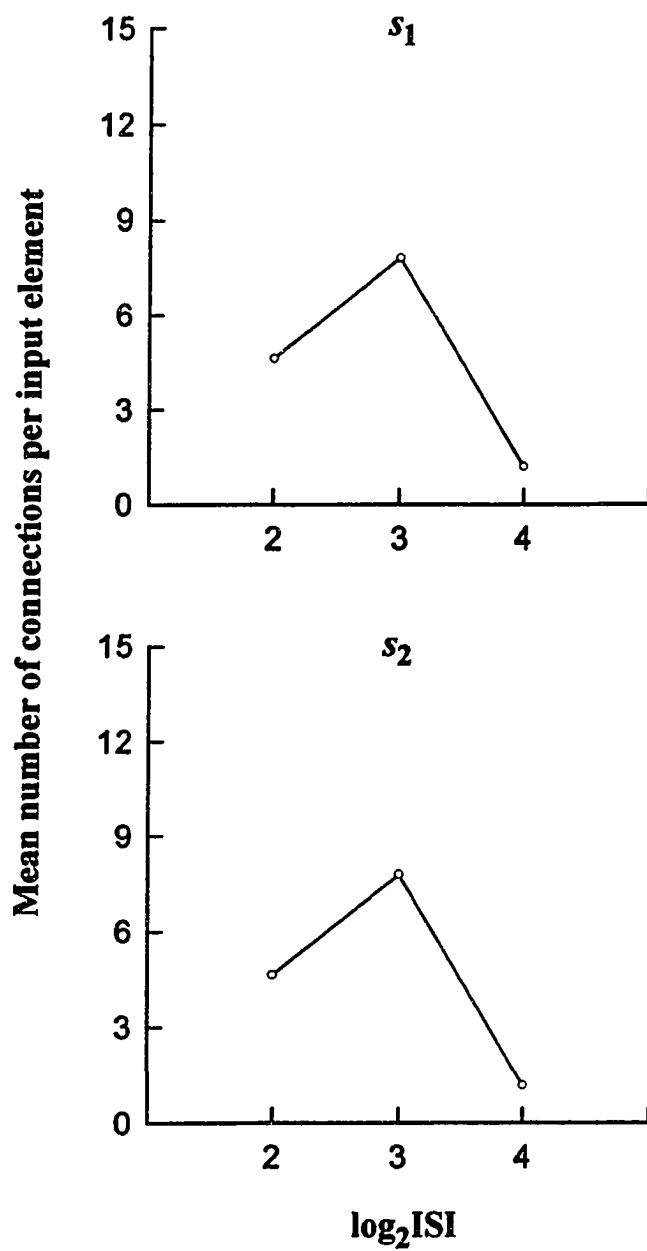
Figure 4.16: Mean number of hidden layers in last generation of each lineage.



**Figure 4.17: Mean number of excitatory and inhibitory NPEs in last generation of each lineage.**

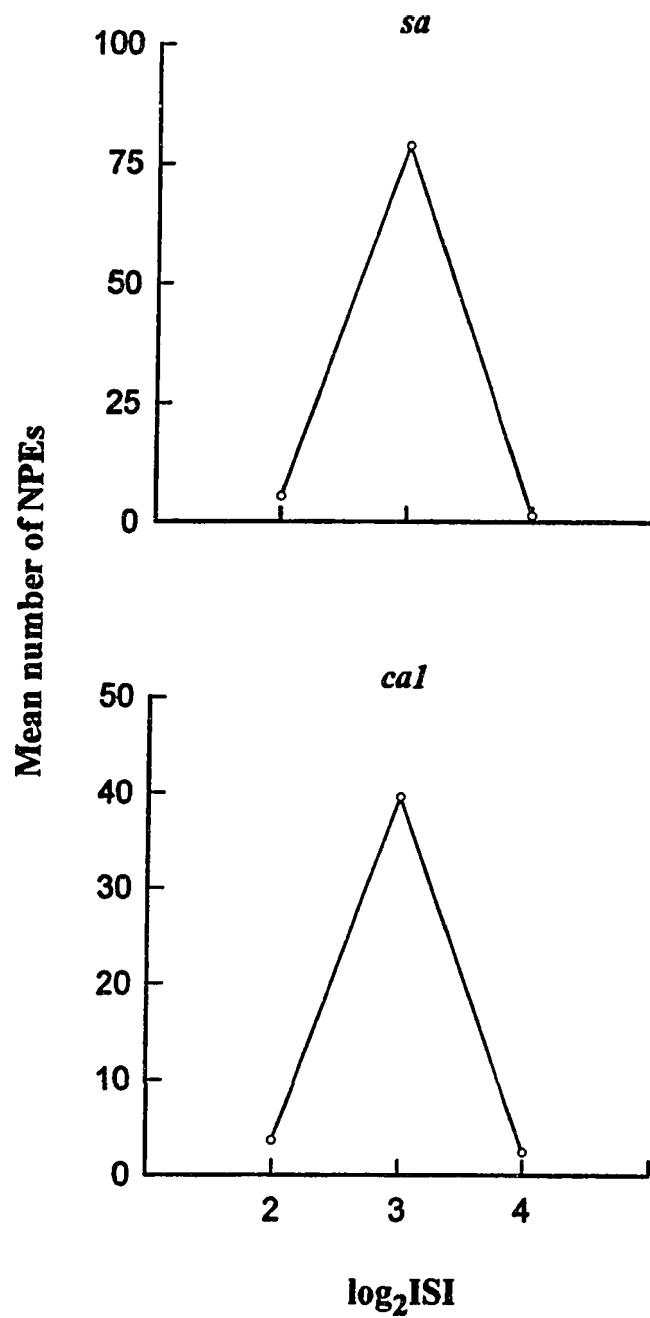


**Figure 4.18: Mean number of input elements in last generation of each lineage.**

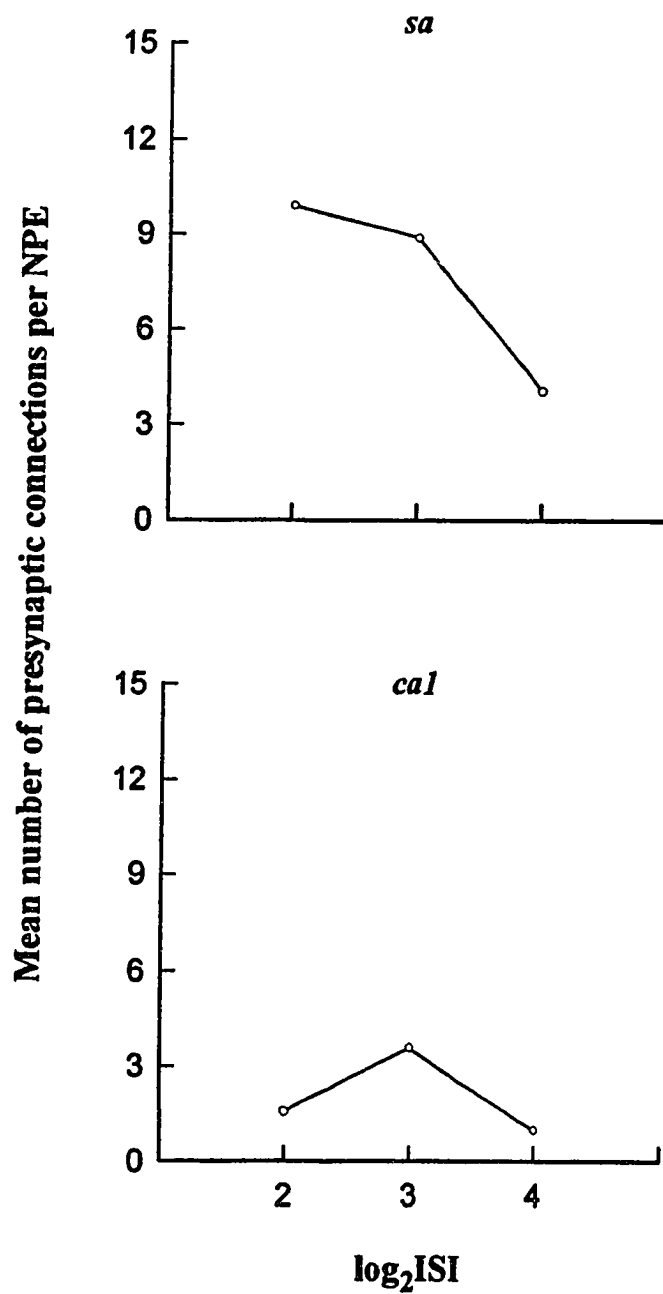


**Figure 4.19: Mean number of connections per input element in last generation of each lineage.**

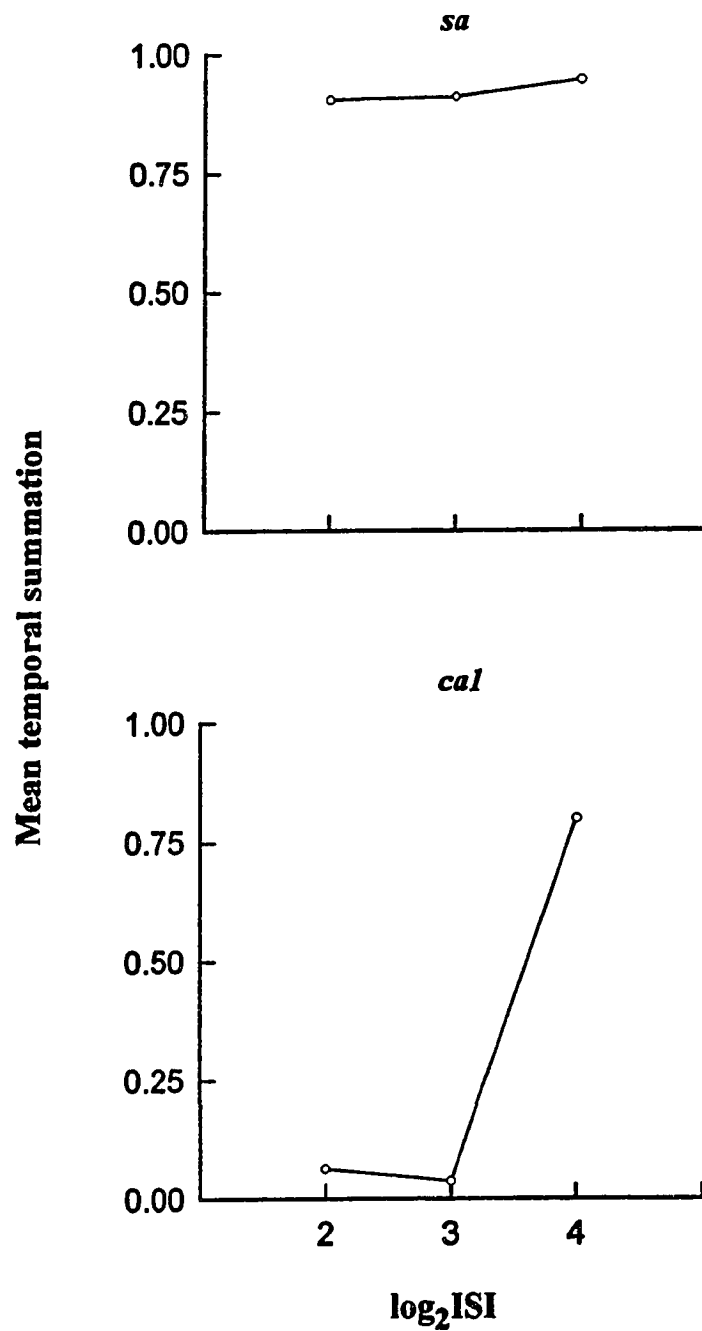




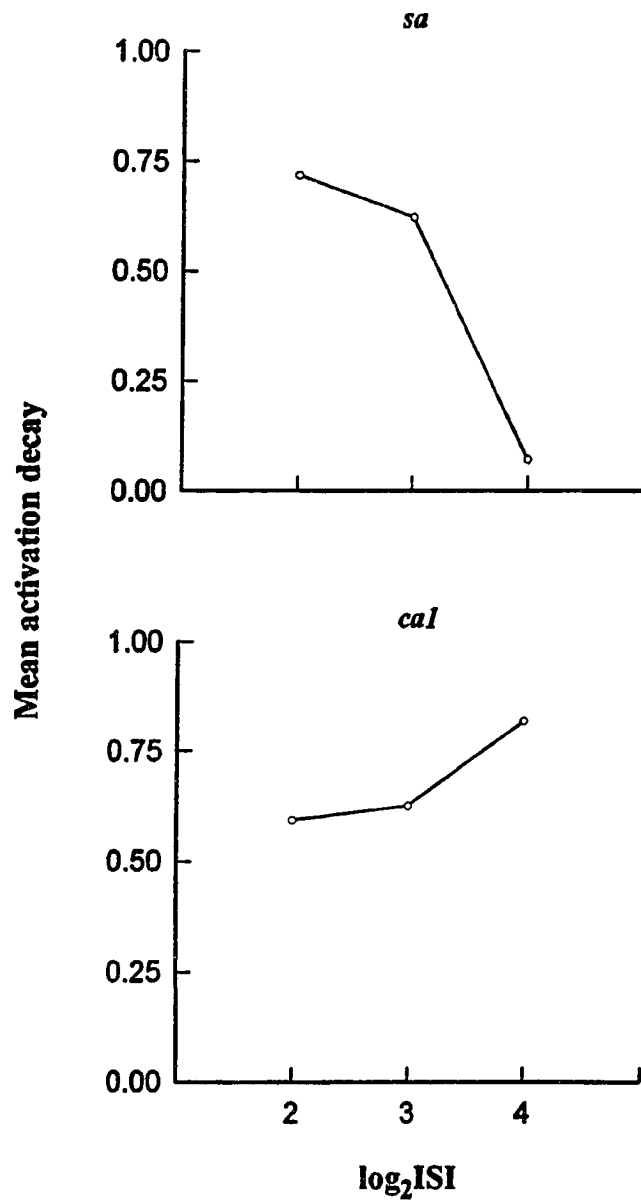
**Figure 4.20: Mean number of *sa* and *cal* NPEs in last generation of each lineage.**



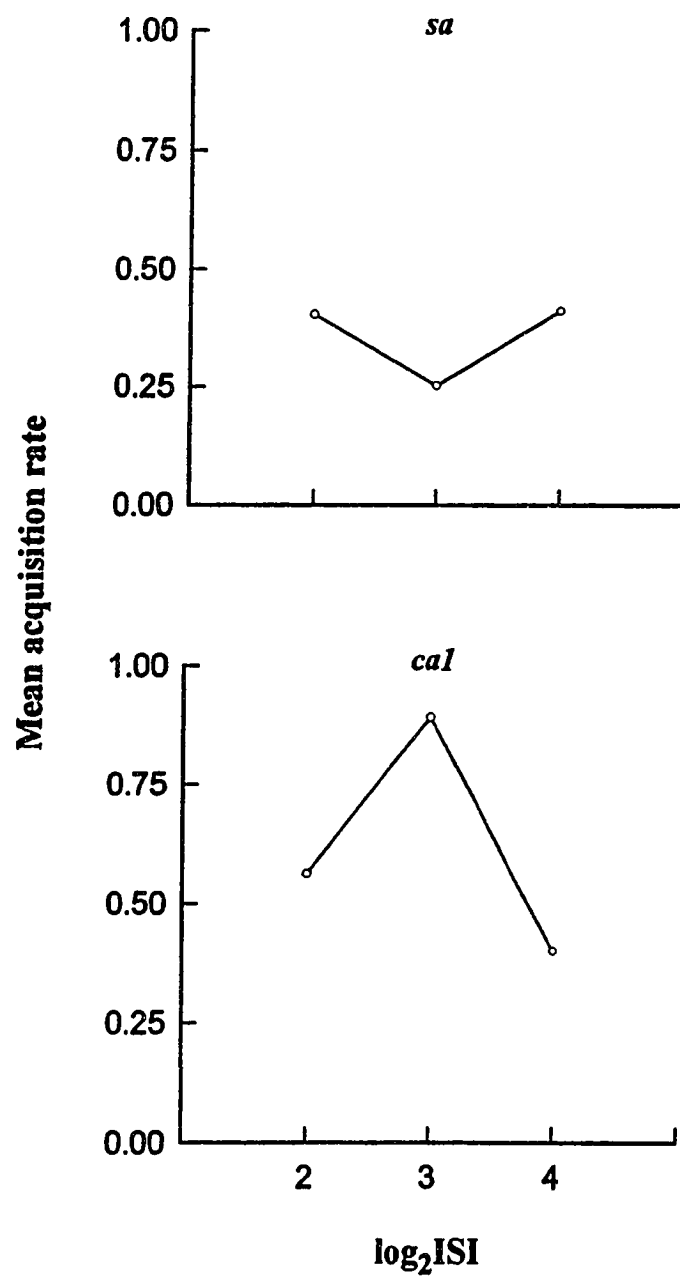
**Figure 4.21: Mean number of presynaptic connections per *sa* and *cal* NPE in last generation of each lineage.**



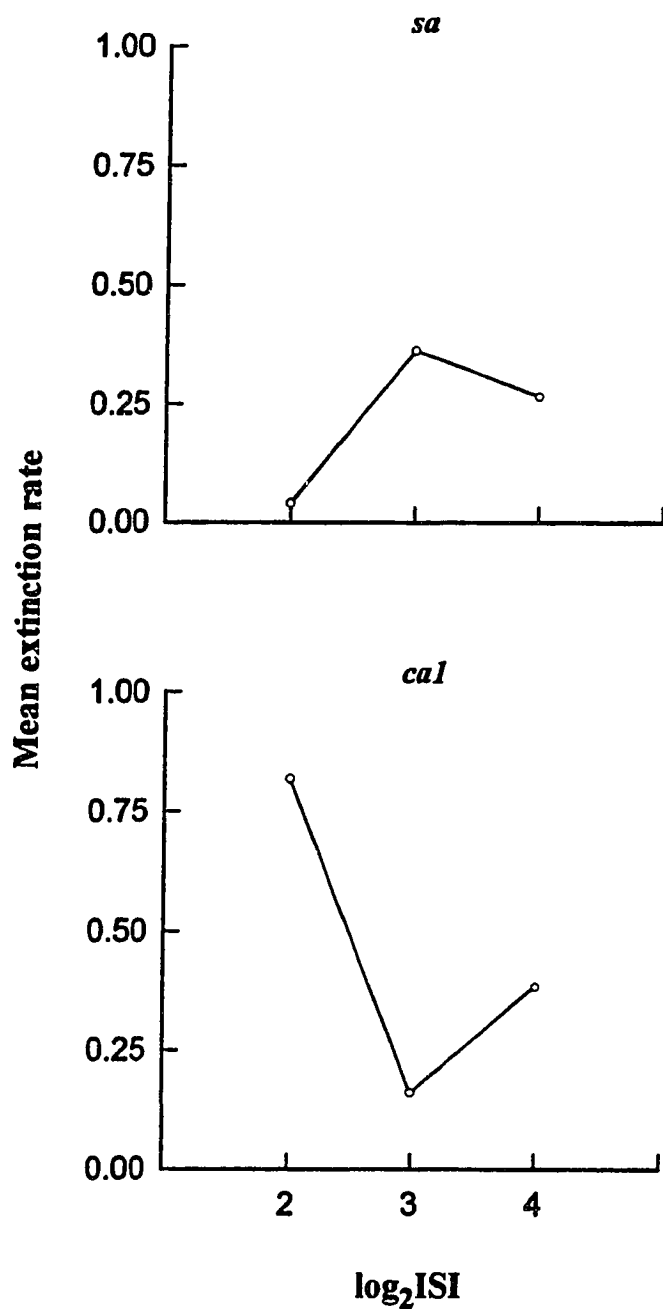
**Figure 4.22: Mean temporal summation in *sa* and *cal* NPEs in last generation of each lineage.**



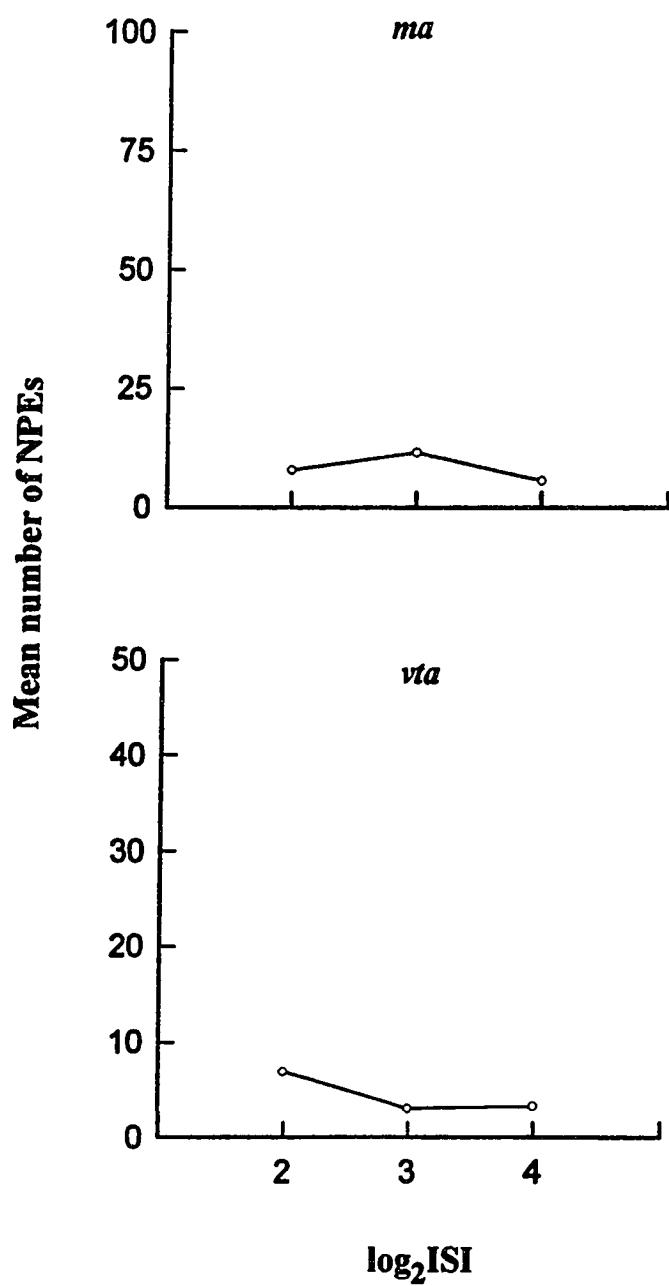
**Figure 4.23: Mean activation decay in *sa* and *cal* NPEs in last generation of each lineage.**



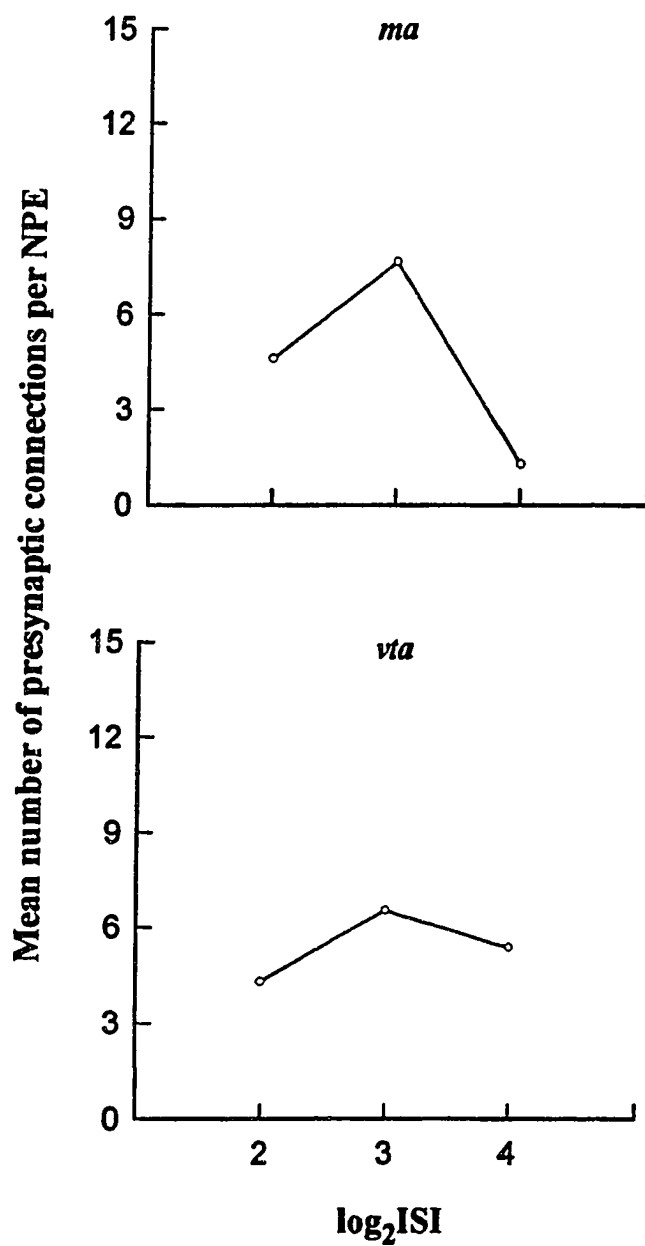
**Figure 4.24: Mean acquisition rate in *sa* and *cal* NPEs in last generation of each lineage.**



**Figure 4.25: Mean extinction rate in *sa* and *cal* NPEs in last generation of each lineage.**

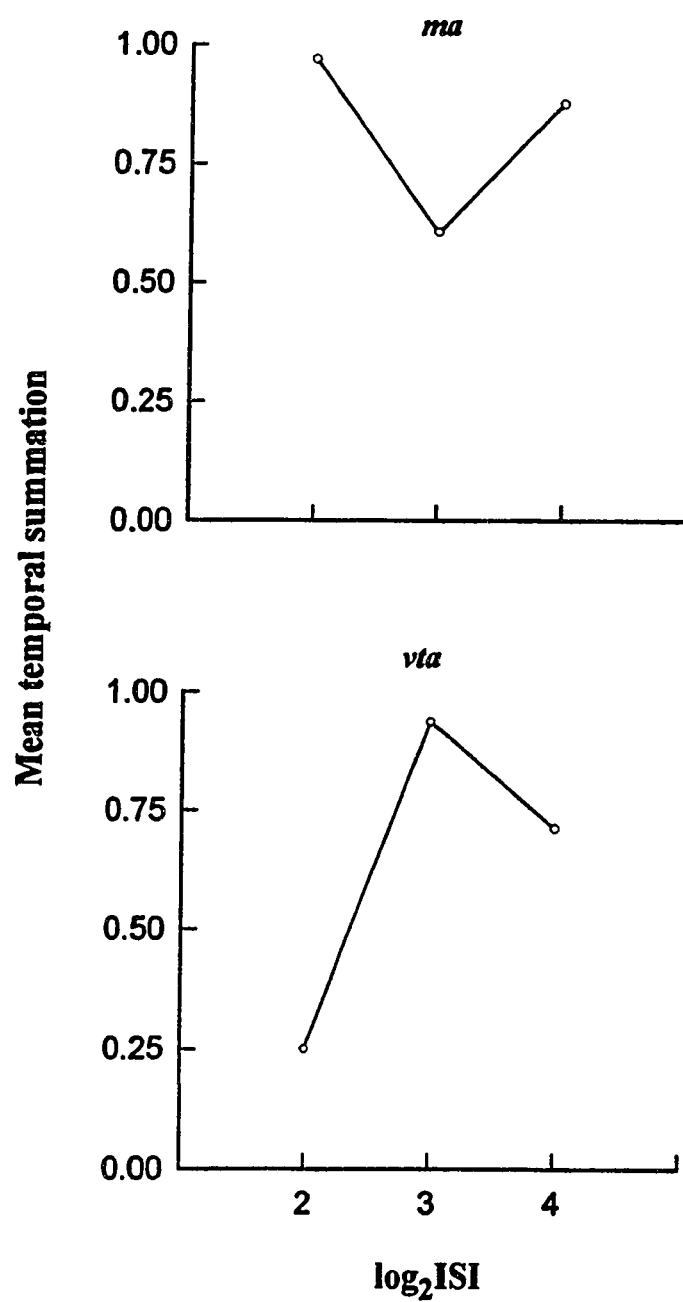


**Figure 4.26: Mean number of *ma* and *vta* NPEs in last generation of each lineage.**



**Figure 4.27: Mean number of presynaptic connections per *ma* and *vta* NPE in last generation of each lineage.**





**Figure 4.28: Mean temporal summation in *ma* and *vta* NPEs in last generation of each lineage.**

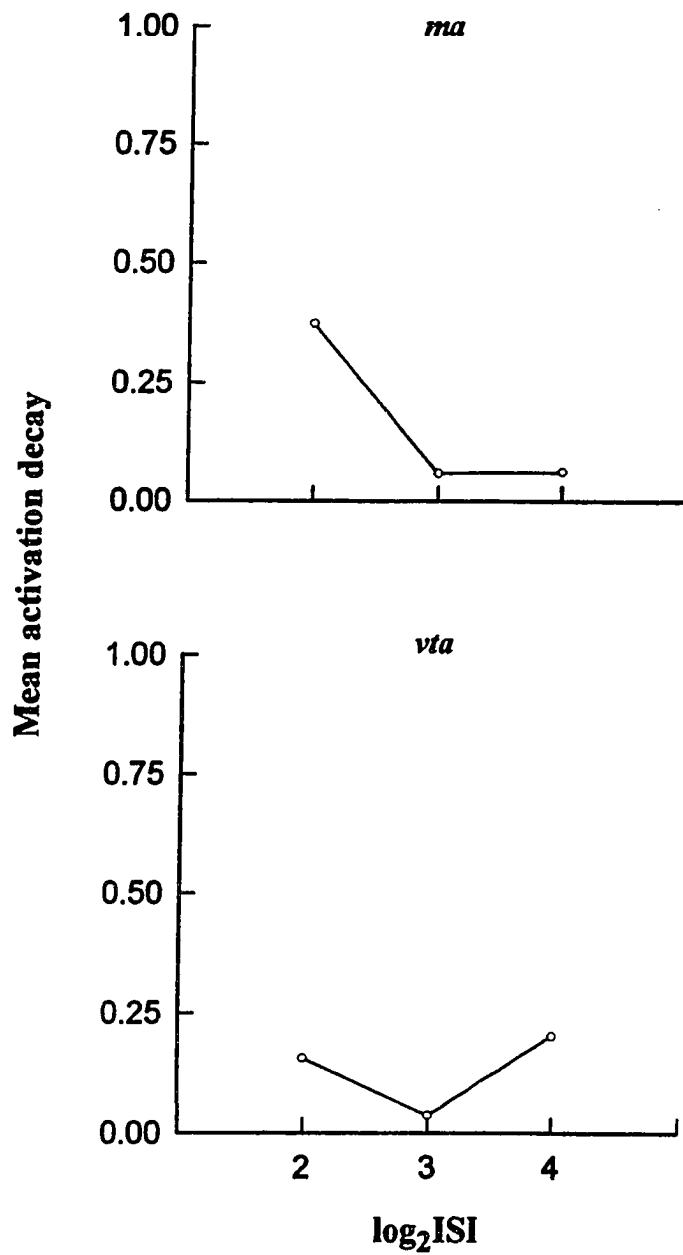


Figure 4.29: Mean activation decay in *ma* and *vta* NPEs in last generation of each lineage.

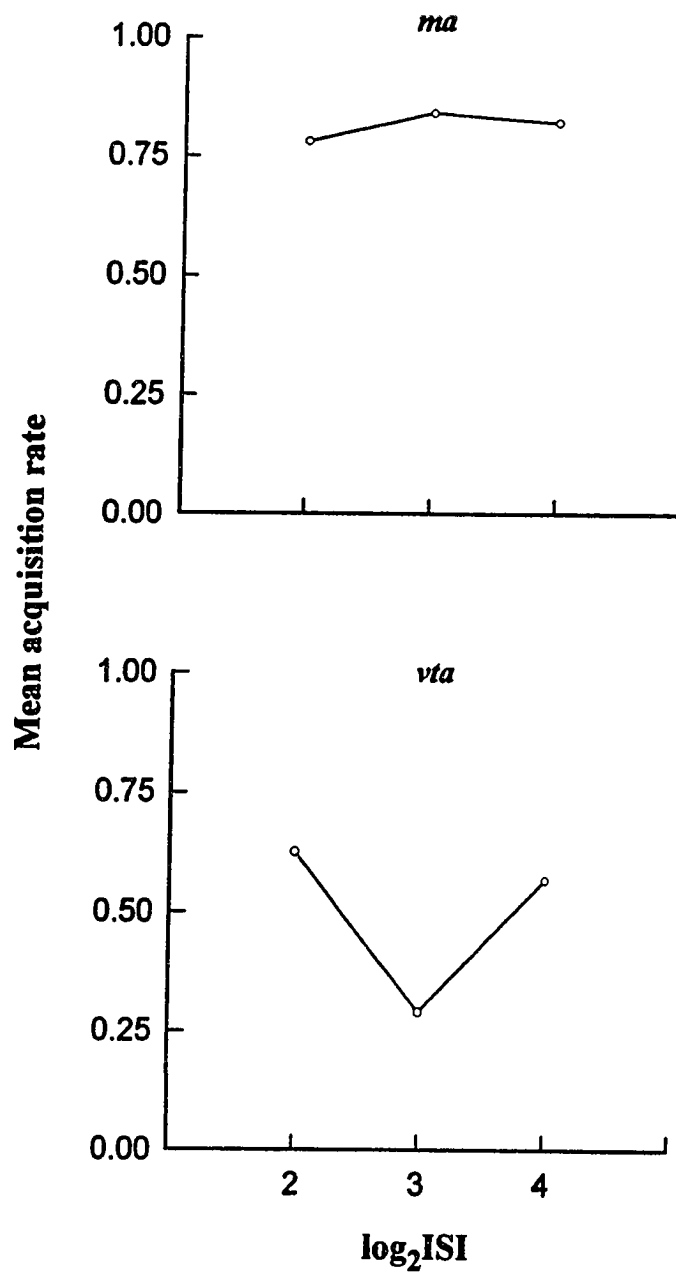
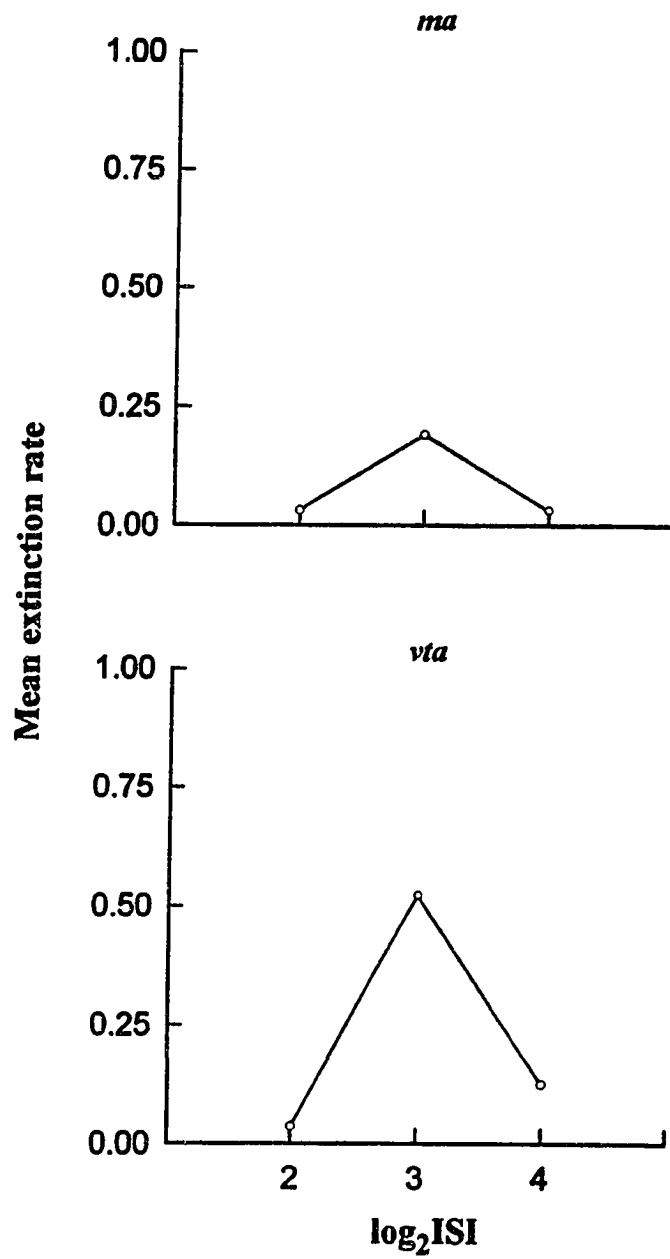


Figure 4.30: Mean acquisition rate in *ma* and *vta* NPEs in last generation of each lineage.



**Figure 4.31: Mean extinction rate in *ma* and *vta* NPEs in last generation of each lineage.**

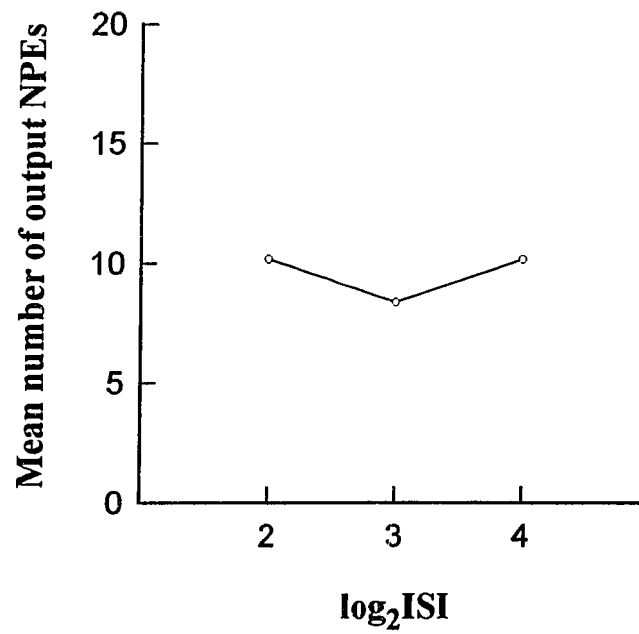
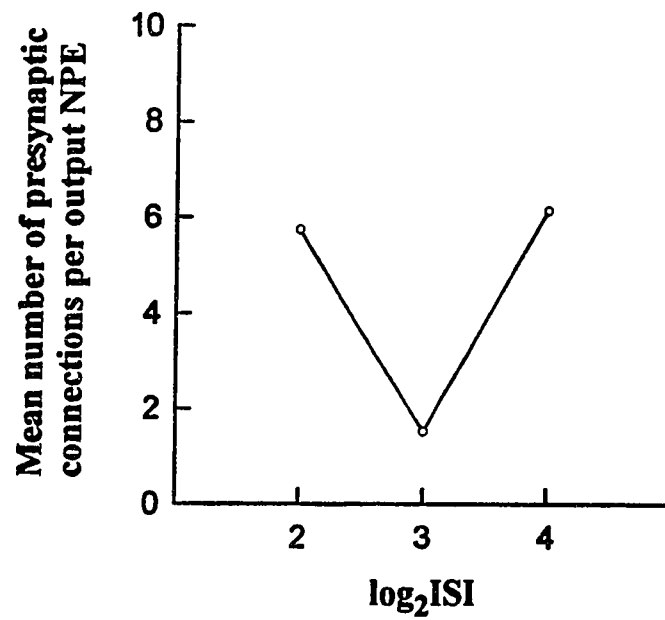
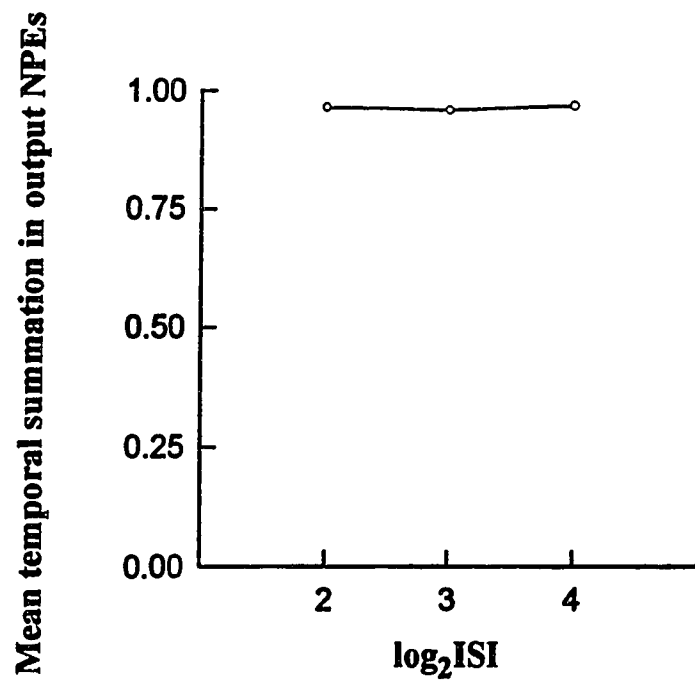


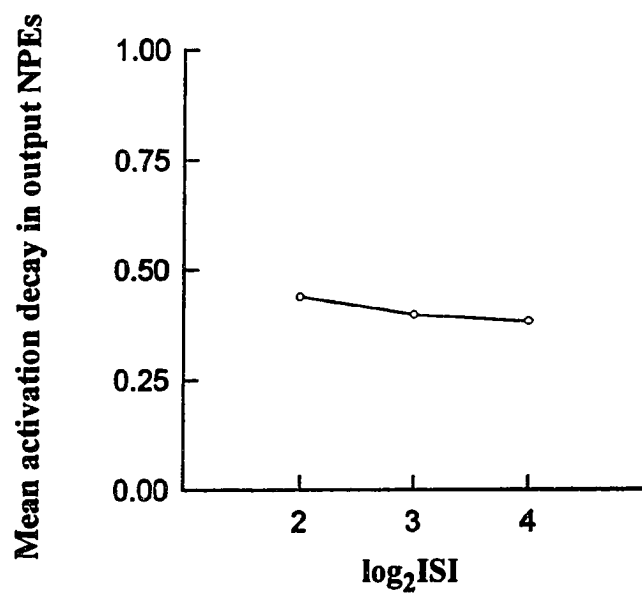
Figure 4.32: **Mean number of output NPEs in last generation of each lineage.**



**Figure 4.33: Mean number of presynaptic connections per output NPE in last generation of each lineage.**

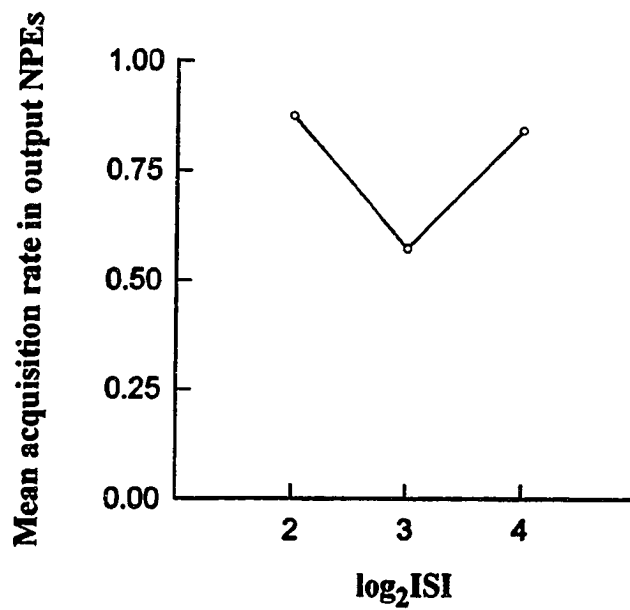


**Figure 4.34: Mean temporal summation in output NPEs in last generation of each lineage.**

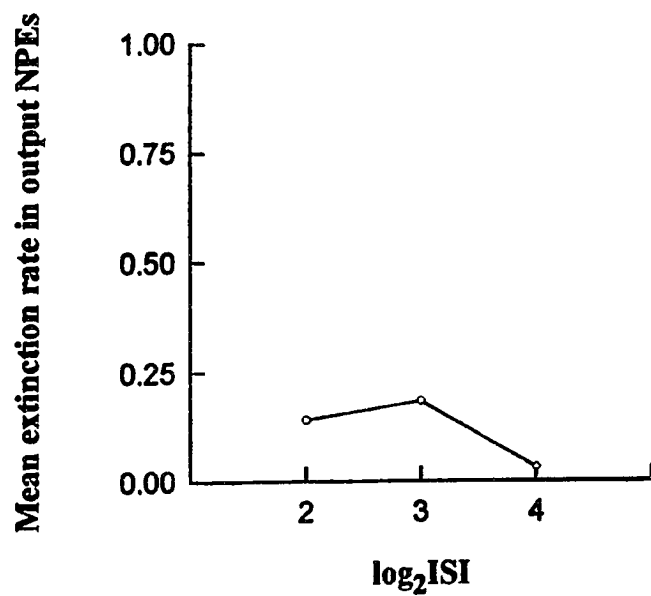


**Figure 4.35: Mean activation decay in output NPEs in last generation of each lineage.**



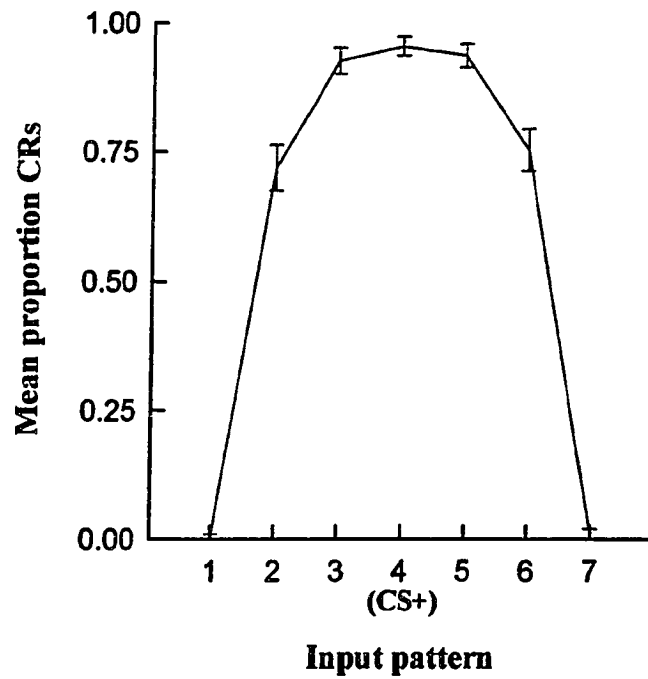


**Figure 4.36: Mean acquisition rate in output NPEs in last generation of each lineage.**

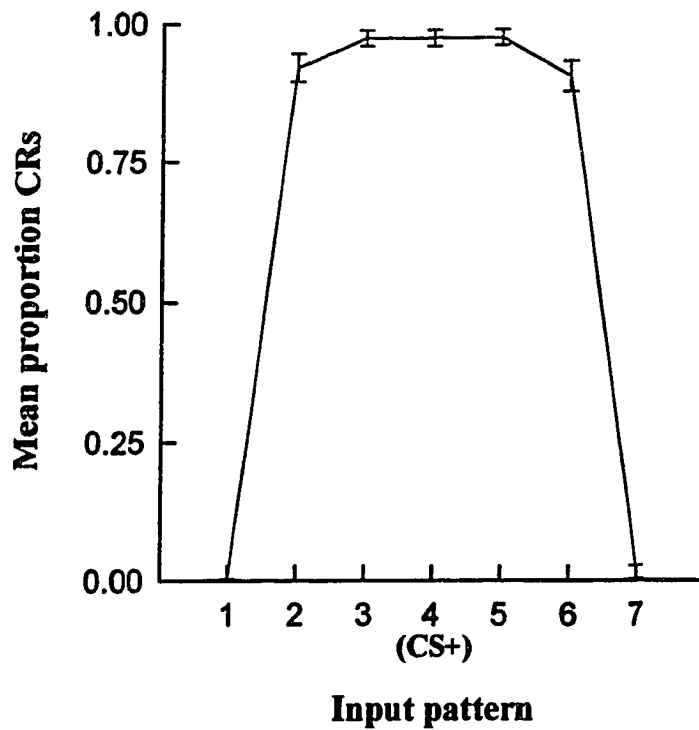


**Figure 4.37: Mean extinction rate in output NPEs in last generation of each lineage.**

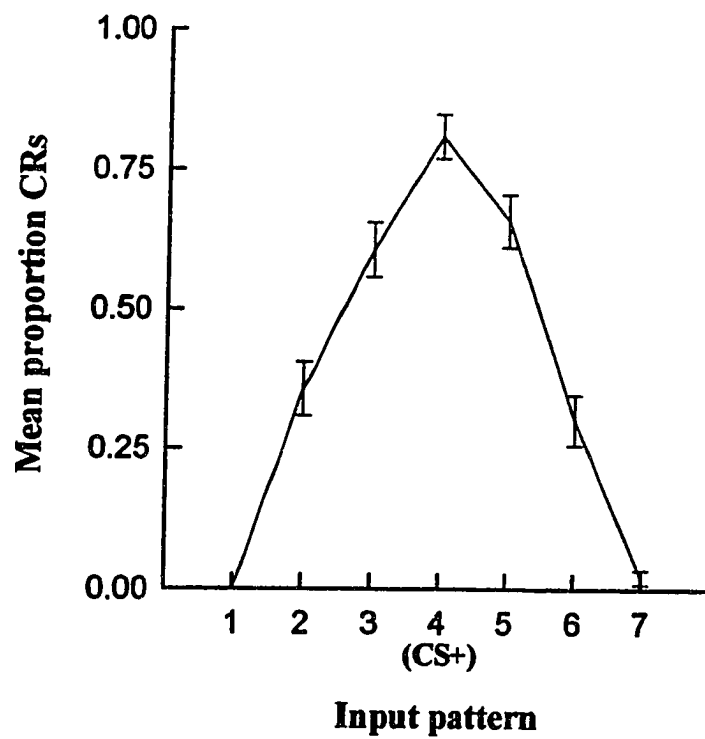




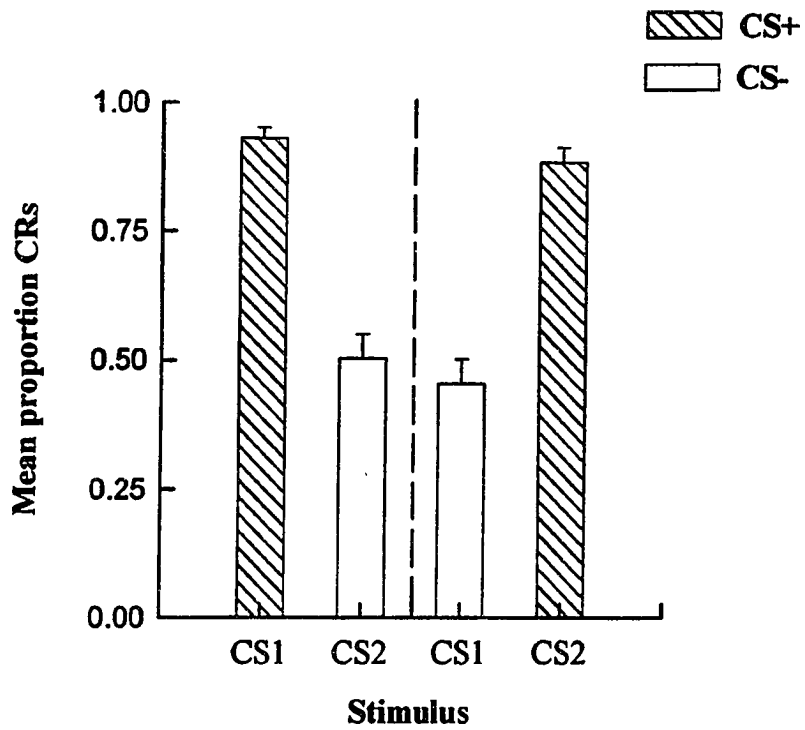
**Figure 4.39: Mean proportion CRs in generalization test of ANNs in last generation of ISI-4. "CS+" indicates the pattern reinforced during training.**



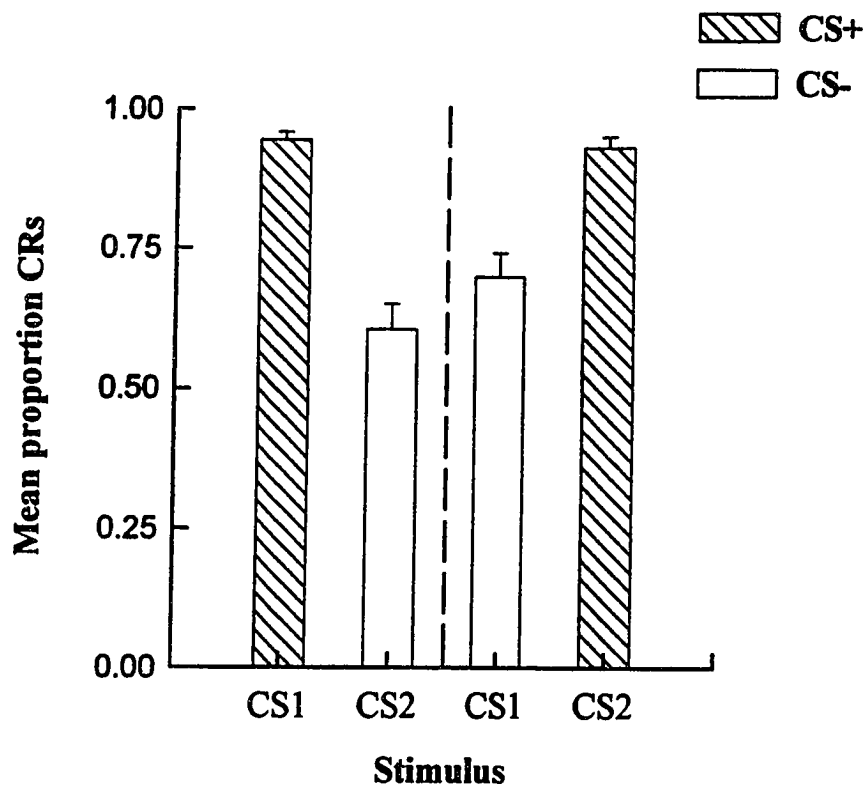
**Figure 4.40: Mean proportion CRs in generalization test of ANNs in last generation of ISI-8. "CS+" indicates the pattern reinforced during training.**



**Figure 4.41: Mean proportion CRs in generalization test of ANNs in last generation of ISI-16. "CS+" indicates the pattern reinforced during training.**

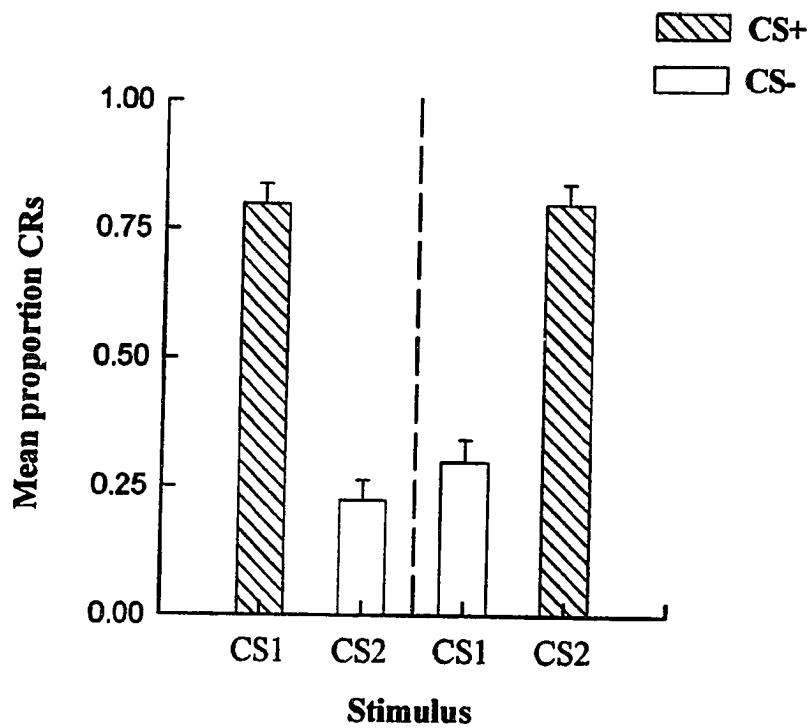


**Figure 4.42: Mean proportion CRs in orthogonal discrimination tests of ANNs in last generation of ISI-4.**

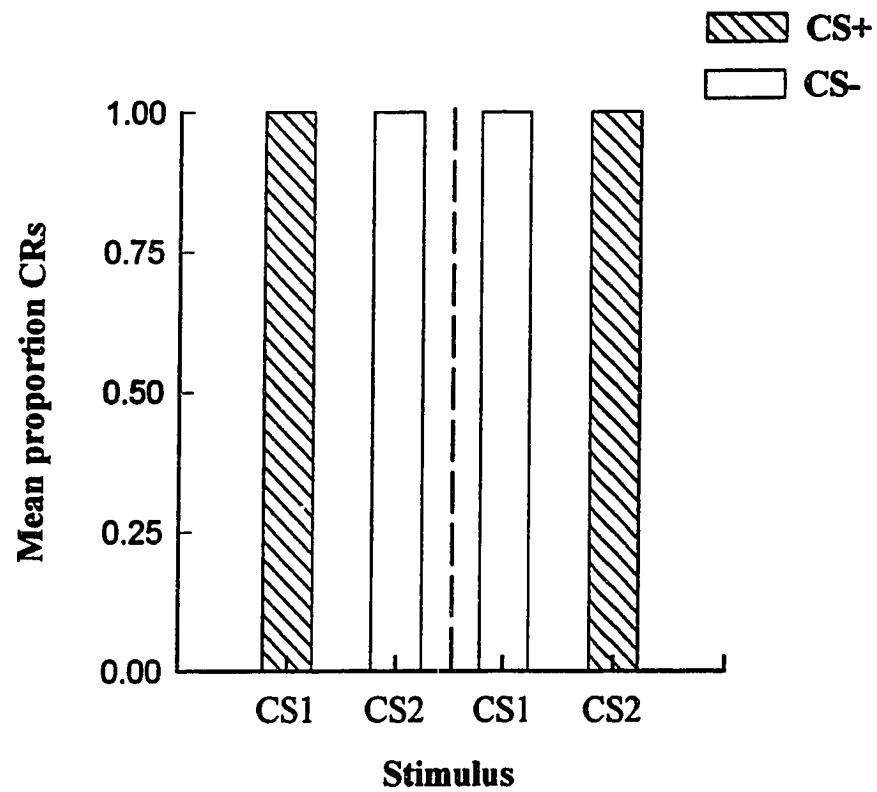


**Figure 4.43: Mean proportion CRs in orthogonal discrimination tests of ANNs in last generation of ISI-8.**

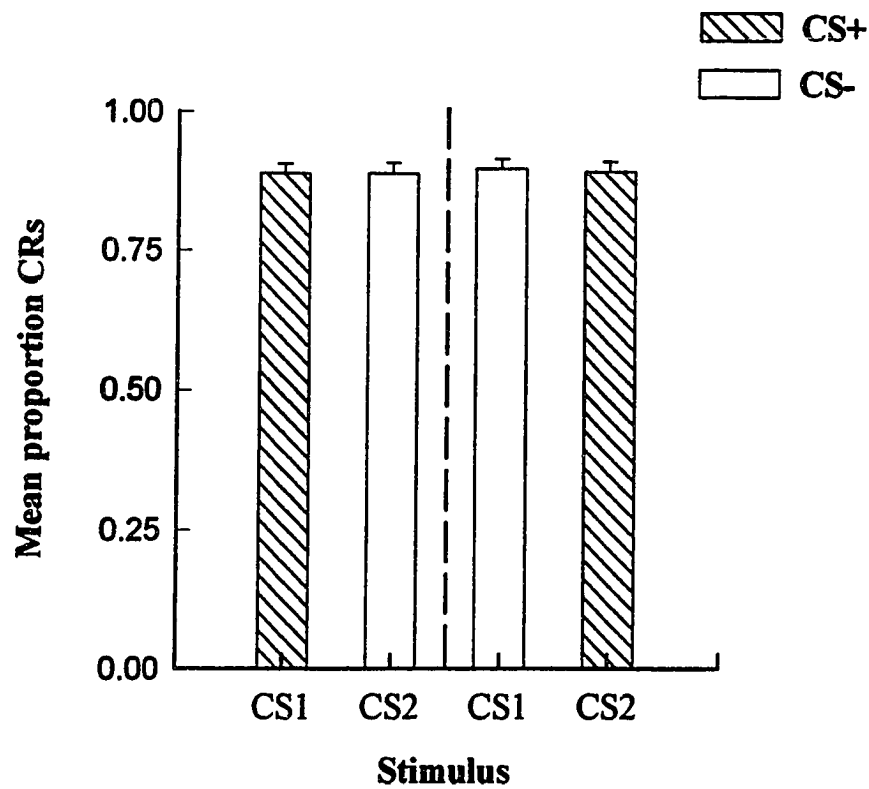




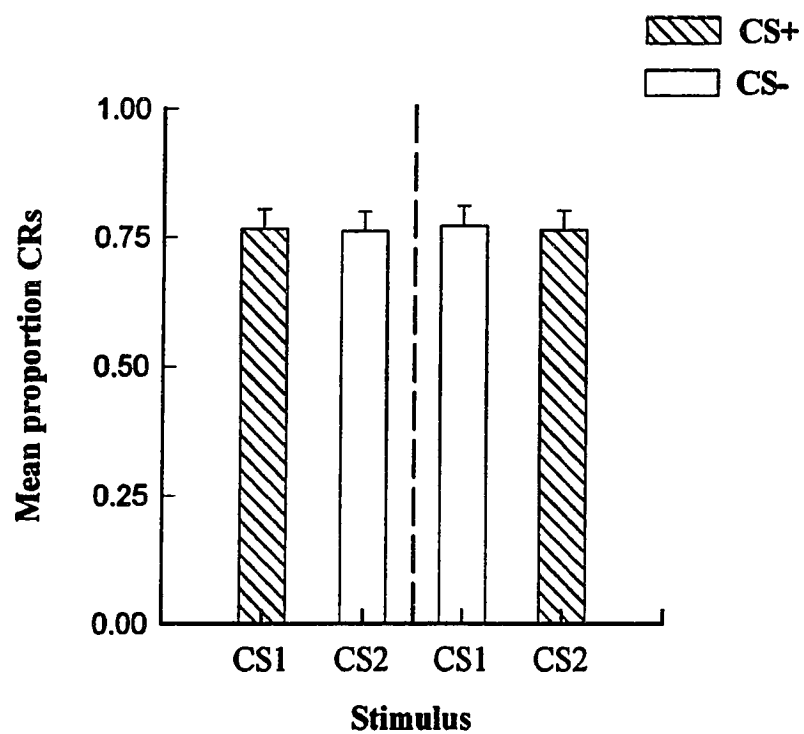
**Figure 4.44: Mean proportion CRs in orthogonal discrimination tests of ANNs in last generation of ISI-16.**



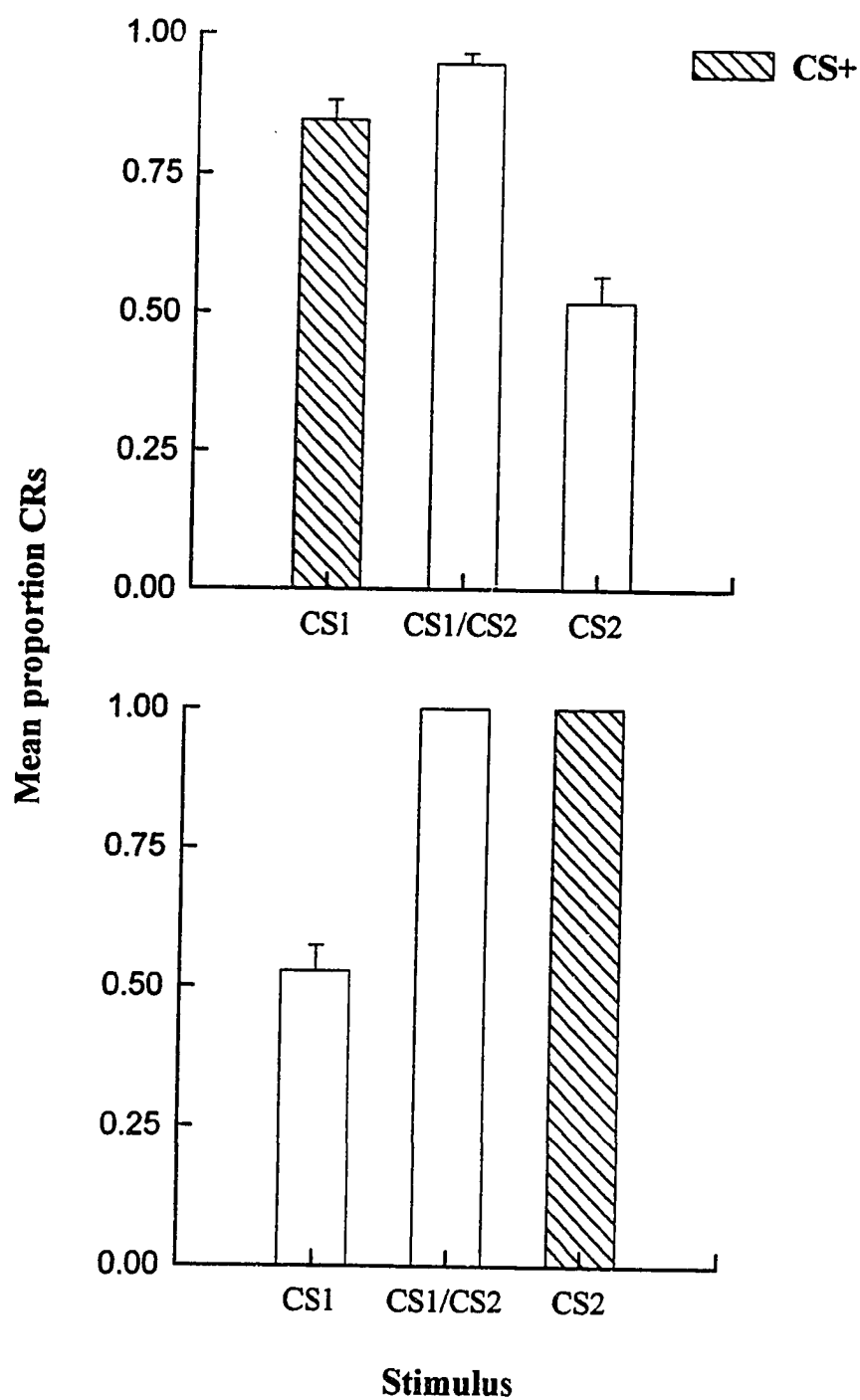
**Figure 4.45: Mean proportion CRs in nonorthogonal discrimination tests of ANNs in last generation of ISI-4.**



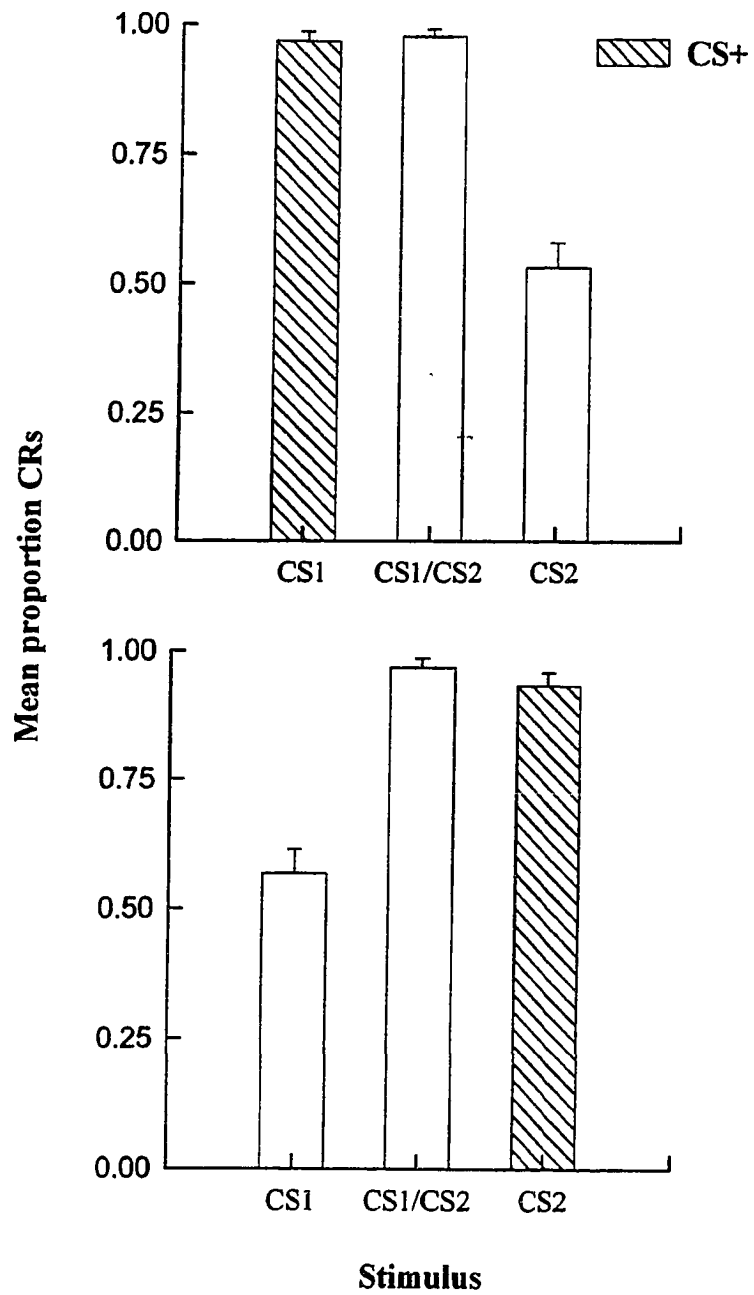
**Figure 4.46: Mean proportion CRs in nonorthogonal discrimination tests of ANNs in last generation of ISI-8.**



**Figure 4.47: Mean proportion CRs in nonorthogonal discrimination tests of ANNs in last generation of ISI-16.**



**Figure 4.48: Mean proportion CRs in blocking tests of ANNs in last generation of ISI-4. "CS+" was the CS reinforced during Phase 1.**



**Figure 4.49: Mean proportion CRs in blocking tests of ANNs in last generation of ISI-8. "CS+" was the CS reinforced during Phase 1.**

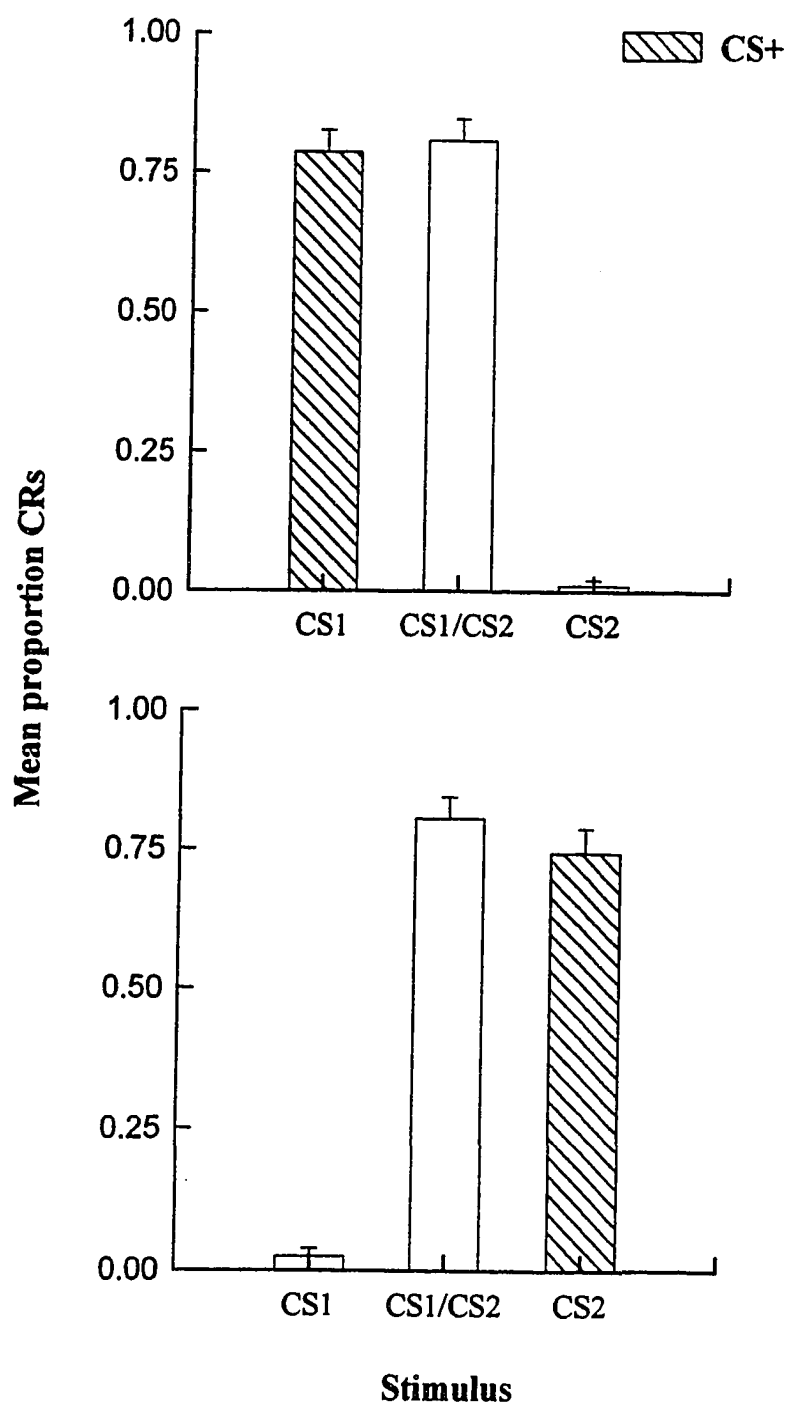


Figure 4.50: Mean proportion CRs in blocking tests of ANNs in last generation of ISI-16. "CS+" was the CS reinforced during Phase 1.

# CHAPTER 5

## EXPERIMENT 3

In Experiment 2, ANNs throughout phylogeny were selected for increased responding to two independently *reinforced* CSs. That is, ancestral environments in that experiment consisted of two CS classes whose instances predicted the *presence* of the US. However, ancestral environments may also consist of CSs that predict the *absence* of a US. One class of Pavlovian procedures involving the latter is referred to as *differential reinforcement*, and the kind of performance generated by them is known as *discrimination*.

In the simplest differential-reinforcement procedure, two CS classes are defined. Instances of one class are paired with the US, whereas instances of the other class are presented alone (i.e., are *not* reinforced). In typical arrangements, instances of each class are presented separately and randomly. The present chapter focuses upon orthogonal differential reinforcement, which simulates arrangements in which the reinforced CS and the nonreinforced CS do not have any input elements in common (see Chapter 4, Footnote 1).

In Experiment 2, orthogonal differential reinforcement was used only in the test simulations. The purpose of the present experiment was to use this procedure in the evolution simulations, first, to determine whether or not phenotypically different types of ANNs would emerge from selection for increased responding under environmental conditions that differed in the kind of reinforced CS and, second, to



determine whether or not emerging functional phenotypes were consistent with generalization, discrimination, and blocking. It was also of interest to determine if selection for orthogonal discrimination would result in ANNs showing improved orthogonal and nonorthogonal discrimination, relative to those observed in Experiment 2.

## 5.1 Evolution Simulations

### 5.1.1 Procedure

G0' was used as the founder chromosome population. Two environmental conditions were defined by two CSs (either CS1 or CS2), and a lineage of generations of ANNs was evolved for each condition. As a first step, two ANNs were developed from each one of the 100 chromosomes in G0', to obtain 200 ANNs, which were divided into groups of 100, one group corresponding to the ANN founder population of a lineage.

As in Experiment 2, all ANNs were trained in a procedure involving CS1 and CS2 trials. However, in the present experiment, trials of *only one* CS class were reinforced. For ANNs in one group, CS1 was reinforced. For ANNs in the other group, CS2 was reinforced. For both groups, reinforced trials involved a forward-delay arrangement with an ISI of 8 ts. This ISI was chosen on the basis of the low orthogonal-discrimination ratios shown by ISI-8 ANNs in the test simulations of Experiment 2.

As in Experiments 1 and 2, CS1 trials involved the activation of all  $s_1$  input elements with a value of 1.0 and no activation of any  $s_2$  input element, whereas CS2 trials involved no activation of any  $s_1$  element and the activation of all  $s_2$  elements with a value of 1.0. All ANNs were given a total of 250 randomly intermixed CS1 and CS2

trials, the last 50 of which served as evaluation trials to determine individual fitness separately for each kind of CS. Fitness for a given CS class was the proportion of evaluation trials of that class that elicited a CR. An individual won a selection tournament if and only if it had the highest CR proportion for the *reinforced* CS.

After evaluation, a new population of chromosomes was created through the selection scheme of the GA. This new population constituted G1 (the next generation). Once the new chromosomes were created, new ANNs were developed and trained in the same procedure as their ancestors. The same operation was repeated generation after generation, until the termination criterion was satisfied. This strategy resulted in two lineages, labeled CS1+ (the lineage for which CS1 was the reinforced CS) and CS2+ (the lineage for which CS2 was the reinforced CS). Both lineages evolved independently of one another.

## **5.1.2 Results and Discussion**

### **5.1.2.1 Genotypic Evolution**

Figure 5.1 shows that the mean chromosomal overlap increased as a negatively accelerated function of generations, for both lineages. Individuals within lineages thus became genetically more similar as generations progressed. The mean overlaps for G0' were close to 0.0, which is consistent with the expected value for a randomly generated population of chromosomes. In contrast, the mean overlaps for the last generation were close to 0.65. Thus, at the end of evolution there still was a substantial amount of genetic variation, perhaps due to the fact the the total number of generations constituting each lineage was relatively small (17 for CS1+ and 13 for CS2+). This result is consistent with that obtained in Experiment 1, suggesting again that a positive relationship may exist between the number of generations in a lineage and the mean

chromosomal overlap in the last generation of that lineage. Finally, no substantial differences were observed between the last-generation overlaps of CS1+ and CS2+.

### **5.1.2.2 Effects on Functional Phenotypes**

#### **5.1.2.2.1 Effects on Fitness**

Figure 5.2 shows that, for both lineages, mean population fitness for the reinforced CS increased as a negatively accelerated function of generations, whereas the mean population fitness for the nonreinforced CS remained close to zero. Indeed, in G0', both lineages started with low-fitness values (close to 0.0) for both CSs, and ended with fitness values between 0.8 and 1.0 for the reinforced CS only. Also, at the end of evolution, ANNs were comparably fit for both lineages.

Figures 5.3 and 5.4 show 3D mesh plots of frequency distributions of fitness values across generations. Each figure shows distributions for CS1 (top panel) and CS2 trials (bottom panel) for a given lineage. All plots show a similar overall tendency for the reinforced CS, that is, a high number of low-fitness individuals at G0', followed by an increase in the number of high-fitness individuals after a number of generations. Indeed, the high frequency peak at the origin of the generation-fitness plane indicates that most individuals in G0' had a fitness between 0.0 and 0.05, for both CSs. As generations advanced, the number of individuals with fitness values between 0.95 and 1.00 increased rapidly for the reinforced CS, in both lineages (see top panel of Figure 5.3 and bottom panel of Figure 5.4). However, the number of individuals with fitness values between 0.0 and 0.05 remained substantially high (close to the population size) for the nonreinforced CS (see bottom panel of Figure 5.3 and top panel of Figure 5.4), *despite the fact that performance to the nonreinforced CS had no effect on the fitness criterion used here* (i.e., fitness was determined only by performance to the reinforced

CS). Finally, functional-phenotypic variability at the end of evolution was relatively low (i.e., the majority of individuals had the same fitness).

#### **5.1.2.2.2 Effects on Learning**

Figures 5.5 through 5.8 show 3D mesh plots of frequency distributions of mean output-NPE activations across the first 75 CS1 and 75 CS2 trials, for the first (top panel) and last (bottom panel) generation of each lineage. In G0', the number of ANNs with high output-NPE activations (between 0.0 and .05) remained low and relatively constant, for both lineages and CSs (see top panels of 5.5 through 5.8). Only a few ANNs responded with activations greater than the CR criterion of 0.5 at the end of training, a condition that was necessary and sufficient to ensure the minimum number of tournament winners required for the GA to generate G1. However, in the last generation, the number of individuals with high output-NPE activations (between 0.95 and 1.0) increased as a function of the reinforced trials (see bottom panel of Figures 5.5 and 5.8, while remaining relatively constant at zero across nonreinforced trials (see bottom panel of Figures 5.6 and 5.7). Moreover, as in Experiments 1 and 2, most ANNs far exceeded the CR criterion towards the end of training, despite the fact that their ancestors did not need to respond with such high activations to win selection tournaments.

Note that, at the end of evolution, many CS1+ ANNs responded with relatively high output-NPE activations (between 0.25 and 0.50) to CS1 early in training (see bottom panel of Figure 5.5). However, the mean output-NPE activations in these ANNs decreased after approximately 5 training trials, after which ANNs with low output-NPE activations were in the vast majority. Then, the number of high-activation ANNs increased again, reaching values close to the population size towards the end of training. Also, note that in the last generation of both lineages, ANNs tended to

respond slightly to nonreinforced CS trials, suggesting that ANNs could *sense* and *react* to the nonreinforced CS.

Finally, Figure 5.9 shows the mean connection weight before and after training for the last generation of each lineage. For all lineages, connection weights were substantially higher after than before training, although the ratio of final to initial weights was higher for CS1+ than CS2+ ANNs. This result suggests that, in the last generation of both lineages, orthogonal differential reinforcement produced a increase in connection weights, regardless of the reinforced CS.

Overall, the above results show that selection for increased responding to the reinforced CS in an orthogonal differential reinforcement procedure generated ANNs whose behavior changed from low to high conditioned responding to the reinforced CS, for both CS1+ and CS2+. And this result was obtained despite the fact that fitness was determined only by responding to the reinforced CS. That is, selection for increased responding to the reinforced CS was sufficient to evolve ANNs that simulated orthogonal discrimination substantially better than ISI-8 ANNs selected for increased responding to two CSs (see Figure 4.42). Selection for *not* responding to the nonreinforced CS was not necessary.

### **5.1.2.3 Effects on Structural Phenotypes**

#### **5.1.2.3.1 General Effects**

Figure 5.10 shows a typical high-fitness ANN from the last generation of each lineage. Both ANNs had a fitness of 1.0 for the reinforced CS and a fitness of zero for the nonreinforced CS. No substantial architectural differences between the ANNs were observed. Indeed, both ANNs had the same numbers of  $s_1$  and  $s_2$  input elements, hidden layers, and output NPEs, and similar numbers of NPEs per hidden layer, and

*cal* and *vta* NPEs. Also, as in previous simulations, both ANNs had substantially more excitatory than inhibitory NPEs. However, in contrast to ISI-8 ANNs in Experiment 2 (see Figure 4.13), the present ANNs had substantially fewer NPEs per hidden layer, and substantially more *ma* than *sa* layers. In fact, this was the only set of simulations in which ANNs possessed more than two *ma* layers. I shall offer an interpretation of this result in the last chapter. Finally, both ANNs had comparable amounts of connectivity.

As in Experiments 1 and 2, some within-population variability was observed among ANN architectures. Indeed, although most ANNs in the last generation of both lineages seemed to be instances of the same basic architecture, a few high-fitness ANNs had architectures different from those shown in Figure 5.10. Figure 5.11 illustrates this result by showing other examples of high-fitness ANNs from CS1+ and CS2+. Both ANNs were considerably larger than their respective conspecifics. The CS1+ ANN had more *sa* than *ma* layers, making it more similar in this respect to the ISI-8 ANN shown in Figure 4.15, although the former had a substantially richer *ma*-output connectivity than the latter. The CS2+ ANN had substantially more NPEs per layer than its conspecifics, although both ANNs were similar in that they both had substantially more *ma* than *sa* NPEs. Again, the selection process generated a class of architectures, rather than a unique architecture.

Also, as in Experiments 1 and 2, not all ANNs in the last generation responded with high CR proportions to the reinforced CS. Figure 5.12 shows one low-fitness ANN from each lineage. Both ANNs had a fitness of zero for both CSs. The last *ma* layer of the CS1+ ANN had only three NPEs, one of which was inhibitory. This inhibitory NPE received one connection from and sent one connection back to the same NPE (the immediately preceding one; these connections are not shown in the diagram). Such a circuit may have been sufficient to prevent the ANN from responding to either CS.

The CS2+ ANN had the same number of hidden layers and inhibitory NPEs, but substantially more NPEs per layer than its conspecific. However, connectivity of the two inhibitory NPEs (not shown in the diagram) was also such that they could have prevented the ANN from responding to either CS.

Finally, a few ANNs showed high responding to both the reinforced *and* the nonreinforced CS. Figure 5.13 shows one ANN example from each lineage. Such a behavior in these ANNs may have arisen from initial weights of connections from the input elements activated by the nonreinforced CS to *sa* NPEs. An examination of these weights indicated that they may have been sufficiently high (around 0.2) to allow innate learning to respond to the nonreinforced CS without external reinforcement, a phenomenon similar to that observed in the ISI-2 ANNs generated in Experiment 1 (see Figure 3.41).

Regarding the number of hidden layers, ANNs in both lineages had substantially more *ma* than *sa* layers (see Figure 5.14). Clearly, the presence of nonreinforced trials made a substantial difference in this respect, relative to Experiments 1 and 2 (ANNs in these experiments tended to have only one *ma* layer). To determine if this result was due to any genetic biases related to input elements, the evolution simulation was repeated with a new chromosome founder population. The new population was generated from G0 by making the alleles that determined the number of  $s_1$  input elements and postsynaptic connections per  $s_1$  element equal to the alleles that determined the number of  $s_2$  input elements and postsynaptic connections per  $s_2$  element. The same effect was observed, that is, a substantial increase in the number of *ma* layers and NPEs. Therefore, the increase in the number of *ma* layers in the present experiment was not dependent upon any genetic biases related to input elements.

Finally, as in previous experiments, ANNs had substantially more excitatory than inhibitory NPEs (see Figure 5.15), which suggests that discrimination under an orthogonal differential reinforcement procedure did not seem to require lateral inhibition.

#### 5.1.2.3.1 Effects on the Sensory Subnetwork

Figure 5.16 shows that ANNs in the last generation of both CS1+ and CS2+ had approximately the same number of  $s_1$  and  $s_2$  input elements, although there was a tendency to have more input elements of the kind activated by the reinforced CS ( $s_1$  for CS1+ ANNs and  $s_2$  for the CS2+ ANNs; see top panel). Also, CS1+ ANNs tended to have more connections per input element than CS2+ ANNs, whereas the latter tended to have more connections per  $s_2$  than  $s_1$  NPE (see bottom panel). Thus, given that ANNs in both lineages had only one *sa* layer and roughly the same number of *sa* NPEs in that layer (see below), competition among  $s_1$  and  $s_2$  elements may have been more balanced in CS1+ ANNs than in CS2+ ANNs. Although this architectural difference did not seem to have any substantial effect upon fitness, it might have affected performance under environmental conditions different from those present when selection took place (see test simulations).

Figure 5.17 shows the mean number of *sa* and *caI* NPEs (top panel) and the number of connections per *sa* and *caI* NPE (bottom panel), for both lineages. ANNs had approximately the same number of *sa* (close to 10) and *caI* NPEs (about five), as well as connections per NPE type, for both lineages. Within lineages, ANNs tended to have more *sa* than *caI* NPEs, and substantially more connections per *sa* NPE than per *caI* NPE. Because most ANNs had only one *sa* layer, the number of presynaptic connections per *sa* NPE is a direct reflection of the number of postsynaptic connections per input NPE. Therefore, both  $s_1$  and  $s_2$  input elements contributed approximately



equally to the number of connections per *sa* NPE for CS1+ ANNs (these ANNs had approximately the same number of connections per  $s_1$  and  $s_2$  element). For CS2+ ANNs,  $s_2$  elements contributed more than  $s_1$  elements to the number of connections per *sa* NPE (these ANNs had more connections per  $s_2$  than connections per  $s_1$  element).

Figures 5.18 and 5.19 show *sa*- and *cal*-NPE activation and synaptic-plasticity parameters, respectively. Temporal-summation parameters for both kinds of NPEs in CS1+ ANNs were around 0.25 (see top panel of Figure 5.18). This result is in contrast to those observed in previous experiments, where *sa* NPEs in ISI-8 ANNs tended to have larger temporal summations, specially in Experiment 2 (see top panel of Figure 4.22). Also, *cal* temporal summations in CS2+ ANNs were relatively low (about 0.25). These temporal summations were considerably smaller than the ones observed in Experiment 1 (which were close to 0.5; see bottom panel of Figure 3.25), and considerably larger than the ones observed in Experiment 2 (which were near zero), for ISI-8 ANNs. In contrast, *sa* NPEs in CS2+ had the highest temporal summations (around 0.85) for NPEs in the sensory subnetwork. Finally, mean *sa* and *cal* activation decays in both lineages were between 0.6 and 0.75 (see bottom panel of Figure 5.18). Activation decays in *sa* NPEs tended to be higher than activation decays in *cal* NPEs in CS1+ ANNs, whereas the opposite was observed in CS2+ ANNs.

Acquisition rates for *sa* NPEs were substantially higher in CS1+ ANNs (around 0.8) than CS2+ ANNs (around 0.25; see top panel of Figure 5.19). This difference may be related to the difference between CS1+ and CS2+ ANNs observed in the parameters for *sa* temporal summations (see top panel of Figure 5.18). That is, *sa* NPEs in CS1+ ANNs had low temporal summations and high acquisition rates, whereas *sa* NPEs in CS2+ had high temporal summations and low acquisition rates. The inverse relationship between the values of these parameters may have tended to

make the average change in connection weights per ts similar, everything else being equal. A similar tendency was observed in *cal* NPEs, that is, low temporal summations corresponded to high acquisition rates. This tendency was also observed in Experiment 2. The adaptive value of this trade-off between temporal summation and acquisition rate in the sensory subnetwork is unclear. Perhaps, keeping one parameter high prevented connection weights from growing too rapidly or too slowly, which may have resulted in an optimal variation range necessary for adapting to an environment involving random occurrences of different kinds of CSs.

Finally, *sa* NPEs in CS2+ ANNs tended to have higher extinction rates than CS1+ ANNs (see bottom panel of Figure 5.19). Acquisition rates also tended to be substantially larger than extinction rates, with the exception of *sa* NPEs in CS2+ ANNs, which tended to have higher extinction than acquisition rates. This feature may have been related to the high *sa* temporal summations observed in these ANNs. Indeed, by having both high *sa* temporal summations (an incremental parameter) and extinction rates (a decremental parameter), the range of variation of *sa*-NPE activations and weights across ts may have been increased. Such an increase may have allowed CS2+ ANNs to achieve the kind of behavioral flexibility that was needed to adapt to an environment whose reinforcement conditions changed over time.

#### **5.1.2.3.2 Effects on the Motor Subnetwork: *ma* and *vta* NPEs**

Figure 5.20 shows the mean number of *ma* and *vta* NPEs (top panel) and connections per NPE type (bottom panel), for both lineages. CS2+ ANNs tended to have more *ma* and *vta* NPEs, as well as connections per NPE type, than CS1+ NPEs. Comparisons with Figure 5.17 show that ANNs in both lineages had substantially more *ma* than *sa* NPEs, as well as substantially more *vta* than *cal* NPEs. This result is consistent with the larger number of *ma* layers observed in Figure 5.14. The mean

number of connections per *ma* NPE was substantially smaller than the mean number of connections per *sa* NPE, which suggests that connectivity in the sensory subnetwork was richer than connectivity in the motor subnetwork. Thus, a trade-off seems to have taken place between number of NPEs and number of connections per NPE. That is, a small number of *sa* NPEs corresponded to a large number of connections per *sa* NPE, whereas a large number of *ma* NPEs corresponded to a small number of connections per *ma* NPE.

Figure 5.21 shows the mean temporal summation (top panel) and activation decay (bottom panel) in *ma* and *vta* NPEs, for both lineages. ANNs in both lineages tended to have approximately the same *ma* temporal summations (about 0.65). However, *vta* NPEs in CS1+ ANNs tended to have larger temporal summations than *vta* NPEs in CS2+ ANNs. This tendency was also observed in *vta* activation decays, although these were smaller than temporal summations, for both lineages and especially in *ma* NPEs. Moreover, a trade-off seems to have occurred between temporal summation and activation decays in *ma* NPEs. That is, large temporal summations corresponded to low activation decays in *ma* NPEs, especially in CS2+ ANNs. This trade-off may have had an adaptive value by driving the range of variation of *ma*-NPE activations toward high values, hence increasing the activation levels of *ma* NPEs in the presence of the reinforced CS.

Finally, Figure 5.22 shows the mean acquisition (top panel) and extinction rates (bottom panel) in *ma* and *vta* NPEs, for both lineages. Acquisition rates were larger in *ma* than in *vta* NPEs (specially for CS2+ ANNs), whereas the opposite tendency was observed in extinction rates. Also, acquisition rates tended to be substantially larger than extinction rates in *ma* NPEs. The same tendency was observed in *vta* NPEs in

CS1+ ANNs. In contrast, extinction rates in *vta* NPEs were larger than acquisition rates, for CS2+ NPEs.

Overall, then, there was a trade-off between acquisition and extinction parameters in the motor subnetwork, for both lineages. Indeed, high acquisition rates in *ma* NPEs (approximately between 0.65 and 0.75) tended to correspond to very low extinction rates (around 0.05). In CS1+ ANNs, relatively high acquisition rates in *vta* NPEs (around 0.65) tended to correspond to relatively low extinction rates (around 0.3). And, in CS2+ ANNs, relatively low acquisition rates (around 0.25) tended to correspond to relatively high extinction rates (around 0.50). This trade-off could have had an adaptive value by allowing for an optimal range of variation of connection weights over time. Indeed, if acquisition and extinction rates are both low, then that range will be too small. In contrast, if acquisition and extinction are both high, the variation range will be too large. By keeping only one parameter high, connection weights are prevented from changing too little or too much over time.

#### **5.1.2.3.3 Effects on the Motor Subnetwork: Output NPEs**

Figures 5.23, 5.24, and 5.25 show structural traits for output NPEs, in both lineages. CS1+ ANNs tended to have somewhat more output NPEs than CS2+ ANNs (see top panel of Figure 5.23), although ANNs in both lineages tended to have a quite similar number of connections per output NPE (see bottom panel of Figure 5.23). The latter effect suggests that the same amount of convergent connectivity in the output layer may have been sufficient to achieve comparable levels of performance under environmental conditions that differed in the kind of reinforced CS. In contrast, output NPEs in CS2+ ANNs tended to have larger temporal summations and activation decays than CS1+ ANNs (see top and bottom panels of Figure 5.24, respectively). Thus, CS2+ ANNs seem to have compensated for the smaller number of output NPEs by

having both larger temporal summations and activation decays. Such a compensation may have had an adaptive value by increasing the range of activation variability of individual output NPEs over time, and hence, increasing the behavioral flexibility needed to respond adaptively to a changing environment. Finally, output NPEs in ANNs across lineages had about the same acquisition (see top panel of Figure 5.25) and extinction rates (see bottom panel of Figure 5.25), although the former were substantially larger than the later, suggesting, again, a trade-off between these parameters.

## 5.2 Test Simulations

A series of simulations were run in which the same ANNs described in Section 5.1.2.3 were tested in generalization, orthogonal discrimination, nonorthogonal discrimination, and blocking. These procedures involved environmental conditions different from those under which the selection process took place. Hence, they determined if those ANNs showed emergent functional phenotypes consistent with the kind of effects of such procedures on animal learning and behavior.

### 5.2.1 Generalization

In a training phase, naïve ANNs were given 100 reinforced trials in which the CS involved the activation of *all* input elements with a value 1.0 (i.e., trials consisted of *compound* CSs). In a testing phase, trained ANNs were given 175 probe trials of seven randomly intermixed input-activation patterns. The patterns used were the same as those used in the generalization test of ISI-8 ANNs in Experiment 2 (see Figure 4.38). To avoid extinction during the testing phase, the synaptic-plasticity function of the NC model was disabled.

Figures 5.26 and 5.27 show generalization gradients obtained during the test phase of ANNs in CS1+ and CS2+, respectively. In general, it can be observed that ANNs responded with higher CR proportions to input patterns that were more similar to the trained pattern (i.e., ANNs showed generalization). However, CS1+ ANNs showed substantially more generalization to Pattern 2 (which involved the activation of all  $s_1$  input elements and no activation of any  $s_2$  element) than CS2+ ANNs. Alternatively, CS2+ ANNs showed substantially more generalization to Pattern 6 (which involved the activation of all  $s_2$  input elements and no activation of any  $s_1$  element) than CS1+ ANNs. Thus, ANNs showed more generalization to the test pattern that was more similar to the CS reinforced throughout phylogeny. This result represents an emergent functional difference between CS1+ and CS2+ ANNs. In this sense, one can argue that Patterns 2 and 6 (i.e., CS1 and CS2) were nonequipotential in the generalization task. These results indicate that selection for increased responding to the reinforced CS in an orthogonal differential reinforcement procedure generated ANNs that were capable of showing generalization as an emergent functional phenotype, that is, as a functional phenotype that was not explicitly selected for throughout phylogenetic history. However, generalization to certain patterns was consistent with that history.

## 5.2.2 Reversal of Orthogonal Discrimination

The results of the evolution simulations show that selection for increased responding to the reinforced CS in an orthogonal differential reinforcement procedure was sufficient to generate ANNs that discriminated between the reinforced and the nonreinforced CS. However, it is not clear whether ANNs would learn a discrimination involving a reversal of the ancestral reinforcement conditions. To address this issue, naïve ANNs were trained in two conditions, each consisting of a

training and a testing phase. In the training phase of a first condition, CS1+ ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials (an average of 100 trials per CS kind). In the training phase of a second condition, CS2+ ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials. In both conditions, CS1 involved the activation of all  $s_1$  input elements with a value of 1.0, whereas CS2 involved the activation of all  $s_2$  input elements, also with a value of 1.0. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 probe trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled.

Figures 5.28 and 5.29 show the CR proportions obtained during the orthogonal discrimination testing phase of CS1+ and CS2+ ANNs, respectively. ANNs showed orthogonal discrimination in that they responded with substantially higher CR proportions to the reinforced CS than to the nonreinforced CS. Moreover, ANNs in both lineages showed comparable discrimination performances. Thus, selection for increased responding to a reinforced CS in an orthogonal differential reinforcement procedure generated ANNs that were capable of simulating, as an emergent functional phenotype, orthogonal discrimination in which the reinforced and nonreinforced CSs were reversed from the relation present throughout phylogeny.

### 5.2.3 Nonorthogonal Discrimination

Naïve ANNs were trained in two conditions, each one consisting of a training and a testing phase. In the training phase of the first condition, ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials (an average of 100 trials per CS kind). In both conditions, CS1 and CS2 were identical to Patterns 3 and 5, respectively, used in Experiment 2 for ISI-8 ANNs (see Figure 4.38). In the training

phase of the second condition, ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled.

Figures 5.30 and 5.31 show the CR proportions obtained during the testing phase of each condition (conditions are separated by a dotted line), for CS1+ and CS2+ ANNs, respectively. ANNs in both lineages did not show appreciable nonorthogonal discrimination in that they responded with approximately equal CR proportions to the reinforced CS and the nonreinforced CS. However, CS1+ ANNs showed somewhat better nonorthogonal discrimination when the reinforced CS involved the activation of all  $s_1$  input elements and only half the  $s_2$  elements. In contrast, CS2+ ANNs showed somewhat better nonorthogonal discrimination when the reinforced CS involved the activation of half the  $s_1$  input elements and all  $s_2$  elements. Thus, ANNs tended to discriminate better when the reinforced CS was similar to that reinforced when selection took place. But, in general, these results indicate that selection for increased responding to a reinforced CS in an orthogonal differential reinforcement procedure generated ANNs that were not capable of simulating nonorthogonal discrimination as an emergent functional phenotype.

### 5.2.4 Blocking

Naïve ANNs were trained in two conditions consisting of three phases each. In Phase 1 (or single-CS phase), ANNs were given 100 reinforced presentations of a CS. Then, in Phase 2 (or compound-CS phase), those trained ANNs were given 100 reinforced concurrent presentations of the previously reinforced CS and another,



previously nonexperienced CS. In one condition, the CS reinforced in the single-CS phase was CS1, whereas the novel CS presented in the compound-CS phase was CS2. Conversely, in the other condition, the CS reinforced in the single-CS phase was CS2, whereas the novel CS presented in the compound-CS phase was CS1. Finally, in Phase 3 (or test phase), ANNs were given 75 randomly intermixed trials of CS1, CS2, and the CS1/CS2 compound (an average of 25 trials per CS kind). For all ANNs, CS1 involved the activation of all  $s_1$  input elements with a value of 1.0, whereas CS2 involved the activation of all  $s_2$  input elements, also with a value of 1.0. The CS1/CS2 compound involved the activation of all input elements. The synaptic-plasticity function of the NC model was disabled during the test phase.

Figures 5.32 and 5.33 show the mean CR proportions per CS kind during Phase 3 of the blocking test of CS1+ and CS2+ ANNs, respectively. The top panels show the results for the first condition, in which CS1 was the CS reinforced in Phase 1 (marked as CS+ in the graphs) and CS2 was the novel CS introduced in Phase 2. The bottom panels show the results for the second condition, in which CS2 was the CS reinforced in Phase 1 and CS1 was the novel stimulus introduced in Phase 2. ANNs in both conditions responded substantially more to the CS reinforced in Phase 1 and the CS1/CS2 compound than to the novel CS introduced in Phase 2. However, blocking in CS1+ ANNs was stronger when CS1 was reinforced in Phase 1 (i.e., blocking to CS2 was stronger than blocking to CS1). In contrast, blocking was stronger in CS2+ ANNs when CS2 was reinforced in Phase 1 (i.e., blocking to CS1 was stronger than blocking to CS2).

These results indicate that selection for increased responding to a reinforced CS in an orthogonal differential reinforcement procedure generated ANNs that were capable of simulating blocking as an emergent functional phenotype. However, CS1 and CS2

were nonequipotential in that CS1+ and CS2+ ANNs performed differently under different blocking conditions. Also, note that the emergence of blocking did not depend on either few *sa* (this experiment) or few *ma* layers (Experiment 2), for it occurred with both.

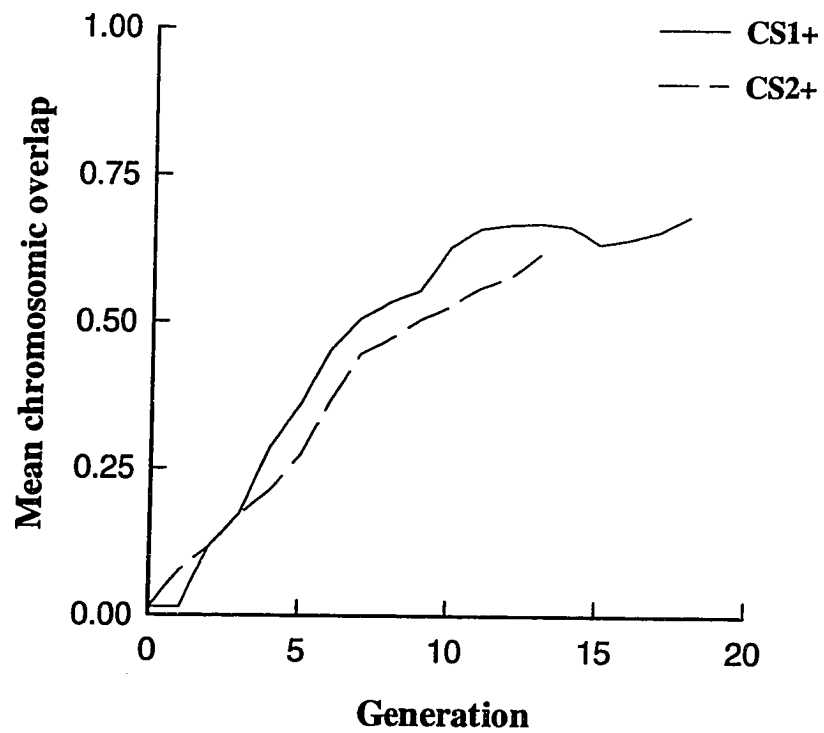


Figure 5.1: Mean chromosomal overlap as a function of generations.

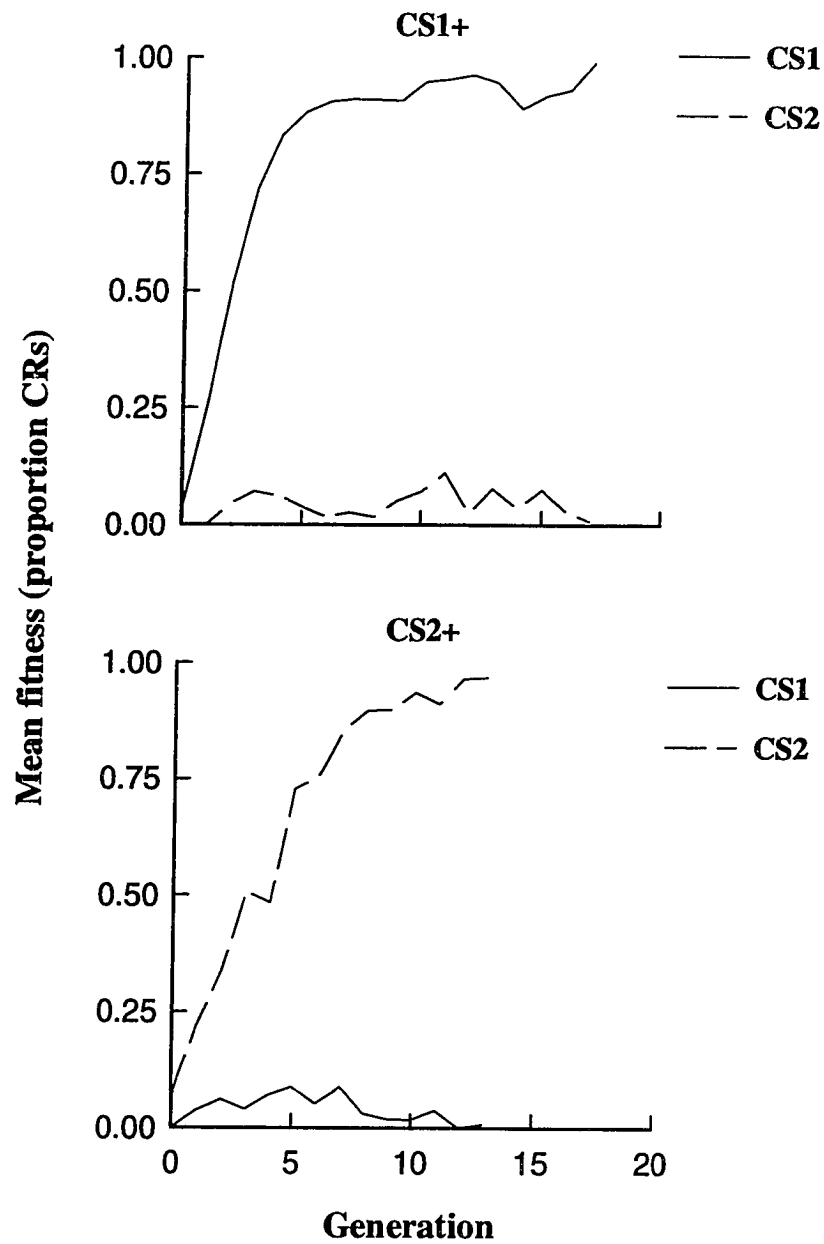


Figure 5.2: Mean population fitness as a function of generations.

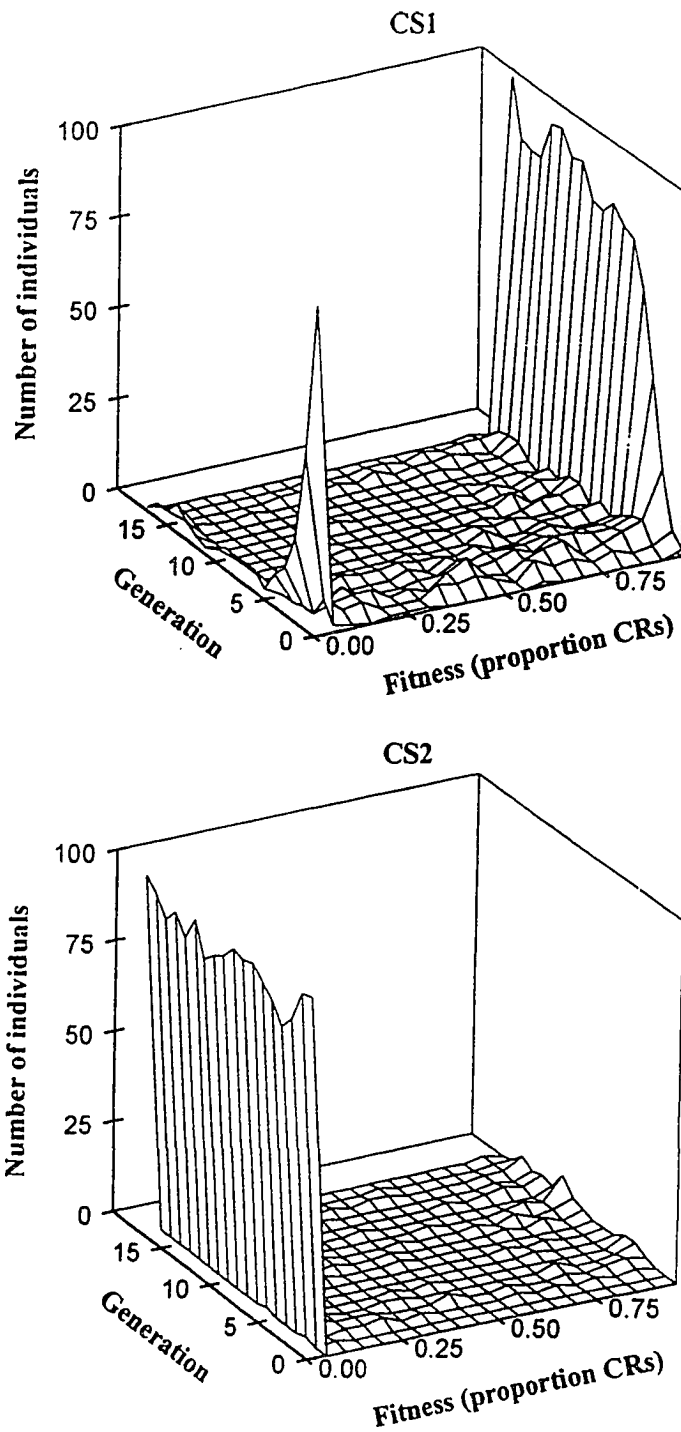
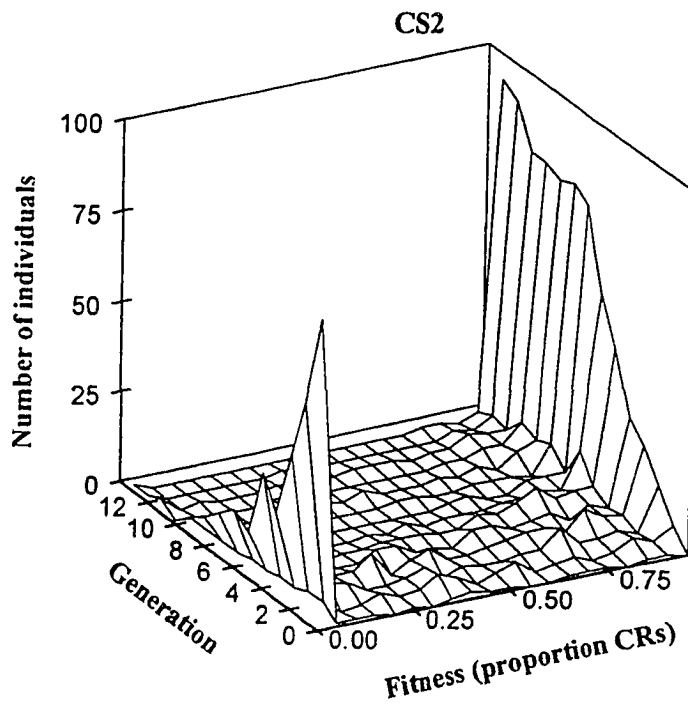
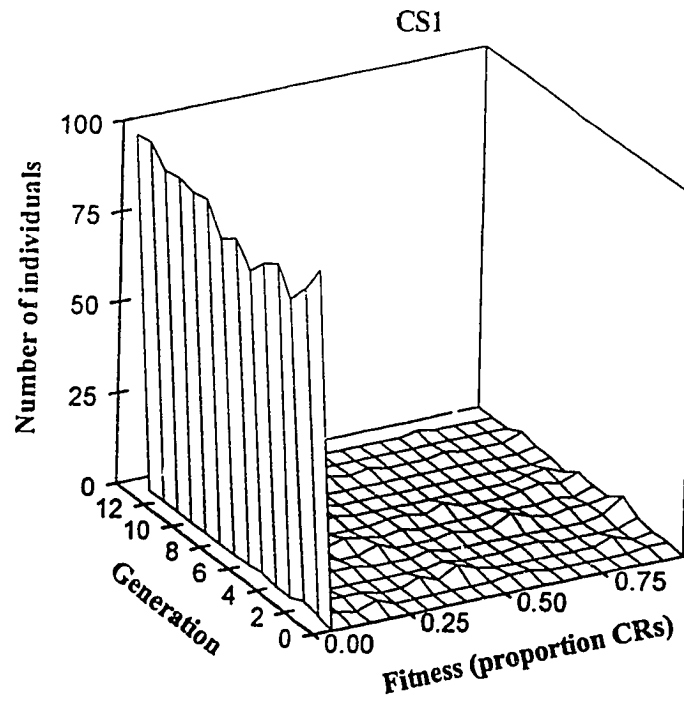
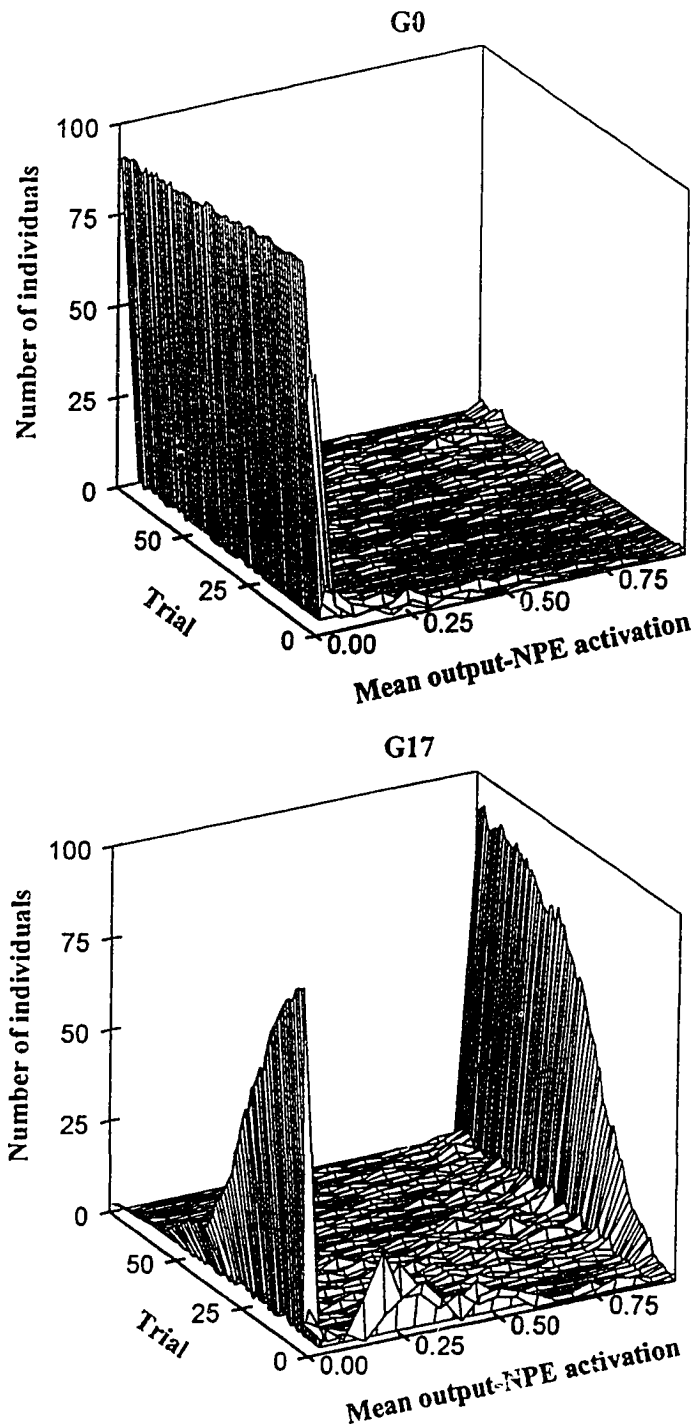


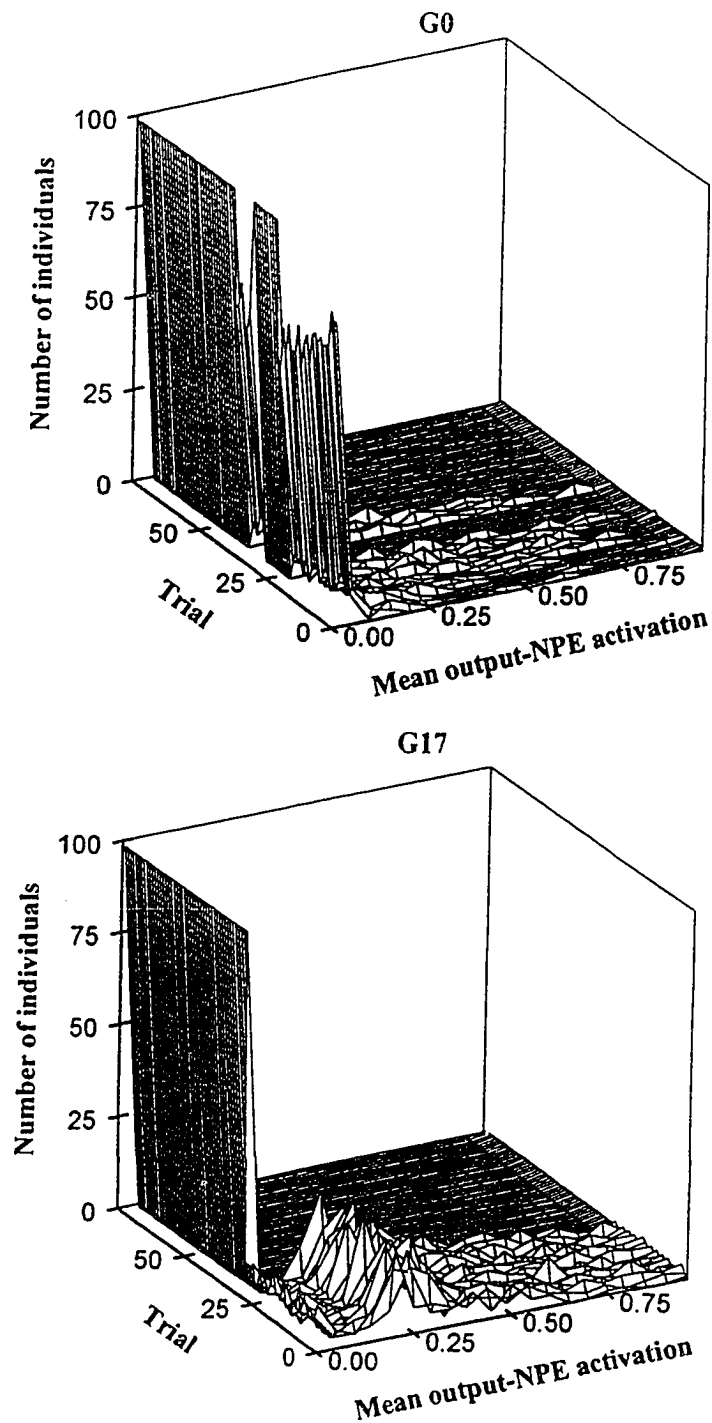
Figure 5.3: Frequency distributions of fitness values over generations, for CS1+.



**Figure 5.4: Frequency distributions of fitness values over generations, for CS2+.**

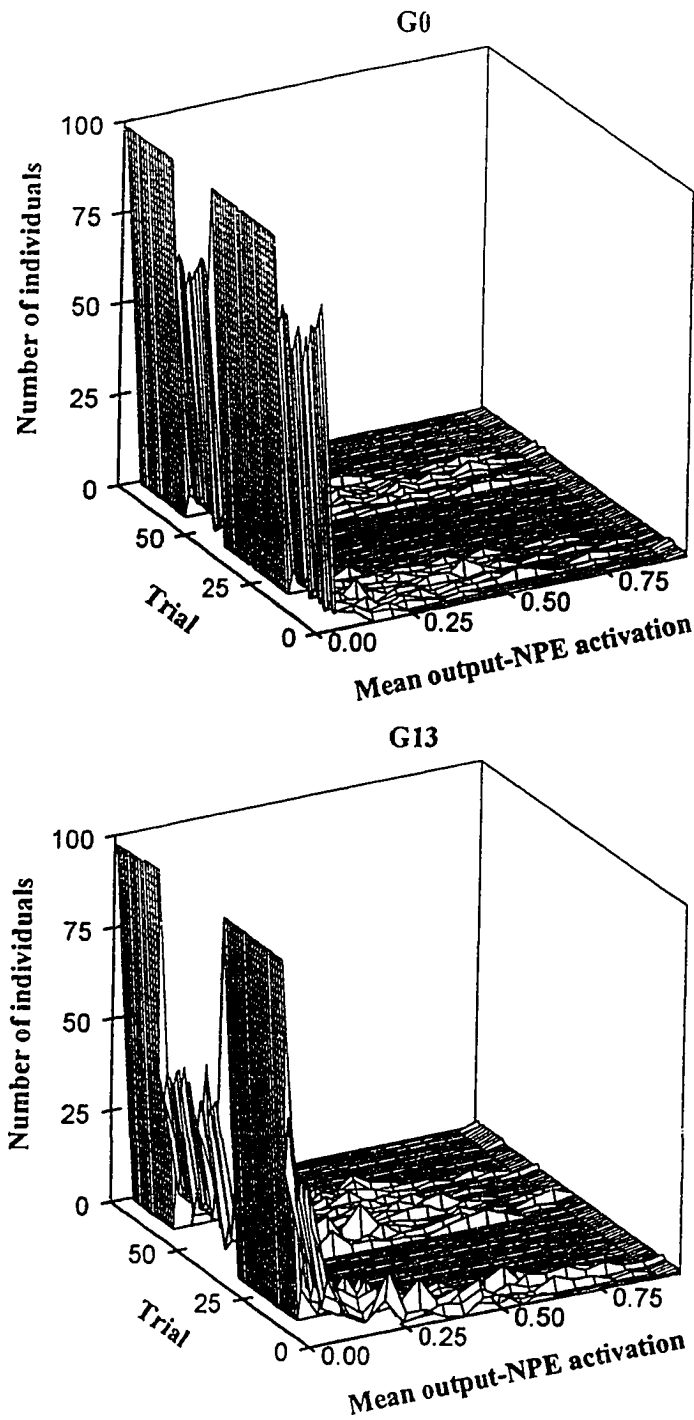


**Figure 5.5: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of CS1+.**

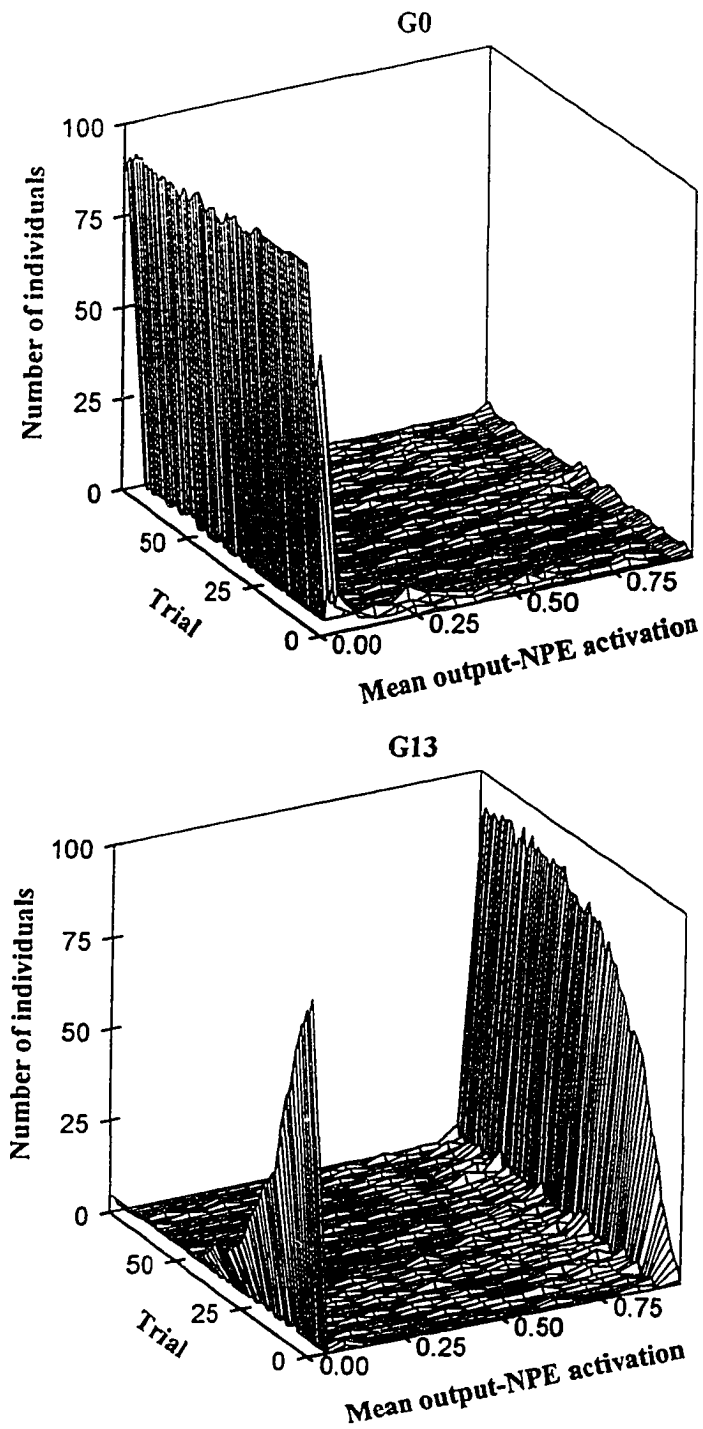


**Figure 5.6: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of CS1+.**

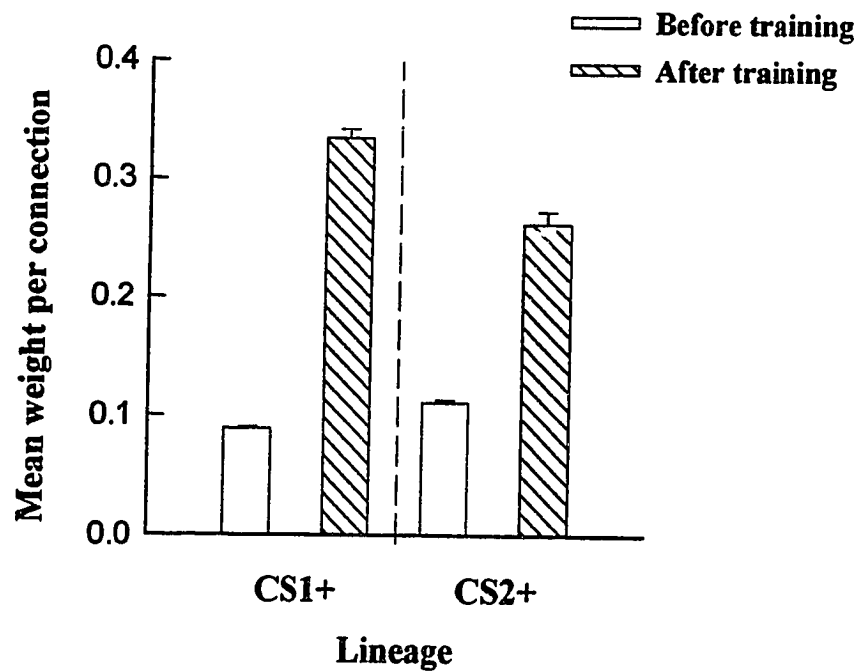




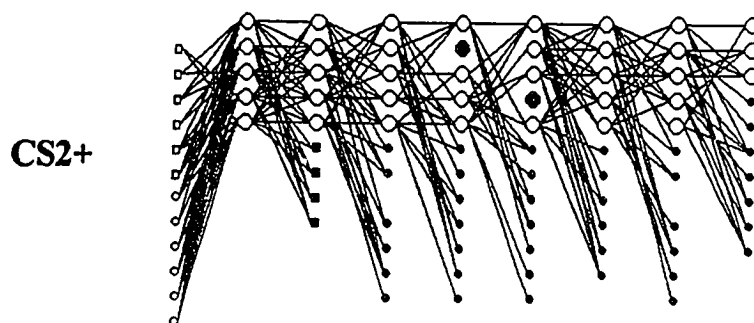
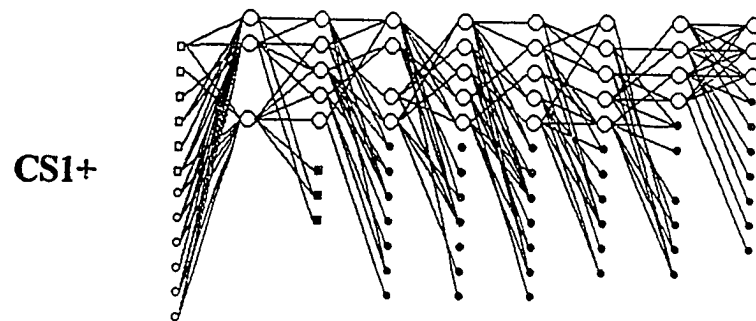
**Figure 5.7: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of CS2+.**



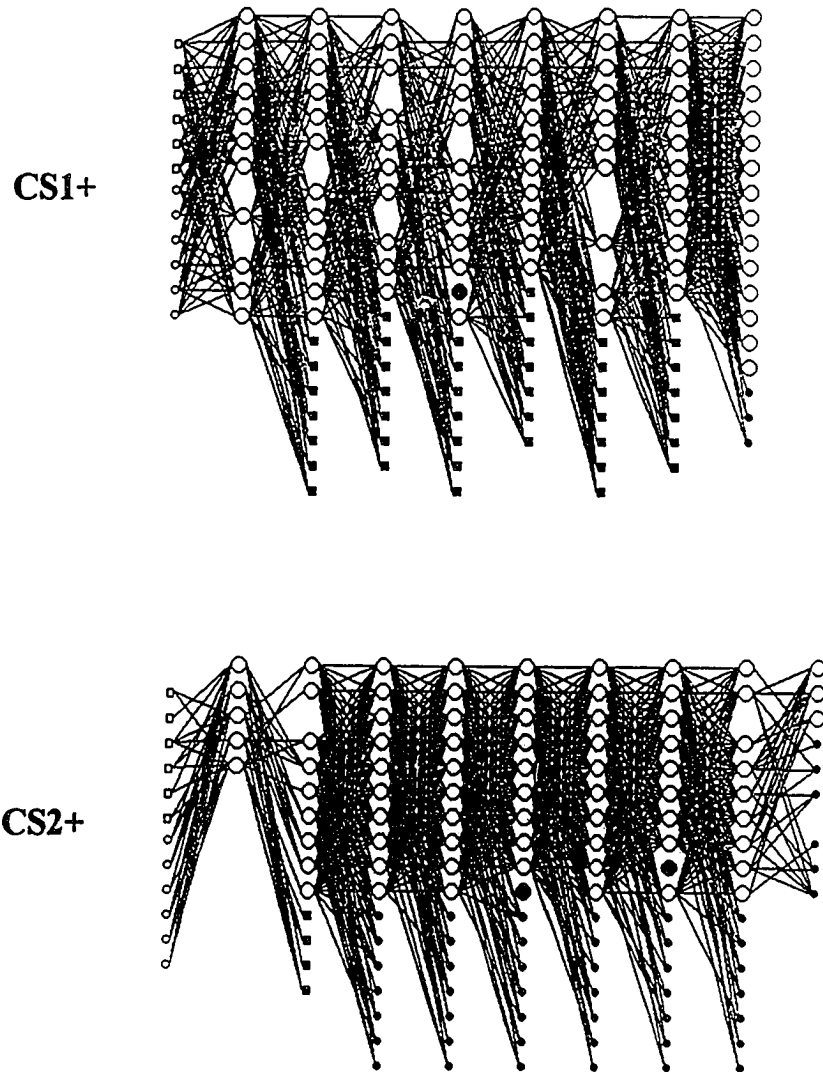
**Figure 5.8: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of CS2+.**



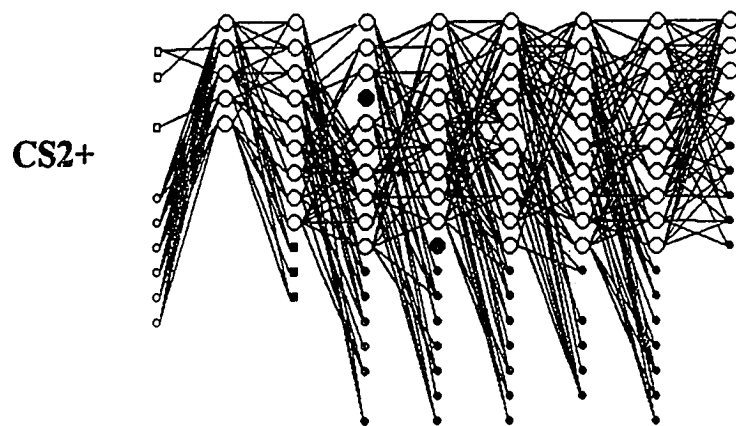
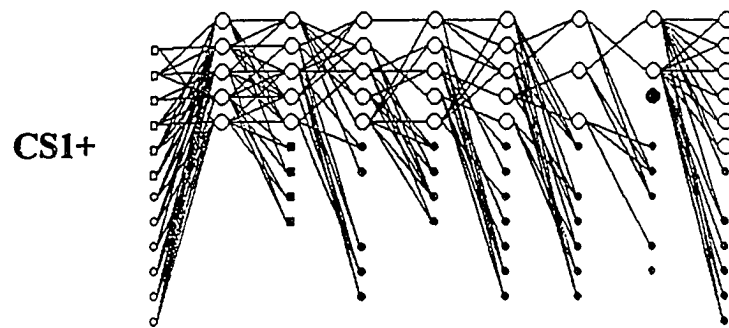
**Figure 5.9: Mean weight per connection in last generation of each lineage, before and after training.**



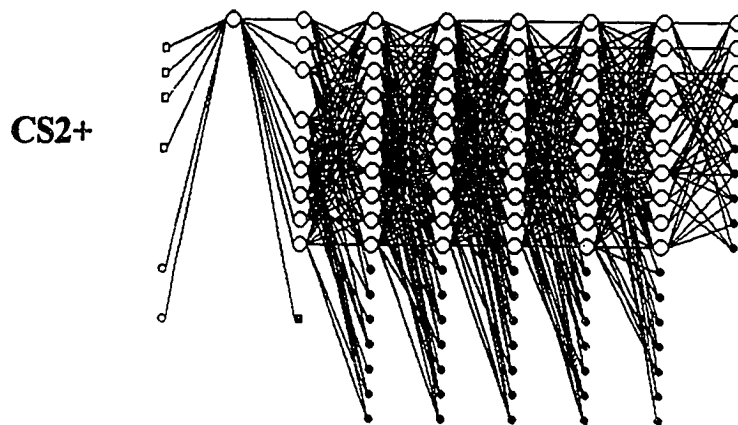
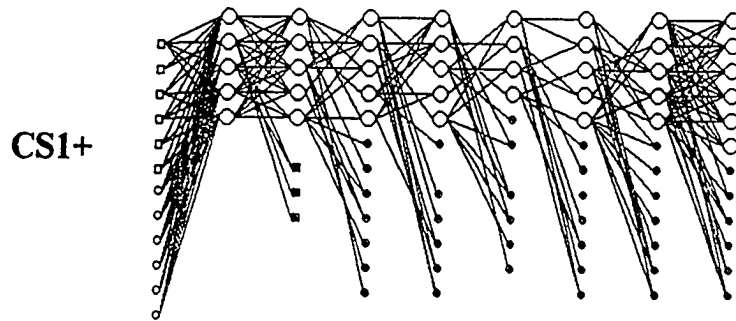
**Figure 5.10: Sample of high-fitness ANNs from the last generation of each lineage.** CR proportions of the CS1+ ANN were 1.00 for CS1 and 0.00 for CS2. CR proportions of the CS2+ ANN were 0.0 for CS1 and 1.00 for CS2.



**Figure 5.11: Another sample of high-fitness ANNs from the last generation of each lineage. CR proportions of the CS1+ ANN were 1.00 for CS1 and 0.00 for CS2. CR proportions of the CS2+ ANN were 0.0 for CS1 and 1.00 for CS2.**



**Figure 5.12: Sample of low-fitness ANNs from the last generation of each lineage. CR proportion of both ANNs was 0.0, for both CSs.**



**Figure 5.13: Sample of high-fitness nondiscriminating ANNs from the last generation of each lineage. CR proportions of both ANNs were above 0.75, for both CSs.**

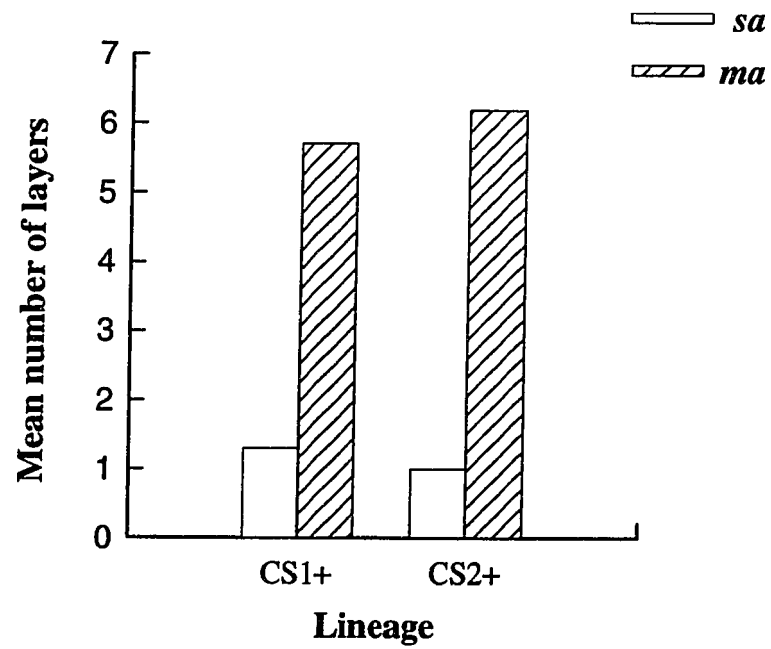
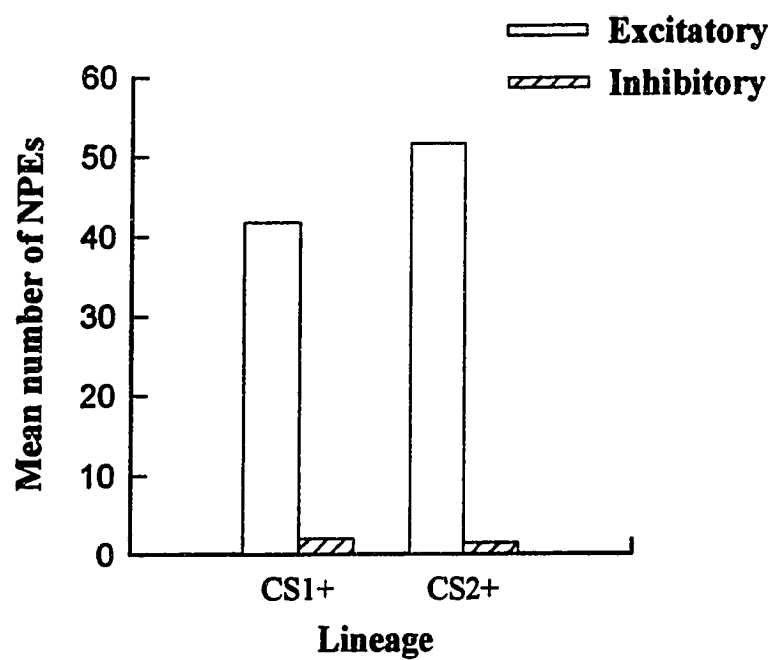
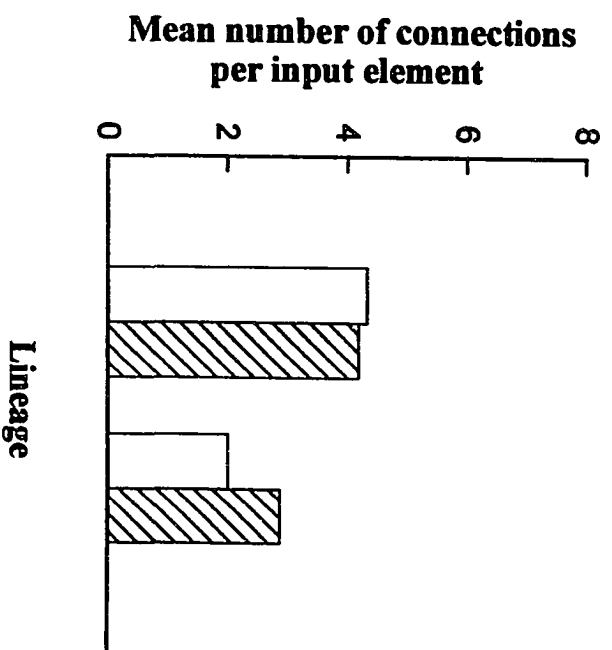
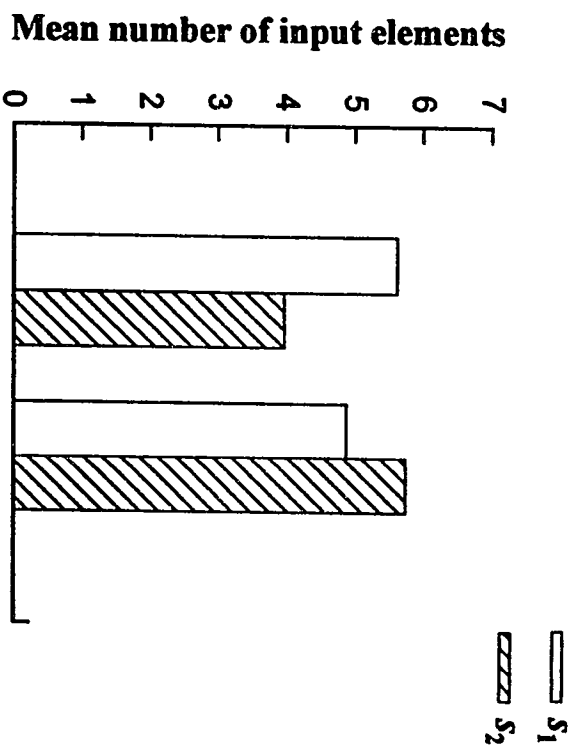


Figure 5.14: Mean number of hidden layers in last generation of each lineage.

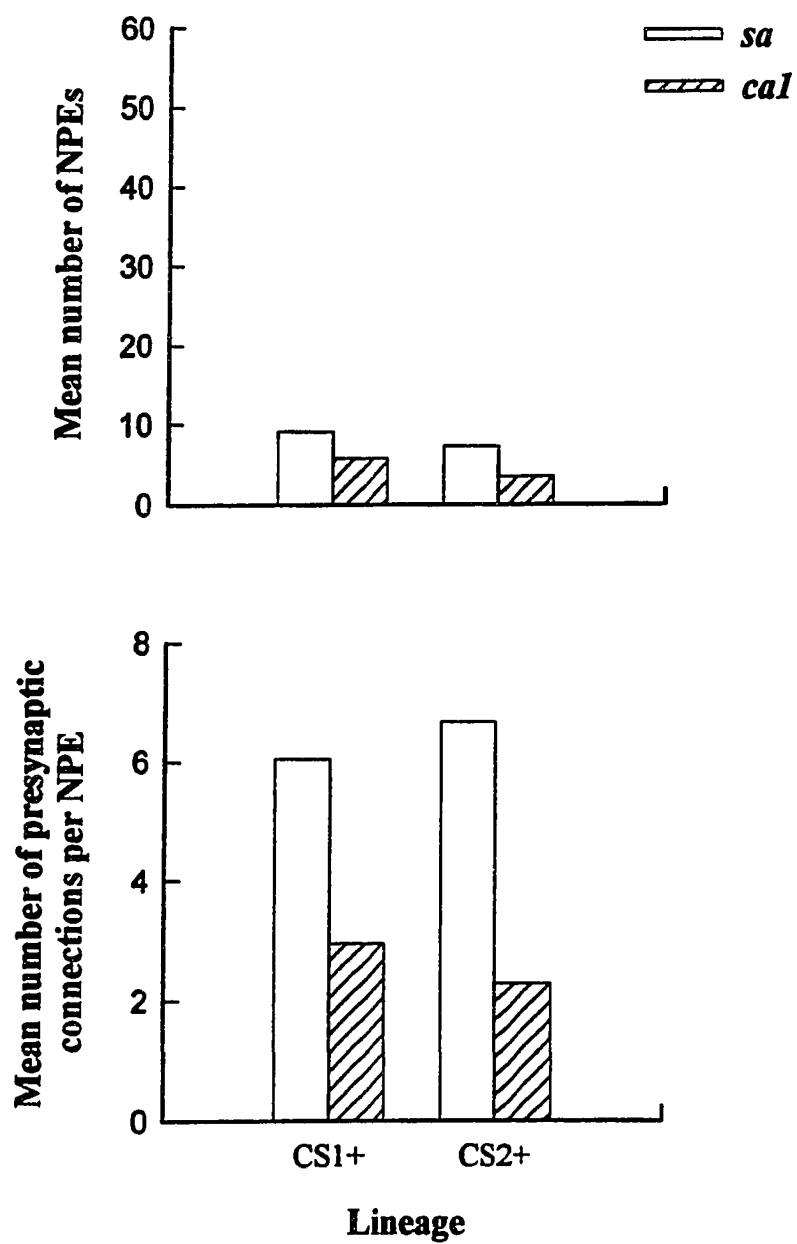




**Figure 5.15: Mean number of excitatory and inhibitory NPEs in last generation of each lineage.**



**Figure 5.16: Mean number of input elements and connections per element in last generation of each lineage.**



**Figure 5.17: Mean number of *sa* and *cal* NPEs and presynaptic connections per *sa* and *cal* NPE in last generation of each lineage.**

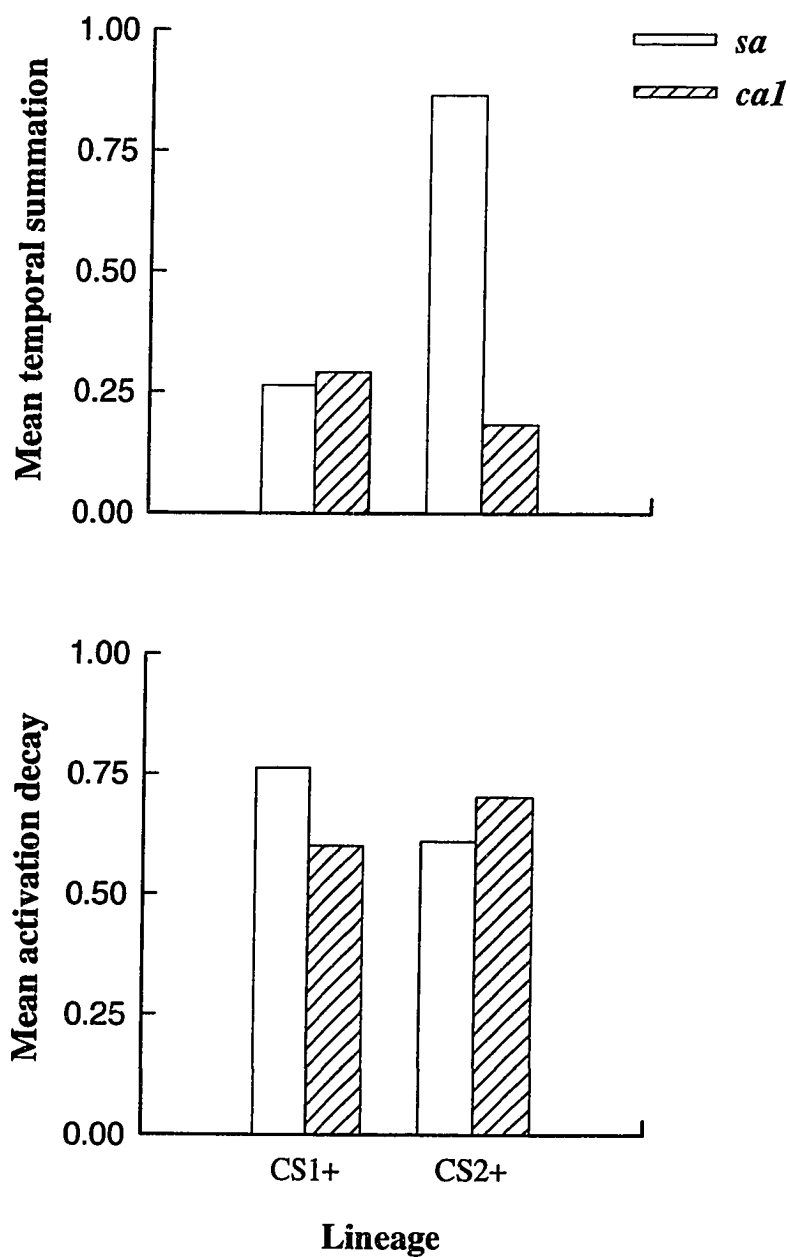


Figure 5.18: Mean activation parameters in *sa* and *cal* NPEs in last generation of each lineage.

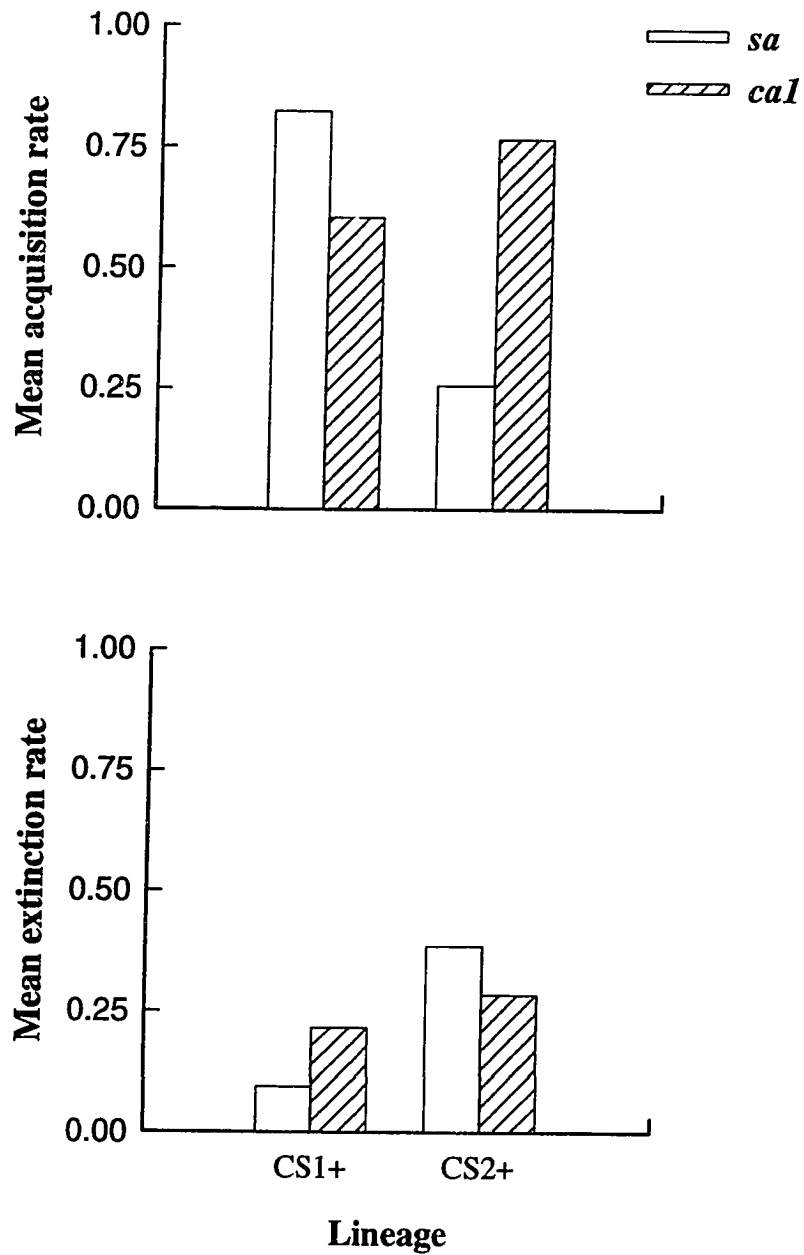


Figure 5.19: Mean synaptic-plasticity parameters in *sa* and *cal* NPEs in last generation of each lineage.

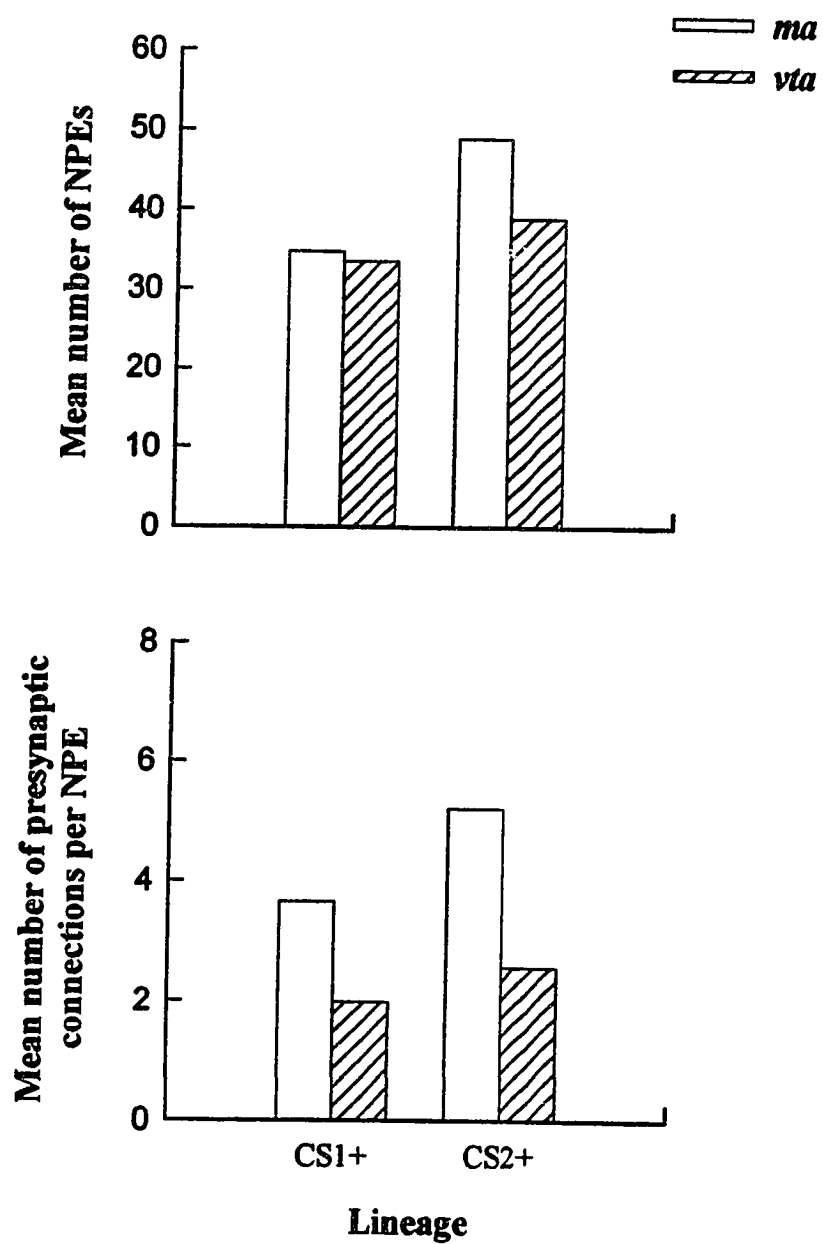


Figure 5.20: Mean number of *ma* and *vta* NPEs and presynaptic connections per *ma* and *vta* NPE in last generation of each lineage.

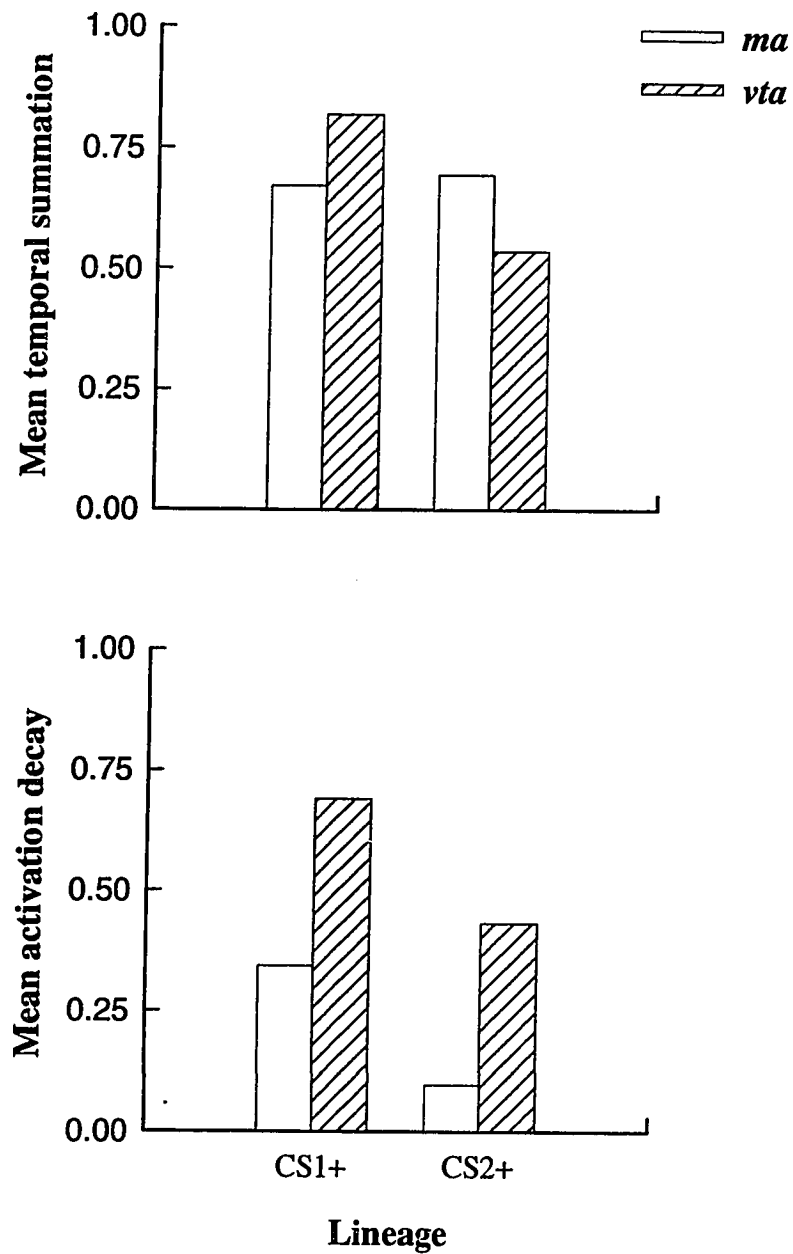


Figure 5.21: Mean activation parameters in *ma* and *vta* NPEs in last generation of each lineage.

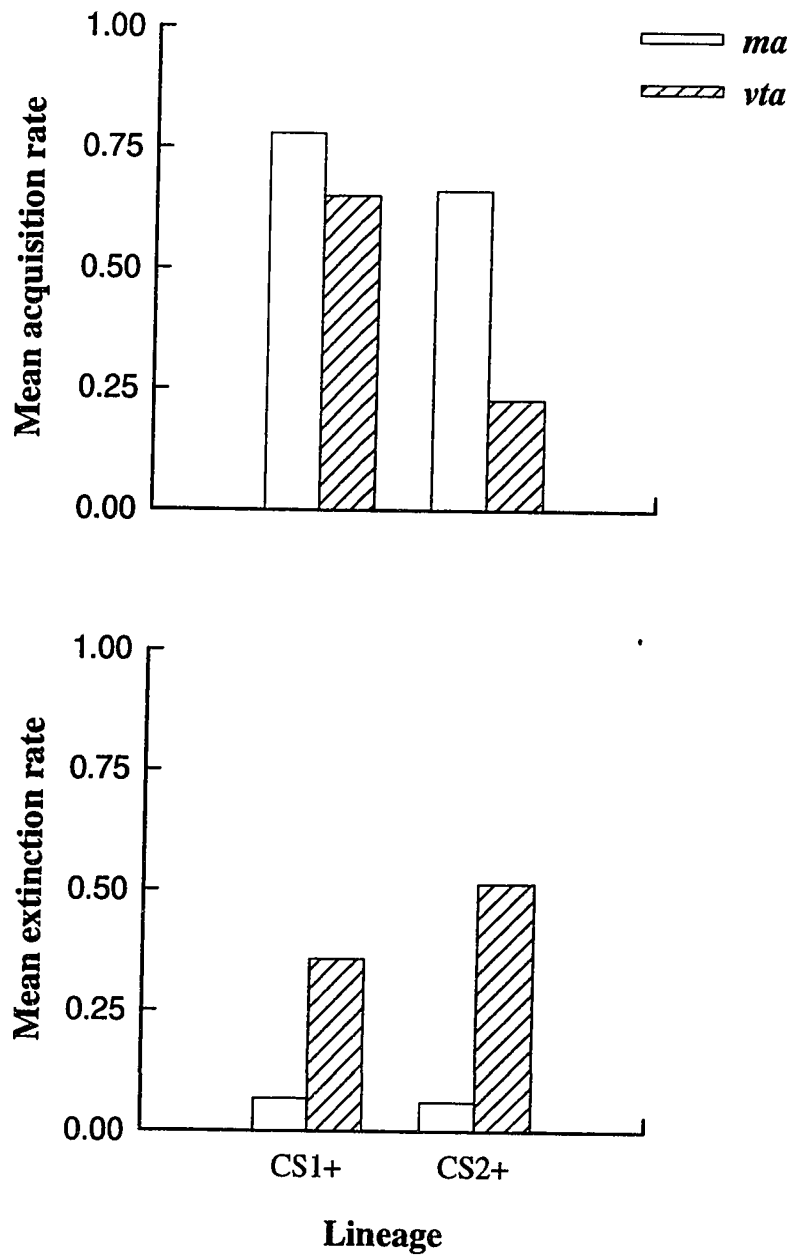
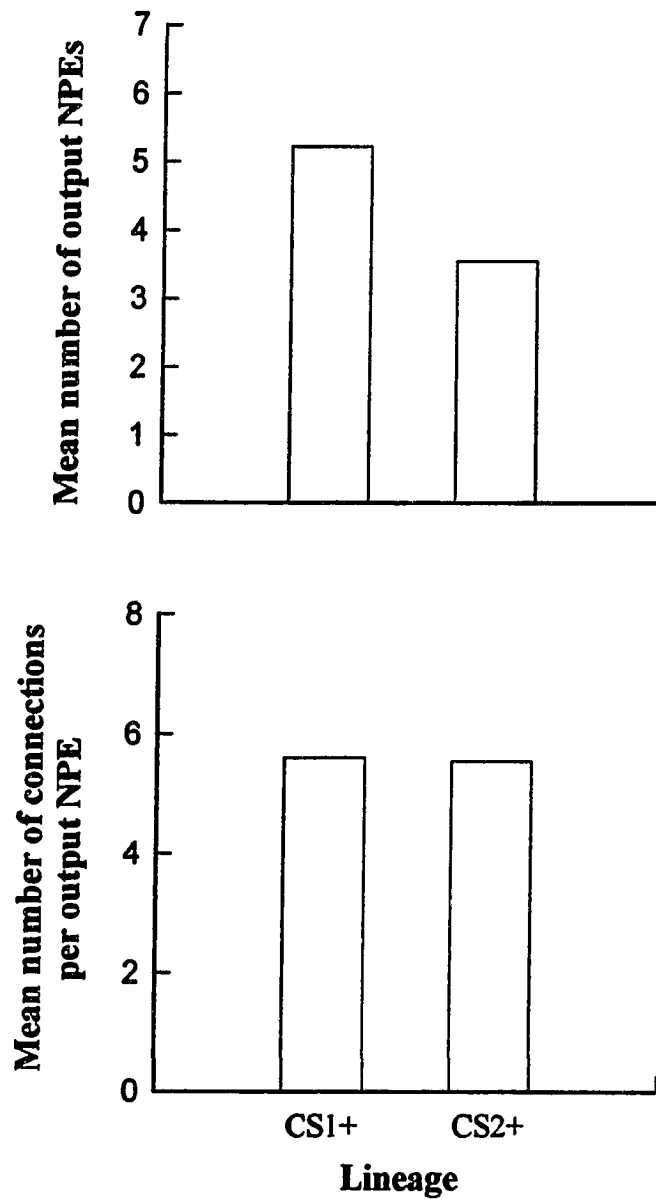
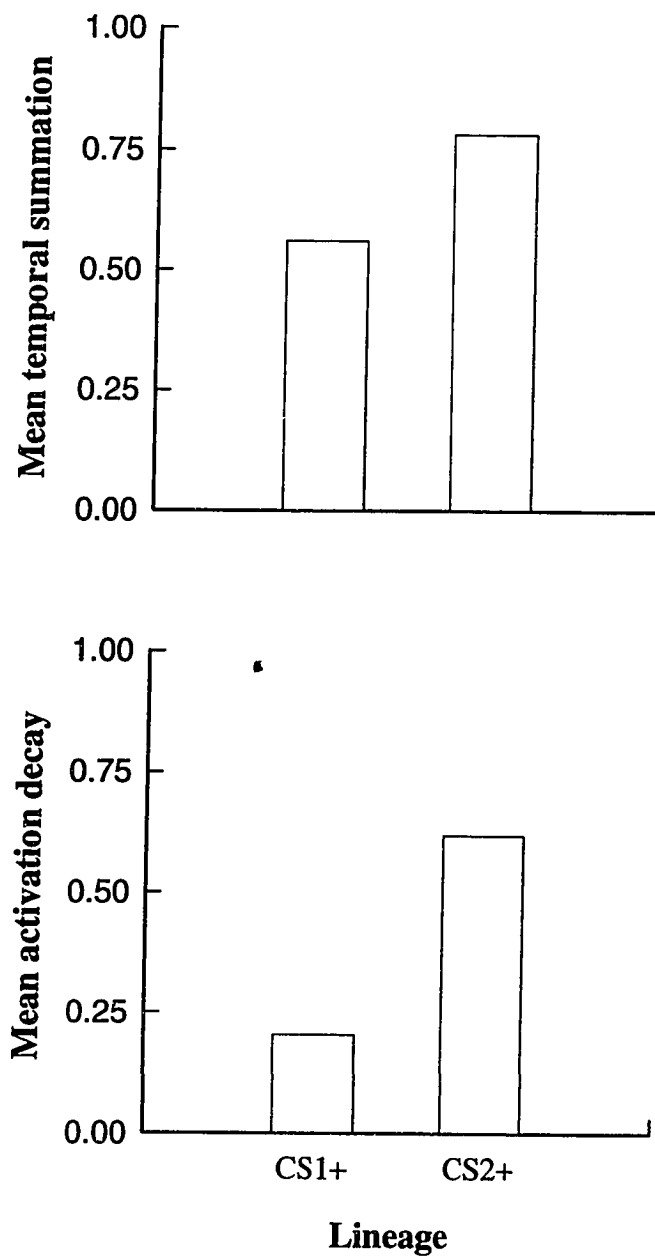


Figure 5.22: Mean synaptic-plasticity parameters in *ma* and *vta* NPEs in last generation of each lineage.

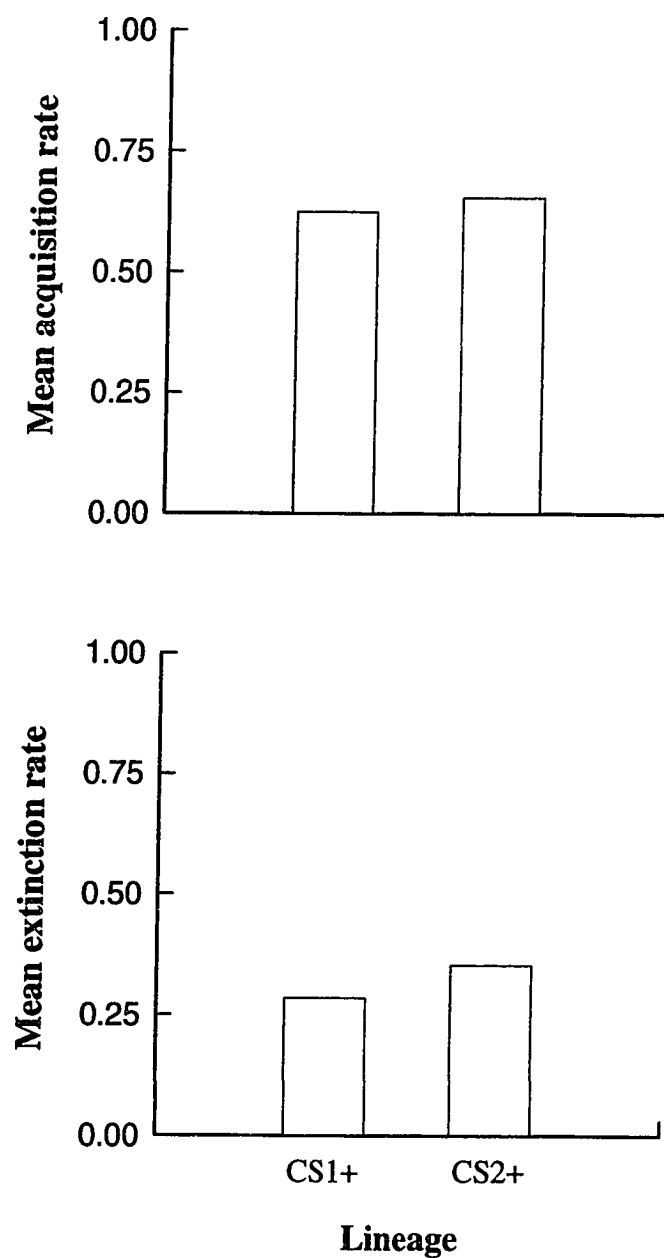




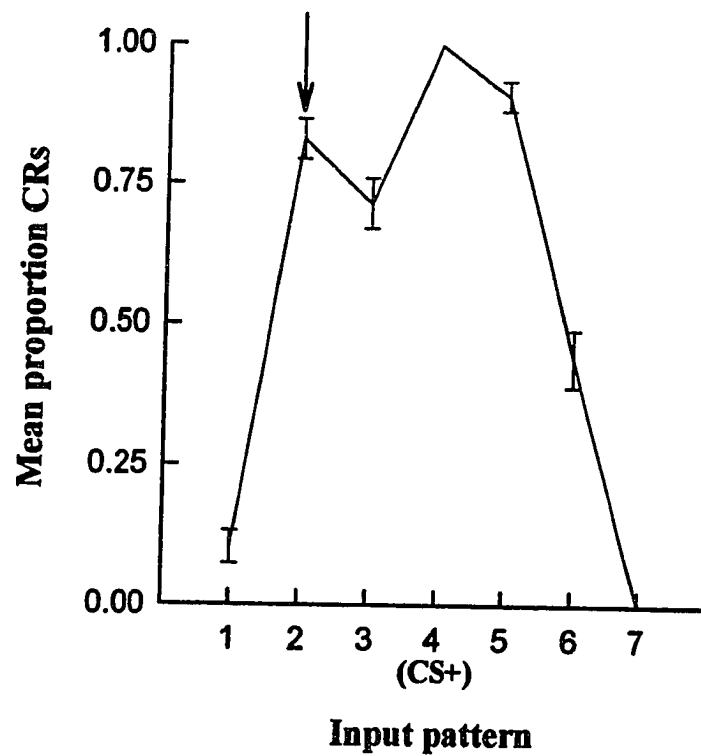
**Figure 5.23: Mean number of output NPEs and connections per output NPE, in last generation of each lineage.**



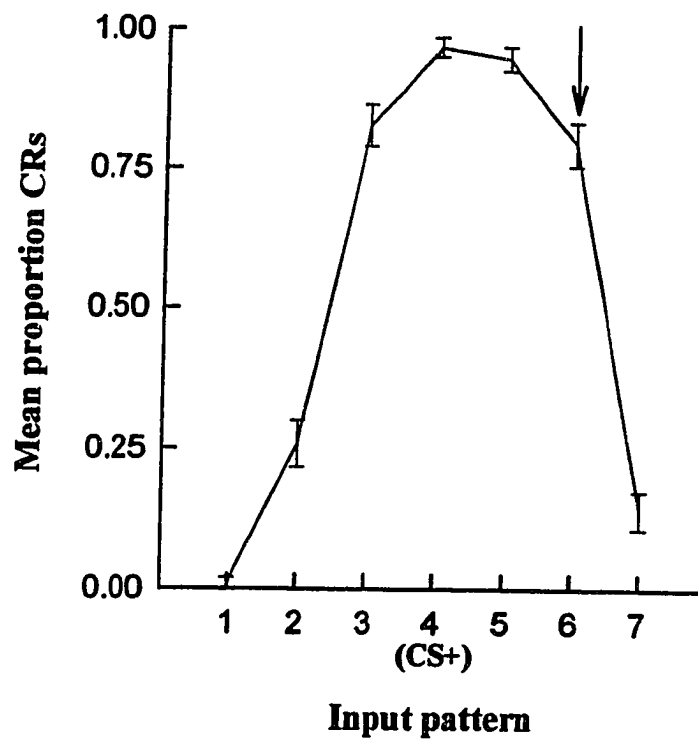
**Figure 5.24: Mean activation parameters in output NPEs in last generation of each lineage.**



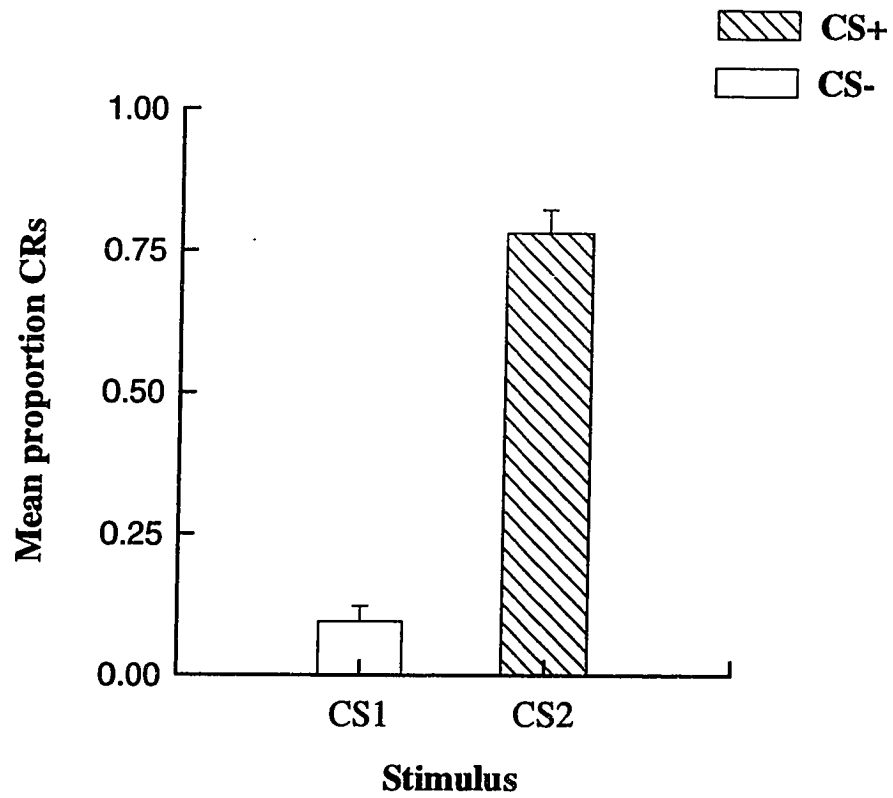
**Figure 5.25: Mean synaptic-plasticity parameters in output NPEs in last generation of each lineage.**



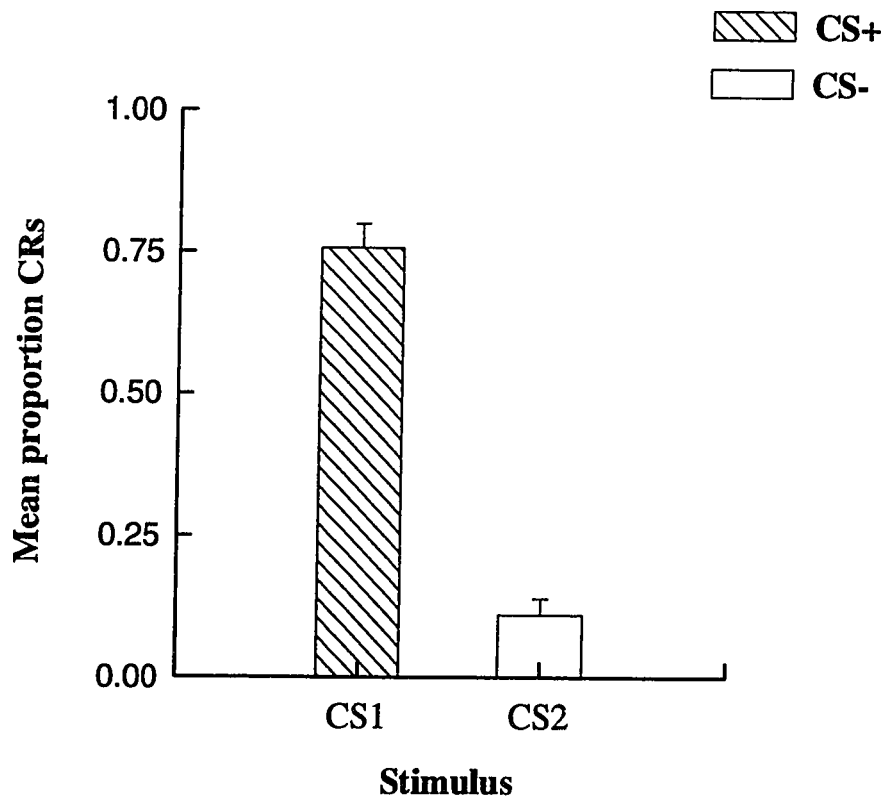
**Figure 5.26: Mean proportion CRs in generalization test of ANNs in last generation of CS1+. "CS+" was the pattern reinforced during training (see Figure 4.38 for patterns). The arrow indicates the pattern reinforced throughout phylogeny.**



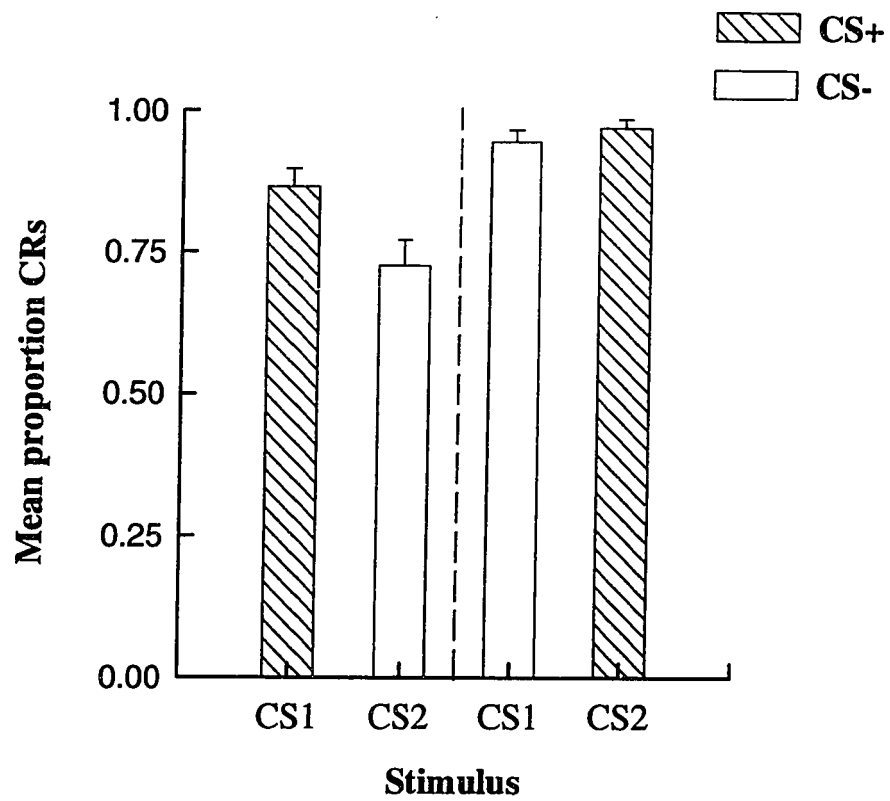
**Figure 5.27: Mean proportion CRs in generalization test of ANNs in last generation of CS2+. "CS+" was the pattern reinforced during training (see Figure 4.38 for patterns). The arrow indicates the pattern reinforced throughout phylogeny.**



**Figure 5.28: Mean proportion CRs in orthogonal discrimination test of ANNs in last generation of CS1+.**

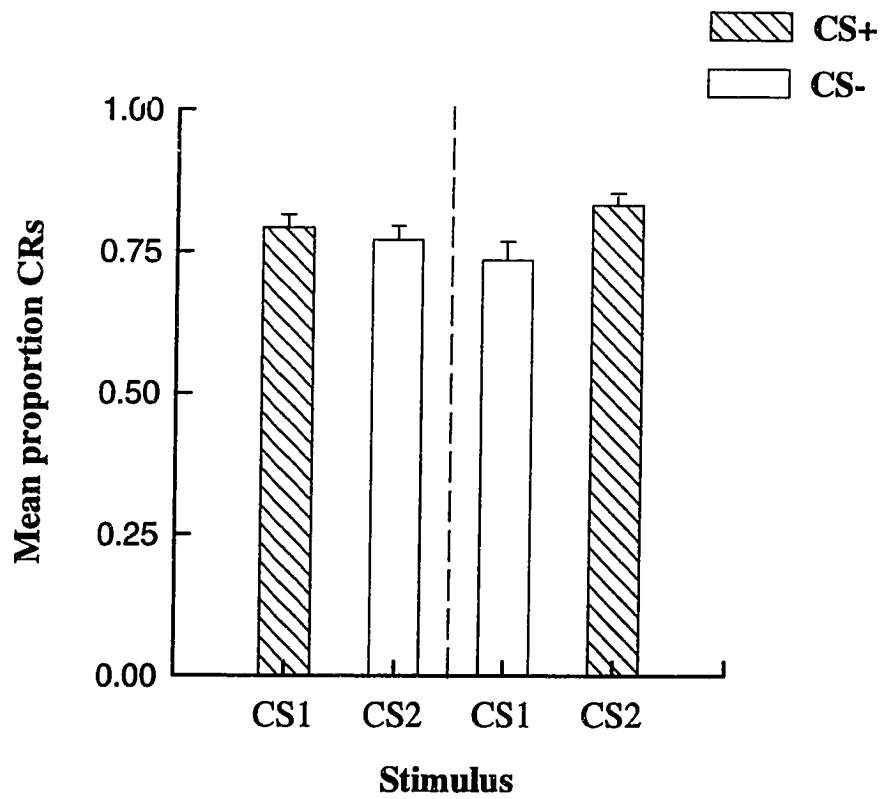


**Figure 5.29: Mean proportion CRs in orthogonal discrimination test of ANNs in last generation of CS2+.**

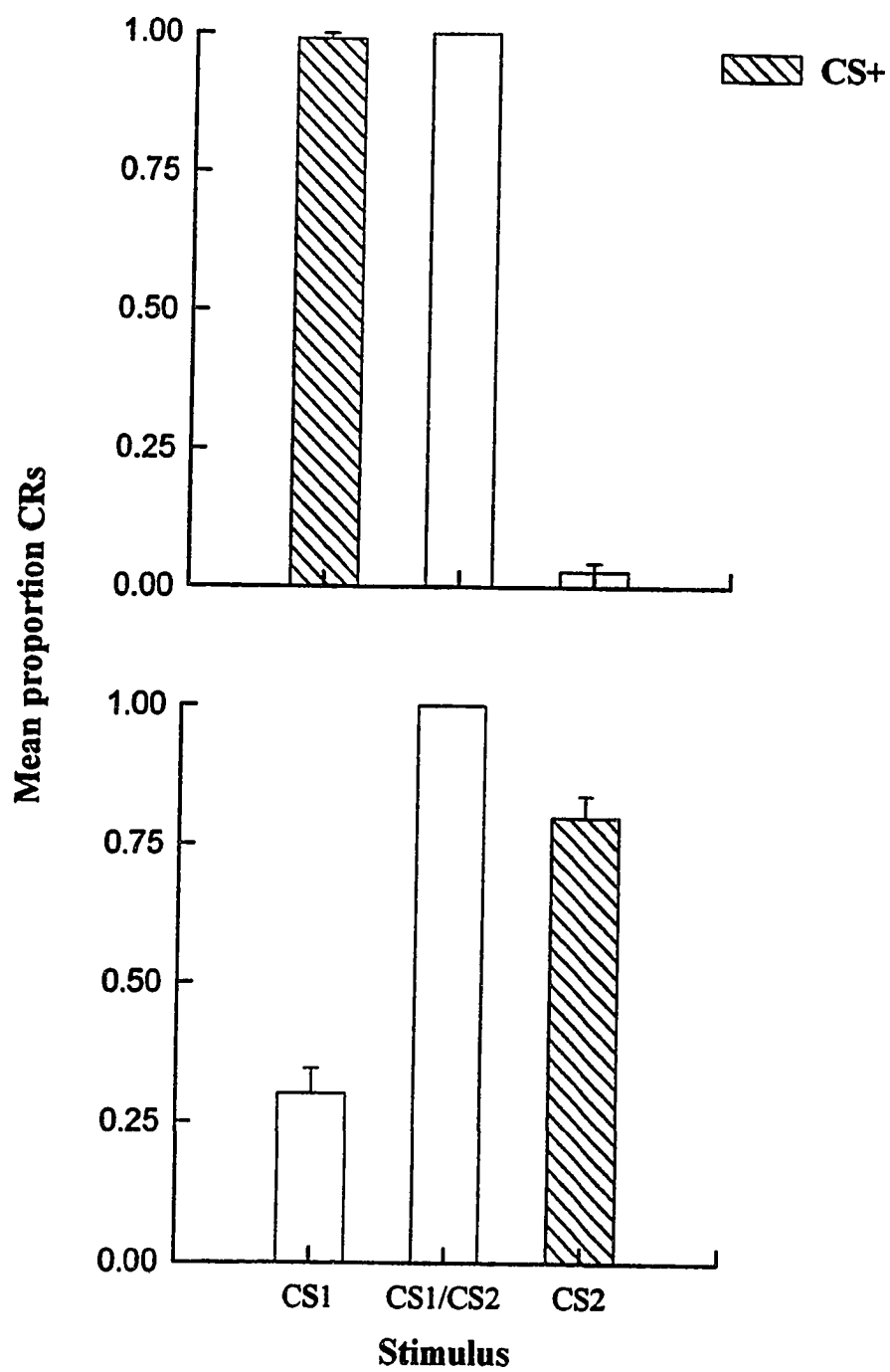


**Figure 5.30: Mean proportion CRs in nonorthogonal discrimination tests of ANNs in last generation of CS1+.**

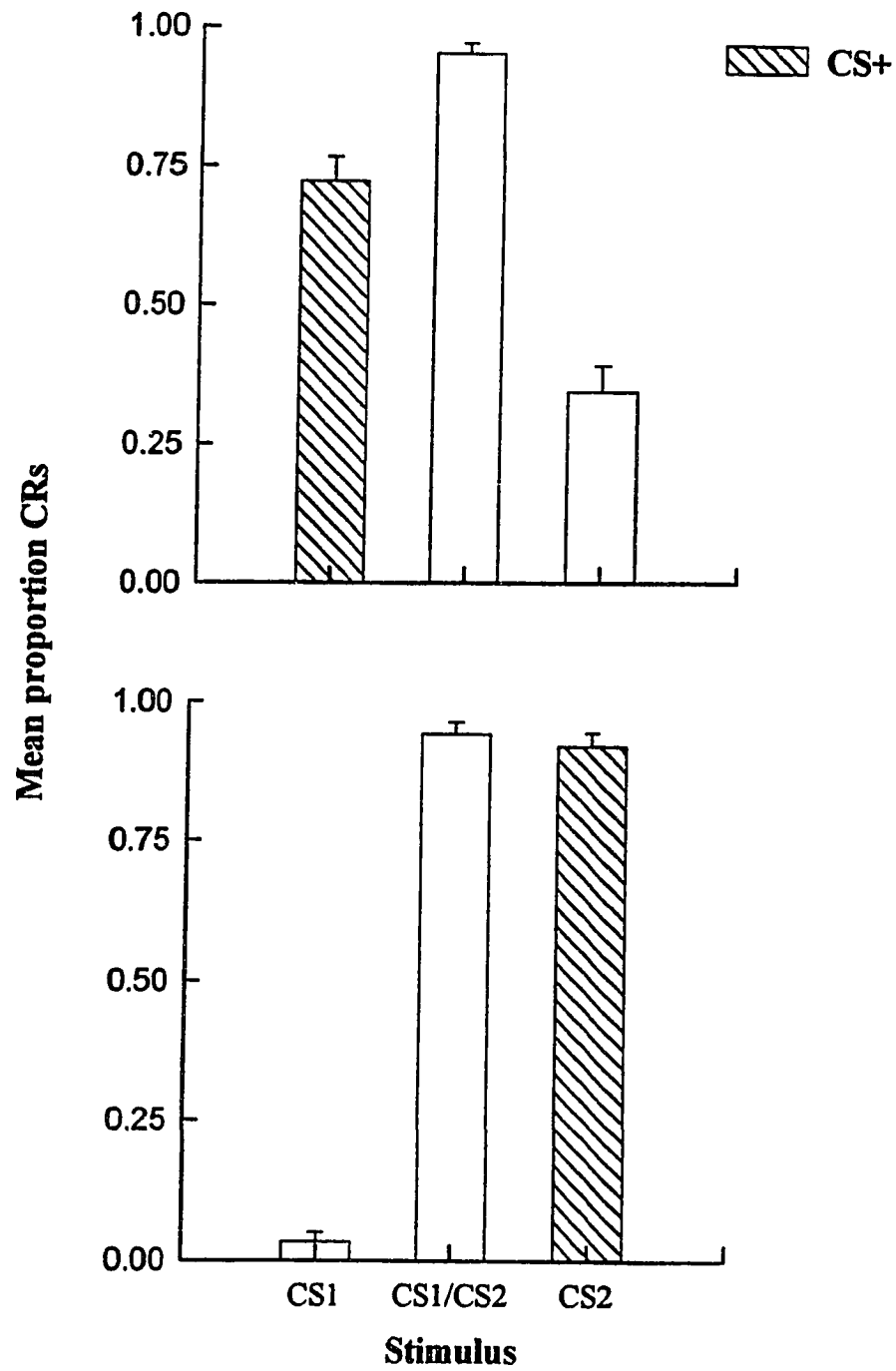




**Figure 5.31: Mean proportion CRs in nonorthogonal discrimination tests of ANNs in last generation of CS2+.**



**Figure 5.32: Mean proportion CRs in blocking tests of ANNs in last generation of CS1+.** "CS+" was the CS reinforced during Phase 1.



**Figure 5.33: Mean proportion CRs in blocking tests of ANNs in last generation of CS2+.** "CS+" was the CS reinforced during Phase 1.

# CHAPTER 6

## EXPERIMENT 4

In Experiment 3, ANNs throughout phylogeny were selected for increased responding to the reinforced CS in an orthogonal differential reinforcement procedure. That is, ancestral environments consisted of two CSs, one correlated with the presence and the other with the absence of the US. A defining formal feature of this arrangement was that the reinforced and nonreinforced CSs involved input activation patterns such that the inner product of their vector representations was equal to zero (i.e., the patterns had no elements in common). This kind of arrangement was analogous to those in which reinforced and nonreinforced CSs involve either different sensory modalities (e.g., lights vs. tones) or the same sensory modality but easily discriminable stimuli (e.g., tones vs. clicks, or yellow light vs. blue light).

However, many learning situations involve differential reinforcement of (and hence discrimination between) nonorthogonal CSs, that is, input patterns involving some overlap between the activations of different kinds of input elements (formally, input patterns whose vector representations result in dot products different from zero). In the present simulations, a CS in this kind of arrangement involved the activation of *both* kinds of input elements. This type of arrangement is analogous to those in which reinforced and nonreinforced CSs involve the same sensory modality

but are relatively difficult to discriminate (e.g., red lights vs. orange lights, or 500-Hz tones vs. 550-Hz tones).

In Experiments 2 and 3, nonorthogonal differential reinforcement was used only during test simulations. However, ANNs did not show nonorthogonal discrimination in any of these simulations. The purpose of the present experiment, then, was to use nonorthogonal differential reinforcement in the evolution simulations to determine if selection under nonorthogonal differential reinforcement was necessary to evolve ANNs showing improved nonorthogonal discrimination, relative to those observed in previous experiments. As in previous experiments, it was also of interest to determine if phenotypically different types of ANNs would emerge from selection for increased responding to different reinforced CSs, and those ANNs would perform under the generalization, discrimination, and blocking tasks.

## **6.1 Evolution Simulations**

### **6.1.1 Procedure**

As in Experiments 2, 3, and 4, G0' was used as the founder chromosome population. Two different environmental conditions were defined by two CSs (either CS1 or CS2). A lineage of generations of ANNs was evolved separately and independently for each condition. As a first step, two ANNs were developed from each chromosome in G0', to obtain 200 ANNs, which were divided into groups of 100, one group representing the ANN founder population of a condition. ANNs were trained in a nonorthogonal differential reinforcement procedure in which the CS was either CS1

or CS2, and whose reinforced trials involved a forward-delay arrangement in which the ISI was 8 ts.

ANNs were given trials of both CS classes, but trials of one CS class were nonreinforced. For ANNs in one condition, CS1 was the reinforced CS. For ANNs in the other condition, CS2 was the reinforced CS. However, in contrast to Experiments 1, 2, and 3, CS1 trials involved the activation of all  $s_1$  input elements and *some*  $s_2$  elements with a value of 1.0, whereas CS2 trials involved the activation of some  $s_1$  elements and all  $s_2$  elements with a value of 1.0. The input patterns used as CS1 and CS2 were the same as Patterns 3 and 5 used in the generalization tests in Experiments 2 and 3 (see Figure 4.38). All ANNs were given a total of 250 randomly intermixed CS1 and CS2 trials, the last 50 of which served as evaluation trials to determine individual fitness separately for each kind of CS. Fitness for a given CS class was the proportion of evaluation trials of that CS that elicited a CR. An individual won a selection tournament if and only if it had the highest *difference* between CR proportion in the presence of the reinforced CS and CR proportion in the presence of the nonreinforced CS. This criterion was used after preliminary simulations showed that the criterion used in Experiment 3 (i.e., selection of the ANN with the highest fitness to the reinforced CS) generated ANNs that did not show nonorthogonal discrimination.

After evaluation, a new population of chromosomes was created through the selection scheme of the GA. This new population constituted G1 (the next generation). Once the new chromosomes were created, new ANNs were developed and trained in the same procedure as their ancestors. The same operation was repeated generation after generation, until the termination criterion (either 100 generations after G0' or fitness stability) was satisfied. This strategy resulted in a total of two lineages, labeled

as CS1+ (the lineage for which CS1 was the reinforced CS) and CS2+ (the lineage for which CS2 was the reinforced CS).

## **6.1.2 Results and Discussion**

### **6.1.2.1 Genotypic Evolution**

Figure 6.1 shows that the mean chromosomic overlap increased as a negatively accelerated function of generations, for both lineages. Individuals within lineages thus became genetically more similar as generations progressed. The mean overlaps for G0' were close to 0.0, which is consistent with the expected value for a randomly generated population of chromosomes. In contrast, the mean overlaps for the last generation were relatively high (between 0.8 and 0.9). Thus, at the end of evolution there still was a small amount of genetic variation. Also, note that overlaps for CS1+ seem to have reached asymptotic levels earlier than overlaps for CS2+, despite the fact that the total number of generations in each lineage was very similar (30 for CS1+ vs. 29 for CS2+). This result suggests that genetic variation in the last generation seemed to depend upon not only upon the number of generations, but also upon the selection conditions present throughout phylogeny. These relations may have arisen from differences in the founder chromosome populations.

### **6.1.2.2 Effects on Functional Phenotypes**

#### **6.1.2.2.1 Effects on Fitness**

Figure 6.2 shows that, for both lineages, mean population fitness for the reinforced CS increased as a negatively accelerated function of generations. Indeed, in G0, the two lineages started with low-fitness values (close to 0.0) for both CSs, and ended with a high-fitness values (between 0.8 and 1.0) for the reinforced CS only. Thus, ANNs in

the last generation were comparably fit. In contrast, the mean population fitness for the nonreinforced CS showed an initial increase (for the first 10 or 15 generations) to approximately 0.25, followed by a decrease to approximately 0.1. The duration of this increase was about five generations longer in CS2+ than in CS1+. Also, CS2+ showed a second, although small increase in responding to the nonreinforced stimulus toward the last generation.

Figures 6.3 and 6.4 show 3D mesh plots of frequency distributions of fitness values across generations, and for CS1 (top panel) and CS2 trials (bottom panel). For the reinforced CS, all plots show a high number of low-fitness individuals at G0, followed by a relatively rapid increase in the number of high-fitness individuals after a number of generations. At the beginning of evolution, most individuals in G0 had a fitness between 0.0 and 0.05, for both CSs. As generations advanced, the number of individuals with fitness values between 0.95 and 1.00 increased rapidly for the reinforced CS, in both lineages (see top panel of Figure 6.3 and bottom panel of Figure 6.4).

In contrast, the number of individuals with fitness values between 0.0 and 0.05 remained substantially high for the nonreinforced CS (see bottom panel of Figure 6.3 and top panel of Figure 6.4). Also, note that there was an initial decrease in the number of individuals with low fitness values, followed by another increase a few generations later, for the nonreinforced CS in both lineages (see bottom panel of Figure 6.3 and top panel of Figure 6.4). This pattern corresponds to the initial rise in mean fitness to the nonreinforced CS showed described before (see Figure 6.2). Finally, functional-phenotypic variability at the end of evolution was relatively low (i.e., the majority of individuals had the same fitness).



#### 6.1.2.2.2 Effects on Learning

Figures 6.5 through 6.8 show 3D mesh plots of frequency distributions of mean output-NPE activations across the first 75 CS1 and 75 CS2 trials, for the first (top panel) and last (bottom panel) generation of each lineage. As in previous experiments, the number of ANNs with low output-NPE activations (between 0.0 and 0.05) in G0 remained relatively constant and close to the population size, for both lineages and CSs (see top panels of Figures 6.5 through 6.8), with only a few individuals having mean output-NPE activations above 0.5. In contrast, the number of individuals with high output-NPE activations (between 0.95 and 1.0) in the last generation of both lineages increased as a function of reinforced trials, remaining relatively constant at values close to the population size until the end of training (see bottom panel of Figures 6.5 and 6.8). In the case of the nonreinforced CS, the number of individuals with high output-NPE activations in the last generation increased at the beginning of training for a few trials after which this number decreased to near-zero values (see bottom panel of Figures 6.6 and 6.7).

This initial rise in responding to the nonreinforced CS is consistent with what is observed in real animals during the formation of a nonorthogonal discrimination in Pavlovian (Gynther, 1957) and operant conditioning (Hanson, 1959). In these experiments, responding to both the reinforced and nonreinforced CSs increases at the beginning of training, after which the former continues increasing whereas the latter decreases. In the present simulations, a similar pattern was observed but throughout *phylogenetic history* (compare Figures 6.2, 6.3, and 6.4 with bottom panel of Figure 6.6 and top panel of Figure 6.7), which suggests a relation between the formation of a

nonorthogonal discrimination at the ontogenetic level and the evolutionary history of nonorthogonal discrimination at the phylogenetic level<sup>1</sup>.

Also, as in previous experiments, most ANNs showed maximally high mean output-NPE activations at the end of training, despite the fact that such activations were not needed to win selection tournaments throughout phylogeny.

Finally, Figure 6.9 shows the mean weight per connection before and after training for the last generation of each lineage. Connection weights were substantially higher after than before training, although the ratio of final to initial weights was higher for CS1+ than CS2+ ANNs. This result suggests that, in the last generation of both lineages, orthogonal differential reinforcement produced a increase in connection weights, regardless of the reinforced CS.

Overall, the above results show that selection for increased responding to the reinforced CS in a nonorthogonal differential reinforcement procedure generated ANNs whose behavior changed from low to high conditioned responding to the reinforced CS, for both CS1+ and CS2+. However, in contrast to Experiment 3, the present result was obtained by defining the tournament-winning criterion in terms of responding to *both* the reinforced and the nonreinforced CS. Therefore, selection for increased responding to the reinforced CS was *not* sufficient to evolve ANNs that simulated nonorthogonal discrimination. Selection for *not* responding to the nonreinforced CS was also necessary.

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<sup>1</sup> This interpretation might be seen as contending that the formation of an nonorthogonal discrimination during ontogeny "recapitulates" the evolution of nonorthogonal discrimination throughout phylogeny. However, I am not proposing an application of the biogenetic law to the evolution of learning and behavior. I am only suggesting that the recapitulation-like pattern described here may have arisen from the close *formal* relationship between the definition of fitness in terms of CR proportion and the definition of CR proportion in terms of mean output-NPE activations. In the present context, then, *recapitulation* refers simply to a general isomorphic relation, without any of the nomological or ontological implications of that law.

### 6.1.2.3 Effects on Structural Phenotypes

#### 6.1.2.3.1 General Effects

Figure 6.10 shows one typical high-fitness ANNs from the last generation of each lineage. Both ANNs had a fitness of 1.0 for the reinforced CS and a fitness of zero for the nonreinforced CS. Some similarities are apparent. Both ANNs were relatively small, in comparison to the ISI-8 ANNs generated in Experiments 1 and 2, although they seem to have the same number of elements as the ANNs that were evolved under orthogonal discrimination. Also, both ANNs had the same number of *sa* and *ma* layers (one layer of each kind), similar numbers of NPEs per hidden layer, and more excitatory than inhibitory NPEs. However, the number of inhibitory NPEs relative to excitatory NPEs was substantially larger than in ANNs described in previous experiments. In fact, between 20% and 25% of hidden NPEs in the present ANNs were inhibitory, whereas between only 2% and 5% of hidden NPEs in comparable ANNs in previous experiments were inhibitory.

Also, each ANN had the same number of  $s_1$  as  $s_2$  input elements. Some differences can be observed as well. For example, the CS1+ ANN had more  $s_1$  and  $s_2$  input elements, substantially more connections per input element, and more *sa* inhibitory NPEs than the CS2+ ANNs. In contrast, the CS2+ ANN had substantially more *cal*, *ma* inhibitory NPEs, output NPEs, and connections per output NPE.

As in previous experiments, within-population variability was observed among ANN architectures. Indeed, although most ANNs in the last generation of both lineages seemed to be instances of the same basic architecture, a few high-fitness ANNs had architectures different from those shown in Figure 6.10. Figure 6.11 illustrates this result by showing another example of a high-fitness ANNs from the CS1+ and CS2+ lineages. The main difference between these ANNs and their respective conspecifics is

the absence of inhibitory NPEs. Also, the CS2+ had substantially more connections per output NPE than its conspecific shown in Figure 6.10. Thus, as in previous experiments, selection for increased responding under certain environmental conditions generated a class of architectures, rather than a unique architecture.

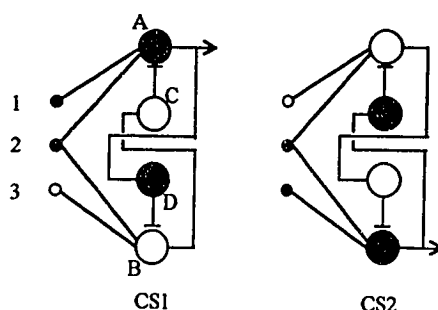
Also, in previous experiments, not all ANNs in the last generation responded with high CR proportions to the reinforced CS. Figure 6.12 shows a low-fitness ANN from each lineage. Both ANNs had a fitness of zero for both CSs. The CS1+ ANN had substantially more inhibitory NPEs than its conspecific (35% of hidden NPEs in this ANN were inhibitory). This excess in inhibitory NPEs may have been sufficient for this ANN to respond poorly to the reinforced CS. In the case of the CS2+ ANN, no critical architectural differences were observed. Rather, the critical differences were observed in the NC parameters. Indeed, output NPEs in this ANN had an acquisition rate of 0.28 and an extinction rate of 0.53 (output NPEs in most ANNs in this population had an acquisition rate of 0.6 and an extinction rate of 0.02; cf. below). Therefore, the rate of decrement of connection weights in output NPEs was substantially higher than the rate of increment, which may have been sufficient to hinder responding to the reinforced CS.

A few ANNs were nondiscriminating in that they showed high responding to the reinforced *and* the nonreinforced CS. Figure 6.13 shows one nondiscriminating ANN from each lineage. As in the nondiscriminating ANNs described in Experiment 3 (see Figure 5.13), the initial weights of the connections from the  $s_1$  input elements to the *sa* NPEs were sufficiently high to allow for relatively high responding to the nonreinforced CS, without the need of external reinforcement.

Also, ANNs in both lineages had only one *sa* and one *ma* layers (see Figure 6.14). In Experiments 1 and 2, ISI-8 ANNs tended to have far more *sa* layers (between 4 and

6), which differs from the present result, but only one *ma* layer, which is consistent with the present result. In contrast, ANNs in Experiment 3 tended to have only one *sa* layer, which is consistent with the present result, but far more *ma* layers (approximately six), which differs from the present result. Thus, selection for nonorthogonal discrimination under a particular ISI generated ANNs that were architecturally different from those generated by selection for increased responding to one reinforced CS (Experiment 1), two reinforced CSs (Experiment 2), or the reinforced CS in an orthogonal differential reinforcement arrangement (Experiment 3), under the same ISI.

Finally, as in previous experiments, ANNs had more excitatory than inhibitory NPEs (see Figure 6.15). However, as mentioned before, the percentage of inhibitory NPEs in the present ANNs was considerably larger than that observed in ANNs from previous experiments. This result indicates that lateral inhibition played a more substantial role in nonorthogonal than in orthogonal discrimination. Learning to discriminate between similar stimuli may have required the kind of *contrast enhancement* that is facilitated by lateral inhibition. One way in which such an enhancement may have been accomplished in the present ANNs is to allow for the activation of NPEs only by input elements that were *not* shared by the reinforced and nonreinforced CSs, as follows:



The diagram shows the activations of elements in a circuit under two CSs, namely CS1 and CS2. Information flows from left to right. The circuit consists of a layer of

three input elements (small circles labeled as 1, 2, and 3), and layer of two excitatory (outermost large circles, labeled as A and B) and two inhibitory NPEs (innermost large circles, labeled as C and D). A receives excitatory connections from 1 and 2, and an inhibitory connection from C. In contrast, B receives excitatory connections from Elements 2 and 3, and an inhibitory connection from NPE D. Finally, C and D receive excitatory connections from B and A, respectively.

CS1 activates 1 and 2, whereas CS2 activates 2 and 3. Hence, CS1 and CS2 are nonorthogonal in that both involve input patterns sharing one element (i.e., Element 2). When CS1 is presented, both A and B activated. A is activated more strongly than B, because the former receives inputs from both 1 and 2. This difference makes the inhibition of B by D stronger than the inhibition of A by C, which enhances the difference between the activations of A and B by decreasing the activation of B and C, thus allowing the activation of A to increase further. When CS2 is presented, again, both A and B are activated, but this time B is activated more strongly than A, because the former receives inputs from both 2 and 3. This difference makes the inhibition of A by C stronger than the inhibition of B by D, which enhances the difference between the activations of A and B by decreasing the activation of A and D, thus allowing the activation of B to increase further. In both cases, the net effect is that only one excitatory NPE is activated.

#### **6.1.2.3.2 Effects on the Sensory Subnetwork: Input Elements**

Figure 6.16 shows quantitative differences and similarities in input elements. ANNs in the last generation of both CS1+ and CS2+ tended to have the same number of  $s_1$  and  $s_2$  input elements, with CS1+ ANNs having substantially more input elements of both the kinds than CS2+ ANNs (see top panel of Figure 6.16). Also, CS1+ ANNs tended to have substantially more connections per input element than CS2+ ANNs,

although the latter tended to have more connections per  $s_2$  than  $s_1$  NPE (see bottom panel of Figure 6.14). Thus, competition among  $s_1$  and  $s_2$  elements was balanced in CS1+ ANNs. In contrast,  $s_2$  connectivity in CS2+ ANNs favored  $s_2$  over  $s_1$  elements in their competition for *sa* connection weights (i.e., on the average, any given *sa* NPE received more connections from  $s_2$  than from  $s_1$  elements. This result is consistent with the evolutionary history of these ANNs, in which the reinforced CS involved the activation of all  $s_2$  and only some  $s_1$  elements. Ancestral selection in this lineage thus tended to favor connectivity from  $s_2$  elements. This interpretation, however, does not apply to CS1+ ANNs. Perhaps, having substantially more input elements in these ANNs comparable activations of *sa* NPEs through a more balanced connectivity from  $s_1$  and  $s_2$  elements.

#### **6.1.2.3.3 Effects on the Sensory Subnetwork: *sa* and *cal* NPEs**

Figure 6.17 shows the mean number of *sa* and *cal* NPEs (top panel) and the number of connections per *sa* and *cal* NPE (bottom panel), for both lineages. ANNs had approximately the same number of *sa* (between 10 and 12), for both lineages. However, CS2+ ANNs tended to have substantially more *cal* NPEs than CS1+ ANNs. Therefore, unless *cal* activations were substantially larger, mean *cal* discrepancies may have been lower in CS2+ than in CS1+ ANNs.

Also, CS1+ ANNs tended to have substantially more presynaptic connections per *sa* NPE than CS2+ NPEs. Because most ANNs had only one *sa* layer, the number of presynaptic connections per *sa* NPE is a direct reflection of the number of postsynaptic connections per input NPE. Both  $s_1$  and  $s_2$  input elements contributed equally to the number of connections per *sa* NPE for CS1+ ANNs, whereas  $s_2$  elements contributed more than  $s_1$  elements to the number of connections per *sa* NPE for CS2+ ANNs. The same tendency was observed in Experiment 3. Finally, CS2+ ANNs tended to have

more connections per *cal* NPE than per *sa* NPE, as well as more connections per *cal* NPE than CS1+ ANNs. This latter effect may have contributed to increase the activation of *cal* NPEs for CS2+ ANNs, and hence compensate for the number of *cal* NPEs in these ANNs.

Figure 6.18 shows temporal summations (top panel) and activation decays (bottom panel) in *sa* and *cal* NPEs, for both lineages. Temporal summation was substantially higher in *sa* than in *cal* NPEs, for both lineages. In contrast, activation decays in *sa* and *cal* NPEs tended to be similar for CS1+ ANNs, whereas *cal* decays were substantially larger than *sa* decays for CS2+ ANNs. Finally, temporal summations tended to be substantially higher than activation decays in *sa* NPEs, while the opposite tendency was observed in *cal* NPEs, for both lineages. As in previous experiments, then, an inverse relation between temporal summations and activation decays was observed. That is, high temporal summations corresponded to low activation decays, and vice versa. These relations may have had an adaptive value by allowing for an intermediate range of activations of *sa* and *cal* NPEs, thus preventing activations from changing too rapidly or too slowly across ts.

Figure 6.19 shows acquisition (top panel) and extinction rates (bottom panel) in *sa* and *cal* NPEs, for both lineages. Acquisition rates tended to be substantially higher in *sa* than in *cal* NPEs. The same tendency was observed in extinction rates for CS2+ ANNs. However, the opposite tendency was observed in extinction rates for CS1+ ANNs. Also, as in previous experiments, acquisition rates tended to be higher than extinction rates in *sa* NPEs, for both lineages. Again, this trade-off may have had an adaptive value by allowing for an optimal variation range in changes in connection weights over time. In contrast, acquisition rates in *cal* NPEs tended to be smaller than extinction rates for CS1+ and similar to extinction rates for CS2+ ANNs. This last



result indicates that a trade-off between acquisition and extinction rates in *cal* may have been less critical for fitness (i.e., was not selected against) than a trade-off in *sa* NPEs (i.e., was selected for).

Comparisons with Figure 6.18 show trade-offs between temporal summations, and acquisition and extinction rates in *cal* NPEs for CS1+ ANNs (i.e., low temporal summations corresponded to high acquisition and extinction rates). Trade-offs can also be observed between activation decays, and acquisition and extinction rates in *cal* NPEs for CS2+ ANNs (i.e., high *cal* activation decays corresponded to low acquisition and extinction rates). Overall, these trade-off relations between activation and synaptic-plasticity parameters suggest that the search for optimal variation ranges in activations may have been coordinated with the search for optimal variation ranges in connection weights.

#### **6.1.2.3.4 Effects on the Motor Subnetwork: *ma* and *vta* NPEs**

Figure 6.20 shows the mean number of *ma* and *vta* NPEs (top panel) and connections per NPE type (bottom panel). CS1+ and CS2+ ANNs tended to have the same number of *ma* NPEs, *vta* NPEs, and connections per NPE type. Also, note that although ANNs had far more *ma* than *vta* NPEs for both lineages, the number of connections per NPE was very similar for both kinds of NPEs. Comparisons with Figure 6.17 indicate that ANNs in both lineages tended to have more *ma* than *sa* NPEs, but about the same number of *vta* as *cal* NPEs. Also, the number of connections per *ma* NPE was substantially larger than the number of connections per *sa* NPE, for CS2+ ANNs. Thus, in these ANNs, connectivity from *sa* and *ma* NPEs was richer than connectivity from input elements to *sa* NPEs. These architectural differences may have arisen from differences in the number and connectivity of input elements across lineages (see Figure 6.16).

Figure 6.21 shows the mean temporal summations (top panel) and activation decays (bottom panel) of *ma* and *vta* NPEs, for both lineages. A substantial trade-off between these parameters was observed only in the *ma* NPEs for CS1+ ANNs, in which high temporal summations corresponded to low activation decays. Such a trade-off may have had an adaptive value by increasing the likelihood of high activations in *ma* NPEs. Also, comparisons with previous effects suggest that the low activation decays observed in *ma* NPEs for CS1+ ANNs may have been inversely related to (and hence compensated for) the relatively high temporal summations, activation decays, and extinction rates observed in *sa* NPEs (see top panel of Figure 6.18).

Activation decays were substantially higher in *vta* than *ma* NPEs, again for CS1+ ANNs, whereas the opposite tendency was observed for CS2+ ANNs. The absence of a trade-off between summation and decay in *vta* NPEs for CS2+ ANNs may have had an adaptive value by increasing the range of variation of *vta* activations, and hence *vta* discrepancies. An increase in the range of variation of *vta* discrepancies may have allowed ANNs to show the kind of behavioral flexibility needed to adapt to a changing environment in which reinforcement and nonreinforcement were associated with similar CSs.

Finally, Figure 6.22 shows the mean acquisition (top panel) and extinction rates (bottom panel) in *ma* and *vta* NPEs, for both lineages. Again, an overall trade-off relation can be observed between these two parameters. Indeed, *ma* NPEs tended to have higher acquisition rates than *vta* NPEs, whereas the opposite tendency was observed in extinction rates, for both lineages. Also, acquisition rates tended to be substantially higher than extinction rates in *ma* NPEs, again for both lineages. Extinction rates were higher than acquisition rates in *vta* NPEs, for CS2+ ANNs.

In general, then, there was a trade-off between acquisition and extinction parameters in *ma* and *vta* NPEs. Indeed, for both lineages, high acquisition rates in *ma* NPEs tended to correspond to low extinction rates. This trade-off could have had an adaptive value by allowing for an optimal range of variation of connection weights over time. Indeed, if acquisition and extinction rates are both low, then that range will be too small. In contrast, if acquisition and extinction are both high, the variation range will be too large. By keeping only one parameter high, connection weights are prevented from changing too little or too much over time. Similar results were observed in Experiment 3 (see Figure 5.22). Thus, selection for orthogonal discrimination and selection for nonorthogonal discrimination generated ANNs that were similar in the synaptic-plasticity of their *ma* and *vta* NPEs.

#### **6.1.2.3.5 Effects on the Motor Subnetwork: Output NPEs**

Figures 6.23, 6.24, and 6.25 show changes in structural phenotypes in output NPEs, for both lineages. CS2+ ANNs tended to have more output NPEs than CS1+ ANNs (see top panel of Figure 6.23). This result differs from that obtained in Experiment 3, in which CS1+ ANNs tended to have more output NPEs than CS2+ ANNs. Also, ANNs in both lineages had between 1 and 3 connections per output NPE (see bottom panel of Figure 6.23).

Output NPEs in CS1+ and CS2+ ANNs tended to have similar temporal summations (close to 1.0) and activation decays (approximately 0.3) (see top and bottom panels of Figure 6.24, respectively). These results also differ from those obtained in Experiment 3, in which output NPEs in CS2+ tended to have substantially higher temporal summations as well as activation decays than output NPEs in CS1+ ANNs. In fact, there seemed to have been a trade-off between temporal summation and activation decay (i.e., high temporal summations corresponded to lower activation

decays). The adaptive value of this trade-off may have been to bias output NPEs toward high activations. Finally, output NPEs tended to have the same acquisition rates (around 0.6; very similar to those observed in Experiment 3) and extinction rates (close to zero; much lower than those observed in experiment 3), for both lineages. Again, such a trade-off between acquisition and extinction rates may have facilitated high output-NPE activations by increasing the speed of growth of connection weights.

## **6.2 Test Simulations**

A series of simulations were run in which the same ANNs described in Section 6.1.2.3 were tested in generalization, orthogonal discrimination, nonorthogonal discrimination, and blocking. These procedures involved environmental conditions that were not present throughout phylogeny. Hence, they allowed for testing whether or not those ANNs showed emergent functional phenotypes consistent with the kind of effects that such procedures have on animal learning and behavior as observed in the laboratory.

### **6.2.1 Generalization**

As in previous generalization tests, the present test consisted of a training and a testing phase. During the training phase, naïve versions of the same ANNs described before were given 100 reinforced trials in which the CS involved the activation of *all* input units with a value 1.0. During the testing phase, the trained ANNs were given 175 probe trials of seven randomly intermixed input-activation patterns (the same patterns used in previous generalization ; see Figure 4.38). As usual, the synaptic-plasticity function of the NC model was disabled to avoid extinction during the testing phase.

Figures 6.26 and 6.27 show generalization gradients obtained during the testing phase of ANNs in CS1+ and CS2+, respectively. As in previous tests, ANNs in both lineages showed gradients consistent with those observed in generalization experiments with real animals. That is, ANNs responded with higher CR proportions to input patterns that were more similar to the trained pattern. CS2+ ANNs showed substantially more generalization to Pattern 6 (which involved the activation of all  $s_2$  input elements and no activation of any  $s_1$  element) than to Pattern 2 (which involved the activation of all  $s_1$  input elements and no activation of any  $s_2$  element). This result is not surprising, for ancestors of these ANNs were selected for high responding to the reinforced CS, which involved the activation of all  $s_2$  elements. In this sense, one can Patterns 2 and 6 were nonequipotential in the generalization task for CS2+ ANNs. In contrast, CS1+ ANNs generalized to Pattern 6 as much as to Pattern 2 (which involved the activation of all  $s_1$  elements). In general, then, CS1+ and CS2+ were functionally different in the extent to which they generalized to patterns similar to the reinforced CS. This result represents an emergent functional difference between CS1+ and CS2+ ANNs.

Overall, these results indicate that selection for increased responding to the reinforced CS in a nonorthogonal differential reinforcement procedure generated ANNs that showed generalization as a functional phenotype that was not explicitly selected for throughout phylogenetic history.

## 6.2.2 Orthogonal Discrimination

Selection for increased responding to the reinforced CS and decreased responding to the nonreinforced CS in a nonorthogonal differential reinforcement arrangement generated ANNs showing nonorthogonal discrimination. However, it is not clear

whether these ANNs would learn an orthogonal discrimination. To address this issue, naïve ANNs were trained in two conditions, each one consisting of a training and a testing phase. In the training phase of the first condition, CS1+ ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials (an average of 100 trials per CS kind). In the training phase of the second condition, CS2+ ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials. In both conditions, CS1 involved the activation of all  $s_1$  input units with a value of 1.0 and no activation of any  $s_2$  input elements, whereas CS2 involved the activation of all  $s_2$  input units, also with a value of 1.0, and no activation of any  $s_1$  input elements. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 probe trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled.

Figures 6.28 and 6.29 show the CR proportions obtained during the orthogonal discrimination testing phase of CS1+ and CS2+ ANNs, respectively. ANNs in both lineages showed orthogonal discrimination in that they responded with substantially higher CR proportions to the reinforced CS than to the nonreinforced CS, regardless of whether the reinforced or nonreinforced CS was CS1 or CS2. Also, ANNs in both lineages showed comparable orthogonal discrimination. Note that CS2+ ANNs showed *errorless* discrimination (i.e., at the end of training) they did not respond to the nonreinforced CS). Thus, selection for nonorthogonal discrimination generated ANNs that showed orthogonal discrimination as an emergent functional phenotype.

### 6.2.3 Reversal of Nonorthogonal Discrimination

In the evolution simulations, selection for nonorthogonal discrimination generated ANNs that showed high responding to the reinforced CS and low responding to the

nonreinforced CS. However, it is not clear whether or not the evolved ANNs would learn a discrimination involving a reversal of the ancestral reinforcement conditions. To address this issue, naïve ANNs were trained in two conditions, each one consisting of a training and a testing phase. In the training phase of the first condition, CS1+ ANNs were given 200 randomly intermixed nonreinforced CS1 and reinforced CS2 trials (an average of 100 trials per CS kind). In the training phase of the second condition, CS2+ ANNs were given 200 randomly intermixed reinforced CS1 and nonreinforced CS2 trials. In the testing phase of both conditions, trained ANNs were given 50 randomly intermixed CS1 and CS2 trials (an average of 25 trials per CS kind), during which the synaptic-plasticity function of the NC model was disabled. For both conditions and ANNs, CS1 and CS2 were the same input patterns used in the evolution simulations (see Patterns 3 and 5 in Figure 4.38).

Figures 6.30 and 6.31 show the CR proportions obtained during the testing phase of each condition, for CS1+ and CS2+ ANNs, respectively. ANNs in both lineages showed substantial nonorthogonal discrimination in that they responded with higher CR proportions to the reinforced CS than to the nonreinforced CS. Therefore, CS1+ ANNs showed nonorthogonal discrimination after training in an arrangement in which CS2 was the reinforced CS, whereas CS2+ ANNs showed nonorthogonal discrimination after training in an arrangement in which CS1 was the reinforced CS. Selection for nonorthogonal discrimination, then, generated ANNs that could learn a nonorthogonal discrimination after training in an arrangement in which the reinforced CS is different from that present when ancestral selection took place. Therefore, selection under nonorthogonal differential reinforcement generated ANNs showing considerably better nonorthogonal discrimination than ANNs generated by selection under orthogonal differential reinforcement.

## 6.2.4 Blocking

Finally, as in other experiments, ANNs were tested in a blocking procedure. Naïve ANNs were trained in two conditions consisting of three phases each. In Phase 1, ANNs were given 100 reinforced presentations of a CS. Then, in Phase 2, those trained ANNs were given 100 reinforced concurrent presentations of the previously reinforced CS and another, previously nonexperienced CS. In one condition, the CS reinforced in the single-CS phase was CS1, whereas the novel CS presented in the compound-CS phase was CS2. Conversely, in the other condition, the CS reinforced in the single-CS phase was CS2, whereas the novel CS presented in the compound-CS phase was CS1. Finally, in Phase 3, ANNs were given 75 randomly intermixed trials of CS1, CS2, and the CS1/CS2 compound (an average of 25 trials per CS kind). For all ANNs and conditions, CS1 and CS2 corresponded to the input patterns used in the evolution simulations and in the nonorthogonal discrimination tests<sup>2</sup>. The CS1/CS2 compound involved the activation of all input elements. Again, the synaptic-plasticity function of the NC model was disabled during the test phase.

Figures 6.32 and 6.33 show the mean CR proportions per CS kind during Phase 3 of the blocking test of CS1+ and CS2+ ANNs, respectively. The top panels show the results for the first condition, in which CS1 was the CS reinforced in Phase 1 (marked CS+ in the graphs) and CS2 was the novel CS introduced in Phase 2. The bottom panels show the results for the second condition, in which CS2 was the CS reinforced in Phase 1 and CS1 was the novel stimulus introduced in Phase 2. No ANNs showed

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<sup>2</sup> Other simulations showed that ANNs responded very little to the reinforced CS in Phase 1 if it involved the activation of only  $s_1$  input elements or only  $s_2$  input elements. This finding is puzzling in view of the results obtained in the generalization and orthogonal discrimination tests. Indeed, CS1+ ANNs, for example, showed substantial generalization to Patterns 2 and 6, after reinforcement of a CS involving the activation of all input units. Also, these ANNs showed substantial responding to these patterns after a differential reinforcement arrangement. However, these ANNs were incapable of responding to a reinforced CS after an acquisition arrangement like the one used in Experiment 1.



blocking. Indeed, CS1+ ANNs showed substantially higher responding to the novel CS introduced in Phase 2 than to the CS reinforced in Phase 1. Similarly, ANNs showed substantial responding to all stimuli. Therefore, selection for nonorthogonal discrimination generated ANNs that did not show blocking. This result is not altogether surprising, for CS1 and CS2 involved nonorthogonal input patterns. Thus, high responding to the novel CS introduced in Phase 2 was due to generalization.

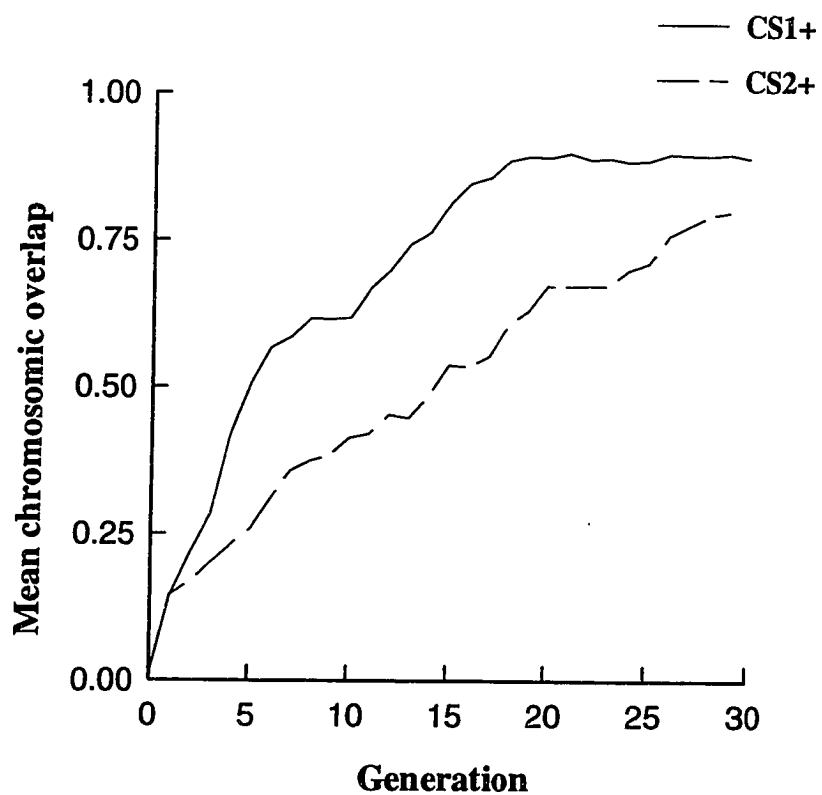


Figure 6.1: Mean chromosomal overlap as a function of generations.

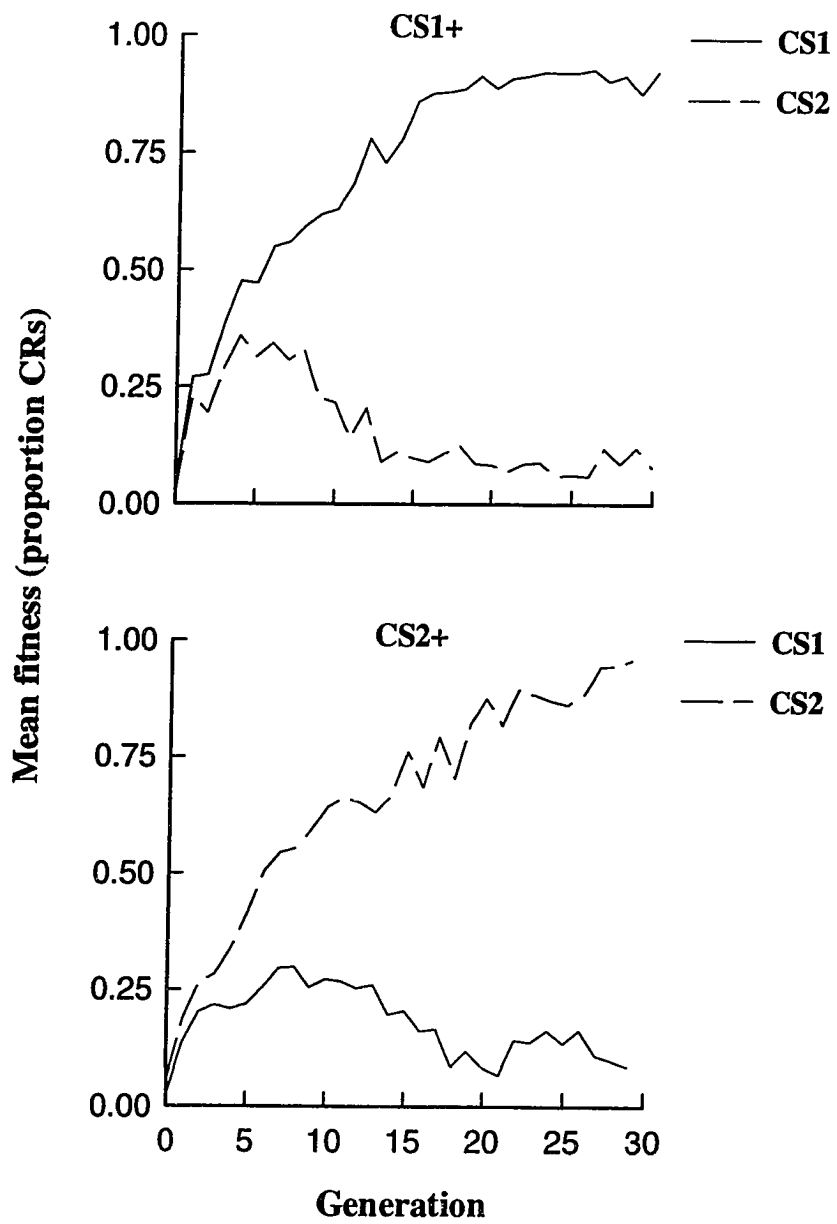


Figure 6.2: Mean population fitness as a function of generations.

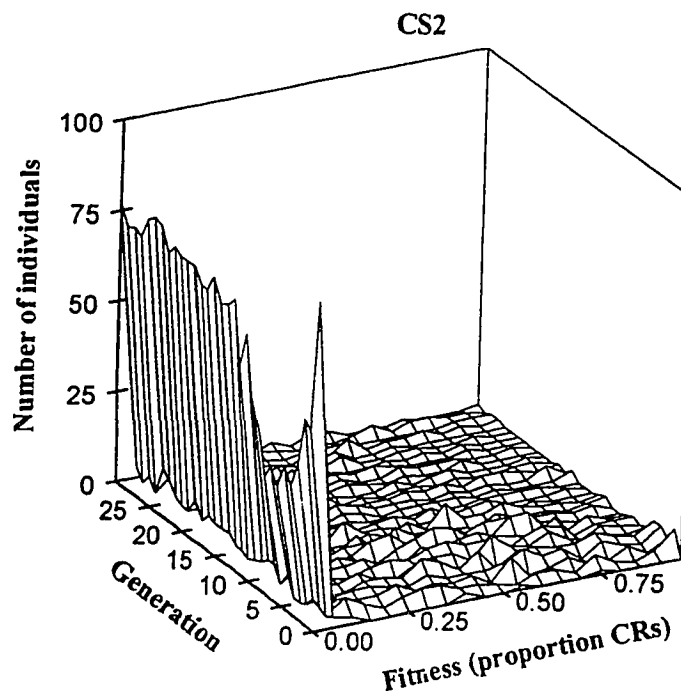
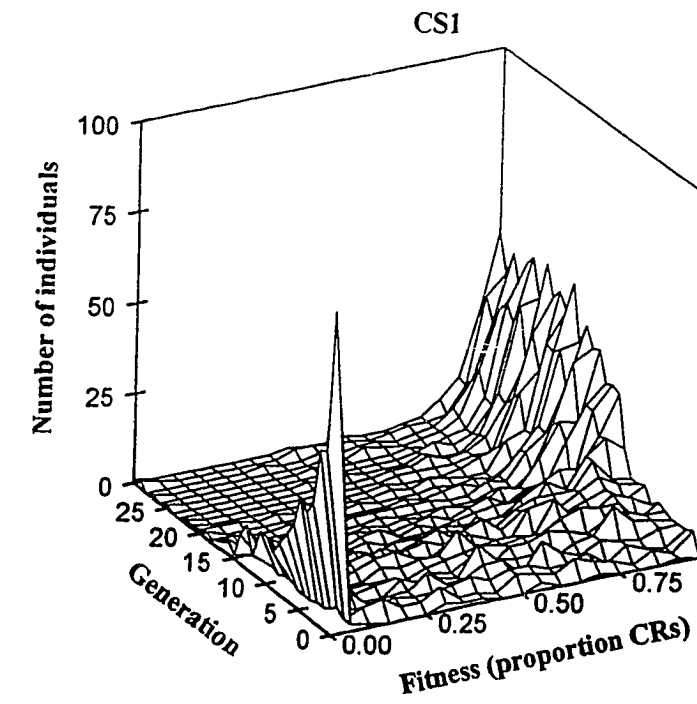


Figure 6.3: Frequency distributions of fitness values over generations, for CS1+.

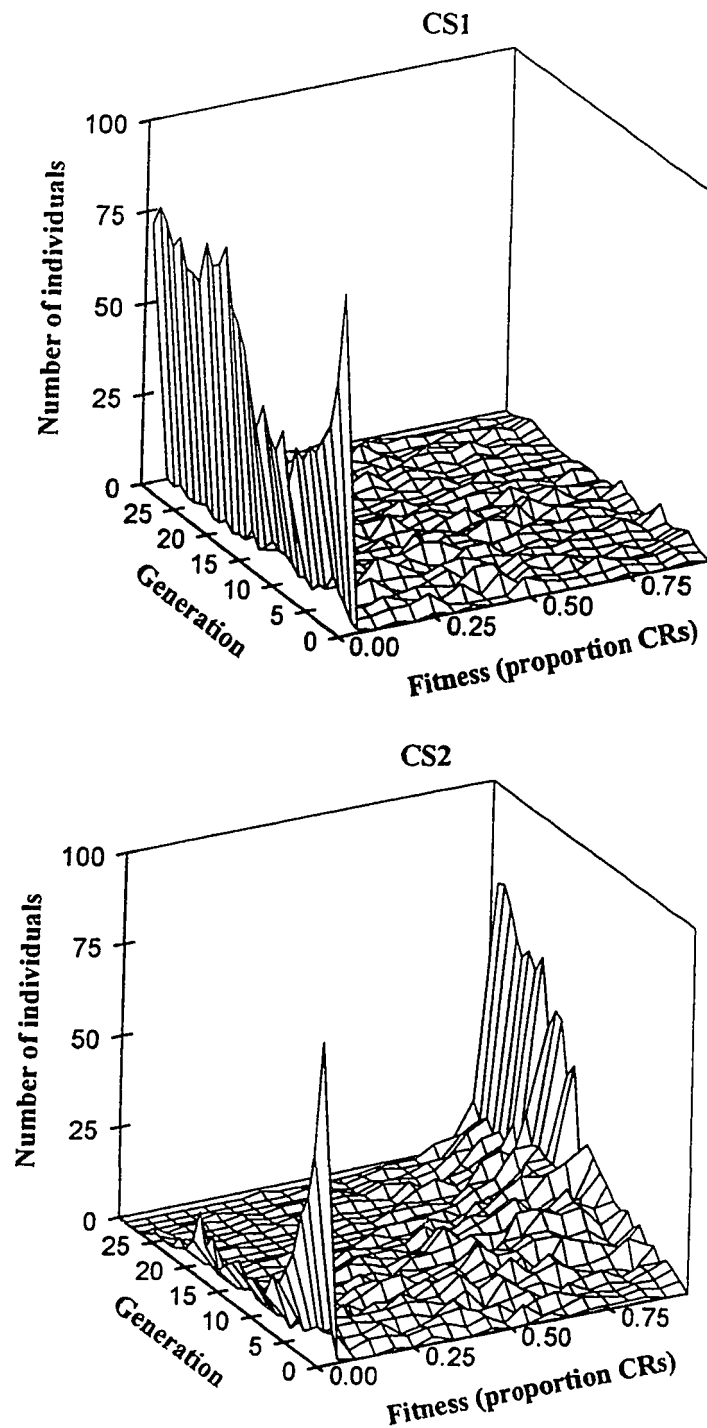
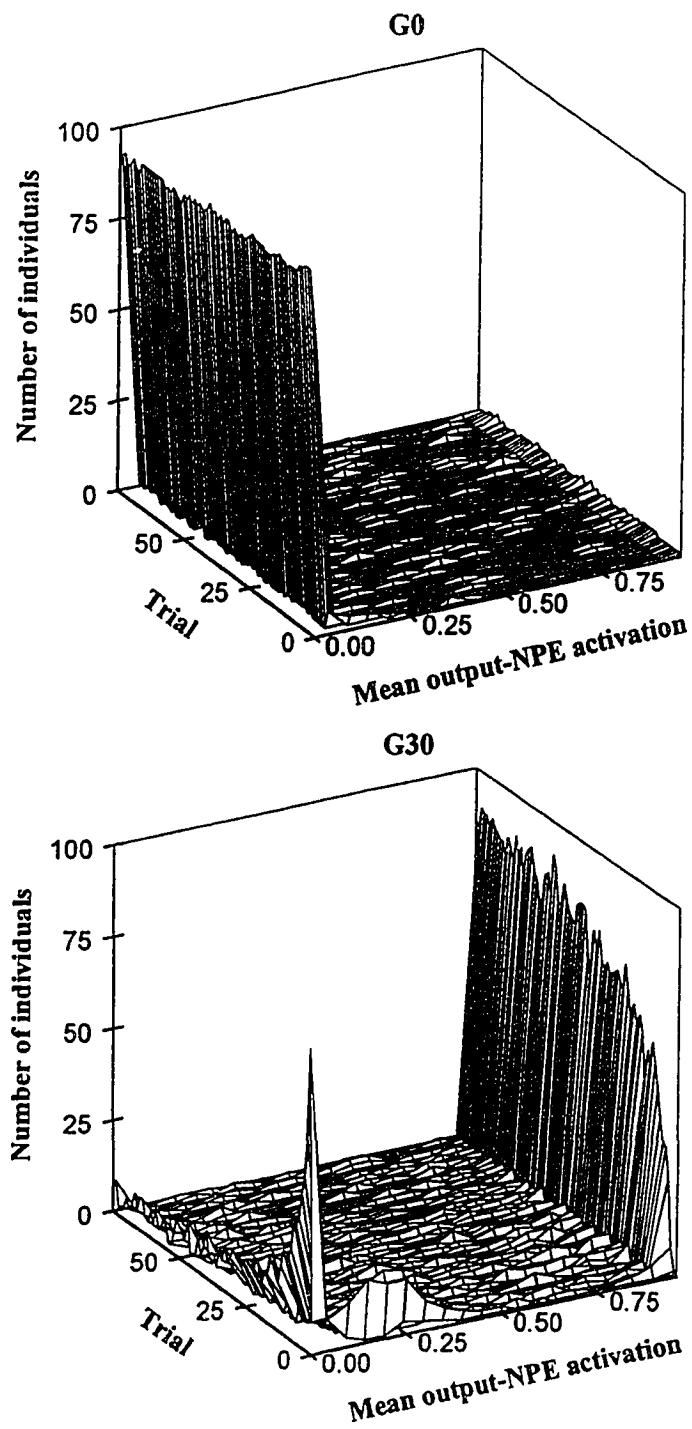
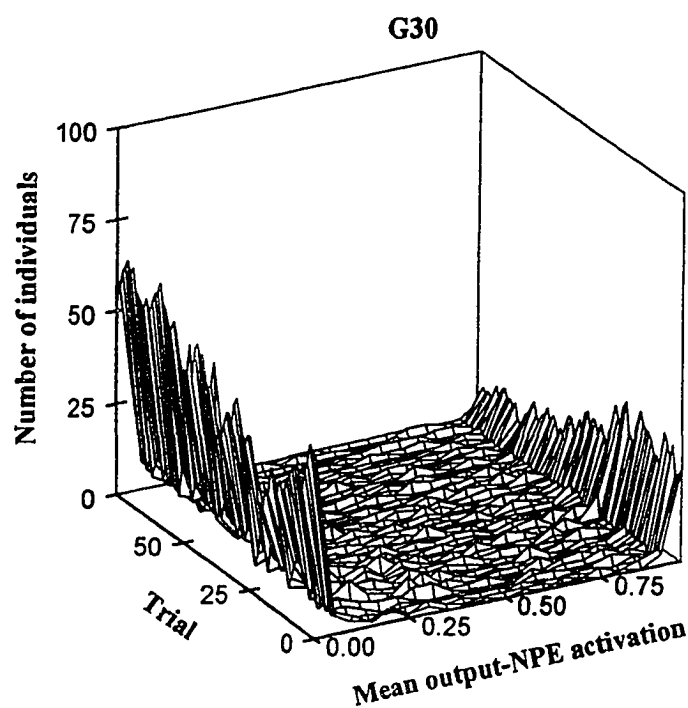
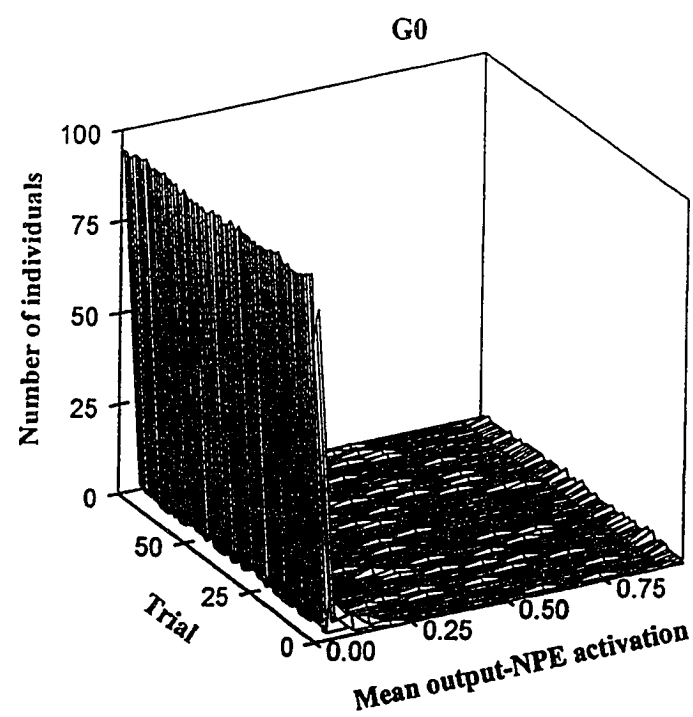


Figure 6.4: Frequency distributions of fitness values over generations, for CS2+.



**Figure 6.5: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of CS1+.**



**Figure 6.6: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of CS1+.**

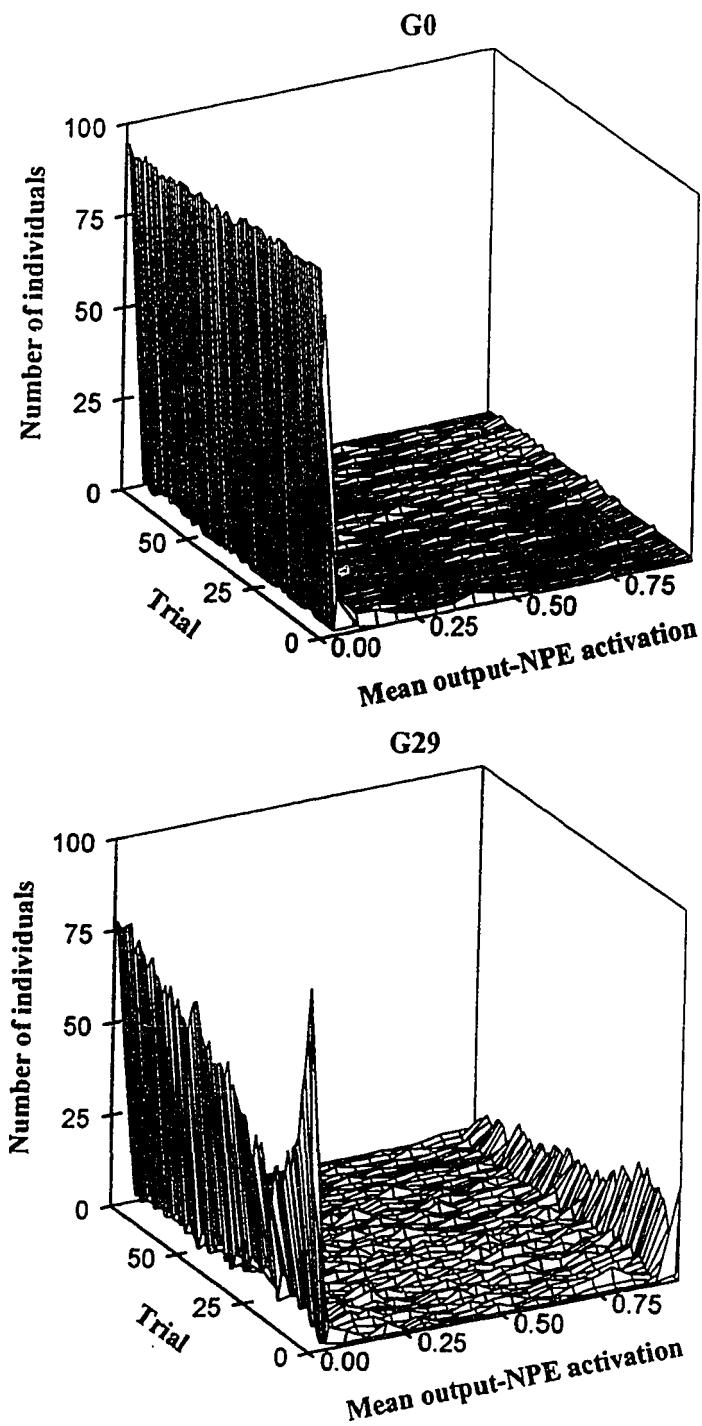
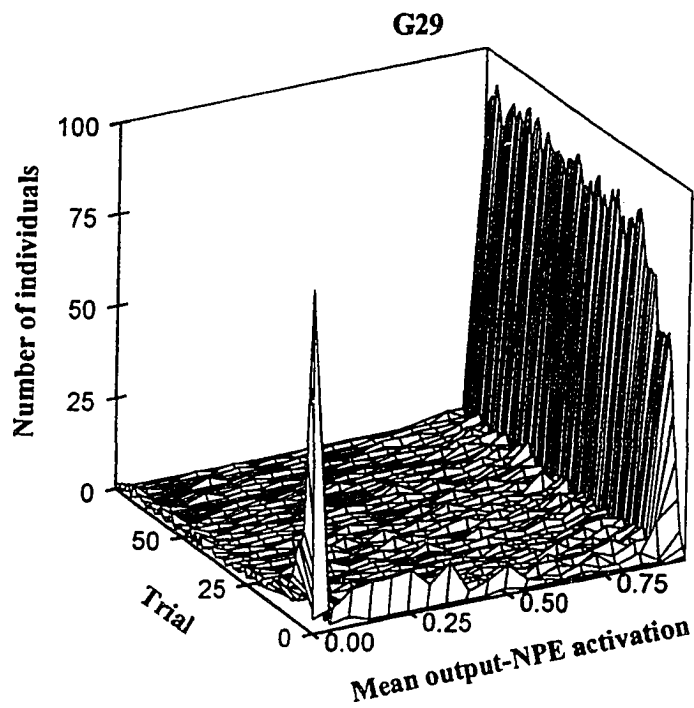
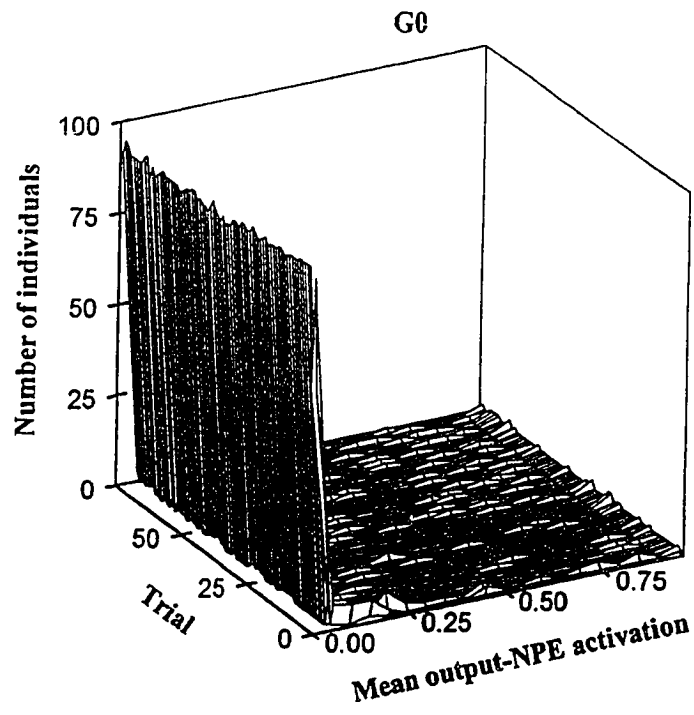
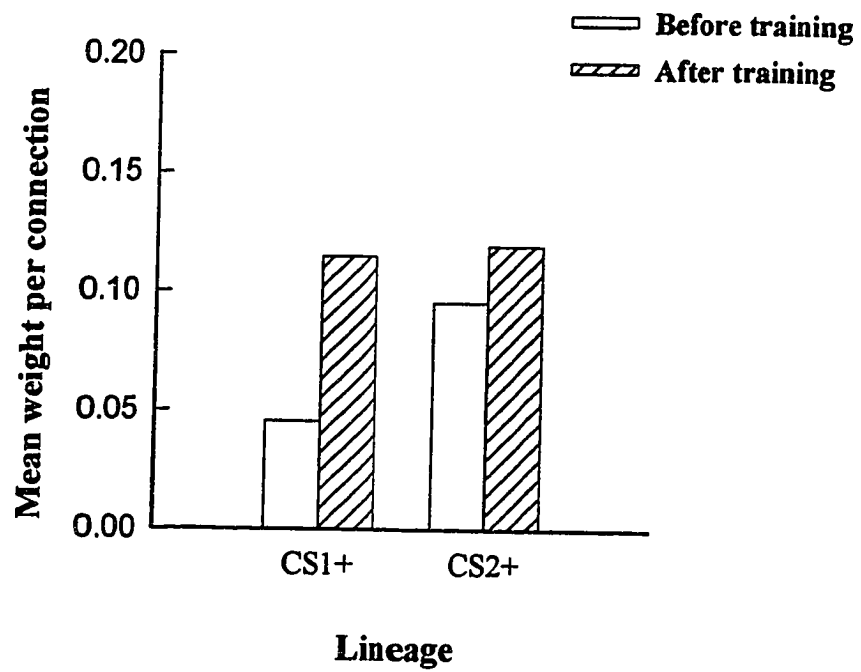


Figure 6.7: Frequency distributions of mean output-NPE activations over CS1 trials, for the first and last generations of CS2+.



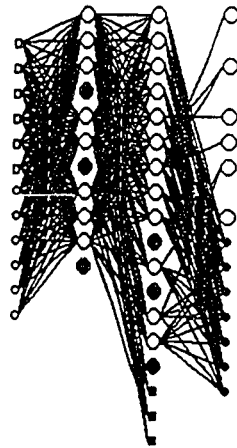


**Figure 6.8: Frequency distributions of mean output-NPE activations over CS2 trials, for the first and last generations of CS2+.**

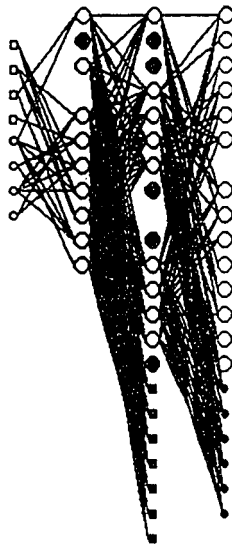


**Figure 6.9: Mean weight per connection in last generation of each lineage, before and after training.**

**CS1+**

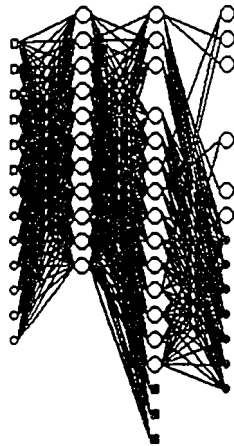


**CS2+**

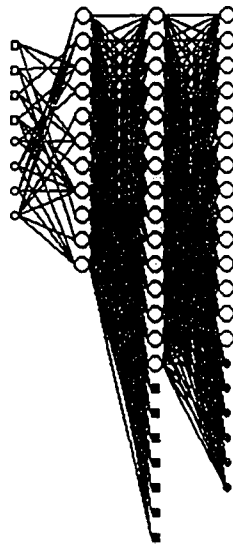


**Figure 6.10: Sample of high-fitness ANNs from the last generation of each lineage.** CR proportions of the CS1+ ANN were 1.00 for CS1 and 0.00 for CS2. CR proportions of the CS2+ ANN were 0.0 for CS1 and 1.00 for CS2.

**CS1+**

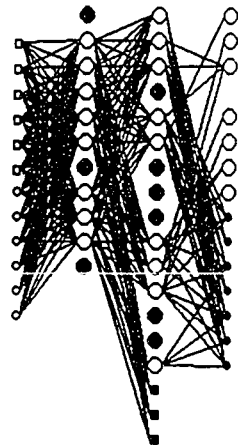


**CS2+**

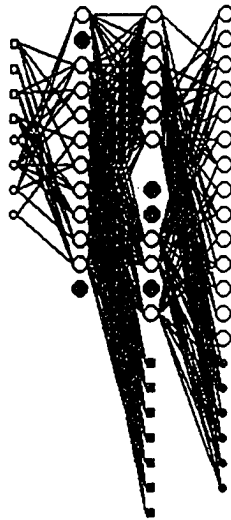


**Figure 6.11: Another sample of high-fitness ANNs from the last generation of each lineage. CR proportions of the CS1+ ANN were 1.00 for CS1 and 0.00 for CS2. CR proportions of the CS2+ ANN were 0.0 for CS1 and 1.00 for CS2.**

**CS1+**

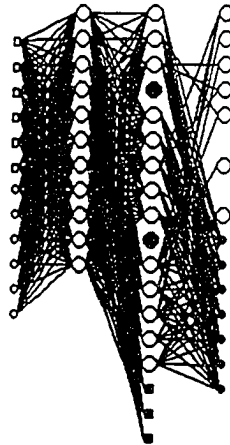


**CS2+**

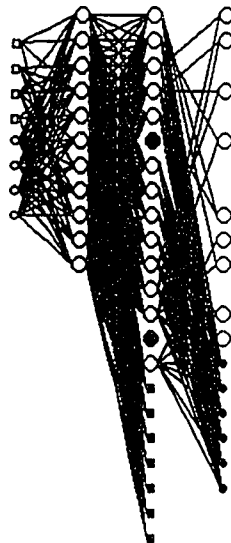


**Figure 6.12: Sample of low-fitness ANNs from the last generation of each lineage.** CR proportion of both ANNs was 0.0, for both CSs.

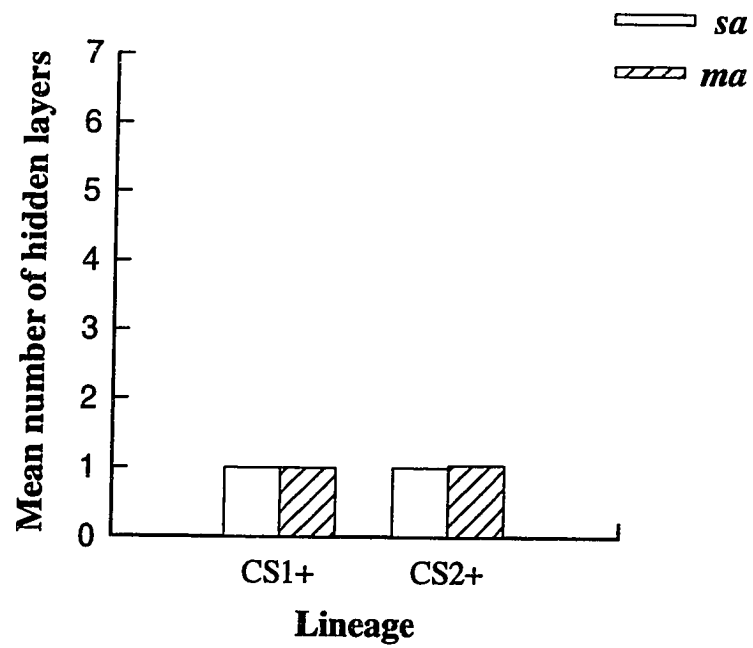
**CS1+**



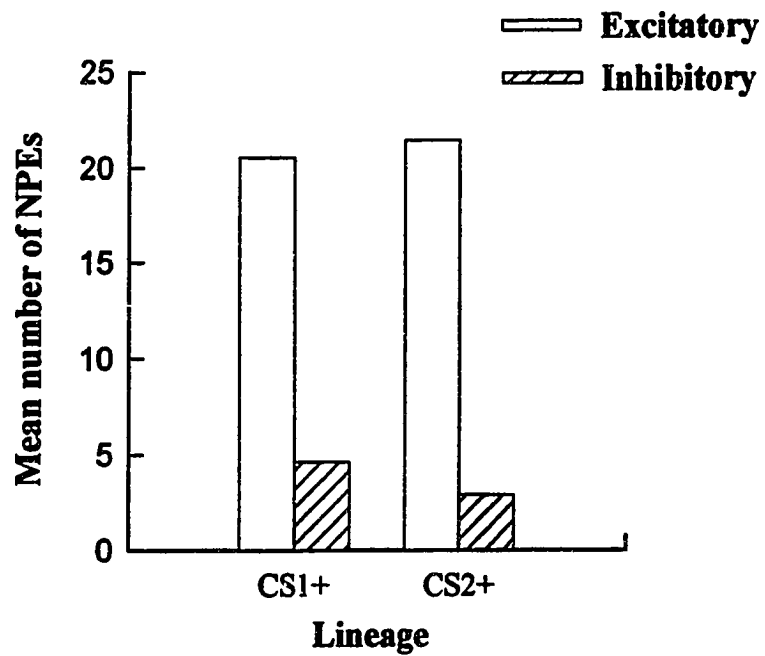
**CS2+**



**Figure 6.13: Sample of high-fitness nondiscriminating ANNs from the last generation of each lineage. CR proportions of both ANNs were above 0.75, for both CSs.**



**Figure 6.14: Mean number of hidden layers in last generation of each lineage.**



**Figure 6.15: Mean number of excitatory and inhibitory NPEs in last generation of each lineage.**



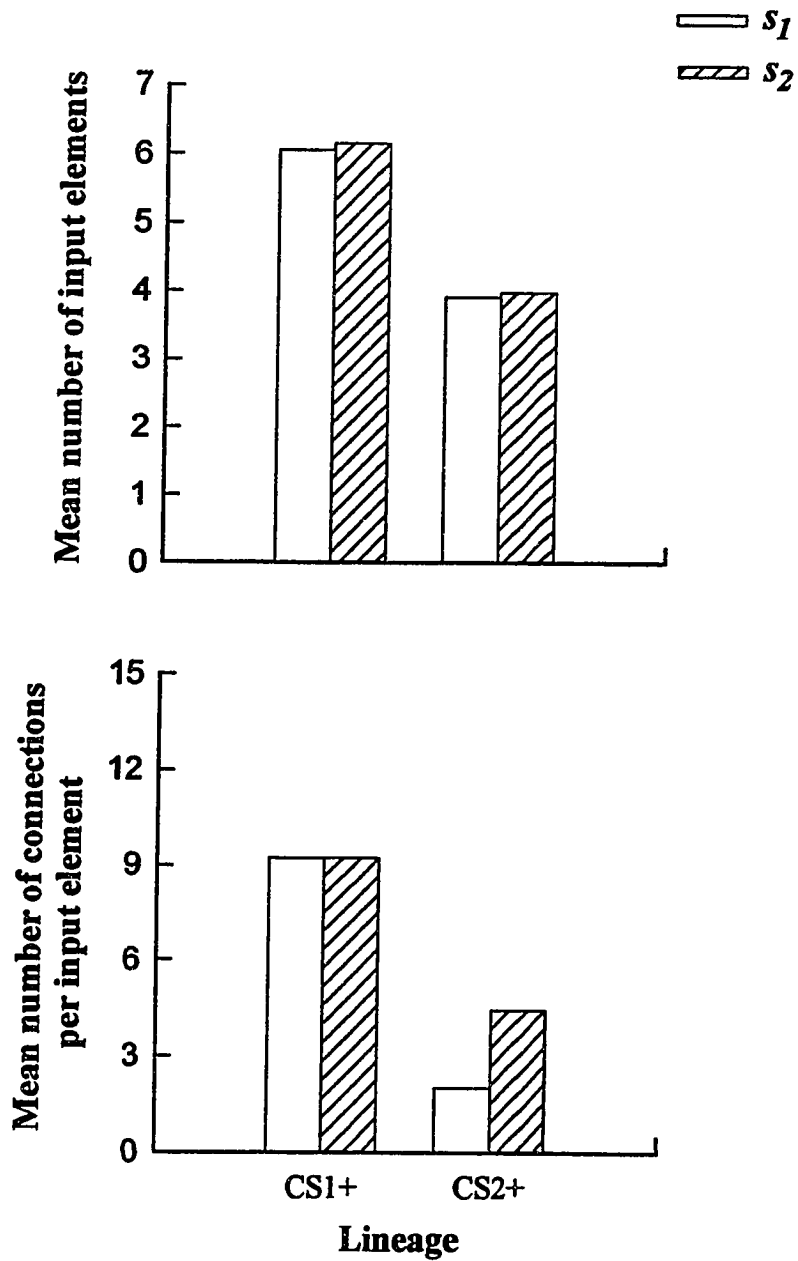
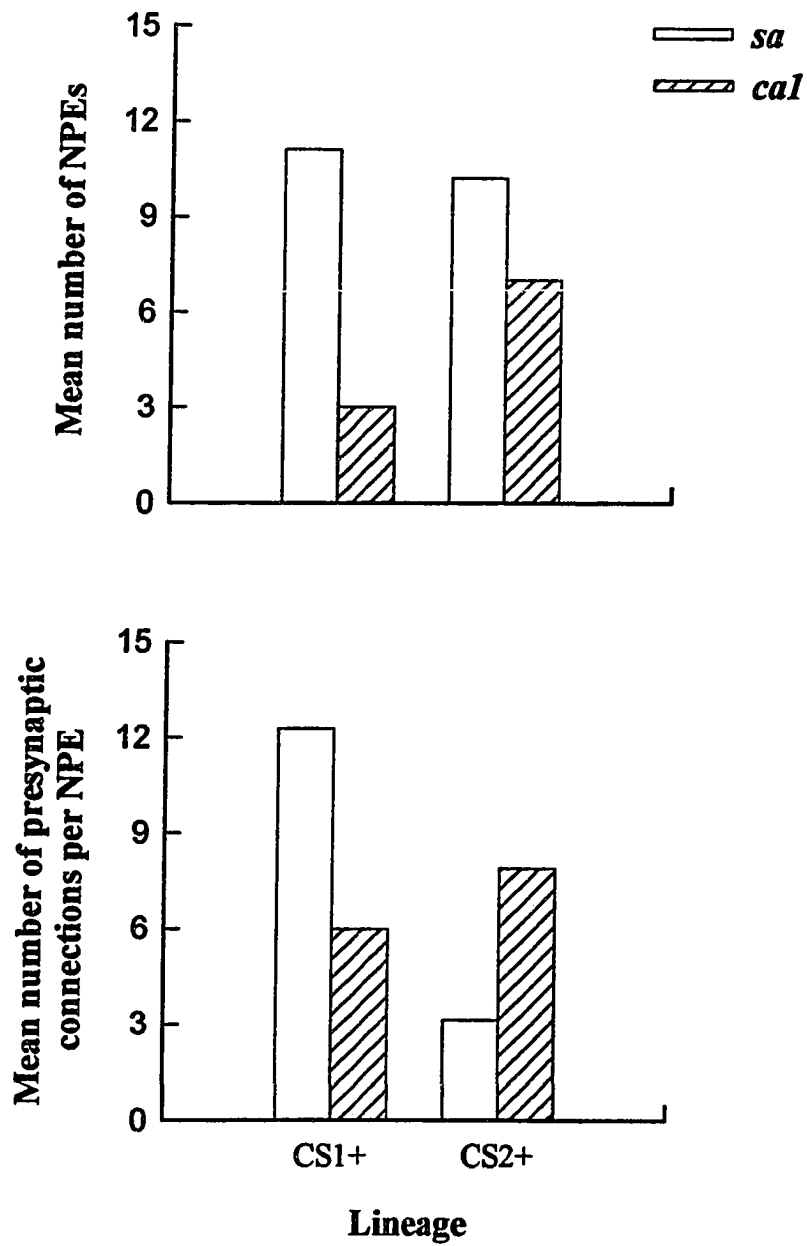


Figure 6.16: Mean number of input elements and connections per element in last generation of each lineage.



**Figure 6.17: Mean number of *sa* and *cal* NPEs and presynaptic connections per *sa* and *cal* NPE in last generation of each lineage.**

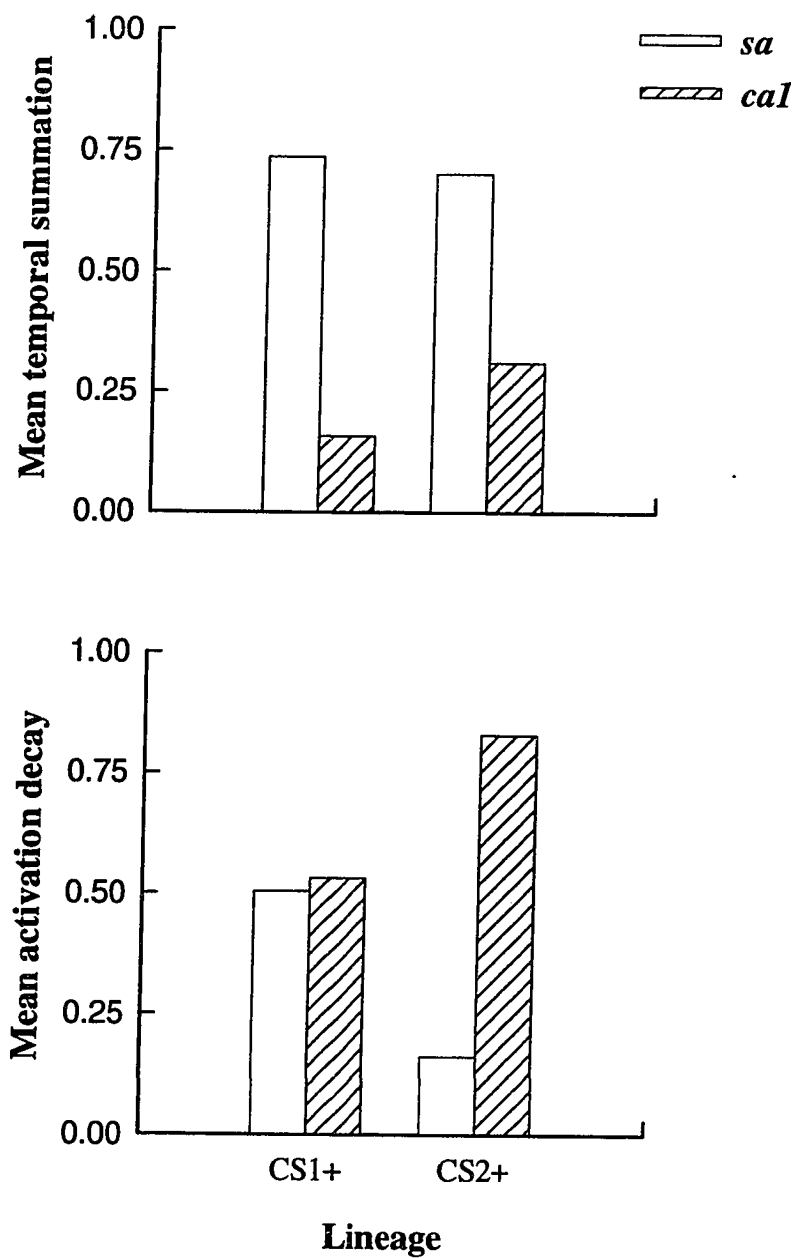
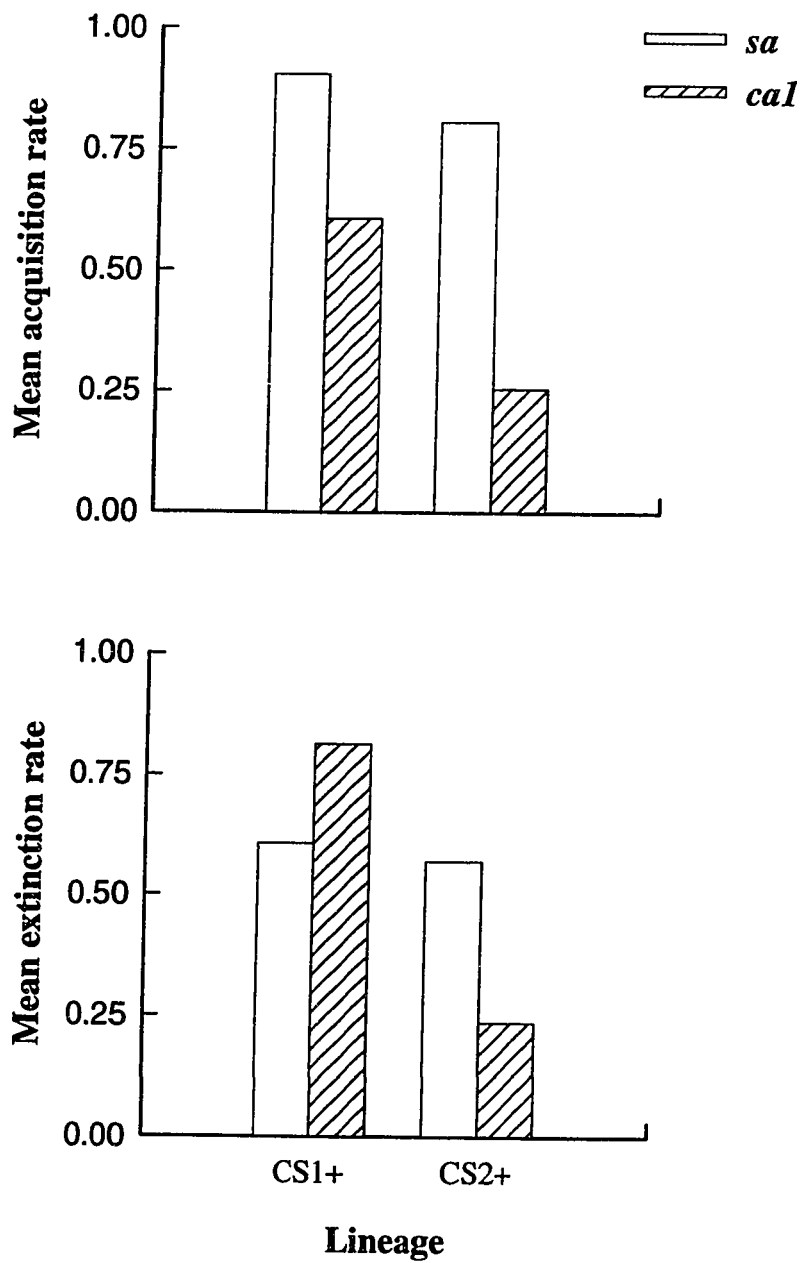


Figure 6.18: Mean activation parameters in *sa* and *cal* NPEs in last generation of each lineage.



**Figure 6.19: Mean synaptic-plasticity parameters in *sa* and *cal* NPEs in last generation of each lineage.**

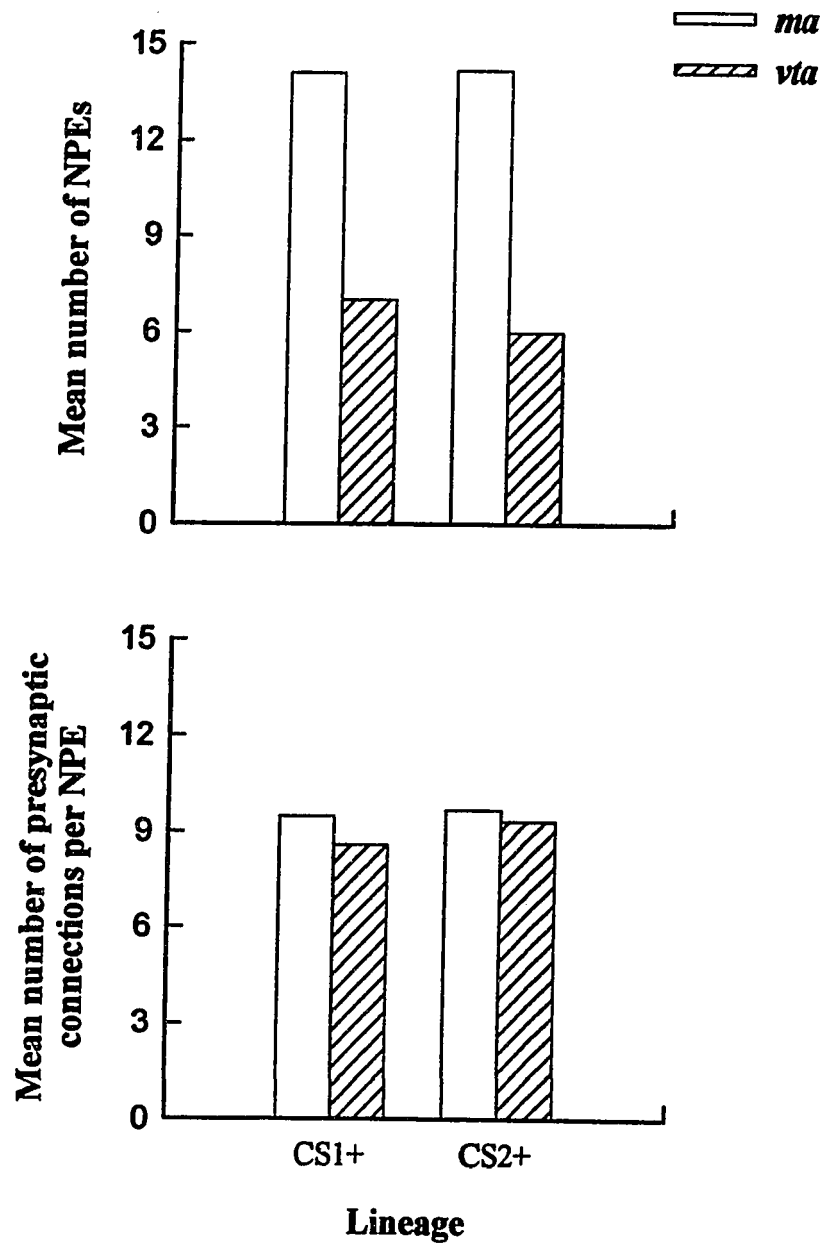


Figure 6.20: Mean number of *ma* and *vta* NPEs and presynaptic connections per *ma* and *vta* NPE in last generation of each lineage.

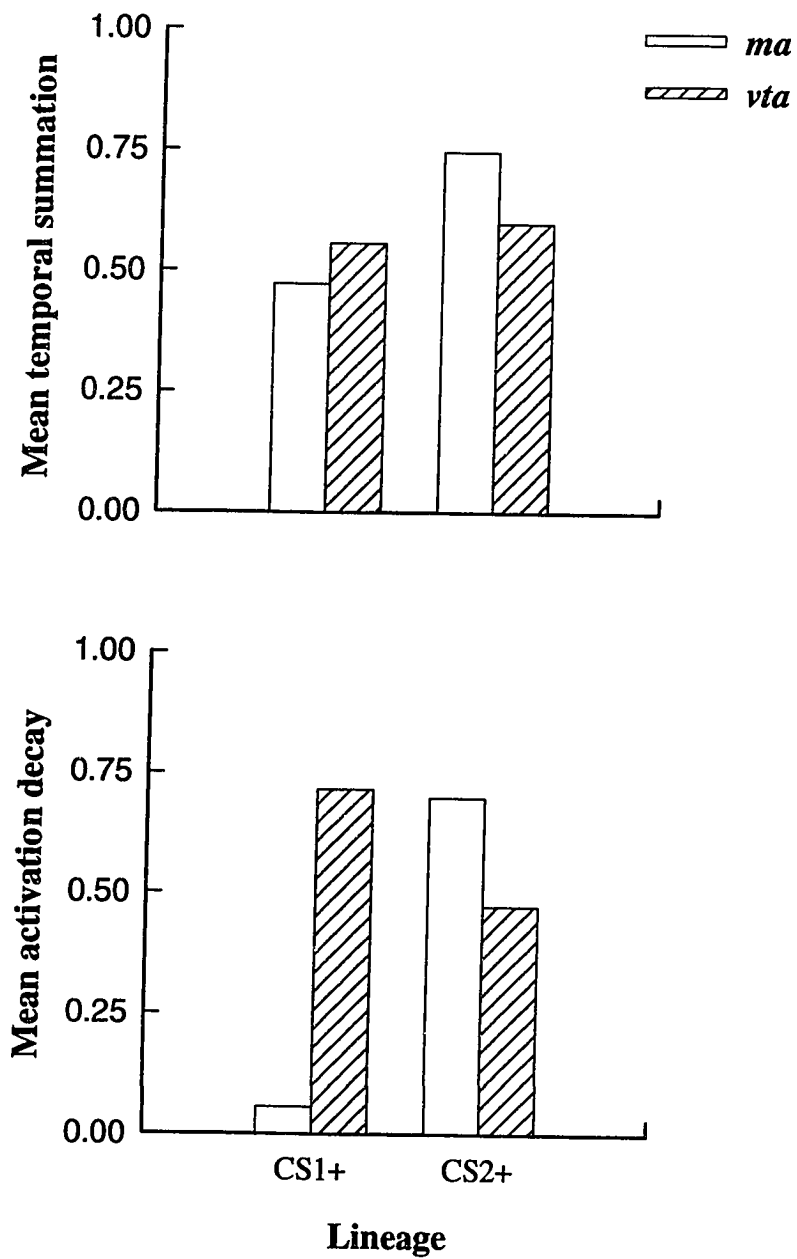


Figure 6.21: Mean activation parameters in *ma* and *vta* NPEs in last generation of each lineage.

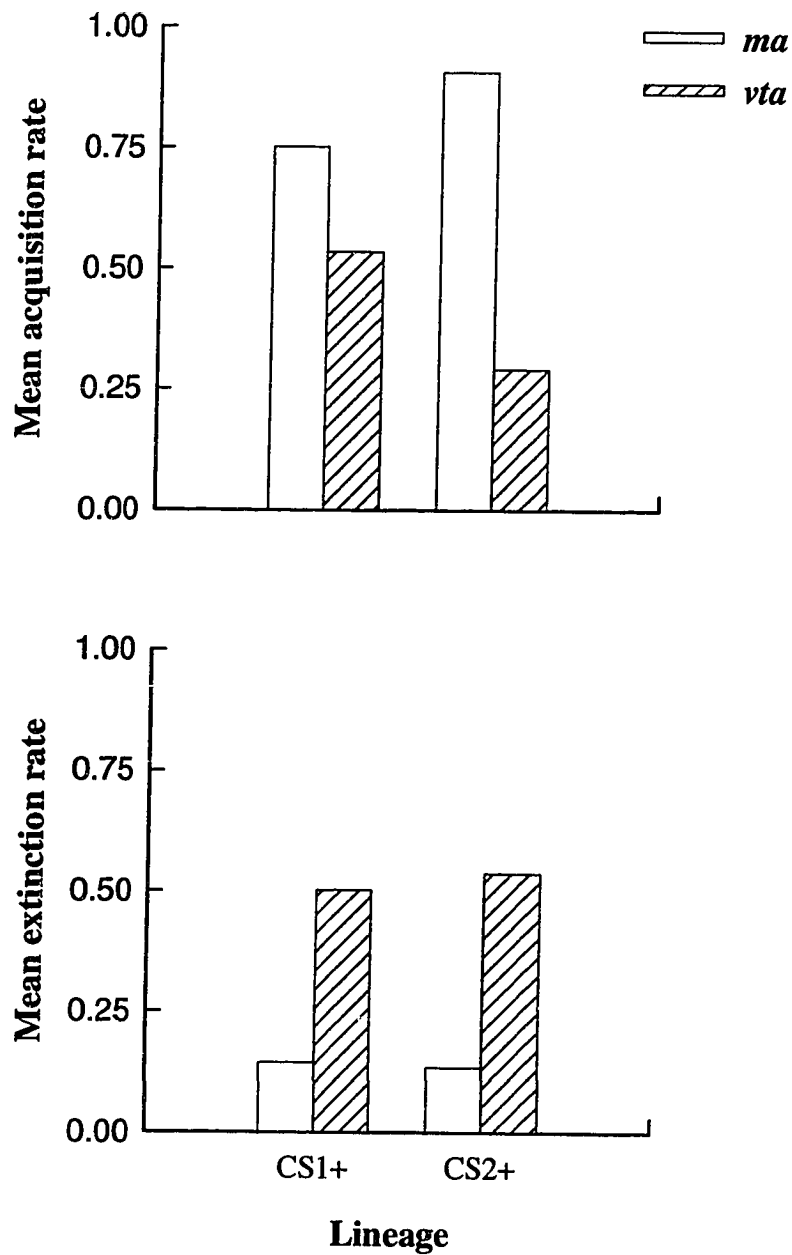
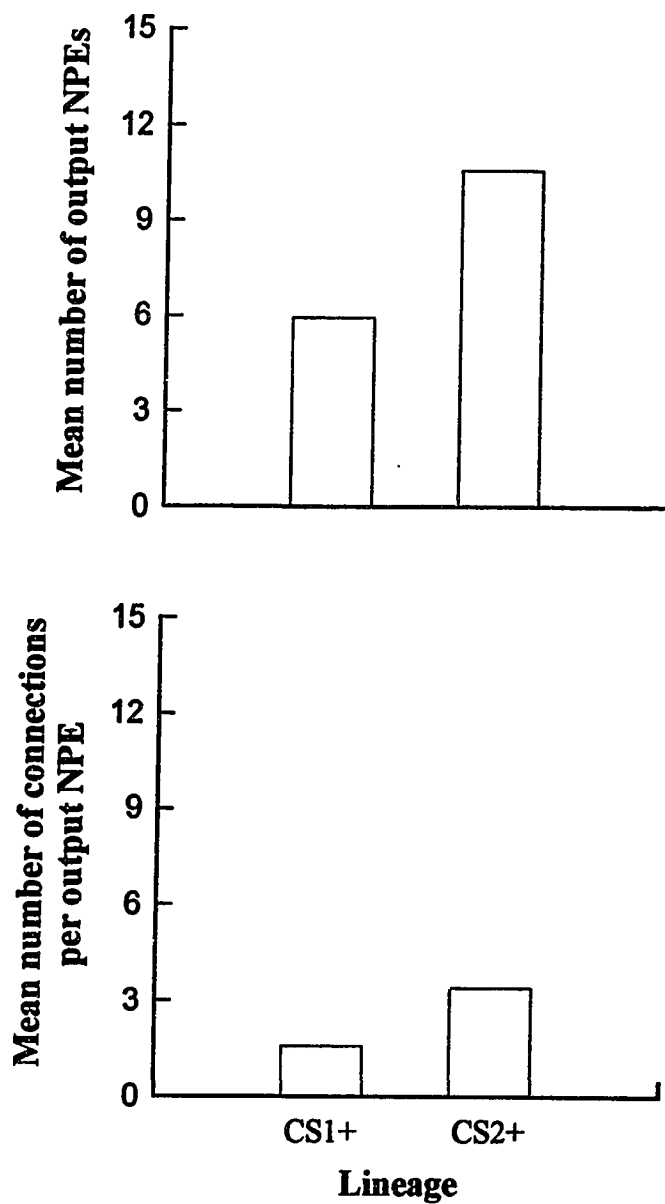
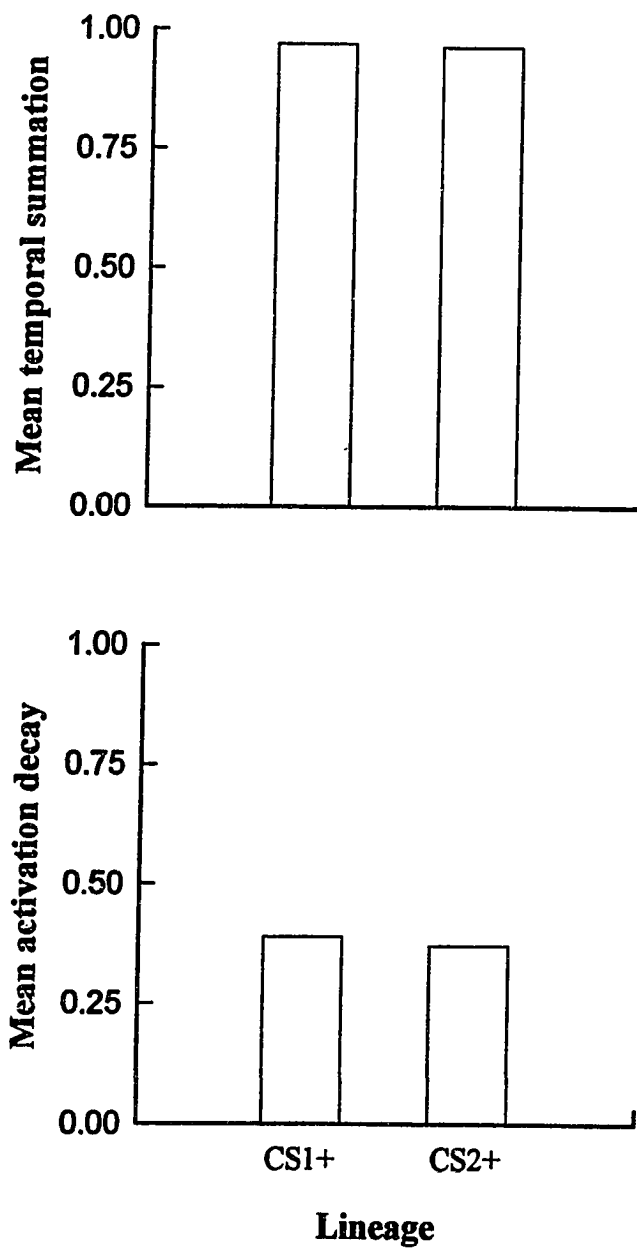


Figure 6.22: Mean synaptic-plasticity parameters in *ma* and *vta* NPEs in last generation of each lineage.

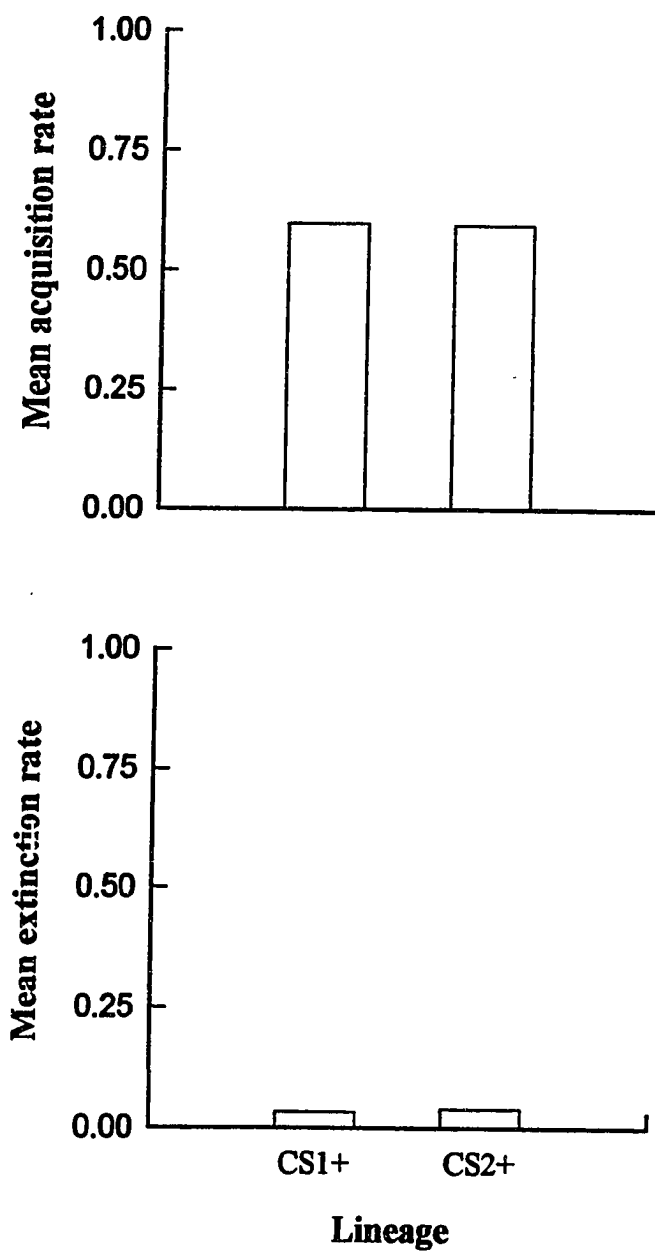


**Figure 6.23: Mean number of output NPEs and connections per NPE in last generation of each lineage.**

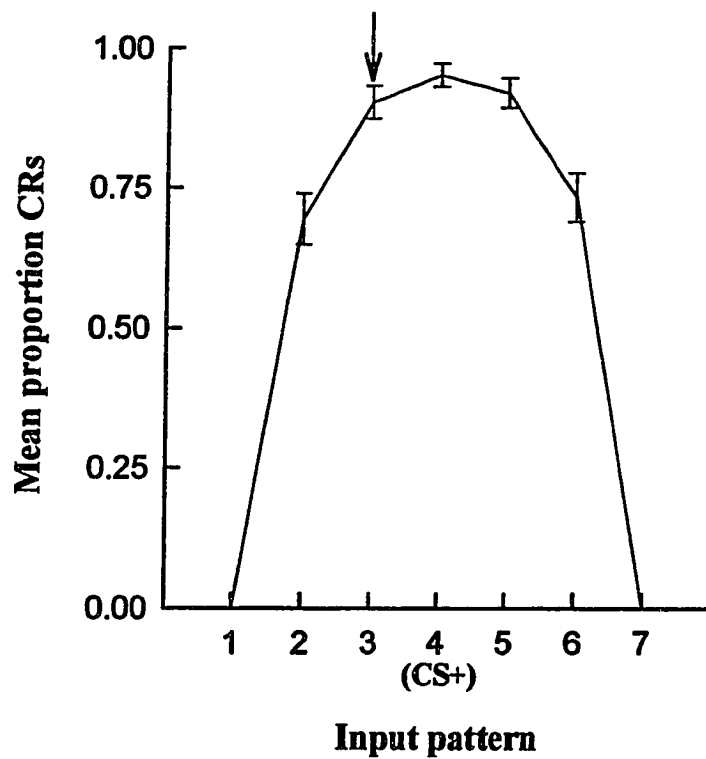




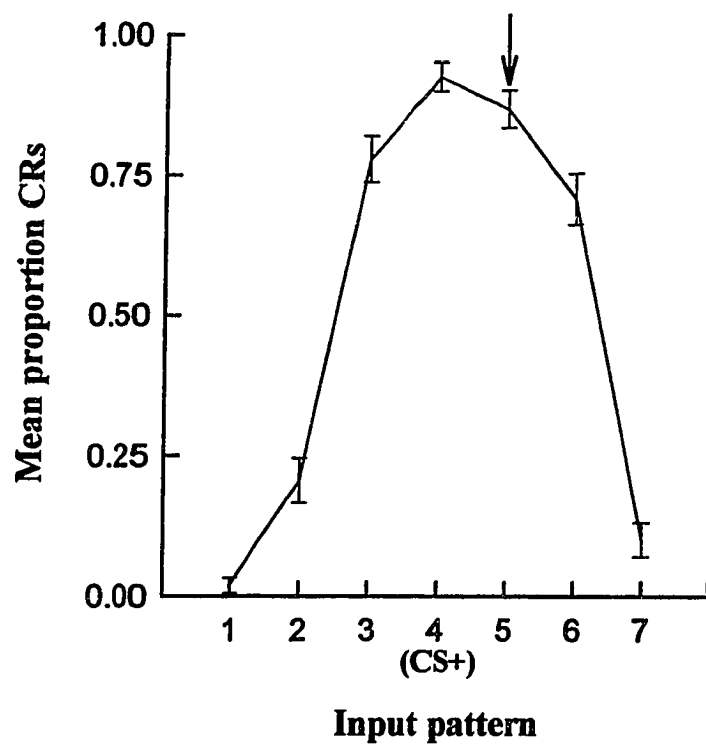
**Figure 6.24: Mean activation parameters of output NPEs in last generation of each lineage.**



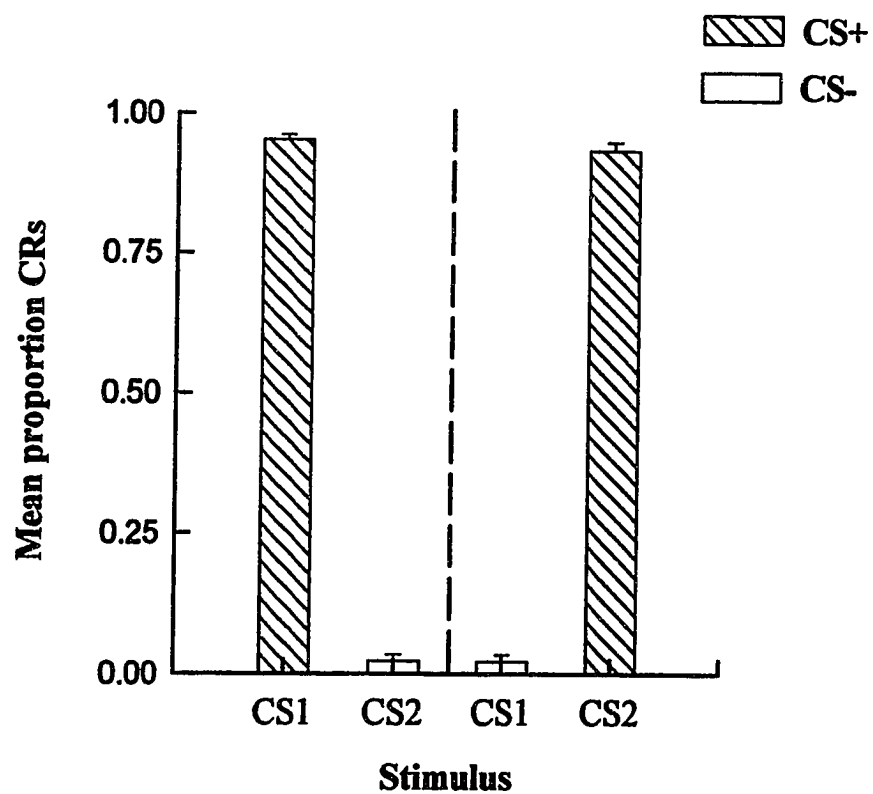
**Figure 6.25: Mean synaptic-efficacy parameters of output NPEs in last generation of each lineage.**



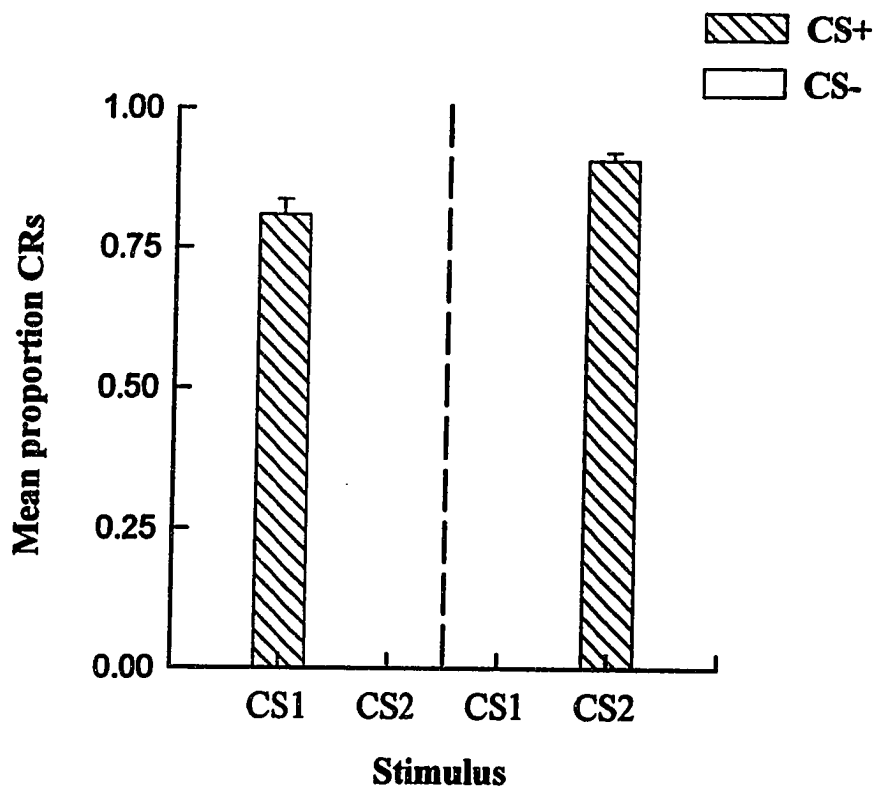
**Figure 6.26: Mean proportion CRs in generalization test of ANNs in last generation of CS1+. "CS+" was the pattern reinforced during training (see Figure 4.38 for patterns). The arrow indicates the pattern reinforced throughout phylogeny.**



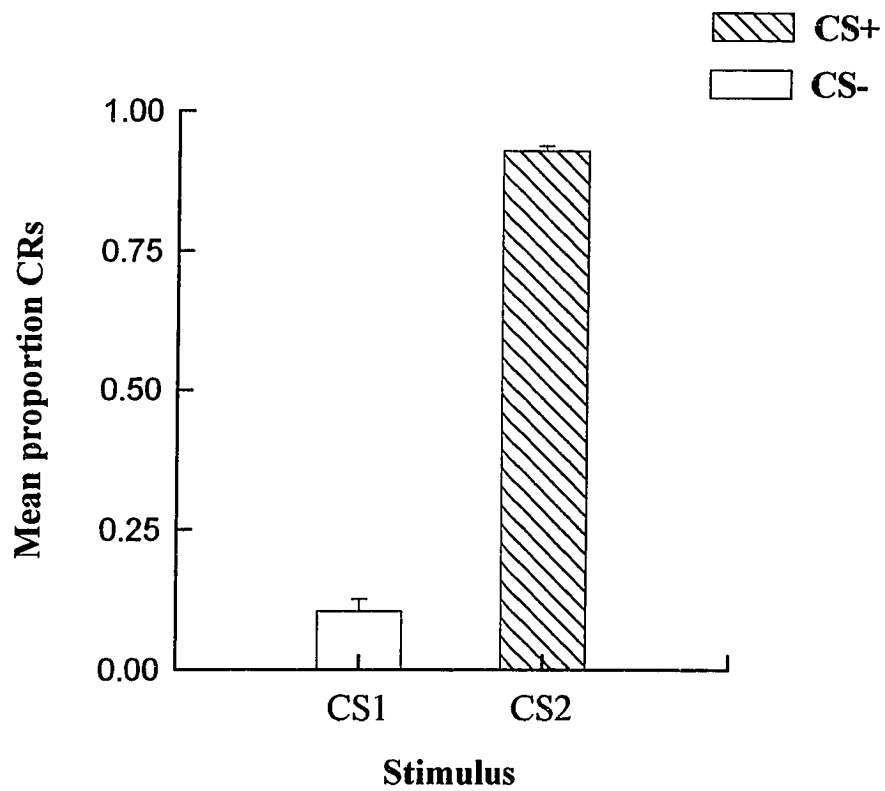
**Figure 6.27: Mean proportion CRs in generalization test of ANNs in last generation of CS2+. "CS+" was the pattern reinforced during training (see Figure 4.38 for patterns). The arrow indicates the pattern reinforced throughout phylogeny.**



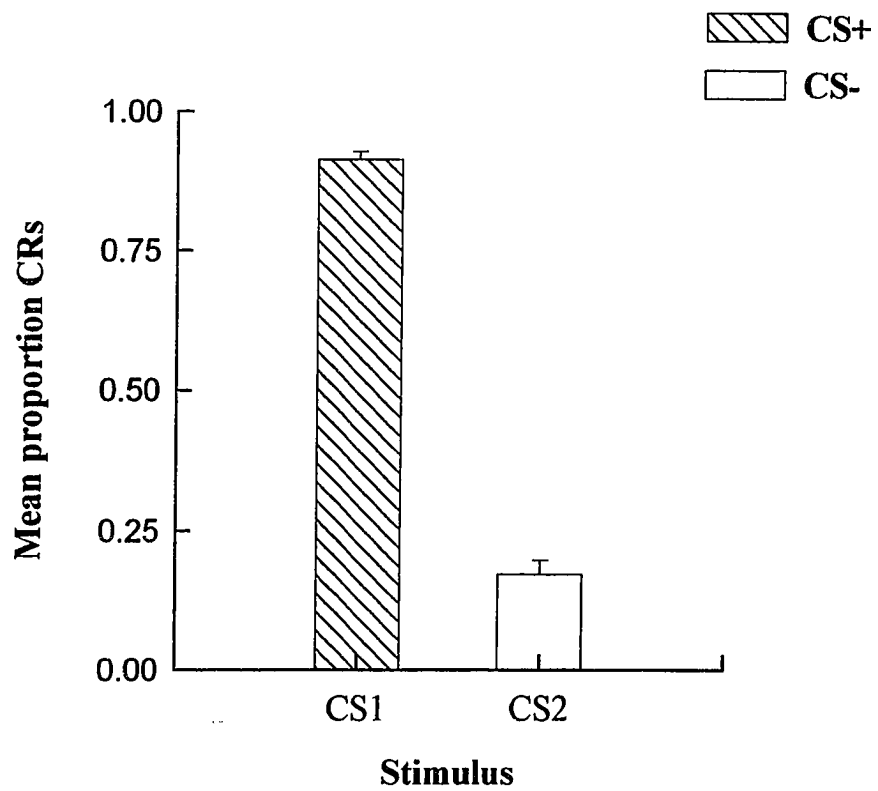
**Figure 6.28: Mean proportion CRs in orthogonal discrimination tests of ANNs in last generation of CS1+.**



**Figure 6.29: Mean proportion CRs in orthogonal discrimination tests of ANNs in last generation of CS2+.**

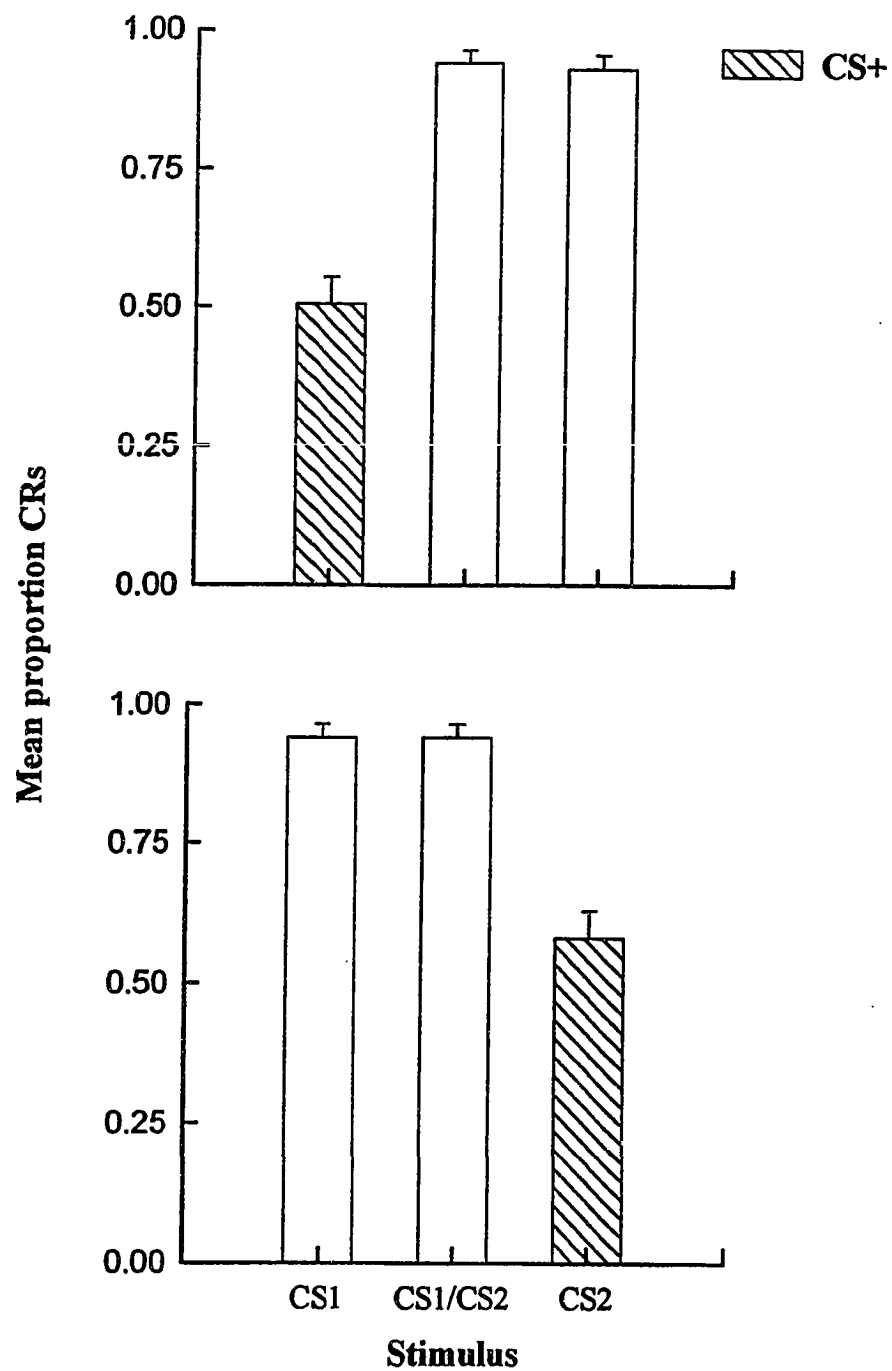


**Figure 6.30: Mean proportion CRs in nonorthogonal discrimination test of ANNs in last generation of CS1+.**

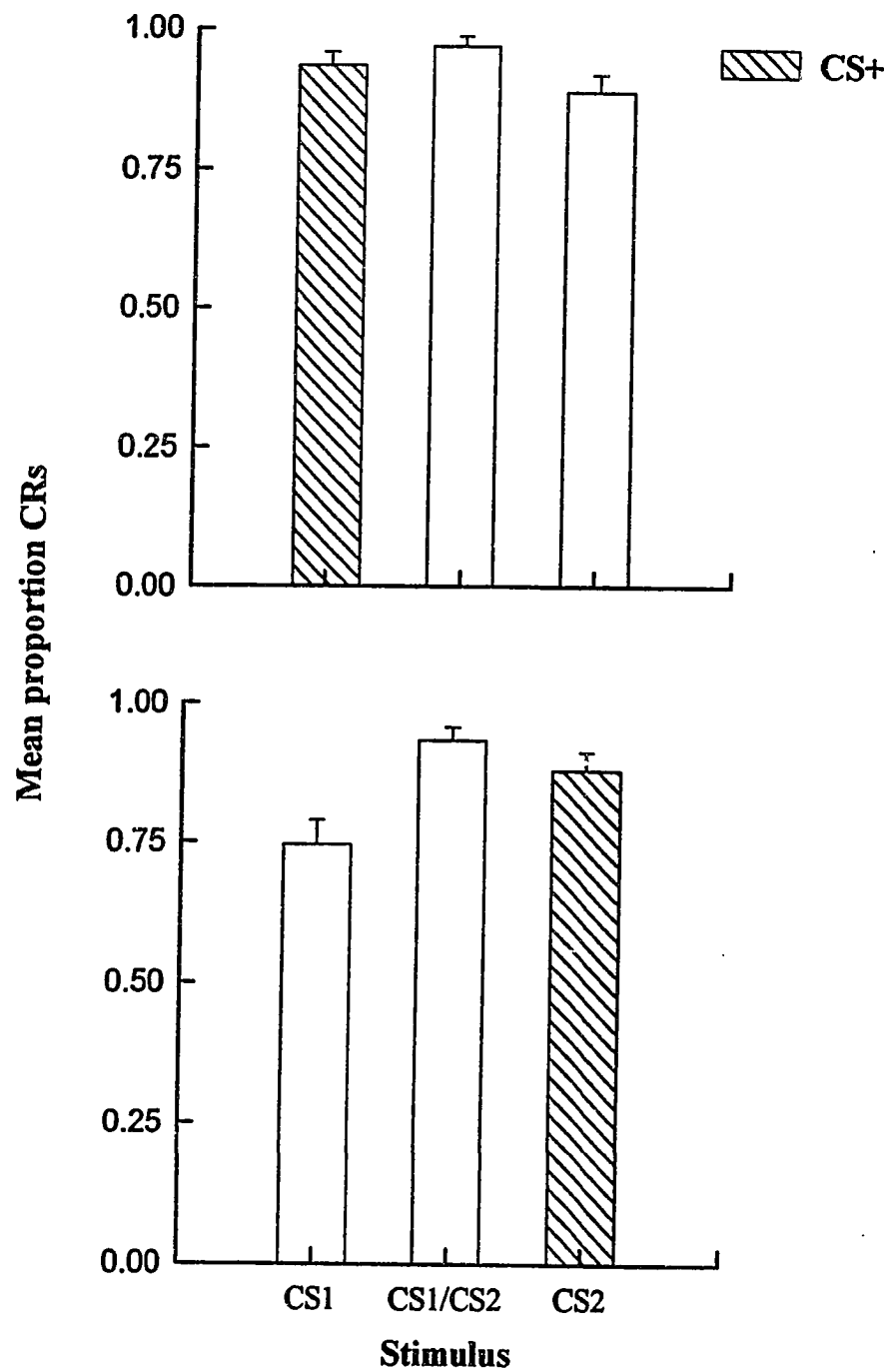


**Figure 6.31: Mean proportion CRs in nonorthogonal discrimination test of ANNs in last generation of CS2+.**





**Figure 6.32: Mean proportion CRs in blocking tests of ANNs in last generation of CS1+. "CS+" was the CS reinforced during Phase 1.**



**Figure 6.33: Mean proportion CRs in blocking tests of ANNs in last generation of CS2+. "CS+" was the CS reinforced during Phase 1.**

# CHAPTER 7

## GENERAL DISCUSSION

The main goal of the present work was to provide a preliminary and mainly empirical characterization of the competence and range of variation of an Artificial-Life, computational approach to the study of the evolution of learning and behavior. The approach was implemented through a model that combined neurocomputational and genetic-algorithm techniques. The model also included a neurodevelopmental program inspired by general concepts from developmental neurobiology. Also, environmental selective pressures were defined in terms of Pavlovian procedures. The model was implemented through computer simulations of the effects of selection for increased responding under a number of environmental conditions.

At the most general level, the present work demonstrated the possibilities of a neurocomputational / genetic-algorithm hybrid model that included a developmental program involving a one-many relation between genotypes and structural phenotypes. The results showed that the model was capable of generating a variety of structural phenotypes from founder populations consisting of *fixed-length* chromosomes that encoded for *the same set of developmental variables*. Also, the same kinds of structural phenotypes were capable of mediating a wide number of functional phenotypes that were consistent with Pavlovian-conditioning phenomena such as ISI functions, optimal-ISI noninvariance, CS nonequipotentiality, generalization, discrimination, and blocking.

Most of these phenomena were *emergent* in that they were observed under environmental conditions that were different from the ancestral ones. Thus, selection for a particular functional phenotype resulted in ANNs that showed that phenotype as well as other functional phenotypes that were not explicitly selected for throughout phylogeny. For example, ANNs whose ancestors were selected for increased responding to two reinforced CSs showed this functional phenotype. However, the *same* ANNs showed other functional phenotypes such as generalization, orthogonal discrimination, and blocking, which were not explicitly selected for throughout phylogeny.

The present work also demonstrated the competence of a single NC model to change NPE activations and connection weights in a variety of ANNs architectures. In fact, different Pavlovian-conditioning phenomena were shown to be emergent properties of structural differences among ANNs. For example, structurally different ANNs performed in ways consistent with ISI functions that differed in the optimal ISI.

ANNs were structurally different in their sizes, connectivity, and NPE parameters. In many cases, functional phenotypes were mediated by different kinds of trade-offs between structural traits. Such trade-offs were observed across structurally different ANNs, as well as across different kinds of NPEs within the same structural phenotype. For example, selection for orthogonal discrimination generated ANNs that tended to have far more *ma* NPEs than ANNs generated by selection for increased responding to two reinforced CSs. Also, the former ANNs had far more *ma* than *sa* NPEs, but more connections per *sa* NPE than per *ma* NPE. In some of these ANNs, *sa* NPEs tended to have low temporal summations, high activation decays, high acquisition rates, and low extinction rates. In contrast, *ma* NPEs in the same ANNs tended to have greater

temporal summations and low activation decays, but far greater acquisition rates than extinction rates.

Structurally different ANNs were functionally equivalent when observed under the kinds of environmental conditions that were present during ancestral selection. This finding is consistent with the thesis of *fitness supervenience* (Rosenberg, 1978, 1985), according to which an asymmetrical relation exists between functional and structural phenotypes. That is, although structurally similar individuals must be functionally similar, functionally similar individuals are not necessarily structurally similar. This relation was illustrated in the present work, for only functionally similar ANNs that were members of the last generation of a *particular lineage* were structurally similar. Functionally similar ANN across different lineages were structurally quite different.

The following three sections present a more detailed discussion of some of these results. Specifically, the discussion concentrates on the effects of the ISI on ANN size in Experiments 1 and 2, the effects of the CS on the number of input elements in Experiment 1, and the effects of reinforcement contingency on the number of *sa* and *ma* NPEs in Experiments 3 and 4. Throughout the discussion, I shall offer conjectures that might account for some of these effects. The last section presents a number of general conceptual issues that bear on the interpretation and relevance of the results.

## 7.1 Effects of the ISI

The ISI was an independent variable in the evolution simulations of Experiments 1 and 2, and in the test simulations of Experiment 1. In the evolution simulations, the most salient systematic effect of the ISI was on the size of ANNs, as measured by the number of NPEs and layers. In the test simulations, the ISI affected responding in

ways similar to those observed in ISI functions, whose peak coincided with the ISI under which ancestral selection took place. Let me discuss each kind of effect in turn, focusing on *sa* NPEs, for most of the systematic ISI effects were observed there.

## 7.1.1 Evolution Simulations

### 7.1.1.1 Effects on ANN Sizes

In Experiments 1 and 2, ANNs generated by selection under longer ISIs tended to have far more *sa* NPEs and layers than ANNs generated by selection under shorter ISIs, although some nonmonotonic relations were observed. Indeed, in both experiments, there was a consistent increase in the number of *sa* NPEs and layers as the ISI increased from either 2 to 8 ts (Experiment 1) or 4 to 8 ts (Experiment 2). However, the number of *sa* NPEs tended to decrease for ISI 16. In contrast, the number of *ma* NPEs remained roughly constant across ISIs, except for the ISI-16 ANNs in Experiment 1, which tended to have more *ma* NPEs.

In general, and ignoring the nonmonotonic relations, an increase in ANNs as a function of the ISI seems to be roughly consistent with the overall size differences found across neural circuits that mediate Pavlovian conditioning in different preparations. Although the present ANNs are too abstract to be related in a one-one fashion to specific neural circuits, it is possible to formulate some general conjectures. For example, the circuitry that mediates acquisition in the NMR preparation involves basically the brain stem and the cerebellum (e.g., Moore, Desmond, & Berthier, 1982; Moore & Blazis, 1989; Rosenfield & Moore, 1985; Steinmetz, Logan, & Thompson, 1988; Yeo, Hardiman, & Glickstein, 1984), with little contribution from the neocortex (Moore *et al.*, 1980) or the hippocampus (Berry & Swain, 1989; Moore & Solomon,

1984). In contrast, the circuitry that mediates acquisition in conditioned suppression involves the auditory thalamus, primary auditory cortex, auditory association cortex, sensory association cortex, hippocampal formation, and amygdala (e.g., Davis, 1989; LeDoux, 1986, 1993). On this basis, it is reasonable to believe that the circuitry that mediates acquisition in NMR is considerably smaller than the circuit that mediates acquisition in conditioned suppression. And, because both preparations require different optimal ISIs, the present results suggest that the corresponding neural circuits are the end-products of selection for increased responding under different ISIs.

### **7.1.1.2 Effects on Connectivity**

Regarding connectivity, different ISI effects were observed in Experiments 1 and 2. In Experiment 1, the number of presynaptic connections per *sa* NPE tended to increase proportionally to the number of *sa* NPEs, so that the mean number of presynaptic connections per *sa* NPE tended to remain relatively constant across ISIs. In Experiment 2, the mean number of connections per *sa* NPE decreased consistently as the ISI increased. Thus, selection for increased responding to one reinforced CS (Experiment 1) versus selection for increased responding to two reinforced CSs (Experiment 2) exerted different pressures upon convergent connectivity in the sensory subnetwork. In Experiment 1, selection pressures arising from the ISI biased architectural phenotypes toward more *sa* NPEs *and* more connections per *sa* NPE. In Experiment 2, such pressures biased architectural phenotypes toward more *sa* NPEs *but fewer* connections per *sa* NPE.

These differences may be interpreted in terms of synaptic competition. Indeed, as the number of presynaptic elements connected to an NPE increases, the competition

among the former for the amount of connection weight available at the latter also increases. Such an increase, in turn, reduces the possibility that an ANN will be able to respond to two independently reinforced CSs. By reducing the number of presynaptic connections per NPE, the selection process simulated in Experiment 2 also reduced the amount of competition among presynaptic elements, thus increasing the possibility that ANNs would learn to respond to two reinforced CSs.

The effects observed in both experiments for ISI 16 can be interpreted in terms of a trade-off between architectural and parametric traits. Indeed, as the ISI increased from 2 to 8 ts, selection biased architectures toward more *sa* NPEs. However, the present NC-GA model imposed a constraint upon the maximum number of *sa* NPEs, which may have shifted the evolutionary search toward the parametric space. Perhaps, many more NPEs may have been needed for ANNs to achieve comparable levels of adaptedness. Because larger ANNs were not available to the NC-GA model, the evolutionary search shifted towards the parametric space. The NC-GA model thus was capable of finding a configuration of NC parameters that allowed smaller ANNs to learn under an ISI of 16 ts.

## 7.1.2 ISI Test Simulations

In the test simulations of Experiment 1, the ISI was used as an independent variable to determine how ANNs that evolved under a particular ISI would perform under other ISIs. The results showed that, overall, 2, 4, 8, and 16 ts represented different optimal ISIs for ANNs in the corresponding lineages. That is, ANNs whose ancestors were selected for increased responding under a particular ISI performed according to inverted U-shaped functions whose peak coincided with that ISI. In this way, the



present ANNs simulated ISI functions and the noninvariance of optimal ISIs as learning phenomena that emerged from evolutionary processes. In this sense, the present results are consistent with the idea of biological constraints on learning.

Indeed, in the present simulations, ISI functions can be interpreted as arising from architectural constraints, for smaller ANNs tended to perform worse under longer ISIs, whereas larger ANNs tended to perform worse under short ISIs, at least for ISIs 2, 4, and 8. That is, structural phenotypes that resulted from the evolution simulations, constrained learning under environmental conditions that differed from the ancestral ones. In the language of biological constraints, we can say that, in general, small ANNs (whose ancestors were selected for increased responding under short ISIs) were *prepared* to learn under short ISIs and *contraprepared* to learn under long ISIs. Conversely, large ANNs (whose ancestors were selected for increased responding under long ISIs) were prepared to learn under long ISIs and contraprepared to learn under short ISIs. These relationships between ANN size and ISIs provide us with a basis for considering architectural traits as one aspect of the notion of preparedness. According to this interpretation, then, being prepared to learn under a particular ISI may signify possessing a circuitry whose architecture is such that it *permits* learning under that ISI.

The decrease in ANN size for ISI 16 suggests that complex interactions between architectural and parametric traits may have contributed to the effects observed here. An analysis of such interactions is beyond the scope of the present work. However, it is clear that the present GA was able to exploit architectural variation, and generate structurally different kinds of ANNs, that showed differential responding consistent with ISI functions and optimal-ISI noninvariance. Also, it is clear that a systematic

relationship existed between ANN size and these two phenomena. The origin and nature of this relationship may provide clues on the neuroanatomical substrates of these phenomena in real organisms, and a starting point for models of Pavlovian conditioning that take into account biological constraints and evolutionary interpretations.

Standard interpretations of the optimal-ISI noninvariances have focused upon differences in *temporal* aspects of neural processing (e.g., latency and conduction speed) across species (e.g., Noble & Adams, 1963) and response systems (e.g., Hilgard & Marquis, 1940; Jones, 1962; Ross, 1961). The present simulations suggest that some of those aspects may be the result of constraints imposed by structural traits. To show this possibility, we can ignore the decrease in ANN size observed for ISI-16 ANNs and focus upon the issue of why larger ANNs tended to perform better under longer ISIs while smaller ANNs tended to perform better under shorter ISIs. Although an adequate resolution of this issue is well beyond the scope of the present work, a general conjecture is possible.

### **7.1.2.1 The Updating Procedure and a Conjecture**

As a starting point, we must consider the possibility that the NPE-updating procedure (i.e., the procedure used to recompute and change activations and connection weights at each ts) may have played a crucial role in the present results. Traditionally, the updating procedure has been regarded as theoretically unimportant (a mere procedural device), under the assumption that robust models should be "insensitive" to it (e.g., Amit, 1989, p. 80). However, authors have recently shown the importance of giving the updating procedure a more theoretical status (e.g., Bersini & Detours, 1994). In the present work, the updating procedure may have interacted with

structural phenotypes and caused the kind of relationships observed between ANN size and the ISI. Let me suggest how.

The updating procedure used in the present simulations was asynchronous with random sequence. That is, for each ANN, a random sequence of *all* the NPEs was generated at each  $t_s$ , and all NPEs were updated (i.e., their activations and weights changed) in the order specified by the sequence. Consequently, at a given  $t_s$ , an NPE could be updated either *before* or *after* all or some of its presynaptic inputs. If at  $t$  an NPE were updated before *any* of its presynaptic inputs, then its state would be determined by their states at  $t-1$  (i.e., by past presynaptic-input signals). In contrast, if the NPE were updated after *all* of its inputs at  $t$ , then its state would be determined by their states at  $t$  (i.e., by present presynaptic-input signals). Finally, if the NPE were updated before *some* of its inputs at  $t$ , then its state would be affected by the states of the unupdated inputs at  $t-1$  and by the states of the updated inputs at  $t$  (i.e., by a mixture of past and present input signals). Hence, the effect of changes in the states of an NPE's presynaptic inputs on the NPE's state at a given  $t_s$  may be *delayed* or *immediate*, depending upon whether the NPE in question was updated before or after its presynaptic inputs, respectively.

To appreciate the importance of these relations, we must take the NC parameters into account. In the present simulations, the NPEs constituting an ANN could differ in two *decremental* parameters (*viz.*, activation decay and extinction rate) and two *incremental* parameters (*viz.*, temporal summation and acquisition rate). Decremental parameters produce decreases, whereas incremental parameters produce increases in NPE activations over time.

If an NPE is updated after all of its presynaptic inputs at a given  $t_s$ , then the effect of their parameters on the NPE's state will be immediate. Conversely, if an NPE is updated before any of its presynaptic inputs at a given  $t_s$ , then the effect of their parameters on the NPE's state will be delayed<sup>1</sup>. Therefore, in general, if an NPE's presynaptic inputs have lower decremental than incremental parameters, then its activation at any given  $t_s$  will be necessarily lower if it is updated before than if it is updated after all of its presynaptic inputs. Conversely, if an NPE's presynaptic inputs have higher decremental than incremental parameters, then the NPE's activation at a given  $t_s$  will be necessarily higher if it is updated before than if it is updated after all of its presynaptic inputs. These differences will increase as the differences between decremental and incremental parameters increase.

The above possibilities have consequences for the magnitude of the discrepancy signals within and between trials. Indeed, for these signals to be positive (and hence, produce an increase in connection weights), the activations of *cal* or *vta* NPEs at a given  $t_s$   $t$  must be larger than their activations at  $t-1$  (see Appendix A for discrepancy equations). This situation will occur whenever either of two conditions is satisfied. In the first condition, the presynaptic decremental parameters are lower than the presynaptic incremental parameters, and the NPEs are updated before their presynaptic inputs at  $t-1$  and after their presynaptic inputs at  $t$ . In the second condition, the presynaptic decremental parameters are higher than the presynaptic incremental parameters, and the NPEs are updated after their presynaptic inputs at  $t-1$  and before

---

<sup>1</sup> The emphasis is on the parameters of the NPE's *presynaptic inputs*, rather than on the NPE's parameters, because the NPE's state is immediately affected by its own parameters at *every*  $t_s$ , regardless of its position within an updating sequence. In the context of asynchronous updating, then, the update position of an NPE within an updating sequence becomes relevant only in relation to changes in the NPE's state as a function of changes in the states of its presynaptic inputs, which in turn depend upon the NC parameters of these inputs.

their presynaptic inputs at  $t$ . Discrepancy signals will be zero or negative none of these two conditions are satisfied<sup>2</sup>.

However, in the present simulations, the before-after conditions throughout training, as well as differences between decremental and incremental parameters *at the beginning of evolution*, were randomly determined. Hence, a *statistical balance* may have been required between the number of ts per trial in which NPEs were updated before and after their presynaptic inputs, to achieve a sufficient number of ts per trial at which discrepancy signals were positive.

I hypothesize that such a balance was determined, at least partially, by a relationship between the total number of NPEs in an ANN, the number of presynaptic inputs per NPE, and the ISI. Again, assuming full feedforward connectivity, as the ratio of presynaptic inputs per NPE to the total number of NPEs decreases (which can be achieved by increasing the number of layers while keeping the number of NPEs per layer constant, which in turn has the effect of increasing the total number of NPEs while keeping the number of presynaptic inputs per NPE constant), the number of ts per trial in which NPEs are updated before or after their presynaptic inputs also decreases. However, this situation can be compensated by increasing the number of trials in which NPEs are updated before or after their presynaptic inputs at least 1 ts, which can be achieved by increasing the ISI. On this basis, I hypothesize that ISI functions arose in the present simulations from a disruption of that balance, while optimal-ISI noninvariances arose from different ways of achieving the balance.

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<sup>2</sup> Clearly, under this interpretation, parametric traits seem to be as critical as architectural ones. The relationships between NPE parameters and ISIs provide us with a basis for considering neuronal individual properties as another aspect of the notion of preparedness. In this sense, then, being prepared to learn under a particular ISI may signify possessing a circuit whose neurons have properties such that their collective functioning permits learning under that ISI.

In the case of ISI functions, the balance was disrupted when large ANNs were trained under short ISIs, and when small ANNs were trained under long ISIs. In the first situation, the number of updating sequences per trial may have been too small to allow for a balanced updating of NPEs before and after their presynaptic inputs. In the second situation, the number of updating sequences per trial may have been too large to allow for a balanced updating of NPEs before and after their presynaptic inputs. In the case of optimal-ISI noninvariances, the balance was achieved when large ANNs were trained under long ISIs and small ANNs were trained under short ISIs. Let me illustrate this conjecture through a simple simulation.

#### 7.1.2.2 An Example

A simulation was run in which two handcrafted ANNs, N1 and N2 (see Figure 7.1), were given 200 reinforced trials, according to a forward-delay procedure in which the ISI was either 2 or 12 ts and the CS involved the activation of one input element with a value of 1.0. Both ANNs had three input elements, and two output NPEs. Also, all NPEs in both ANNs had the same NC parameters (i.e., a spontaneous-activation, temporal-summation, activation-decay, and extinction rates of 0.1, and an acquisition rate of 0.5). Based upon the results obtained in Experiments 1 and 2, the main difference between the ANNs was in the number of *sa* hidden layers. N1 had one while N2 had three *sa* layers. Both ANNs had one *ma* layer. All hidden layers in both ANNs consisted of three NPEs. Also, NPEs in each layer were connected to one discrepancy element. Thus, N1 had only one while N2 had three *ca*/ NPEs. Finally, ANNs had full feed-forward connectivity, for which the number of presynaptic connections per NPE was three for all NPEs in both ANNs.

Performance was measured in terms of CR proportion across the last 50 trials of training. Consistent with the results of the test simulations of Experiment 1, and as shown in the top panel of Figure 7.2, N1 performed substantially better under an ISI of 2 ts than under an ISI of 12 ts, whereas the N2 performed substantially better under an ISI of 12 ts than under an ISI of 2 ts. The bottom panel shows the values of  $U$  for each ANN-ISI combination.  $U$  was the mean number of ts per trial that any given NPE was updated at any given position within updating sequences. The best performance corresponded to the ANN-ISI combination for which the value of  $U$  was closest to 1.0 (i.e., N2-12). The worst performance corresponded to the ANN-ISI combination for which  $U$  was above 1.0 (i.e., N1-12).

These results suggest that, in the N2-12 combination, a balance was achieved between the number of ts NPEs were updated before and after their presynaptic inputs, by updating NPEs an average of 1 ts per trial at any position within updating sequences. NPEs in N1, when trained under the long ISI, tended to be updated more than 1 ts per trial at certain positions. NPEs in N2, when trained under the short ISI, tended not to be updated at certain positions.

### 7.1.2.2 A More Precise Formulation: Latin Squares

The above account can be formulated more precisely by examining the *ideal* case, in which trial structure is such that *all* NPEs are updated *one and only one* ts at *each position*, within each trial. Generalizing to all NPEs in an ANN, we can characterize this case in terms of a *Latin square* whose rows represent time-steps (or updating sequences) within a trial, and whose columns represent NPEs in an ANN. A Latin square is a matrix in which each element appears one and only one time in each row

and each column. This concept can be useful as a tool for describing more formally the nature of the balance in the numbers of times NPEs are updated before and after their presynaptic inputs. Although a complete formalization is beyond the scope of the present work, some speculative, yet intriguing ideas are worth mentioning here.

In the ideal case, the number of NPEs in an ANN is identical to the ISI, and the updating procedure is such that trials constitute *only* Latin squares. In this case, each NPE will be updated at each position one and only one ts per trial. In this arrangement, the probabilities that an NPE will be updated before and after its presynaptic inputs become as similar as possible. Of course, this ideal case does not apply to any of the ANNs generated in the present work, for their sizes tended to be much larger than the largest ISI. Therefore, it is clear that other factors must be taken into account, if the conjecture is to be applicable to those ANNs.

For example, the randomness of the updating procedure, combined with the developmental constraints on minimum ANN size, make the likelihood of generating Latin-square trials (i.e., trials on which update sequences determine Latin squares) vanishingly small. Given these two factors, then, Latin *rectangles* may have represented a better arrangement. And, if we take connectivity into account, then Latin *cubes*, or even Latin *hypercubes* may have represented better arrangements. Finally, the fact that different kinds of NPEs could have different kinds of NC parameters is also relevant, for, as mentioned before, the importance of a balance between the number of ts NPEs are updated before and after their presynaptic inputs will depend upon differences in their NC parameters.

It is clear that the above conjecture needs considerable further elaboration (let alone experimental verification). However, the notion of a Latin square seems intuitively



promising as a conceptual basis for a unified formal description of the relationships between Pavlovian conditioning and nervous systems at the ontogenetic and the phylogenetic level. Both levels can be viewed as searches in a *Latin-square space*. A Latin-square space is the class of all possible Latin squares that can be defined given certain genotypic, phenotypic, and environmental constraints.

Ontogenetically, we can view Pavlovian conditioning as involving, *at least partially*, a search in a Latin-square space determined by an ANN's size and the ISI imposed by the environment of this ANN. Ideally, the more the relationship between the ANN's size and the ISI *approaches* a Latin square, the larger the number of trials constituting Latin squares, and the higher the ANN's performance. Optimal ISIs, as observed in the present simulations, can thus be seen as environmental conditions that lead to an increase in the likelihood of Latin-square trials, for a particular ANN. On this basis, we can view ISI functions as arising from a decrease in the number of Latin-square trials due to an increase or a decrease in the ISI from the optimal ISI, and optimal-ISI noninvariances as arising from balances achieved through Latin squares of different orders, as determined by different optimal ISIs *and* different ANN sizes.

Phylogenetically, we can view the evolution of Pavlovian conditioning as a search in a Latin-square space determined by the number of different ANNs that can be developed from a genotype (i.e., the structural-phenotype space) and the different ISIs imposed by the environment. That is, the phylogenetic Latin-square space is the class of all ontogenetic Latin-square spaces that are determined by genotypic and environmental constraints. Indeed, given the selection pressures imposed by the ISI, the evolution of Pavlovian conditioning, as simulated in the present study, can be seen, *at least in part*, as a search for genotypes that developed into ANNs such that the

relationship between their sizes and the ISI *approached* a Latin square *of a particular order*.

A conceptualization of the ontogeny and phylogeny of Pavlovian conditioning in terms of Latin-square spaces seems advantageous in at least two ways. First, *both* the environment *and* the organism are given *equally* critical (in fact, *defining*) roles in *both* the ontogenetic and the phylogenetic level. This property facilitates contact with biological considerations about the ontogeny and phylogeny of behavior. Second, Latin squares represent a relatively well characterized mathematical object (Dénes & Keedwell, 1974), which facilitates further formalizations. Such formalizations may even include relations to probability theory, which may be necessary in the context of asynchronous-random updating procedures. These relations could involve, for example, the identification (or definition) of a rule for computing the probability that a trial will be a Latin square, given the ISI and ANN architecture.

## 7.2 Effects of the CS

The CS was an independent variable in the evolution simulations of Experiments 1, 3, and 4, and in the test simulations of Experiment 1. In the evolution simulations of Experiment 1, ANNs were selected for increased responding to one reinforced CS, which involved the activation of all input elements of one kind (either  $s_1$  or  $s_2$ ). In the evolution simulations Experiments 3 and 4, ANNs were selected for increased responding to the reinforced CS in an orthogonal and nonorthogonal differential-reinforcement arrangement, respectively.

In Experiment 1, ANNs at the end of evolution tended to have more input elements of the kind activated by the CS present when ancestral selection occurred. The

functional significance of these architectural differences was observed in the test simulations, where CS1 and CS2 were shown to be nonequipotential for ANNs in some lineages. More specifically, ANNs in the last generation of some lineages showed substantially more responding to the same kind of CS that was experienced by their ancestors. This result is consistent the phenomenon of CS nonequipotentiality, and hence with the notion of biological constraints on learning.

Again, using the language of this notion, we can say that ANNs possessing more  $s_1$  than  $s_2$  input elements (and whose ancestors were selected for high responding to CS1), were prepared to learn to respond to CS1 and contraprepared to learn to respond to CS2. Conversely, ANNs possessing more  $s_2$  than  $s_1$  input elements (and whose ancestors were selected for high responding to CS2), were *prepared* to learn to respond to CS2 and contraprepared to learn to respond to CS1.

In Section 7.1.2, I interpreted ISI functions and optimal-ISI noninvariance in terms of the notion of preparedness, on the basis of considering the relationships between architectural traits, parametric traits, and ISIs as aspects of that notion. In the present results, the relationships between CS and input-element architectural traits provide us with a basis for considering such traits as another aspect of the notion of preparedness. According to this interpretation, then, being prepared to learn to respond to a particular CS may signify possessing an input-element circuitry whose architecture is such that it permits such learning. These considerations can be neuroanatomically interpreted in terms of the kinds of circuits found in structures such as the peripheral nervous system, the thalamus, or primary sensory cortex. In any case, the above interpretations indicate that the notion of preparedness is a complex one, possessing a number of interrelated aspects.

## 7.3 Effects of Reinforcement Contingencies

A third kind of effect observed across experiments relates to the reinforcement contingency, that is, to the *statistical* (as opposed to temporal) relations between CSs and the US. In Experiment 1, ANNs were presented with *reinforced* instances of only one kind of CS, and were selected for increased responding to that CS. In Experiment 2, ANNs were presented with reinforced instances of two different kinds of CSs, and were selected for increased responding to *each* CS. These two experiments were similar in that *all* trials of all CSs kinds were *reinforced*. In contrast, Experiments 3 and 4 involved presenting ANNs with instances of two CS classes. However, instances of one class were *nonreinforced*, and ANN were selected for increased responding to the reinforced CS.

Three general effects of reinforcement contingency were observed. First, ANNs whose ancestors were selected for increased responding to two-reinforced CSs tended to have an equal number of input elements of both kinds more frequently than ANNs whose ancestors were selected for increased responding to one reinforced CS. Second, ANNs whose ancestors were selected for increased responding to two reinforced CSs tended to be smaller than ANNs whose ancestors were selected for increased responding to one reinforced CS, with the exception of ISI-8 ANNs, which tended to be substantially larger (in terms of number layers and number of NPEs per layer) in Experiment 2 than in Experiment 1. And third, differences in ANN size across Experiments 1 and 2 were mostly due to increases in the number of *sa* layers and/or NPEs. ANNs in these experiments tended to have only one *ma* layer, with the exception of 16-CS1 ANNs in Experiment 1, which tended to have two *ma* layers. ANNs in Experiment 4 (selection for nonorthogonal discrimination) also tended to have

only one *ma* layer. However, in contrast to ANN in other experiments, ANNs in Experiment 3 (selection for orthogonal discrimination) tended to have only one *sa* and between five and seven *ma* layers. This third effect seems to be the most puzzling one, so let me propose some general conjectures about it.

The increase in the number of *sa* layers across ISIs can be accounted for in terms of an interaction between how reinforcing signals from *cal* and *vta* NPEs were determined in the present model, and architectural constraints on how *cal* and *vta* NPEs were served by *sa* and *ma* NPEs, respectively. Indeed, the sources of reinforcing signals (i.e., signals that increased connection weights) in ANNs were the activations of *cal* and *vta* NPEs across ts. Activations of *cal* NPEs served as reinforcing signals for connection weights in *sa* and *cal* NPEs, whereas activations of *vta* NPEs served as reinforcing signals for connection weights in *ma*, *vta*, and output NPEs. Activations of *vta* NPEs also served to *amplify cal* signals. Hence, if activations of *cal* and *vta* NPEs were comparable *before the presence of the US* (and an examination of these activations verified this assumption), then *effective cal* signals were necessarily larger than *vta* signals (see Appendix A for equations). Consequently, an increase in the number of *cal* NPEs allowed for an increase in the magnitude of the overall reinforcing signal within trials before the US. And an increase in the number of *cal* NPEs was possible only through an increase in the number of *sa* layers, because each *sa* layer served its own set of *cal* NPEs. This strategy may have been selected for because fitness was determined by performance in the ts immediately before the US.

The opposite process *at the moment of reinforcement* may have caused the increase in the number of *ma* layers in Experiment 3. Indeed, orthogonal differential reinforcement may have produced a situation in which an increase in the overall

magnitude of reinforcing signals was accomplished at the moment of reinforcement by increasing the number of *vta* NPEs, rather than the number of *cal* NPEs. Let me show how this is possible.

Nonreinforced trials caused a decrease in connection weights due to extinction, which in turn caused a decrease in NPE activations. When a nonreinforced trial was followed by a reinforced trial, connection weights and activations continued to decrease until the last ts, the moment of reinforcement. This decrease produced a situation in which *cal* activations continued to decrease from the penultimate to the last ts, whereas *vta* activations suddenly increased due to the presence of the US. This situation, in turn, caused the *vta* reinforcing signal to be larger than the *cal* reinforcing signal, which favored an increase in *vta* NPEs and hence *ma* layers.

Of course, the difference between the *cal* and *vta* reinforcing signals at the moment of reinforcement must have been substantially higher than that produced in nondifferential reinforcement procedures. Otherwise, an increase in the number of *vta* NPEs would not have been justified. However, that must have been the case in the orthogonal arrangement, precisely because of the presence of nonreinforced trials, which caused larger *vta* discrepancies. In the nondifferential arrangements, there was less extinction and hence smaller discrepancies at the moment of reinforcement. Finally, the increase in the magnitude of the overall reinforcing signal due to *vta* activity at the moment of reinforcement must have been substantially larger than the increase in the magnitude of the overall reinforcing signal due to *cal* activity before reinforcement. Otherwise, an increase in the number of *cal* NPEs would have been as justified as an increase in the number of *vta* NPEs. But that must have been the case in the orthogonal arrangement, again due to the presence of nonreinforced trials.

This conjecture, of course, does not explain why selection for nonorthogonal discrimination under ISI 8 generated ANNs having only one *sa* and one *ma* layer. Clearly, the fact that reinforced and nonreinforced trials involved CSs that are not easily discriminable was the critical factor here. However, it is not clear how this factor contributed to the differences between reinforcing signals due to *cal* activity and reinforcing signals due to *vla* NPEs.

## 7.4 Concluding Remarks

The main contribution of the present work is that it initiates a research program on computational, AL approaches to evolutionary accounts of the kinds of behavioral phenomena observed in animal learning and conditioning. However, one could ask for a justification for the present work, in light of the results discussed above. Why are these results important? What do they mean? What do they tell us about real nervous systems? How can they be useful for understanding the relationships between learning and evolution? What do they tell us about Pavlovian conditioning? These and other questions point at philosophical issues that are extremely difficult and deserve special treatment, beyond the confines of a preliminary and tentative work like the present one. However, one could mention a few general issues that must be resolved before attempting to answer any of these questions.

For example, in interpreting the possible functional significance of interrelations among structural traits, I have adopted an adaptationist approach, in which I implicitly assumed that *all* interrelations were adaptations (i.e., conferred a fitness advantage throughout phylogeny). However, another possibility is that some of those interrelations *were not selected against*, and hence, were adaptively neutral. In this

way, the issue of adaptationism bears directly on the significance of the present results for an understanding of the relationships between learning and evolution.

Another issue is whether architectural differences observed between ANNs across different lineages correspond to neuroanatomical differences underlying response systems across different species or within the same species. In this respect, it is important to clarify that experimental subjects across different Pavlovian-conditioning preparations are typically members of different species. In this case, the present results would imply a correspondence between structural differences and neuroanatomical differences across species. However, this situation is not a logical necessity. For example, it is possible to use *Rattus norvegicus* for very different preparations, such as CER and taste-aversion, which involve very different optimal ISIs. In this case, the present results would imply that structural differences correspond to neuroanatomical differences across *response systems, within the same species*. In this sense, different kinds of ANNs can be seen as representing structurally different circuits within a kind of nervous system in a species, rather than structurally different nervous systems across different species. Under this interpretation, then, differences among ANNs can be understood in terms of *mosaic* evolution, a process through which different parts of nervous systems may evolve at different rates. In this sense, the evolution of nervous systems may have involved a *distributed* process of change in different circuits at different points throughout evolutionary time, rather than a unitary process of phylogenetic change in *all* circuits at all points throughout evolutionary time.

A third issue relates to the meaning of the term *adaptive*. I already mentioned one aspect of this issue in Chapter 1, regarding usages of the term in current Pavlovian conditioning research. There, I proposed that we must give a strictly evolutionary



definition of that term, based upon the concept of fitness. However, there is another aspect to the issue, which relates to the application of the term to the present simulations. It is clear that such an application is consistent with the above prescription if we are referring to ANNs that were selected for increased responding under a particular ISI, CS, and reinforcement arrangement. However, what about responding in the test simulations? In what sense are ISI functions, CS nonequipotentiality, blocking, and generalization adaptive? Clearly, these phenomena cannot be referred to as adaptive, because they did not confer any fitness advantage to ancestral ANNs, nor did they confer any adaptive advantage to present ANNs.

This issue can be clarified in terms of the distinction between traits that are adaptive and traits that are adaptations (Sober, 1993), introduced in Chapter 1. Indeed, increased responding under conditions similar to those under which ancestral selection took place represents an adaptation, for its current presence is explainable in terms of an ancestral selection process. If the current environmental conditions were such that that responding conferred a fitness advantage to individual ANNs by allowing them to win selection tournaments, then such a trait would be adaptive as well. However, high responding to environmental conditions different from the ancestral ones does not represent an adaptation, because its current existence is not explainable (at least directly) in terms of selection *for* such responding.

These considerations raise the issue of what kinds of Pavlovian conditioning phenomena we should refer to as being adaptations. The present simulations suggest that some basic Pavlovian conditioning phenomena may not be adaptations, even if they are currently adaptive. That is, the fact that present organisms show blocking, orthogonal discrimination, or generalization does not necessarily mean that these

phenomena conferred a fitness advantage in the phylogenetic history of those organisms.

A fourth issue relates to *structuralistic explanations*. It is clear that many of the kinds of explanations offered in the present work are structuralistic in that they appeal to structural phenotypes as a source of differences in ANN performance. For example, ISI functions and optimal-ISI noninvariances were explained by arguing that ANN size and connectivity play a critical role. This form of explanation gets dangerously close to the kind of structuralism exemplified by mainstream cognitive psychology. The present structuralist interpretation, however, can be distinguished from cognitive-psychology structuralism in two ways. First, the specific nature of the structures is unclear in the case of cognitive-psychology structuralism. In the present work, the structure can be analyzed, teased apart, operationally defined, and interpreted neuroanatomically, which in turn allows for existential verification. Second, in the present work, the *origin* of the structure is clearly interpretable from a strictly neo-Darwinian evolutionary perspective. Thus, in the present approach, structure is operationally definable, neuroanatomically interpretable, and evolutionarily explainable.

A fifth issue arises from the fact that the present simulations are consistent with the notion of biological constraints. Yet, such a consistency does not preclude the possibility of pursuing general-process approaches to learning. The present simulations suggest that such approaches can still be pursued, but along lines different from those imposed by the optimal-ISI invariance and CS equipotentiality premises. Indeed, we can still assume that the same kinds of phylogenetic and ontogenetic processes underlie learning in all organisms past, present, and future, artificial and natural, in the laboratory as well as in nature. This assumption does not imply generality in the sense

specified by the optimal-ISI invariance and CS equipotentiality premises. Rather, the assumption implies that generality is in the underlying *mechanisms*, not in the specific kinds of environmental conditions that may or may not affect an organism's behavior in ordered ways (i.e., a particular ISI or CS). In this sense, a notion of biological constraints on learning is consistent with a *revised* general-process approach that does not depend upon those premises. This possibility is exemplified by the present work, in which phenotypic diversity was observed among ANNs, despite the fact their performances were determined by the *same* NC-GA model.

A sixth issue relates to the reliability of the present results. We must ask to what extent repeating a simulation would produce similar results. To address this issue, the 4-CS1 condition of Experiment 1 was repeated, with the same chromosome founder population. The results obtained in this new simulation were quite similar to those obtained in the original one. Indeed, the new simulation ended after 17 generations, while the original simulation ended after 20. The mean chromosomic overlaps toward the end of evolution in the new simulation tended to be around .4, while overlaps for the original simulation were around .6. Mean fitness at the end of both simulations tended to be close to 1.0. Architectural traits of representative ANNs in the last generation of both simulations were quite similar. In effect, ANNs in both simulations tended to have substantially more  $s_1$  than  $s_2$  input elements, 2 *sa* layers consisting of approximately 6 to 10 NPEs each, 7 *cal* NPEs, 1 *ma* layer consisting of approximately 6 to 10 NPEs, between 5 and 6 *via* NPEs, and between 10 and 15 output NPEs. Also, representative ANNs in both simulations tended to have similar connectivity (between 3 and 6 connections per NPE). In sum, representative ANNs in the last generation of the new simulation tended to be comparable to the 4-CS1 ANN shown in Figure 3.16.

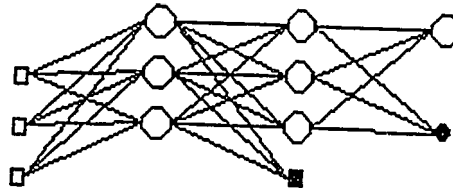
Regarding parametric traits, the results were equally consistent from one simulation to the other. Indeed, a *t*-test showed no significant differences between the mean values of the NC parameters obtained in the two simulations [ $t(19) = 0.43, p > 0.1$ ].

Clearly, one replication is insufficient to arrive at definite conclusions. However, these results, as well as casual observations made during the course of the investigation, suggest that the effects reported throughout the present work are reliable. It is important to remember that both simulations were run using the same chromosome founder population. It is unclear whether or not a systematic replication using a different founder population would produce similar results. This issue clearly requires further study.

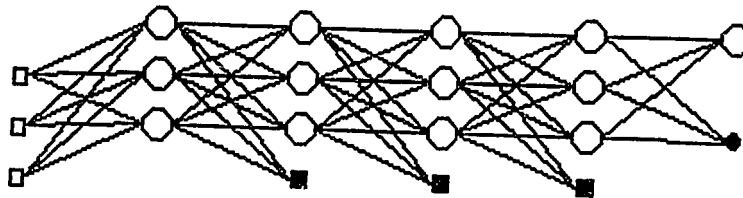
Finally, one more consideration on biological plausibility. The effects observed in the present simulations on structural traits are not clearly interpretable in terms of real nervous systems. Some of those effects may even be neurobiologically implausible (e.g., NC parameters close to 0.0 or close to 1.0). However, the *existence* of such effects seems more important than their specific directions, for the latter are very likely to depend upon the initial conditions as determined by the founder population. That is, simulations starting with a different founder population could produce CS effects different from those described above. This possibility (not verified in the present work) exemplifies the critical difficulty underlying the idea of "rewinding the tape of evolution" (e.g., Gould, 1989), which implies that actual biological entities (present as well as ancestral) are largely the result of very improbable sequences of historical accidents, shaped by natural selection and self-organization. Thus, actual phylogenies represent only one among many possible, although *equally likely* evolutionary paths.

An intriguing implication of this idea is that consistency with real nervous systems is eliminated as a criterion for evaluating models of the evolution of nervous systems. Moreover, given the evolutionary improbability of real nervous systems, as implied by that idea, models of the evolution of nervous systems become more useful if they generate structures that *depart* from real nervous systems. Such a feature would be consistent with the notion of AL as a discipline advocated to the study of *possible* biological entities. Evolutionary models that generated only artificial nervous systems that are similar to real nervous systems would be useless for the study of lawful regularities underlying the evolution of nervous systems. In contrast, evolutionary models that generate artificial nervous systems that are very different from real ones seem most useful to determine the generality of a neo-Darwinian account of the evolution of actual nervous systems. It is here where some of the virtues of the present approach reside.

On the basis of the above argument, then, it is likely that the kinds of structural differences observed in the present work do not simulate any real nervous system. However, this inconsistency does not necessarily indicate that the present model is inadequate. Perhaps what is inadequate is the initial conditions from which the simulations started. But then again, there exists no empirical evidence about the real initial conditions. Rather, a more promising approach would be to ask about the meaning for the *evolution* of nervous systems *in general*, artificial and real.



N1



N2

Figure 7.1: **Two handcrafted ANNs.** The main difference between N1 and N2 was the number of *sa* and *cal* NPEs. N1 had three *sa* and one *cal* NPEs, whereas N2 had nine *sa* and three *cal* NPEs.

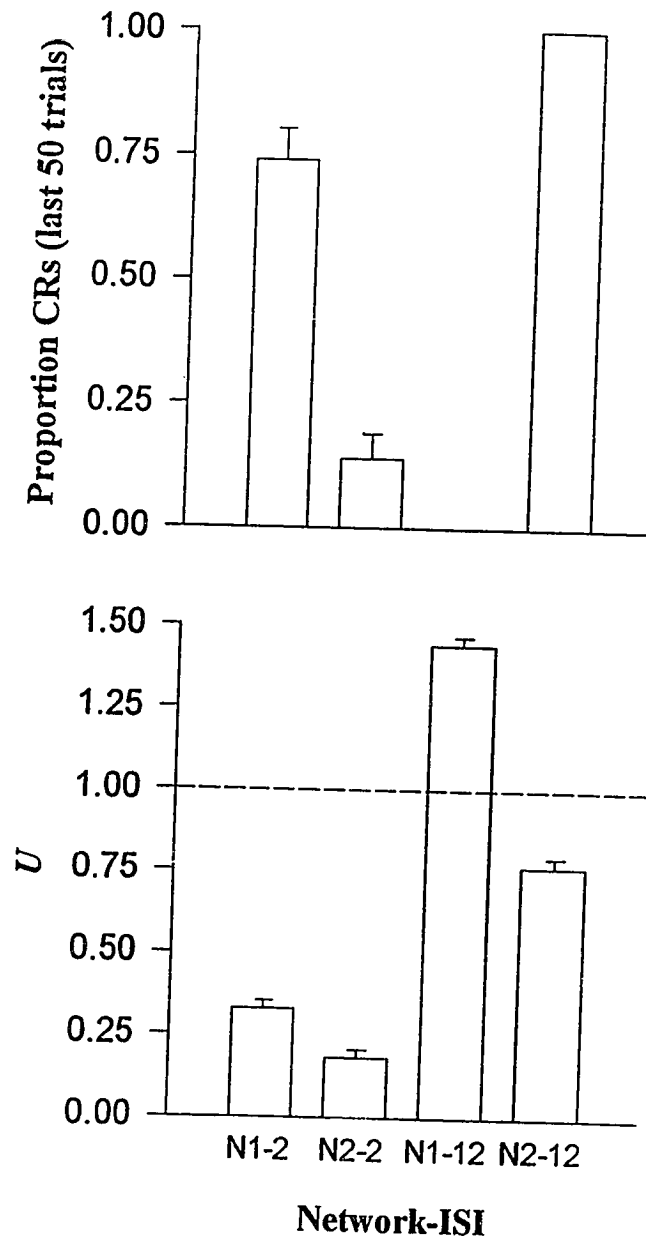


Figure 7.2: CR proportion and  $U$  for a small and a large ANN, under a short and a long ISI.  $U$  represents the mean number of times per trial any given NPE was updated at any given position within a time-step.

# APPENDIX A

## DEVELOPMENT

### A.1 Encoding Structure and Decoding Functions

The encoding structure was a set of variables whose values determined the course of a neural development program. Such values were encoded in bit strings of fixed length  $l$ . Each bit string represented a chromosome. Thus,  $\mathcal{C} = \{0,1\}^l$  defined the chromosome space. Each chromosome  $C_i \in \mathcal{C}$  was divided into  $f$  building segment. Each segment  $F_j \subset C_i$  encoded a value of a neural development variable. Segments could encode natural or real numbers. Thus, two fragment decoding functions were used. A binary code interpretation of the chromosomes was used for decoding purposes. The segment decoding function for natural-valued variables was the mapping  $g : \{0,1\}^{l_i} \rightarrow \mathbb{N}$ , where  $l_i$  was the length of segment  $F_j$  and

$$g^j(a_{i,1}, \dots, a_{i,l_i}) = \sum_{j=1}^{l_i} a_{i,j} 2^j, \quad (1)$$

where  $a \in \{0,1\}$  was an allele, and  $l$  was either 3 or 4. The segment decoding function for real-valued variables was  $h : \{0,1\}^{l_i} \rightarrow [0.0, 1.0]$ , where

$$h^j(a_{i,1}, \dots, a_{i,l_i}) = u_i + [(b - a) / 2^{l_i}] \sum_{j=1}^{l_i} a_{i,j} 2^j. \quad (2)$$

where  $l = 5$ .



## A.2 ND Program

### A.2.1 Proliferation Routine

Chromosome segments  $F_1$  to  $F_8$  encoded the following architectural traits: maximum number of  $s_1$  input elements ( $F_1$ ), maximum number of  $s_2$  input elements ( $F_2$ ), number of proliferative units for producing *sa* NPEs ( $F_3$ ), maximum number of *cal* NPEs per *sa* layer ( $F_4$ ), number of proliferative units for producing *ma* NPEs ( $F_5$ ), maximum number of *vta* NPEs per *ma* layer ( $F_6$ ), maximum number of output NPEs ( $F_7$ ), and the probability that a new element was produced at the outset of proliferation ( $F_8$ ). The maximum possible number of  $s_1$ ,  $s_2$ , *cal* NPEs per *sa* layer, or *vta* NPEs per *ma* layer, and *sa* and *ma* layers was 7. Hence, the maximum possible number of hidden layers was 14). The maximum number of output NPEs was 15. Also, a viable ANN could have a minimum of 1  $s_1$  element, 1  $s_2$  element, 1 *sa* NPE, 1 *cal* NPE, 1 *vta* NPE, 1 *ma* NPE, and 1 output NPE. All ANNs had only one US input element.

Proliferation consisted in a sequence of  $t$  proliferation epochs. A proliferation epoch consisted of a sequence of  $b$  proliferation trials at each of which the probability of producing a new element  $p$  remained constant. Thus, proliferation was conceptualized as a series of  $t$  sequences of  $b$  Bernoulli trials, where success was defined as the production of a new element. The total number of new elements produced by a given proliferative unit at a proliferation trial, then, was given by the mean of a member of the binomial-distribution family with parameters  $b$  and  $p$ . Because  $p$  decayed at a rate  $\delta$  from one proliferation epoch to the other, proliferation consisted in a series of  $t$  members of the binomial-distribution family, occurring at each proliferative unit. The initial value of  $p$  was encoded by  $F_8$ . In the present simulations,

$b = 3$ ,  $t = 5$ , and  $\delta = .1$ , for all proliferative units and developing ANNs. Thus, the maximum possible number of *sa* or *ma* NPEs per hidden layer was 15, for which the maximum possible number of *sa* and *ma* NPEs was 210.

### A.2.2 Migration Routine

Migration also consisted in a sequence of  $t$  migration epochs. A migration epoch consisted of a sequence of  $m$  migration trials at each of which the probability of migration of a newly element  $p$  remained constant. Thus, like proliferation, migration was conceptualized as a series of  $t$  sequences of  $m$  Bernoulli trials, where success was defined as the migration of a newly produced element. The total number of elements that migrated from a given proliferative unit at a migration trial, then, was given by the mean of a member of the binomial-distribution family with parameters  $m$  and  $p$ . In the present simulations,  $m = 3$  and  $t = 5$ . Separate values of  $p$  were encoded for migration of *sa* and *ma* NPEs by  $F_9$  and  $F_{10}$ , respectively. The migration of elements from a proliferative unit ceased either at  $t$  or when the elements produced by the unit were depleted, whatever occurred first. If the former occurred first, then the remaining elements in the unit eliminated (i.e., they "died" either during migration or before their opportunity to migrate was available).

### A.2.3 Differentiation Routine

The first step of the differentiation routine was to determine whether an NPE was excitatory or inhibitory, for each *sa* and *ma* NPE. This step consisted in a single epoch in which each *sa* and *ma* NPE represented a time step at which the current NPE had a probability  $p$  of being excitatory, or a probability  $1-p$  of being inhibitory.  $F_{11}$  and  $F_{12}$

encoded for values of  $p$  for *sa* and *ma* NPEs, respectively. All input, *cal*, *vta*, and output NPEs were assumed to be excitatory.

The second step of the differentiation routine was to assign the NC parameters (see Appendix B) to each NPE. Each kind of NPE was assigned a total of five NC parameters: standard deviation of the logistic function (which determined spontaneous-activation levels), temporal summation, activation decay, acquisition rate, and extinction rate. The first parameter was used as encoded during synaptogenesis only (see below), and then was fixed to 0.1 for all NPEs and adult ANNs, before training. Table A.1 below shows the relation of parameters to segments.

**Table A.1:** Relation of chromosome segments to NC parameters per NPE type.

| Segment              | NPEs                 |
|----------------------|----------------------|
| $F_{13}$ to $F_{17}$ | <i>sa</i> excitatory |
| $F_{18}$ to $F_{22}$ | <i>sa</i> inhibitory |
| $F_{23}$ to $F_{27}$ | <i>cal</i>           |
| $F_{28}$ to $F_{32}$ | <i>ma</i> excitatory |
| $F_{33}$ to $F_{37}$ | <i>ma</i> inhibitory |
| $F_{38}$ to $F_{42}$ | <i>vta</i>           |
| $F_{43}$ to $F_{47}$ | Output               |

#### A.2.4 Synaptogenesis Routine

The synaptogenesis routine involved generating a list of all possible pairs of pre- and postsynaptic elements that could be formed between adjacent layers, and connecting the pre- to the postsynaptic element within each pair with a probability  $p$ .

Different values of  $p$  were encoded for different kinds of pre- and postsynaptic elements. Table A.2 on next page shows the relation of segments to connection probabilities per type of pre- and postsynaptic NPEs.

If a synaptogenesis trial was successful (i.e., if a connection was formed), then the initial connection weight for the newly formed connection was given by  $a_i a_j$ , where  $a_i$  was the spontaneous activation level of the presynaptic element and  $a_j$  was the spontaneous activation level of the postsynaptic element. If the presynaptic element was an input element, then the spontaneous activation was given by a random number between 0.0 and 1.0. For NPEs, spontaneous activations were given by the logistic function with an argument of zero (see Appendix B).

**Table A.2:** Relation of chromosome segments to connection probabilities per connection type.

| Segment  | Type of presynaptic element                  | Type of postsynaptic element                  |
|----------|----------------------------------------------|-----------------------------------------------|
| $F_{48}$ | $s_1$                                        | <i>sa</i> excitatory in first <i>sa</i> layer |
| $F_{49}$ | $s_2$                                        | <i>sa</i> excitatory in first <i>sa</i> layer |
| $F_{50}$ | <i>sa</i> excitatory                         | <i>sa</i> excitatory in next <i>sa</i> layer  |
| $F_{51}$ | <i>sa</i> excitatory                         | <i>cal</i>                                    |
| $F_{52}$ | <i>sa</i> excitatory                         | <i>sa</i> inhibitory in same layer            |
| $F_{53}$ | <i>sa</i> inhibitory                         | <i>sa</i> excitatory in same layer            |
| $F_{54}$ | <i>sa</i> excitatory                         | <i>ma</i> excitatory in first <i>ma</i> layer |
| $F_{55}$ | <i>ma</i> excitatory                         | <i>ma</i> excitatory in next layer            |
| $F_{56}$ | <i>ma</i> excitatory                         | <i>vta</i>                                    |
| $F_{57}$ | <i>ma</i> excitatory                         | <i>ma</i> inhibitory in same layer            |
| $F_{58}$ | <i>ma</i> inhibitory                         | <i>ma</i> excitatory in same layer            |
| $F_{59}$ | <i>ma</i> excitatory in last <i>ma</i> layer | Output                                        |

# APPENDIX B

## NC MODEL

### B.1 Activation function

Let  $N$  be the set of neural processing elements (NPEs),  $T$  be the set of points in time, and  $R$  be the  $[0.0, 1.0]$  interval. The activation function was the mapping  $a : N \times T \rightarrow R$ .

The rule for computing  $a$  was defined as follows. Let  $\mathbf{e}_{j,t}$  be the vector of activations of the excitatory presynaptic inputs to NPE  $j$  at  $t$ ,  $\mathbf{i}_{j,t}$  be the vector of activations of the inhibitory inputs,  $\mathbf{w}_{j,t}$  be the vector of weights of the excitatory inputs, and  $\mathbf{w}'_{j,t}$  be the vector of weights of the inhibitory inputs to  $j$  at  $t$  ( $j \in N, t \in T$ ). Assuming that  $\mathbf{e}_{j,t}, \mathbf{i}_{j,t}, \mathbf{w}_{j,t}, \mathbf{w}'_{j,t} \in R^n$ , the amounts of excitation (*exc*) and inhibition (*inh*) produced on  $j$  at  $t$  are defined as

$$exc_{j,t} = \mathbf{e}_{j,t} \cdot \mathbf{w}_{j,t} \quad (1),$$

$$inh_{j,t} = \mathbf{i}_{j,t} \cdot \mathbf{w}'_{j,t} \quad (2).$$

The activation  $a$  of  $j$  at  $t$  is the firing probability, defined as

$$a_{j,t} = \begin{cases} p(epsp_{j,t}) + \tau_j p(epsp_{j,t-1}) [1 - p(epsp_{j,t})] - p(ipsp_{j,t}) & \text{if } exc_{j,t} \geq \theta_{j,t} \text{ \& } exc_{j,t} > inh_{j,t} \\ a_{j,t-1} - \kappa_j a_{j,t-1} (1 - a_{j,t-1}) & \text{if } exc_{j,t} < \theta_{j,t} \text{ \& } exc_{j,t} > inh_{j,t} \\ 0.0 & \text{if } exc_{j,t} \leq inh_{j,t} \end{cases} \quad (3),$$

$p(eps_{j,t}) = L(exc_{j,t})$ ,  $\tau_j$  is the temporal-summation parameter,  $p(ips_{j,t}) = L(inh_{j,t})$ ,  $\theta_{j,t}$  is a random excitation threshold generated according to a Gaussian distribution with parameters  $\mu = 0.2$  and  $\sigma = 0.15$ , and  $\kappa_j$  is the activation-decay parameter. All parameters could take values greater than zero but less than 1.0. The term  $p(eps_{j,t})$  can be interpreted as the probability of occurrence of an excitatory postsynaptic potential (*eps*) on  $j$  at  $t$ . The term  $p(ips_{j,t})$  can be interpreted as the probability of occurrence of an inhibitory postsynaptic potential (*ips*) on  $j$  at  $t$ .  $L$  was the logistic function with parameter  $0.0 < \delta_j < 1.0$ , which determined the value of the function when  $x = 0.0$ . Such a value represented the level of spontaneous activation of an NPE.  $L$  was given by  $1 / (1 + \exp[(-x + .5)/\delta_j])$ .

## B.2 Synaptic-Plasticity Function

Let  $r : E \rightarrow \{0,1\}$  be the function "releases", where "0" signifies "inhibitory transmitter" (e.g., GABA) and "1" signifies "excitatory transmitter" (e.g., glutamate).  $E^+ = \{x \in E \mid r(x) = 1\}$  is the set of excitatory elements,  $E^- = \{x \in E \mid r(x) = 0\}$  is the set of inhibitory elements,  $\mathcal{E}^+ = E^+ \times N$  is the set of excitatory connections, and  $\mathcal{E}^- = E^- \times N$  is the set of inhibitory connections in an ANN.  $w : \mathcal{E}^+ \times T \rightarrow R$  is the learning function for excitatory presynaptic inputs, and  $w' : \mathcal{E}^- \times T \rightarrow R$  is the learning function for inhibitory presynaptic inputs.

The rules for computing functions  $w$  and  $w'$  are defined as follows. The connection weight  $w$  between presynaptic element  $i \in E^+$  and postsynaptic element  $j \in N$  at  $t$  is given by the rule

$$w_{i,j,t} = \begin{cases} w_{i,j,t-1} + \alpha_j a_{j,t} d_t p_{i,t} r_{j,t} & \text{if } d_t > 0 \\ w_{i,j,t-1} - \beta_j w_{i,j,t-1} a_{i,t} a_{j,t} & \text{otherwise} \end{cases} \quad (4),$$

where  $\alpha_j$  and  $\beta_j$  are the acquisition and extinction rates for excitatory connections respectively, ( $0.0 < \alpha_j, \beta_j < 1.0$ ).  $d_t$  is the value of the reinforcing signal.

$$p_{i,t} = (a_{i,t} w_{i,j,t-1}) / exc_{j,t} \text{ and } r_{j,t} = 1 - \sum_{i=1}^m w_{j,t-1}, \text{ where } m \text{ is the number of}$$

presynaptic excitatory NPEs connected to  $j$ . These two last terms introduced a form of synaptic competition among presynaptic inputs for the available weight. If  $j$  was an *sa* NPE,  $d_t = d_s = \phi_t + \psi_t (1 - \phi_t)$ , where  $\phi_t = |avg(\mathbf{h}_t - \mathbf{h}_{t-1})|$  and  $\psi_t = avg(\mathbf{v}_t - \mathbf{v}_{t-1})$ . In the last two equations,  $\mathbf{h}$  is a vector of activations of all the hippocampal NPEs and  $\mathbf{v}$  is a vector of activations of all the VTA NPEs. If  $j$  is a motor-association NPE, then  $d_t = d_m = \psi_t$ . The use of the absolute value in  $\phi$  was based on the assumption that both the onset and offset of hippocampal CA1 NPEs influence sensory learning (i.e., the output of these NPEs is sensitive to sensory stimulus change).

The connection weight  $w'$  between a presynaptic element  $i \in E^-$  and a postsynaptic element  $j \in N$  at  $t$  is given by

$$w'_{i,j,t} = \begin{cases} w'_{ij,t-1} + \alpha'_j a_{j,t} d_t p'_{i,t} r'_{j,t} & \text{if } d_t > 0 \\ w'_{i,j,t-1} - \beta'_j w'_{i,j,t-1} a'_{i,t} a_{j,t} & \text{otherwise} \end{cases} \quad (5),$$

where  $\alpha'_j$  and  $\beta'_j$  are the acquisition and extinction rates for inhibitory connections

$$(0.0 < \alpha'_j, \beta'_j < 1.0), p'_{i,t} = (a_{i,t} w'_{i,j,t-1}) / inh_{j,t}, \text{ and } r'_{j,t} = 1 - \sum_{i=1}^n w'_{j,t-1}, \text{ where } n$$

is the number of presynaptic inhibitory NPEs connected to  $j$ . The term  $d_t$  is defined as above.



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